



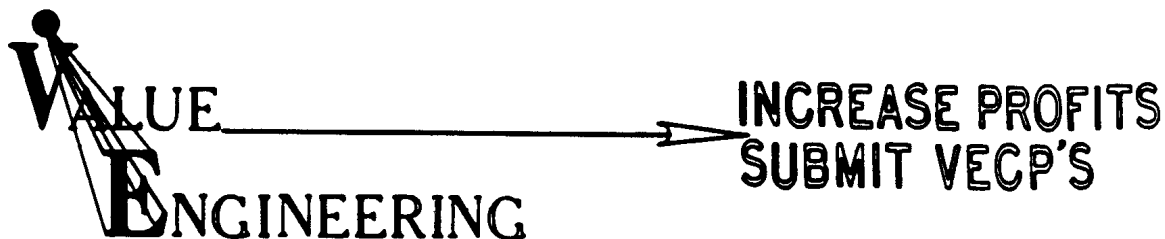
US Army Corps
of Engineers®

Jacksonville District

Tampa Harbor, Maintenance Dredging Upper Channels and Cut-G (PT)

Hillsborough County, Florida

Construction Solicitation and Specifications Volume 2 of 2



See Volume 1 of 2 for issue date

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Designation: THFO ARP (TAMPA HYDRO FIELD OFFICE ARP)

PID: BBGK81

Stamping: N/A

Stability: Monuments of questionable or unknown reliability

Setting: Unspecified setting

Description: Station is in St. Petersburg, Florida at US Coast Guard Station St. Petersburg. Station is more specifically located at the US Army Corps of Engineers, Jacksonville District, Tampa Hydrographic Field Office. Station is operated by the USACE Hydrographic Survey Section. The Antenna is mounted to the NW corner of the boat storage building above the roof.



SECTION 00 31 21 CONTROL MONUMENT DESCRIPTIONS

DESIGNATION - EDD 60

PID - AG5298
TIDAL BM - This is a Tidal Benchmark.
STATE/COUNTY- FL/PINELLAS

MARKER: SURVEY DISK
SETTING: SET IN A RETAINING WALL OR CONCRETE LEDGE
SP_SET: SEAWALL
STAMPING: EDD 60 1935 8.242 M.S.L.
MARK LOGO: USE
MAGNETIC: NO MAGNETIC MATERIAL
STABILITY: MAY HOLD, BUT OF TYPE COMMONLY SUBJECT TO SURFACE MOTION
SATELLITE: THE SITE LOCATION WAS REPORTED AS NOT SUITABLE FOR SATELLITE
OBSERVATIONS - August 22, 2009

STATION DESCRIPTION

AT ST. PETERSBURG, NEAR THE SOUTHWEST CORNER OF THE ALBERT WHITTED
MUNICIPAL AIRPORT, NEAR THE NORTHEAST CORNER OF THE HOUSING AREA OF THE
UNIVERSITY OF SOUTH FLORIDA BAY CAMPUS, IN THE TOP NEAR THE NORTHEAST
CORNER OF CONCRETE SEA WALL, 3.6 FEET EAST OF THE WEST END OF THE SEA
WALL, 1.1 FEET NORTH OF THE NORTH END OF THE NORTH-SOUTH SEA WALL AND
ABOUT 1 FOOT ABOVE THE PARKING AREA.

STATION RECOVERY (2009)

RECOVERED AS DESCRIBED.

SECTION 00 31 21 CONTROL MONUMENT DESCRIPTIONS

DESIGNATION - 872 6668 4 RESET

PID - AG6071

TIDAL BM - This is a Tidal Benchmark

STATE/COUNTY- FL/HILLSBOROUGH

MARKER: BENCHMARK DISK SET IN ABUTMENT/PIER.

STAMPING: BM NO 4 RESET 1963

MARK LOGO: CGS

MAGNETIC: NO MAGNETIC MATERIAL

STABILITY: PROBABLY HOLD POSITION/ELEVATION WELL

SATELLITE: THE SITE LOCATION WAS REPORTED AS NOT SUITABLE FOR SATELLITE
OBSERVATIONS - November 01, 2008

STATION DESCRIPTION

DESCRIBED BY COAST AND GEODETIC SURVEY 1966

AT TAMPA, ON HOOKER POINT AT THE NORTHEAST END OF THE RICHARD E. KNIGHT
PIER, IN THE TOP OF THE CONCRETE FLOOR OF THE PIER, 4.0 FEET SOUTHWEST OF
THE NORTHEAST END, 0.5 FOOT NORTHWEST OF THE SOUTHEAST EDGE AND ABOUT
LEVEL WITH THE FLOOR.

STATION RECOVERY (2008)

RECOVERY NOTE BY FL DEPT OF ENV PRO 2008 (BPJ)

RECOVERED AS DESCRIBED.



SECTION 00 31 21 CONTROL MONUMENT DESCRIPTIONS

Designation: TPA BERTH 227 BASE ARP

PID: BBHH77

Stamping: N/A

Stability: Monuments of questionable or unknown reliability

Setting: Mark is an operational GNSS/GPS station

Description: Station is in Tampa, Florida inside the Tampa Port Authority Facility. The antenna is mounted on top of a 12-foot-high 4"x4" steel post at the Southwest end of Tampa Port Authority berth 227. The post is 8.0' North of the Northwest corner of a metal guard shack, 16.0' North West of the end of an 8' high chain link fence, and 2.5' South of the south end of a rack of electrical boxes mounted on 3 concrete posts. Station is operated by Tampa Port Authority Survey Division.



SECTION 00 33 50 WEATHER, WATER STAGE AND TIDE DATA

1. The following water stage, wind, wave, and weather data are provided for inclusion in the subject project's P&S. The following water level datums are provided for information only and are not for use in conjunction with any contract-related hydrographic surveying.

a) **Water Stages.** Water levels in the project area are mainly affected by wind and mixed tidal fluctuations in the Gulf of America (diurnal and semi-diurnal). The project area is subject to storm surge from tropical systems from June through November. Surge from extratropical storms may minimally affect the area any time of the year.

Elevations of tidal datums in the vicinity of the project area were gathered using the National Oceanic and Atmospheric Administration's (NOAA) National Ocean Service (NOS) tide stations shown in Figure 1 and Tables 1 – 2. These data are referred to Mean Lower-Low Water (MLLW) from the tidal epoch period of 1983 – 2001. Additional tidal data on stations in the vicinity of the subject project area can be found at the following website: <https://tidesandcurrents.noaa.gov/stations.html?type=Datums#Florida>.

The elevations in Table 1 were taken from NOAA's Tide Station 8726607 (Old Port Tampa, FL) and are based on a controlling tide station near St. Petersburg, Florida (NOAA Tide Station 8726520). These datums are based on periodic analyses from January 1999 – December 2000, January 2003 – December 2003, December 2005 – November 2008, April 2010 – March 2011, and January 2012 – December 2016 from a Tampa Bay tide gauge roughly 5.0 miles north of Cut G. These values are applicable to dredging operations in the vicinity of Cut G within Tampa Harbor. Table 2 presents datums taken from NOAA's Tide Station 8726674 (East Bay, FL) and are based on a controlling tide station near St. Petersburg, Florida (NOAA Tide Station 8726520). These elevations are based on a three-month analysis from October 2019 – December 2019 from a tide gauge roughly 1.6 miles east of the Upper Channels. These values are applicable to dredging operations in the vicinity of the Upper Channels within Tampa Harbor.

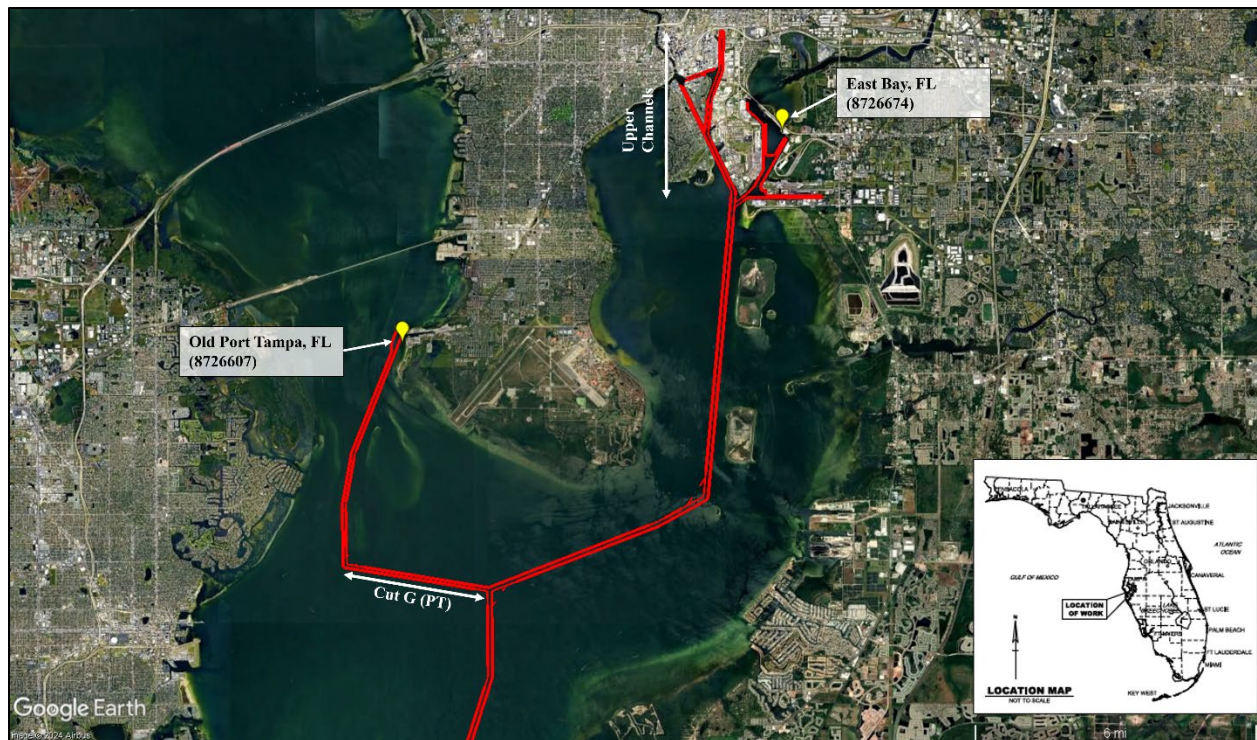


Figure 1. NOAA Tide Stations in the Vicinity of Tampa Harbor, FL

SECTION 00 33 50 WEATHER, WATER STAGE AND TIDE DATA

Table 1. Datums from NOAA Station 8726607 (Old Port Tampa, FL)

Location: 27° 51.5' N, 82° 33.2' W Analysis Period: Jan. 1999 – Dec. 2000, Jan. 2003 – Dec. 2003, Dec. 2005 – Nov. 2008, Apr. 2010 – Mar. 2011, Jan 2012 – Dec. 2016	
Datum	Elevation (ft-MLLW)
Highest Observed (9/27/2024)	9.31
Mean Higher-High Water (MHHW)	2.48
Mean High Water (MHW)	2.16
North American Vertical Datum of 1988 (NAVD88)	1.69
Mean Sea Level (MSL)	1.30
Mean Tide Level (MTL)	1.30
Mean Low Water (MLW)	0.44
Mean Lower-Low Water (MLLW)	0.00
Lowest Observed (9/29/2022)	-4.31

Table 2. Datums from NOAA Station 8726674 (East Bay, FL)

Location: 27° 55.4' N, 82° 25.3' W Analysis Period: Oct. 2019 – Dec. 2019	
Datum	Elevation (ft-MLLW)
Highest Observed (9/27/2024)	9.88
Mean Higher-High Water (MHHW)	2.68
Mean High Water (MHW)	2.34
North American Vertical Datum of 1988 (NAVD88)	1.82
Mean Sea Level (MSL)	1.46
Mean Tide Level (MTL)	1.43
Mean Low Water (MLW)	0.53
Mean Lower-Low Water (MLLW)	0.00
Lowest Observed (9/29/2022)	-5.62

Daily tidal predictions at locations along the coastline of North and South America, including locations in the vicinity of the project area, are included in the publication East Coast of North and South America Tide Tables, United States (U.S.) Department of Commerce, National Oceanic and Atmospheric Administration, National Ocean Service. In addition to daily tidal predictions, this publication provides mean tide levels, diurnal tide ranges, and some astronomical data (sunrise, sunset, moonrise, and moonset). Of note, tidal data provided in this publication is also listed at the following website:
https://tidesandcurrents.noaa.gov/tide_predictions.html.

b) Wind and Wave Data. The publications in this section include wind and wave information in the vicinity of the project area. They are available for review at the websites or agencies indicated.

SECTION 00 33 50 WEATHER, WATER STAGE AND TIDE DATA

i. U.S. Coast Pilot Volume 5, Gulf of America, Puerto Rico, and Virgin Islands, U.S. Department of Commerce, National Oceanic and Atmospheric Administration, National Ocean Service is a publication that supplements the navigational information shown on NOAA's NOS nautical charts and provides miscellaneous meteorological data. This publication is available through NOAA or at the following website: <https://nauticalcharts.noaa.gov/publications/coast-pilot/index.html>.

ii. U.S. Army Corps of Engineers' Coastal & Hydraulics Laboratory, Wave Information Study (WIS) Database is a wave hindcast database for various stations located along the U.S. Atlantic and Gulf of America coastlines. WIS Station 73272 is in 5.0 meters-MSL (16.4 ft-MSL) of water at 27.50° N, 82.80° W. This station is located approximately 26 miles southwest of the Cut G and Gadsden Point Cut within Tampa Harbor. Available data includes a hindcast time series of wave height, wave spread, wave period, wave and wind direction, and wind speed at 1-hour intervals for the 44-year period of January 1980 – January 2024. The following website provides links to WIS summary tables and plots, as well as other sources of wave data: <https://wisportal.erdc.dren.mil/>.

iii. National Data Buoy Center (NDBC) Website provides a wide range of real-time and archived meteorological and oceanographic data collected at offshore and nearshore stations worldwide. Data provided on this website include wind speed, wind gusts, atmospheric pressure, air temperature, sea temperature, wave height, and wave period. Gauge readings are updated hourly, and archived data is available for most stations. The website address for data relevant to Florida projects is as follows: <https://www.ndbc.noaa.gov/>.

c) Weather Conditions. The project area is subject to tropical systems June through November and to windy and rainy weather any time of the year. The climate of the area is essentially tropical, as temperatures below freezing are seldom. During the wet season (May through October), rainfall events are usually short and sporadic and closely associated with convective activity. Winter months generally constitute the dry season, where rainfall is typically associated with mid-latitude, or extratropical, systems and distributed in a more spatially uniform pattern. Daily rainfall events in the dry season are occasionally heavy when mid-latitude systems impact southwest Florida. Notably, hazardous thunderstorms can occur in this area at any time of the year.

It is the contractor's responsibility to obtain information concerning weather conditions in the project area. The Climate Data Online website (<https://www.ncdc.noaa.gov/cdo-web/>) published by NOAA's National Centers for Environmental Information (formerly the National Climatic Data Center) in Asheville, North Carolina contains climatological and meteorological data relevant to this region. Available data includes quality-controlled daily, monthly, seasonal, and yearly measurements of temperature, precipitation, wind, and degree days as well as radar data and 15- to 30-year climate normals.

NOAA's Climate Data Online website includes Monthly Normals (average values) for temperature, precipitation, and other climatological information over three periods: 1981-2010, 1991-2020, and 2006-2020. NOAA's Tampa International Airport, FL gauge (ID USW00012842) is closest to the project area and lies roughly 12.0 miles north of Cut G and 6.5 miles northwest of the Upper Channels at 27° 57.7' N, 82° 32.4' W. The most applicable period for data at this site is likely 1991-2020 and is found at the following website: <https://www.ncei.noaa.gov/maps/normals/>.

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SECTION 01 11 00

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SECTION 01 11 00

SUMMARY OF WORK

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

GDR SECTION 00 31 32, Geotechnical Data Report
for Tampa Harbor Maintenance Dredging,
Upper Channels and Cut-G (PT),
Hillsborough County, Florida

EM 385-1-1 (2024) Safety and Occupational Health
(SOH) Requirements

1.2 WORK COVERED BY CONTRACT DOCUMENTS

1.2.1 Project Description

Project work consists of maintenance dredging Tampa Harbor at identified dredging areas indicated on the drawings. Dredge areas will incorporate 1 foot of allowable overdepth. All dredged material is to be disposed of in DMPF-2D. Additional work includes turbidity monitoring, sea turtle trawling and relocation, bird monitoring, and environmental species monitoring.

1.2.2 Location

The project is located in Tampa Harbor, Hillsborough County, Florida.

1.2.3 Safety Requirements

See Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS and EM 385-1-1.

1.2.4 Work Performed by Others

See paragraph PROJECT COORDINATION of Section 01 30 00 ADMINISTRATIVE PROCEDURES.

1.2.5 Dredging Commencement

Schedule commencement of dredging to start on a weekday, Monday through Friday, between the hours of 07:00 to 12:00 Eastern Timezone, excluding Federal Holidays, in order to allow for Government personnel to perform the necessary inspections on-board the dredge prior to excavation of the initial dredge loading cycle. Alternative dredging start times will be approved by the Contracting Officer prior to commencement of dredging.

1.3 COMMENCEMENT, PROSECUTION AND COMPLETION OF WORK

Read this paragraph in conjunction with the Clause COMMENCEMENT, PROSECUTION, AND COMPLETION OF WORK (FAR 52.211-10).

1.4 LIQUIDATED DAMAGES - CONSTRUCTION

Refer to the Clause LIQUIDATED DAMAGES - CONSTRUCTION (FAR 52.211-12).

1.5 PHYSICAL DATA

Read this paragraph in conjunction with the Clause PHYSICAL DATA (FAR 52.236-4).

1.5.1 Physical Conditions

The indications of physical conditions on the drawings and in the specifications are the result of site investigations by surveys and/or by core borings. When the indicated physical conditions are the result of site investigations by core borings, the core boring logs and laboratory data are in Section 00 31 32 GEOTECHNICAL DATA REPORT (GDR) and the core boring locations are shown on the drawings. Also, see the GDR for instructions concerning availability of core borings for inspection.

1.5.2 Weather, Water Stage and Tide Data

See Section 00 33 50 WEATHER, WATER STAGE AND TIDE DATA.

1.5.3 Contractor Investigation

Refer to the Clause SITE INVESTIGATION AND CONDITIONS AFFECTING THE WORK (FAR 52.236-3).

1.5.4 Maritime Traffic

Marine Traffic in the project area consists of Naval, cruise ships, commercial, pleasure, and small recreational vessels of all types and sizes which can be accommodated by existing depths.

1.5.5 Obstruction of Channel

The Government will not undertake to keep the channel free from vessels or other obstructions, except to the extent of such regulations, if any, as may be prescribed by the Secretary of the Army, in accordance with the provisions of Section 7 of the River and Harbor Act approved 8 August 1917. Conduct the work in such manner as to obstruct navigation as little as possible, and in case the Contractor's plant so obstructs the channel as to make difficult or endanger the passage of any vessels, promptly move said plant on the approach of any vessel to such an extent as may be necessary to afford a practicable passage. Upon completion of the work, promptly remove the plant, including ranges, buoys, piles, and other marks placed under the Contract in navigable waters or on shore.

1.5.6 Hydrographic Surveys

Hydrographic surveys are provided on the ehydro web site:
<https://www.arcgis.com/apps/opstdashboard/index.html#/4b8f2ba307684cf597617bf1b6d2f85d>
Survey numbers are indicated on the drawings.

1.6 LAYOUT OF WORK

Read this paragraph in conjunction with the Clause LAYOUT OF WORK (FAR 52.236-17).

1.6.1 Established Project Control

The Government has established monuments along the work site(s). Each monument is represented in the contract drawings by a Designation and in most cases, an alphanumeric Permanent Identifier (PID). Information concerning the location and elevations can be found within the Survey Notes on the plan sheets. The Survey Notes for each included field survey will cite a particular survey number, and specify the control used to conduct that survey. Additional information is included in the Survey Notes that must be duplicated if an attempt to recreate the subject survey is desired.

For navigation channels, the Government has established tidal calibration sites for projects constructed and maintained to a tidal datum. Each tidal calibration site is represented in the plan sheet drawings by Station ID, and is cited in the Survey Notes. Calibrate all projects constructed and maintained to a tidal datum to the Station ID specified in the plan sheet drawings. Information regarding the location and associated benchmarks of the referenced Station ID is included in the Monument Description section of these specifications.

See Section 00 31 21 CONTROL MONUMENT DESCRIPTIONS.

1.6.2 Layout

From the monuments, control data and elevations established by the Government, complete the layout of the work and obtain all measurements that may be required for the execution of the work to the location and limit marks prescribed in the specifications or on the contract drawings, subject to such modifications as the Contracting Officer may require to meet changed conditions or as a result of necessary modifications to the contract work.

1.6.3 Survey

Provide equipment, tools, material, labor for laying out any part of the work from the monuments, control data, and elevations established by the Government. Maintain and preserve all monuments, stakes and other marks established by the Contracting Officer until authorized to remove them, and if such marks or monuments are destroyed by the Contractor or through negligence, prior to their authorized removal, they may be replaced by the Contracting Officer, at their discretion, and the expense of replacement will be deducted from any amounts due or to become due the Contractor. The Contracting Officer may require that work be suspended at any time when location and limit marks established by the Contractor are not reasonably adequate to permit checking of the work.

1.7 DAMAGE TO WORK

The responsibility for damage to any part of the permanent work will be as set forth in Clause PERMITS AND RESPONSIBILITIES. However, if any part of the permanent work performed by the Contractor is damaged by flood, earthquake, hurricane, or tornado, which damage is not due to the failure of the Contractor to take reasonable precautions or to exercise sound

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engineering and construction practices in the conduct of the work, make the repairs as ordered by the Contracting Officer and full compensation for such repairs will be made at the applicable Contract unit or job prices as fixed and established in the Contract. If, in the opinion of the Contracting Officer, there are no contract unit or job prices applicable to any part of such work, an equitable adjustment pursuant to Clause CHANGES will be made as full compensation for the repairs of that part of the permanent work for which there are no applicable contract unit or job prices. Except as herein provided, repair damage to all work (including temporary construction), utilities, materials, equipment and plant at the Contractor's expense, regardless of the cause of such damage.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

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 - 1.5.4.2 Measurement
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PART 2 PRODUCTS (NOT USED)

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PART 3 EXECUTION

3.1 PAYMENT PROCEDURES

3.1.1 Requesting Progress Payment

3.1.2 Options and Modification CLINS

-- End of Section Table of Contents --

SECTION 01 20 00

PRICE AND PAYMENT PROCEDURES

PART 1 GENERAL

1.1 SUMMARY

This section describes how Line Items will be measured and paid for when making progress payments. Work to be measured is described in specification sections listed for each Line Item. Measurement procedures for payment, required quantity survey or procurement documentation and payment restrictions are described in applicable specification sections. Allocate costs for work not specifically mentioned to the Line Item most closely associated with work involved. Unless there is a specific Line Item for administrative costs, such as Quality Control and Safety, allocate such costs proportionally across all Line Items.

1.1.1 Reference

The publication listed below forms a part of this specification to the extent referenced. This publication is referred to in the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 1110-2-1003

(2013) Hydrographic Surveying

1.2 RESIDENT MANAGEMENT SYSTEM (RMS)

1.2.1 Definition

The terms Contract Line Item Number (CLIN) and Line Item are interchangeable herein (e.g.: CLIN 0001 is Line Item 0001). The term CLIN is a contracting term used in the Resident Management System (RMS) payment data base.

1.2.2 Instructions

See Section 01 45 00 RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE (RMS CM) for more information on CLINs, schedule of values of pay activities and construction schedule, and payment procedures. The Contractor mode user manual with in-depth instructions can be downloaded from the RMS web site indicated in Section 01 45 00 RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE (RMS CM).

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Transmit submittal items in accordance with Section 01 33 00 SUBMITTAL PROCEDURES. Submit the following preconstruction submittal items no later than 30 calendar

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days after Notice to Proceed:

SD-01 Preconstruction Submittals

Schedule of Values; G, R0

SD-07 Certificates

Request for Progress Payment; G, R0

1.4 JOB PAYMENT ITEMS

Payment items for the work of this contract for which contract lump sum payments will be made are listed in the CLIN SCHEDULE with the unit JOB and described below. All costs for items of work, which are not specifically mentioned to be included in a particular job or unit price payment item, will be included in the listed job item most closely associated with the work involved. The job price and payment made for each item listed will constitute full compensation for furnishing all plant, labor, materials, and equipment, and performing any associated Contractor quality control, environmental protection, meeting safety requirements, tests and reports, and for performing all work required for which separate payment is not otherwise provided.

Progress payments for job CLINs will be made in accordance with the Payments Under Fixed-Price Construction Contracts clause. Submit [Schedule of Values](#) for Government approval with the baseline schedule submittal (see Section [01 32 01](#) PROJECT SCHEDULE). Include a list of pay activities to breakdown bid for each job CLIN as part of the RMS Schedule. Schedule of Values will become basis for CLIN and Pay Activity data in the RMS payment data base. An unbalanced Schedule of Values and Pay Activity Schedule will be returned for revision. If this contract contains either the Continuing Contracts clause or the Continuing Contracts (Alternate) clause, take into account the amount reserved for contract payments when preparing the construction schedule.

1.4.1 Mobilization and Demobilization (Line Item 0001)

1.4.1.1 Payment

Payment will be made for costs associated with and incidental to mobilization and demobilization of the Contractor's plant and associated equipment, and establishment of initial project management and coordination. See Clause PAYMENT FOR MOBILIZATION AND DEMOBILIZATION.

1.4.1.2 Unit of Measure

Unit of measure: Job.

1.4.2 Turbidity Monitoring (Line Item 0004)

1.4.2.1 Payment

Payment will be made for costs associated with and incidental to obtaining, analyzing, and reporting the results of monitoring for turbidity. See Section [01 57 25](#) TURBIDITY AND DISPOSAL MONITORING.

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1.4.2.2 Unit of Measure

Unit of measure: Job.

1.4.3 Environmental Species Monitoring (Line Item 0005)

1.4.3.1 Payment

Payment will be made for costs associated with and incidental to environmental species observers. See Section 01 57 20 ENVIRONMENTAL PROTECTION.

1.4.3.2 Unit of Measure

Unit of measure: Job.

1.4.4 Bird Monitoring (Line Item 0006)

1.4.4.1 Payment

Payment will be made for costs associated with or incidental to furnishing a Bird Monitor and providing deterrence at Dredged Material Placement Facility 2D (DMPF-2D) during the designated monitoring period. See Section 01 57 20 ENVIRONMENTAL PROTECTION.

1.4.4.2 Unit of Measure

Unit of measure: Job.

1.4.5 Mobilization and Demobilization for Sea Turtle Trawling (Hopper Dredge Only, Line Item 0007)

1.4.5.1 Payment

Payment will be made for costs associated with and incidental to mobilization and demobilization for sea turtle trawling (Hopper Dredge Only). See Section 01 57 20 ENVIRONMENTAL PROTECTION. If performance of this line item is not required or implemented per the NOTES in the CLIN SCHEDULE, no payment will be made to the Contractor for this line item.

1.4.5.2 Unit of Measure

Unit of measure: Job.

1.5 UNIT PRICE PAYMENT ITEMS

Payment items for the work of this contract on which the contract unit price payments will be made are listed in the CLIN SCHEDULE and described below. The unit price and payment made for each item listed will constitute full compensation for furnishing all plant, labor, materials, and equipment, and performing any associated Contractor quality control, environmental protection, meeting safety requirements, tests and reports, and for performing all work required for each of the unit price items.

Each Unit Price CLIN may be a single pay activity item or may be broken down into pay activities with smaller quantities equal to CLIN total. Contract unit price multiplied by agreed quantity is full compensation.

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1.5.1 Excavation, Unclassified (Line Items 0002 and 0003)

1.5.1.1 Payment

- a. Payment will be made for costs associated with and incidental to excavation, transportation, and placement of materials; providing and maintaining access to the work site(s) and disposal area(s); noise control; and debris removal. See Sections 35 20 23 DREDGING and 01 57 20 ENVIRONMENTAL PROTECTION.
- b. Insofar as consistent with the paragraph CONTINUITY OF WORK of Section 35 20 23 DREDGING, monthly partial payments will be based on approximate quantities determined by soundings or sweepings performed by the Contractor behind the dredge. The term "area designated by the Contracting Officer" is also defined as "acceptance section" in paragraph CONTINUITY OF WORK of Section 35 20 23 DREDGING.
- c. Hydrographic surveys for final measurement and payment and for acceptance will be made by the Government. All surveys will be performed in accordance with USACE EM 1110-2-1003.

1.5.1.2 Measurement

- a. The maps and/or drawings already prepared (clause CONTRACT DRAWINGS AND SPECIFICATIONS) are believed to represent accurately average existing conditions, but the depths shown thereon may be verified and corrected by soundings taken before dredging. Once quantities within a specified reach or section have been determined through the use of in-place measurement, they will not be reopened except on evidence of collusion, fraud, or obvious error.
- b. The total amount of material removed, and to be paid for under the contract, will be measured by the cubic yard in place. The volume computed will be between the bottom surface shown by soundings taken within 3 weeks before dredging and the bottom surface shown by the soundings taken within 2 weeks after the work specified in each acceptance section has been completed. Give 10 days advance notice, in writing, to the Contracting Officer of the need for a pre-dredging survey or after-dredging survey for final acceptance for each acceptance section. The quantity will include the volume within the limits of the side slopes described in subparagraph SIDE SLOPES of paragraph REQUIRED DEPTH, ALLOWABLE OVERDEPTH, AND SIDE SLOPES of Section 35 20 23 DREDGING, less any deductions that may be required for misplaced material described in subparagraph MISPLACED MATERIALS of paragraph DISPOSAL OF EXCAVATED MATERIAL of Section 35 20 23 DREDGING.

1.5.1.3 Unit of Measure

Unit of measure: Cubic yard.

1.5.2 Sea Turtle Trawling and Relocation (Hopper Dredge Only, Line Item 0008)

1.5.2.1 Payment

Payment will be made for costs associated with and incidental to the sea turtle trawling and relocation. This price includes all costs for furnishing labor, equipment, fuel, oil, materials, and supplies required

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for trawling to clear and/or relocate sea turtles from in the vicinity of the dredge, and the cost of reporting as specified. See Section 01 57 20 ENVIRONMENTAL PROTECTION. If performance of this line item is not required or implemented per the NOTES in the CLIN SCHEDULE, no payment will be made to the Contractor for this line item.

1.5.2.2 Measurement

Sea turtle trawling will be measured for payment based upon the duration of performance on a daily basis. A day of trawling is defined as 24 hours of continuous trawling. Conduct trawling as described in Section 01 57 20 ENVIRONMENTAL PROTECTION.

1.5.2.3 Unit of Measure

Unit of measure: Day.

1.5.3 Standby Time (Line Item 0009)

1.5.3.1 Payment

Payment will be made for costs associated with or incidental to standby of dredge plant only upon implementation of this line item. Standby time is defined as when, at the direction of the Contracting Officer, the dredge is put out of operation for reasons outside the requirements in the contract.

1.5.3.2 Measurement

Standby time, when exercised by the Contracting Officer, will be measured for payment by each hour, or part thereof, that the plant is not in operation.

1.5.3.3 Unit of Measure

Unit of measure: Hour.

1.5.4 Partnering Regroup (Line Item 0010)

1.5.4.1 Payment

Payment will be made for costs associated with and incidental to conducting a partnering regroup. The Contractor will only be paid for the actual number of partnering regroups conducted. See Section 01 30 00 ADMINISTRATIVE PROCEDURES. The initial partnering meeting will not be paid for under this line item as it is a distributed cost mandatory administrative action.

1.5.4.2 Measurement

The basis for payment will be for the completion of each partnering regroup meeting and Government receipt of all completed work products.

1.5.4.3 Unit of Measure

Unit of measure: Each.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 PAYMENT PROCEDURES

Upon receiving initial Resident Management System import file, go to PAY ACTIVITIES and establish a link between bid breakdown schedule of values of PAY ACTIVITIES to contract CLINS using SCHEDULE ACTIVITIES data entry page.

3.1.1 Requesting Progress Payment

For progress payments, ensure ACTIVITY SCHEDULE, FEATURE SCHEDULE, submittal register, and punchlists are all up to date. Use PROGRESS PAYMENTS to REQUEST ACTIVITY for both ACTIVITY EARNINGS data entry page and OTHER EARNING. Provide hard copies of supporting invoices and quantity measurements to support all requested earnings. Ensure that sums of payment activities do not exceed contract award CLIN funding amounts, or unbalanced CLINS error will prevent processing the payment. Submit a [request for progress payment](#) monthly in accordance with Clause 52.232-5 PAYMENTS UNDER FIXED-PRICE CONSTRUCTION CONTRACTS. In addition to the items listed in Clause 52.232-5(b)(1), submit the following in the request for progress payment:

- a. Approved construction schedule.
- b. Up-to-date payroll records.
- c. Current submittal register.
- d. Professional Surveyor and Mapper (PSM) or Registered Land Surveyor (RLS) report and supporting data.
- e. Approved submittals on installed property.
- f. QA/QC deficiency tracking list.
- g. Environmental reports.
- h. Progress photographs.
- i. Working/current as-built drawings.

Once the complete pay application package is received to include the Contractor's signed ENG 93, the 14 calendar days will commence.

3.1.2 Options and Modification CLINS

When additional work is added by modification, existing CLINS funding amounts must be updated, or new CLINS for modification will be created. If contract has option CLINS not yet awarded, option CLINS will appear as zero dollar CLINS until option is awarded by modification. No payment may be requested for Options or Modification CLINS until contract modification has been funded and signed.

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SECTION 01 30 00

ADMINISTRATIVE PROCEDURES

PART 1 GENERAL

1.1 SUMMARY

Manage project and coordinate activities of own employees, subcontractors, suppliers and offsite fabricators. Use computers, E-mail, and Internet resources for administrative work. Notify Contracting Officer of important meetings, schedule events and activities. Furnish labor and materials and equipment required to plan and execute project management functions and coordination. Coordinate activities and manage resources to construct project conforming to contract, on time and within budget. Related Sections are 01 32 01 PROJECT SCHEDULE, and 01 45 05 DREDGING/BEACH FILL PLACEMENT - QUALITY CONTROL.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

APL 95-1-1 (2022) Aviation Policy Letter 95-1-1;
USACE Aviation Policies and Standards
(Separate Document)

EM 385-1-1 (2024) Safety and Occupational Health
(SOH) Requirements

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Transmit submittal items in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.

SD-01 Preconstruction Submittals

View Location Map; G, R0

Progress And Completion Pictures

SD-03 Product Data

Small Unmanned Aircraft Systems (SUAS)

SD-07 Certificates

Partnering Meeting(s) Written Synopsis; G, R0

Signature Authority

Drug-Free Work Place Record

Insurance Requirements

1.4 PROJECT COORDINATION

1.4.1 Resident Management System (RMS)

Use Corps of Engineers' Resident Management System for construction information management. See Section 01 45 00 RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE (RMS CM).

1.4.2 Coordination with Other Contracts

An active construction project may impact Contractor work at DMPF-2D. Coordinate activities to avoid impact to adjacent Contractors and prevent problems from adjacent contracts. Assume responsibility for operating the weirs at DMPF-2D upon arrival.

1.4.3 Access by Others

Maintain access to power poles and other items owned or operated by utility companies. Coordinate with utility companies as required for line marking, hookups, and relocations as needed to perform work. Incoming electrical utilities become the responsibility of Contractor at transformers. Coordinate emergency incident response with local law enforcement and fire rescue authorities. Notify Contracting Officer in writing of coordination problems encountered affecting work.

1.4.4 Inspections by Others

During construction period, inspection or maintenance work will be performed by others within the construction limits. The Contractor is responsible for neatly mowing all grass and vegetation (not to include established trees or other protected vegetation designated as wetland areas) within the construction limits for the duration of the Contract. Areas not accessible to mowers must be edged or trimmed neatly. Mowing or trimming must be performed every three weeks or when vegetation exceeds twelve inches in height.

1.5 VIEW LOCATION MAP

Submit, prior to or with the first digital photograph submittals, a sketch or drawing indicating the required photographic locations. Update as required if the locations are moved.

1.6 PROGRESS AND COMPLETION PICTURES

Photographically document site conditions prior to start of construction operations. Provide monthly, and within one month of the completion of work, digital photographs, 1600x1200x24 bit true color minimum resolution in JPEG file format showing the sequence and progress of work. Take a minimum of 20 digital photographs to include 10 aerial photographs monthly throughout the entire project from a minimum of ten different viewpoints selected by the Contractor unless otherwise directed by the Contracting Officer. Submit with the monthly invoice two sets of digital photographs, each set on a separate compact disc (CD) or data versatile disc (DVD),

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cumulative of all photos to date. Indicate photographs demonstrating environmental procedures. Provide photographs for each month in a separate monthly directory and name each file to indicate its location on the view location sketch. Also provide the view location sketch on the CD or DVD as a digital file. Include a date and time stamp on each photograph as well as a date designator in the file name. Photographs provided are for unrestricted use by the Government.

1.7 SMALL UNMANNED AIRCRAFT SYSTEMS (SUAS)

Comply with all applicable aspects of USACE Aviation Policy Letter 95-1-1 (APL 95-1-1, <https://www.saj.usace.army.mil/About/Divisions-Offices/Engineering/Design-Branch/Specifications-Section/>) for the collection of aerial data by Small Unmanned Aircraft Systems (SUAS). Further, comply with location specific requirements for SUAS operations within the confines of critical infrastructure or DoD facility. Coordinate any activity associated with APL 95-1-1 through the ATPM and Contracting Office to ensure proper procedures and reporting to the USACE Aviation Office. Submit Mission Plan documentation. Submit product data on SUAS used. SUAS provided will be listed as either USACE ETP Approved in the exact configuration authorized or as a DIU Blue Cleared Aircraft in the exact authorized configuration (<https://www.diu.mil/blue-uas-cleared-list>).

1.8 SUPERVISION

1.8.1 Superintendent Qualifications

Provide project superintendent with a minimum of 5 years experience in construction with at least 3 of those years as a superintendent on projects similar in size and complexity. The individual must be familiar with the requirements of EM 385-1-1 and have experience in the areas of hazard identification and safety compliance. The individual must be capable of interpreting a critical path schedule and construction drawings. The qualification requirements for the alternate superintendent are the same as for the project superintendent. The Contracting Officer may request proof of the superintendent's qualifications at any point in the project if the performance of the superintendent is in question.

1.8.2 Minimum Communication Requirements

Have at least one qualified superintendent, or competent alternate, capable of reading, writing, and conversing fluently in the English language, on the job-site at all times during the performance of Contract work. In addition, if a Quality Control (QC) representative is required on the contract, then that individual must also have fluent English communication skills.

1.8.3 Duties

The project superintendent is primarily responsible for managing subcontractors and coordinating day-to-day production and schedule adherence on the project. The superintendent is required to attend Red Zone meetings, partnering meetings, and quality control meetings. The superintendent or qualified alternative must be on-site at all times during the performance of this contract until the work is completed and accepted.

1.8.4 Non-Compliance Actions

The Project Superintendent is subject to removal by the Contracting Officer for non-compliance with requirements specified in the contract and for failure to manage the project to ensure timely completion. Furthermore, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such stop orders is acceptable as the subject of claim for extension of time for excess costs or damages by the Contractor.

1.9 PROJECT MEETINGS

Contracting Officer requires following types of project meetings:

- Preconstruction Conference
- Partnering Meeting
- Partnering Regroup Meetings
- Coordination Meeting
- Preparatory and Initial Phase meeting for each feature of work
- Project Progress Meetings

Project meetings are described in detail in subparts below.

1.10 PROJECT MEETINGS FORMAT

All project meetings will occur face-to-face or virtually, the default format will be face-to-face. The Contracting Officer will approve or request virtual meeting when a face-to-face meeting format can not be used. Meetings at Contractor's facility will have systems capabilities for both audio and video. The audio and video system must provide for full coverage of the conference room.

1.11 PRECONSTRUCTION CONFERENCE

Contracting Officer will conduct a Preconstruction Conference for this project in accordance with Clause PRECONSTRUCTION CONFERENCE. Preconstruction Conference will be held within 30 to 60 calendar days after Notice to Proceed (NTP). (Refer to subparagraph PRECONSTRUCTION CONFERENCE SUBMITTALS below.) Contracting Officer will notify Contractor of time, place, and agenda. At a minimum, Contractor attendees must include the Project Manager, Superintendent, Site Safety and Health Officer (SSHO), Quality Control Manager and key subcontractors and suppliers. Contracting Officer will discuss contract ground rules and general issues including:

- Lines of Contracting Officer authority
- Lines of Contractor authority
- Contract Clauses
- Special Contract Requirements
- Contract Administration
- Progress Payment
- Correspondence Procedures
- Project Schedule
- Submittal Register
- Labor requirements
- General Site Safety
- Permit Requirements
- Joint Risk Management Processes

1.11.1 Preconstruction Conference Minutes and Phone

The Contracting Officer will take detailed minutes of Preconstruction Conference discussions and may record the meeting. Copies of typed minutes will be provided to the Contractor to review for accuracy, sign and return. Signed minutes become part of the contract file. Recordings, if used, will be made available for Contractor to review or copy.

1.11.2 Preconstruction Conference Submittals

The timing of submission of submittals and completion of the Preconstruction Conference is intended to allow the Contractor and the Government adequate time to prepare for commencement of work. However, if the Contractor fails to submit required items and receive acceptance and/or approval within the times stated, the Contractor will not be permitted to commence work until these requirements have been satisfied. Any delays attributable to the Contractor's failure to comply with these pre-work requirements must be at the Contractor's expense and may be cause for remedial action by the Contracting Officer. Submittals required by this Section are described in paragraph SUBMITTALS above.

1.11.2.1 Other Division 01 Submittals

The Contracting Officer reviews submitted draft plans to prepare Coordination Meeting agenda and understand Contractor procedures. Submit the following preconstruction submittal items no later than 30 calendar days after Notice to Proceed unless specifically identified otherwise in the following specifications:

- Project Schedule - See Section 01 32 01 PROJECT SCHEDULE
- Accident Prevention Plan - See Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS
- Quality Control Plan - See Section 01 45 05 DREDGING/BEACH FILL PLACEMENT - QUALITY CONTROL
- Temporary Facilities Plan - See Section 01 50 02 TEMPORARY CONSTRUCTION FACILITIES
- Environmental Protection Plan - See Section 01 57 20 ENVIRONMENTAL PROTECTION

1.11.2.2 Divisions 02 through 48 Submittals

In addition to the above, bring submittal items for materials, workmanship, plans or events required early in project schedule which are ready for transmittal to Contracting Officer. Prepare transmittal of submittal items in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.

1.11.3 Joint Risk Register

Request a sample template from Contracting Officer or receive Government approval for a Contractor's template form prior to submission. The draft template must be project oriented with potential joint risks that will become topics of discussions for the initial partnering meeting.

1.11.4 List of Subcontractors

Documentation of compliance with Clauses SUBCONTRACTS (LABOR STANDARDS) and LIMITATIONS ON SUBCONTRACTING, and, if included, paragraph LIMITATIONS ON SUBSTITUTIONS FOR CERTAIN POSITIONS AND/OR SUBCONTRACTORS. As part of

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the RMS Schedule, provide a list of subcontractors with company name, person to contact, street address, mail address, phone number, type of specialty and estimated subcontract quote no later than 30 calendar days after Notice to Proceed.

1.12 PARTNERING

Contracting Officer wants to have a bilateral project partnership with Contractor that draws on strengths of both organizations to identify and achieve common goals. Typical partnering goals include:

- Effective and efficient contract performance
- Project completion on time and within budget
- Construction completed according to plans and specifications
- Development of cooperative management teams
- Project success with customer (stakeholders) satisfaction
- Improved Project communication
- Joint Risk Management Plan and Register

Provide a facilitator from an approved list of vendors located on the website

<https://www.saj.usace.army.mil/About/Divisions-Offices/Construction/> or propose a vendor as indicated below.

1.12.1 Initial Partnering Meeting

Contractor, key subcontractors and suppliers must attend a one-day project Partnering Meeting. Attendants will be determined by the Contracting Officer. At a minimum, the Contractor's Project Manager, Quality Control Manager, Site Superintendent, and Site Safety and Health Officer (SSHO) must attend the Partnering Meeting. The Partnering Meeting is a project team building meeting facilitated by qualified persons. Contracting Officer attendees and invited persons may include:

- Contracting Officer (Area Engineer/Resident Engineer)
- Construction Quality Assurance Representatives
- Jacksonville District - Construction Branch persons
- Jacksonville District - Engineering Division Designers
- Local Sponsor (Owner) - Tampa Harbor Port Authority

Goal of initial Partnering Meeting is to build trust, identify common goals, develop an initial Joint Risk Plan and Register, and understand individual project members' expectations and organizational values. Usual results include better communication between contract parties, shortened project learning curve and a cohesive two party contract partnership. Participants usually sign a voluntary partnering agreement generally describing mutual obligation to cooperate to achieve project goals and maintain effective communication. Initial Partnering Meeting is mandatory. The expected duration for the initial partnering session is one day (at least eight hours).

1.12.1.1 Contractor Responsibility

Provide a facilitator for all partnering meetings from an approved list of vendors on the web site

<https://www.saj.usace.army.mil/About/Divisions-Offices/Construction/> under the 'Partnering Vendors' column to the bottom right of the page. The Contractor may propose a vendor not included on the approved vendor list for Contracting Officer approval. Pay for all associated facilitator

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costs including travel. Provide a meeting venue adequate to comfortably accommodate a minimum of 40 attendees. Provide all office supplies (i.e., easels, projectors, markers, etc.) necessary to facilitate the meeting. The Contractor will be responsible for all costs associated with and incidental to the partnering meeting.

1.12.1.2 Tasks and Deliverables

The facilitator must provide a written agenda and appropriate work products for the Initial Partnering Meeting. Upon completion of the meeting, submit a [partnering meeting\(s\) written synopsis](#) of the meeting to Government. At a minimum the written synopsis must include the partnering charter, goals, attendance register, lines of communication, working lists, items of discussion, action items, risk register, and any other information addressed in the meeting as requested by the Government in agreement with partnering team. All deliverables must be provided in editable Microsoft Office Suite (i.e., Word, Excel) and PDF formats to the Government for review within 10 business days after completion of the meeting.

1.12.2 Partnering Regroups

After initial Partnering Meeting, participation will be voluntary in accordance with a signed Partnering Agreement. Partnering Agreements include provisions for Contracting Officer and Contractor to request Partnering Regroups when needed. A Partnering Regroup is usually a partnering session of one day (at least four hours) where senior level Contractor and Contracting Officer project members and other interested parties review project issues of concern to include an update to the Joint Risk Register. Issues of conflict are reviewed, attempt to affirm common project goals and understand concerns of other partners.

1.12.2.1 Contractor Responsibility (Regroups)

Provide a facilitator for all partnering meetings from an approved list of vendors on the web site indicated in paragraph CONTRACTOR RESPONSIBILITY above. The Contractor may propose a vendor not included on the approved vendor list for Contracting Officer approval. The Contractor must pay for all associated facilitator costs including travel. Provide a meeting venue adequate to comfortably accommodate a minimum of 25 attendees. Provide all office supplies (i.e., easels, projectors, markers, etc.) necessary to facilitate the meeting. The Contractor will be responsible for all costs associated with and incidental to the partnering meeting.

1.12.2.2 Tasks and Deliverables

The facilitator must provide a written agenda and appropriate work products for the Regroup Partnering Meeting. Upon completion of the meeting, a written synopsis of the meeting must be provided to the Contractor and Government. At a minimum the written synopsis must include updates of the partnering charter, goals, attendance register, lines of communication, working lists, items of discussion, action items, risk register, and any other information addressed in the meeting as requested by the Government in agreement with partnering team. All deliverables must be provided in editable Microsoft Office Suite (i.e., Word, Excel) and PDF formats, and provided to the Government for review within 10 business days after completion of the meeting.

1.12.3 Cancellation of Partnering Meeting or Regroup

Partnering meetings may be postponed or cancelled by the Contracting Officer at no cost to the Government when the Government notifies the Contractor in writing with at least 72 hours prior to the scheduled meeting. No payment will be made for a postponed or cancelled Partnering Meeting.

1.13 NOTICE TO PROCEED

Notice to proceed (NTP) will be issued 15 to 30 calendar days after award or after all required bonds are in place, whichever is later.

1.14 COORDINATION MEETING

Coordination Meeting is scheduled, convened and conducted by Contracting Officer after a Preconstruction Conference and prior to starting physical construction. Plans submitted after NTP (i.e., Quality Control Plan, Environmental Protection Plan and Accident Prevention Program) will have been reviewed. Coordination Meeting is primarily for on-site Contractor Quality Control staff, including subcontractor and supplier employees performing quality control, to meet and discuss the project in detail with Contracting Officer's Quality Assurance Representatives. Attendants will be determined by the Administrative Contracting Officer. At a minimum, the Contractor's Quality Control Manager, Site Safety and Health Officer, Site Superintendent, and major subcontractor's foreman must attend the Coordination Meeting. Purposes of Coordination Meeting are:

- Achieve mutual understanding with Contractor of required Quality Control
- Jointly review submitted draft plans; resolve issues of concern
- Discuss project plans and specifications, schedule, documentation
- Discuss project Joint Risk Management Plan and Register
- Establish a good working relationship between the Contractors Quality Control Staff and Quality Assurance Representatives
- Discuss the number of hard copies submittals requirements

1.15 PROGRESS MEETINGS

Schedule, convene and preside over progress meetings as required. As project activities increase, a minimum of one progress meeting per week is typical of a project of this scope. Convene additional meetings as required, or when requested by Contracting Officer. Notify persons needed to be present to discuss agenda issues. Contracting Officer may direct attendance by key Contractor suppliers, or fabricators as needed. A sample meeting agenda is provided in paragraph GENERAL MEETING REQUIREMENTS below. Take minutes of the meetings and distribute copies of the minutes within three business days after the meetings. Provide a professional grade teleconference system with a conference phone with speakers for adequate audio coverage during the meeting. Provide a conference phone line to accommodate up to 25 individuals. Provide a meeting room to accommodate up to 15 people for weekly progress meetings.

1.15.1 Progress Meeting Participants

Typical participants include:

- Contracting Officer
- Construction Quality Assurance Representatives

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- Contractor's Site Superintendent
- Contractor's Quality Control Manager
- Contractor's Safety Coordinator
- Subcontractors, as appropriate to the agenda
- Suppliers, as appropriate to the agenda
- Others as appropriate to the agenda

1.16 CONTRACTOR PERFORMANCE EVALUATIONS

Contractors are given an opportunity to provide and/or enter performance evaluation comments directly into the Contractor Performance Assessment Reporting System (CPARS). The system is a web-enabled application that supports the completion, distribution, and retrieval of contract performance evaluations. Provide to the responsible area office the name and email address of the contractor user that will be accessing CPARS.

1.16.1 Training

Contractor online training is available at the following web site: <https://www.cpars.gov/lc.htm>. Training gives the student an understanding of the Contractor Representative and Contractor Corporate Senior Management levels of access, the policies and regulations governing CPARS, and the electronic workflow with particular attention to the Contractor comment function.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 GENERAL MEETING REQUIREMENTS

See Section 01 45 05 DREDGING/BEACH FILL PLACEMENT - QUALITY CONTROL for Preparatory and Initial Phase meeting checklists. Contractor is responsible for phase and progress meetings to include:

- Meeting notification to participants
- Prepare agenda for meetings
- Use phase checklists for Preparatory and Initial Phase meetings
- Physical arrangements for meetings
- Preside at meetings
- Record minutes recording proceedings and decisions
- Copy and send minutes to:
 - Meeting participants
 - Project parties affected by decisions
 - Contracting Officer (No later than 3 working days)

3.2 PROGRESS MEETING AGENDA

Modify agenda as needed for on-going work.

- Review minutes from previous progress meetings
- Review RMS Contractor Action Item Report
- Review work progress since previous meeting
- Review current definable features of work:
 - Identify phases of current features of work
 - Identify pending phase changes
 - Identify features for discussion in next scheduled meeting
- Discuss problem prevention:
 - Field observations

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- Deficiencies and tracking
- Procedures working well
- Problems, conflicts
- Methods to improve

Review construction schedule:

- Identify delays
- Discuss proposed corrective actions to regain schedule

Submittals and Requests for Information (design interpretation):

- Review submittal register
- Identify submittals to expedite as required

Review off-site activities:

- Fabrications
- Material and equipment delivery schedule

Review Testing:

- Type, Schedule
- Received Results

Review changes to construction schedule:

- Planned progress during succeeding work period
- Coordination of various schedules
- Effect of changes on construction and completion date

Review site safety

Review and update Joint Risk Management Register

Discuss maintaining contract quality for materials and workmanship

Discuss pending modifications, changes and substitutions

Discuss other business, as appropriate

3.3 SIGNATURE AUTHORITY

Submit a power of attorney or a notarized letter of authority identifying local representatives authorized to sign contract documents.

3.4 DRUG-FREE WORK PLACE RECORD

Submit documentation of compliance with Clause DRUG-FREE WORKPLACE (see subparagraphs (b)(1) through (b)(7)).

3.5 MINIMUM INSURANCE REQUIREMENTS

Provide the minimum insurance coverage required by FAR 28.307-2 Liability during the entire period of performance under this contract. Provide other insurance coverage as required by state law.

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SECTION 01 32 01

PROJECT SCHEDULE (Alternate)

PART 1 GENERAL

1.1 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.

SD-01 Preconstruction Submittals

Project Scheduler Qualifications; G, R0

Initial Project Schedules; G, R0

SD-07 Certificates

Periodic Schedule Update; G, R0

1.2 PROJECT SCHEDULER QUALIFICATIONS

Designate an authorized representative to be responsible for the preparation of the schedule and all required updating and production of reports. The authorized representative will have a minimum of 2-years experience scheduling construction projects similar in size and nature to this project with scheduling software that meets the requirements of this specification. Representative will have a comprehensive knowledge of CPM scheduling principles and application. The authorized representative may also serve as the Project Manager or other staff assigned to the project.

1.3 CONSTRUCTION SCHEDULES

Construction schedules consist of Project Schedules or Periodic Schedule Updates, Bar Charts, and S Curve. Use work breakdown structure of CLINs, Features and Pay Activities. CLINs are Contract Line Item Numbers from bid schedule. Features are quality control grouping of like Pay Activities requiring similar quality control. Pay Activities are scheduled work activities or events. Most Pay Activities have resources (time, money, labor, material and equipment). Use Pay Activities with no dollars allocated to schedule administrative work activities or events.

1.3.1 Initial Project Schedules

See Section 01 45 05 DREDGING/BEACH FILL PLACEMENT - QUALITY CONTROL. Create the Initial Activity Schedule in RMS, in accordance with the terms of FAR 52.236-15 SCHEDULES FOR CONSTRUCTION CONTRACTS. Submit Initial Project Schedules not later than 30 calendar days after receipt of Notice to Proceed. Data can be entered manually or Contractor has option to use

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standard data exchange format (SDEF.txt file). Cost allocate each Pay Activity to an assigned CLIN. Sum of Pay Activities will equal CLIN totals so there is no variance. Features duration is early start of first Pay Activity to late finish date of last Pay Activity. Primavera P6 v6.x and later versions (through 18.x) allow export of a file for conversion to standard data exchange format (SDEF.txt) file format. Primavera provides a utility that will convert the export file to an SDEF.txt file. An SDEF.txt file can then be used to directly import the Primavera schedule into RMS. Provide all programs and hardware necessary to prepare initial and progress construction schedules. The required Primavera P6 Activity Code Structure is as follows:

Field #	Code	Length	Field Description
1	WRKP	3	Workers per day
2	RESP	4	Responsibility
3	AREA	4	Area
4	MODF	6	Modification or Claim #
5	BIDI	6	Bid Item
6	PHAS	2	Phase
7	CATW	1	Category of Work
8	FOW	20	Feature of Work

1.3.2 Bar Chart with S Curve

Contractor may create a manual Bar Chart and S curve or use computer software of choice. Scheduling software allows for easier updates and changes. Submit the electronic schedule file in the file format used to prepare the schedule. The Bar Chart will include an appropriate level of detail including activities, submittals, Government review periods, material/equipment delivery, utility outages, on-site construction, inspection, testing, and closeout activities. Contractor's submissions will follow the direction of the Contracting Officer regarding reasonable activity durations. Reasonable durations are those that allow the progress of activities to be accurately determined between payment periods. All planned durations will take into account weather and climate data, work schedules, and contract restrictions on work periods. No single on-site construction activity can exceed 30 days in duration without further breakdown in detail.

Contractor has an option to schedule an early finish date with float until Contract required completion date. Schedules with early finish are for the benefit of the Contractor, and the Contracting Officer may elect to continue to make balanced progress payments based on funds availability. Neither Contractor nor Contracting Officer own float on early finish schedules. Contractor cannot claim added costs for delay for float used. When unit price CLINs have quantity variations and impact weighted percentage of a Pay Activity by five percent or more, revise bar chart and S Curve to accurately reflect impact of variation of estimated quantities.

1.3.3 Project Milestones

1.3.3.1 Project Start Date Milestone and Constraint

The first activity in the project schedule is a start milestone titled NTP ACKNOWLEDGED, which has a start on constraint date equal to the date that the NTP is acknowledged.

1.3.3.2 End Project Finish Milestone and Constraint

The last activity in the schedule is a finish milestone titled END PROJECT.

The project schedule is constrained to the Contract completion date in such a way that if the schedule calculates an early finish, then the float calculation for the END PROJECT milestone reflects a positive float on the longest path. If the project schedule calculates a late finish, then the END PROJECT milestone float calculation reflects a negative float on the longest path.

1.4 PERIODIC SCHEDULE UPDATE

Update the project schedule on a regular basis, monthly at a minimum. Provide a draft periodic schedule update for review as prescribed in the paragraph PERIODIC PROGRESS MEETINGS. These updates enable the Government to assess the Contractor's progress. Submit Bar Chart, S Curve schedule when schedule is revised. Reflect any changes occurring since the last update. Submit Periodic Schedule Updates not later than 4 working days after the periodic progress meeting.

When work is delayed by causes beyond the Contractor's control, notify the Contracting Officer within 10 days of the beginning of the delay. Provide a description of the event(s) that caused the delay or impact to the work. As part of the description, identify all schedule activities impacted. Show that the event that caused the delay or impact was the responsibility of the Government. Provide an analysis that demonstrates the effects of the delay or impact on the project completion date or interim completion date(s). Evaluate multiple impacts chronologically, each with its own justification of delay. Evaluate the delay or impact for concurrence with delays caused by the Contractor's own actions. With multiple impacts, consider any concurrence of delay. No time extension will be granted unless a delay occurs which consumes all available project float and extends the projected finish beyond the contract completion date. No delay damages will be paid unless the event is shown to be the responsibility of the Government.

1.4.1 Narrative Report

Provide a narrative report with each schedule submission which is expected to communicate to the Government the thorough analysis of the schedule output and the plans to compensate for any problems, either current or potential, which are revealed through that analysis. Include at a minimum the following information in the narrative report:

- a. The work scheduled to start in the next update period.
- b. A description of activities along the two most critical paths where the total float is less than or equal to 20 work days.
- c. A description of current and anticipated problem areas or delaying factors and their impact and an explanation of corrective actions taken or required to be taken.
- d. The reasons activities, based on their calculated late dates, were planned to have either started or finished during the update period but did not.
- e. All schedule changes by activity ID and activity name, including what

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specifically changed, and why the change was needed. Include at a minimum new and deleted activities, logic changes, duration changes, calendar changes, lag changes, resource changes, and actual start and finish date changes.

f. Out-of-sequence work.

1.4.2 Periodic Progress Meetings

Progress meetings to discuss payment will include an on-site meeting at regular intervals mutually agreed to at the Preconstruction Conference. During the meeting, the Contractor will describe, on an activity by activity basis, all proposed revisions and adjustments to the Project Schedule which are required to reflect the current status of the project. The Contracting Officer will approve activity progress, proposed revisions, and adjustments as appropriate.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

-- End of Section --

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PART 3 EXECUTION

ATTACHMENTS:

ENG Form 4025-R

Attachment A - Submittal Register

-- End of Section Table of Contents --

SECTION 01 33 00
SUBMITTAL PROCEDURES

PART 1 GENERAL

1.1 SUMMARY

1.1.1 Submittal Information

The Contracting Officer may request submittals in addition to those specified when deemed necessary to adequately describe the work covered in the respective sections. Each submittal is to be complete and in sufficient detail to allow ready determination of compliance with contract requirements.

Units of weights and measures used on all submittals are to be the same as those used in the contract drawings.

1.1.2 Project Type

The Contractor's Quality Control (CQC) System Manager are to check and approve all items before submittal and stamp, sign, and date indicating action taken. Proposed deviations from the contract requirements are to be clearly identified. Include within submittals items such as: Contractor's, manufacturer's, or fabricator's drawings; descriptive literature including (but not limited to) catalog cuts, diagrams, operating charts or curves; test reports; test cylinders; samples; O&M manuals (including parts list); certifications; warranties; and other such required submittals.

1.1.3 Submission of Submittals

Schedule and provide submittals requiring Government approval before acquiring the material or equipment covered thereby. Pick up and dispose of samples not incorporated into the work in accordance with manufacturer's Safety Data Sheets (SDS) and in compliance with existing laws and regulations.

1.2 DEFINITIONS

1.2.1 Submittal Descriptions (SD)

Submittal requirements are specified in the technical sections. Examples and descriptions of submittals identified by the Submittal Description (SD) numbers and titles follow:

SD-01 Preconstruction Submittals

Submittals that are required prior to or at the start of construction (work) or the next major phase of the construction on a multiphase contract.

Government approved Division 01 preconstruction submittals that are required prior to or commencing with the start of work must be submitted

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within 30 calendar days of contract award unless specified elsewhere in the specifications. Contractor approved Division 01 submittals that are required prior to or commencing with the start of work must be submitted within 45 calendar days of contract award unless specified elsewhere in the specifications.

Preconstruction Submittals include schedules and a tabular list of locations, features, and other pertinent information regarding products, materials, equipment, or components to be used in the work.

Certificates Of Insurance

Surety Bonds

List Of Proposed Subcontractors

List Of Proposed Products

Baseline Network Analysis Schedule (NAS)

Submittal Register

Schedule Of Prices Or Earned Value Report

Accident Prevention Plan

Work Plan

Quality Control (QC) plan

Environmental Protection Plan

SD-02 Shop Drawings

Drawings, diagrams and schedules specifically prepared to illustrate some portion of the work.

Diagrams and instructions from a manufacturer or fabricator for use in producing the product and as aids to the Contractor for integrating the product or system into the project.

Drawings prepared by or for the Contractor to show how multiple systems and interdisciplinary work will be coordinated.

SD-03 Product Data

Catalog cuts, illustrations, schedules, diagrams, performance charts, instructions and brochures illustrating size, physical appearance and other characteristics of materials, systems or equipment for some portion of the work.

Samples of warranty language when the contract requires extended product warranties.

SD-04 Samples

Fabricated or unfabricated physical examples of materials, equipment or workmanship that illustrate functional and aesthetic characteristics of a material or product and establish standards by

which the work can be judged.

Color samples from the manufacturer's standard line (or custom color samples if specified) to be used in selecting or approving colors for the project.

Field samples and mock-ups constructed on the project site establish standards ensuring work can be judged. Includes assemblies or portions of assemblies that are to be incorporated into the project and those that will be removed at conclusion of the work.

SD-05 Design Data

Design calculations, mix designs, analyses or other data pertaining to a part of work.

SD-06 Test Reports

Report signed by authorized official of testing laboratory that a material, product or system identical to the material, product or system to be provided has been tested in accord with specified requirements. Unless specified in another section, testing must have been within three years of date of contract award for the project.

Report that includes findings of a test required to be performed on an actual portion of the work or prototype prepared for the project before shipment to job site.

Report that includes finding of a test made at the job site or on sample taken from the job site, on portion of work during or after installation.

Investigation reports

Daily logs and checklists

Final acceptance test and operational test procedure

SD-07 Certificates

Statements printed on the manufacturer's letterhead and signed by responsible officials of manufacturer of product, system or material attesting that the product, system, or material meets specification requirements. Must be dated after award of project contract and clearly name the project.

Document required of Contractor, or of a manufacturer, supplier, installer or Subcontractor through Contractor. The document purpose is to further promote the orderly progression of a portion of the work by documenting procedures, acceptability of methods, or personnel qualifications.

Confined space entry permits

Text of posted operating instructions

SD-08 Manufacturer's Instructions

Preprinted material describing installation of a product, system or

material, including special notices and (SDS)concerning impedances, hazards and safety precautions.

SD-09 Manufacturer's Field Reports

Documentation of the testing and verification actions taken by manufacturer's representative at the job site, in the vicinity of the job site, or on a sample taken from the job site, on a portion of the work, during or after installation, to confirm compliance with manufacturer's standards or instructions. The documentation must be signed by an authorized official of a testing laboratory or agency and state the test results; and indicate whether the material, product, or system has passed or failed the test.

Factory test reports.

SD-10 Operation and Maintenance Data

Data provided by the manufacturer, or the system provider, including manufacturer's help and product line documentation, necessary to maintain and install equipment, for operating and maintenance use by facility personnel.

Data required by operating and maintenance personnel for the safe and efficient operation, maintenance and repair of the item.

Data incorporated in an operations and maintenance manual or control system.

SD-11 Closeout Submittals

Documentation to record compliance with technical or administrative requirements or to establish an administrative mechanism.

Submittals required for Guiding Principle Validation (GPV) or Third Party Certification (TPC).

Special requirements necessary to properly close out a construction contract. For example, Record Drawings and as-built drawings. Also, submittal requirements necessary to properly close out a major phase of construction on a multi-phase contract.

1.2.2 Approving Authority

Office or designated person authorized to approve the submittal.

1.2.3 Work

As used in this section, on-site and off-site construction required by contract documents, including labor necessary to produce submittals, construction, materials, products, equipment, and systems incorporated or to be incorporated in such construction. In exception, excludes work to produce SD-01 submittals.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification

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identifies the office that will review the submittal for the Government.
Submit the following in accordance with Section 01 33 00 SUBMITTAL
PROCEDURES:

SD-01 Preconstruction Submittals

Submittal Register; G

1.4 SUBMITTAL CLASSIFICATION

1.4.1 Government Approved (G)

Government approval is required for extensions of design, critical materials, variations, equipment whose compatibility with the entire system must be checked, and other items as designated by the Government.

Within the terms of the Contract Clause SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION, submittals are considered to be "shop drawings."

1.4.2 For Information Only

Submittals not requiring Government approval will be for information only. Within the terms of the Contract Clause SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION, they are not considered to be "shop drawings."

1.5 PREPARATION

1.5.1 Transmittal Form

Use the ENG Form 4025-R transmittal form for submitting both Government-approved and information-only submittals. Submit in accordance with the instructions on the reverse side of the form. These forms or similar forms are included in the RMS CM software that the Contractor is required to use for this contract. Properly complete this form by filling out all the heading blank spaces and identifying each item submitted. Exercise special care to ensure proper listing of the specification paragraph and sheet number of the contract drawings pertinent to the data submitted for each item.

1.5.2 Submittal Format

1.5.2.1 Format of SD-01 Preconstruction Submittals

When the submittal includes a document that is to be used in the project, or is to become part of the project record, other than as a submittal, do not apply the Contractor's approval stamp to the document itself, but to a separate sheet accompanying the document.

Provide data in the unit of measure used in the contract documents.

1.5.2.2 Format for SD-02 Shop Drawings

Provide shop drawings not less than 8 1/2 by 11 inches nor more than 30 by 42 inches, except for full-size patterns or templates. Prepare drawings to accurate size, with scale indicated, unless another form is required. Ensure drawings are suitable for reproduction and of a quality to produce clear, distinct lines and letters, with dark lines on a white background.

a. Include the nameplate data, size, and capacity on drawings. Also

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include applicable federal, military, industry, and technical society publication references.

- b. Dimension drawings, except diagrams and schematic drawings. Prepare drawings demonstrating interface with other trades to scale. Use the same unit of measure for shop drawings as indicated on the contract drawings. Identify materials and products for work shown.

Submit an electronic copy of drawings in PDF format and native electronic format.

1.5.2.2.1 Drawing Identification

Include on each drawing the drawing title, number, date, and revision numbers and dates, in addition to information required in paragraph IDENTIFYING SUBMITTALS.

Number drawings in a logical sequence. Each drawing is to bear the number of the submittal in a uniform location next to the title block. Place the Government contract number in the margin, immediately below the title block, for each drawing.

Reserve a blank space, no smaller than 2 inches on the right-hand side of each sheet for the Government disposition stamp.

1.5.2.3 Format of SD-03 Product Data

Present product data submittals for each section. Include a table of contents, listing the page and catalog item numbers for product data.

Indicate, by prominent notation, each product that is being submitted; indicate the specification section number and paragraph number to which it pertains.

1.5.2.3.1 Product Information

Supplement product data with material prepared for the project to satisfy the submittal requirements where product data does not exist. Identify this material as developed specifically for the project, with information and format as required for submission of SD-07 Certificates.

Provide product data in units used in the Contract documents. Where product data are included in preprinted catalogs with another unit, submit the dimensions in contract document units, on a separate sheet.

1.5.2.3.2 Standards

Where equipment or materials are specified to conform to industry or technical-society reference standards of such organizations as the American National Standards Institute (ANSI), ASTM International (ASTM), National Electrical Manufacturer's Association (NEMA), Underwriters Laboratories (UL), or Association of Edison Illuminating Companies (AEIC), submit proof of such compliance. The label or listing by the specified organization will be acceptable evidence of compliance. In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. State on the certificate that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference

standard.

1.5.2.3.3 Data Submission

Collect required data submittals for each specific material, product, unit of work, or system into a single submittal that is marked for choices, options, and portions applicable to the submittal. Mark each copy of the product data identically. Partial submittals will not be accepted for expedition of the construction effort.

Submit the manufacturer's instructions before installation.

1.5.2.4 Format of SD-04 Samples

1.5.2.4.1 Sample Characteristics

Furnish samples in the following sizes, unless otherwise specified or unless the manufacturer has prepackaged samples of approximately the same size as specified:

- a. Sample of Equipment or Device: Full size.
- b. Sample of Materials Less Than 2 by 3 inches: Built up to 8 1/2 by 11 inches.
- c. Sample of Materials Exceeding 8 1/2 by 11 inches: Cut down to 8 1/2 by 11 inches and adequate to indicate color, texture, and material variations.
- d. Sample of Linear Devices or Materials: 10 inch length or length to be supplied, if less than 10 inches. Examples of linear devices or materials are conduit and handrails.
- e. Sample Volume of Nonsolid Materials: Pint. Examples of nonsolid materials are sand and paint.
- f. Color Selection Samples: 2 by 4 inches. Where samples are specified for selection of color, finish, pattern, or texture, submit the full set of available choices for the material or product specified. Sizes and quantities of samples are to represent their respective standard unit.
- g. Sample Panel: 4 by 4 feet.
- h. Sample Installation: 100 square feet.

1.5.2.4.2 Sample Incorporation

Reusable Samples: Incorporate returned samples into work only if so specified or indicated. Incorporated samples are to be in undamaged condition at the time of use.

Recording of Sample Installation: Note and preserve the notation of any area constituting a sample installation, but remove the notation at the final clean-up of the project.

1.5.2.4.3 Comparison Sample

Samples Showing Range of Variation: Where variations in color, finish,

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pattern, or texture are unavoidable due to nature of the materials, submit sets of samples of not less than three units showing extremes and middle of range. Mark each unit to describe its relation to the range of the variation.

When color, texture, or pattern is specified by naming a particular manufacturer and style, include one sample of that manufacturer and style, for comparison.

1.5.2.5 Format of SD-05 Design Data

Provide design data and certificates on 8 1/2 by 11 inch paper.

1.5.2.6 Format of SD-06 Test Reports

By prominent notation, indicate each report in the submittal. Indicate the specification number and paragraph number to which each report pertains.

1.5.2.7 Format of SD-07 Certificates

Provide design data and certificates on 8 1/2 by 11 inch paper.

1.5.2.8 Format of SD-08 Manufacturer's Instructions

Present manufacturer's instructions submittals for each section. Include the manufacturer's name, trade name, place of manufacture, and catalog model or number on product data. Also include applicable federal, military, industry, and technical-society publication references. If supplemental information is needed to clarify the manufacturer's data, submit it as specified for SD-07 Certificates.

Submit the manufacturer's instructions before installation.

1.5.2.8.1 Standards

Where equipment or materials are specified to conform to industry or technical-society reference standards of such organizations as the American National Standards Institute (ANSI), ASTM International (ASTM), National Electrical Manufacturer's Association (NEMA), Underwriters Laboratories (UL), or Association of Edison Illuminating Companies (AEIC), submit proof of such compliance. The label or listing by the specified organization will be acceptable evidence of compliance. In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. State on the certificate that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard.

1.5.2.9 Format of SD-09 Manufacturer's Field Reports

By prominent notation, indicate each report in the submittal. Indicate the specification number and paragraph number to which each report pertains.

1.5.2.10 Format of SD-10 Operation and Maintenance Data (O&M)

Comply with the requirements specified in sections where O&M Data is

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required for O&M Data format.

1.5.2.11 Format of SD-11 Closeout Submittals

When the submittal includes a document that is to be used in the project or is to become part of the project record, other than as a submittal, do not apply the Contractor's approval stamp to the document itself, but to a separate sheet accompanying the document.

Provide data in the unit of measure used in the contract documents.

1.5.3 Source Drawings for Shop Drawings

1.5.3.1 Source Drawings

The entire set of source drawing files (DWG) will not be provided to the Contractor. Request the specific Drawing Number for the preparation of shop drawings. Only those drawings requested to prepare shop drawings will be provided. These drawings are provided only after award.

1.5.3.2 Terms and Conditions

Data contained on these electronic files must not be used for any purpose other than as a convenience in the preparation of construction data for the referenced project. Any other use or reuse is at the sole risk of the Contractor and without liability or legal exposure to the Government. The Contractor must make no claim, and waives to the fullest extent permitted by law any claim or cause of action of any nature against the Government, its agents, or its subconsultants that may arise out of or in connection with the use of these electronic files. The Contractor must, to the fullest extent permitted by law, indemnify and hold the Government harmless against all damages, liabilities, or costs, including reasonable attorney's fees and defense costs, arising out of or resulting from the use of these electronic files.

These electronic source drawing files are not construction documents. Differences may exist between the source drawing files and the corresponding construction documents. The Government makes no representation regarding the accuracy or completeness of the electronic source drawing files, nor does it make representation to the compatibility of these files with the Contractor hardware or software. The Contractor is responsible for determining if any conflict exists. In the event that a conflict arises between the signed and sealed construction documents prepared by the Government and the furnished source drawing files, the signed and sealed construction documents govern. Use of these source drawing files does not relieve the Contractor of the duty to fully comply with the contract documents, including and without limitation the need to check, confirm and coordinate the work of all contractors for the project. If the Contractor uses, duplicates or modifies these electronic source drawing files for use in producing construction data related to this contract, remove all previous indication of ownership (seals, logos, signatures, initials and dates).

1.5.4 Electronic File Format

Provide submittals in electronic format, with the exception of material samples required for SD-04 Samples items. In addition to electronic submittal, provide additional hard copies of submittals when requested by the Contracting Officer. Up to 2 additional hard copies of any submittal

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may be requested at the discretion of the Contracting Officer, at no additional cost to the Government. Compile the submittal file as a single, complete document, to include the Transmittal Form described within, and also separately attach the native files which were used to create PDF. The attached files should include the original digital files used to create the submittal. Name the electronic submittal file specifically according to its contents, and coordinate the file naming convention with the Contracting Officer. Electronic files must be of sufficient quality that all information is legible. Use PDF as the electronic format, unless otherwise specified or directed by the Contracting Officer. Generate PDF files from original documents with bookmarks so that the text included in the PDF file is searchable and can be copied. If documents are scanned, optical character resolution (OCR) routines are required. Index and bookmark files exceeding 30 pages to allow efficient navigation of the file. When required, the electronic file must include a valid electronic signature or a scan of a signature.

E-mail electronic submittal documents smaller than 10MB to an e-mail address as directed by the Contracting Officer. Provide electronic documents over 10 MB on an optical disc or through an electronic file sharing system such as the DOD SAFE Web Application located at the following website: <https://safe.apps.mil/>.

1.6 QUANTITY OF SUBMITTALS

1.6.1 Number of SD-01 Preconstruction Submittal Copies

Unless otherwise specified, submit two sets of administrative submittals.

1.6.2 Number of SD-04 Samples

- a. Submit two samples, or two sets of samples showing the range of variation, of each required item. One approved sample or set of samples will be retained by the approving authority and one will be returned to the Contractor.
- b. Submit one sample panel or provide one sample installation where directed. Include components listed in the technical section or as directed.
- c. Submit one sample installation, where directed.
- d. Submit one sample of nonsolid materials.

1.7 INFORMATION ONLY SUBMITTALS

Submittals without a "G" designation must be certified by the QC manager and submitted to the Contracting Officer for information-only. Provide information-only submittals to the Contracting Officer a minimum of 14 calendar days prior to the Preparatory Meeting for the associated Definable Feature of Work (DFOW). Approval of the Contracting Officer is not required on information only submittals. The Contracting Officer will mark "receipt acknowledged" on submittals for information and will return only the transmittal cover sheet to the Contractor. Normally, submittals for information only will not be returned. However, the Government reserves the right to return unsatisfactory submittals and require the Contractor to resubmit any item found not to comply with the contract. This does not relieve the Contractor from the obligation to furnish material conforming to the plans and specifications; will not prevent the

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Contracting Officer from requiring removal and replacement of nonconforming material incorporated in the work; and does not relieve the Contractor of the requirement to furnish samples for testing by the Government laboratory or for check testing by the Government in those instances where the technical specifications so prescribe.

1.8 PROJECT SUBMITTAL REGISTER

A sample Project Submittal Register showing items of equipment and materials for when submittals are required by the specifications is provided as "Attachment A - Submittal Register."

1.8.1 Submittal Management

Prepare and maintain a submittal register, as the work progresses. Do not change data that is output in columns (c), (d), (e), and (f) as delivered by Government; retain data that is output in columns (a), (g), (h), and (i) as approved. As an attachment, provide a submittal register showing items of equipment and materials for which submittals are required by the specifications. This list may not be all-inclusive and additional submittals may be required. Maintain a submittal register for the project in accordance with Section 01 45 00 RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE(RMS CM). The Government will provide the initial submittal register in electronic format

Column (c): Lists specification section in which submittal is required.

Column (d): Lists each submittal description (SD Number. and type, e.g., SD-02 Shop Drawings) required in each specification section.

Column (e): Lists one principal paragraph in each specification section where a material or product is specified. This listing is only to facilitate locating submitted requirements. Do not consider entries in column (e) as limiting the project requirements.

Thereafter, the Contractor is to track all submittals by maintaining a complete list, including completion of all data columns and all dates on which submittals are received by and returned by the Government.

1.8.2 Preconstruction Use of Submittal Register

Submit the submittal register. Include the QC plan and the project schedule. Verify that all submittals required for the project are listed and add missing submittals. Coordinate and complete the following fields on the register submitted with the QC plan and the project schedule:

Column (a) Activity Number: Activity number from the project schedule.

Column (g) Contractor Submit Date: Scheduled date for the approving authority to receive submittals.

Column (h) Contractor Approval Date: Date that Contractor needs approval of submittal.

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Column (i) Contractor Material: Date that Contractor needs material delivered to Contractor control.

1.8.3 Contractor Use of Submittal Register

Update the following fields in the Government-furnished submittal register program or equivalent fields in the program used by the Contractor with each submittal throughout the contract.

Column (b) Transmittal Number: List of consecutive, Contractor-assigned numbers.

Column (j) Action Code (k): Date of action used to record Contractor's review when forwarding submittals to QC.

Column (l) Date submittal transmitted.

Column (q) Date approval was received.

1.8.4 Approving Authority Use of Submittal Register

Update the following fields:

Column (b) Transmittal Number: List of consecutive, Contractor-assigned numbers.

Column (l) Date submittal was received.

Column (m) through (p) Dates of review actions.

Column (q) Date of return to Contractor.

1.8.5 Action Codes

1.8.5.1 Contractor Action Codes

DESIGN BID BUILD SUBMITTALS			
Submittal Classifications shown in UFGS Sections	Submittal Classification	Corresponding SpecsIntact Submittal Register Code which is populated in the SI Submittal Register. Software Limitations: (The software shows one character delineation in the SpecsIntact Submittal Register)	RMS - The following Submittal Classifications are populated in RMS when the SpecsIntact Submittal Data File is pulled into RMS)
G	Submittal requires Government Approval	G	GA

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DESIGN BID BUILD SUBMITTALS			
BLANK	Submittal is For Information Only (FIO)	BLANK	FIO
S	Submittal is for documentation of Sustainable requirements	S	S/FIO

1.8.6 Delivery of Copies

Submit an updated electronic copy of the submittal register to the Contracting Officer with each invoice request. Provide an updated Submittal Register monthly regardless of whether an invoice is submitted.

1.9 VARIATIONS

Variations from contract requirements require Contracting Officer approval pursuant to contract Clause FAR 52.236-21 Specifications and Drawings for Construction, and will be considered where advantageous to the Government.

1.9.1 Considering Variations

Discussion of variations with the Contracting Officer before submission of a variation submittal will help ensure that functional and quality requirements are met and minimize rejections and resubmittals. For variations that include design changes or some material or product substitutions, the Government may require an evaluation and analysis by a licensed professional engineer hired by the contractor.

Specifically point out variations from contract requirements in a variation submittal. Failure to point out variations may cause the Government to require rejection and removal of such work at no additional cost to the Government.

1.9.2 Proposing Variations

When proposing variation, deliver a submittal, clearly marked as a "VARIATION" to the Contracting Officer, with documentation illustrating the nature and features of the variation including any necessary technical submittals and why the variation is desirable and beneficial to Government. If lower cost is a benefit, also include an estimate of the cost savings. In addition to documentation required for variation, include the submittals required for the item. Clearly mark the proposed variation in all documentation.

The Contracting Officer will indicate an approval or disapproval of the variation request; and if not approved as submitted, will indicate the Government's reasons therefore. Any work done before such approval is received is performed at the Contractor's risk."

Specifically point out variations from contract requirements in a variation submittal. Failure to point out variations may cause the Government to require rejection and removal of such work at no additional cost to the Government.

Check the column "variation" of ENG Form 4025 for submittals that include

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variations proposed by the Contractor. Set forth in writing the reason for any variations and note such variations on the submittal. The Government reserves the right to rescind inadvertent approval of submittals containing unnoted variations.

1.9.3 Warranting that Variations are Compatible

When delivering a variation for approval, the Contractor warrants that this contract has been reviewed to establish that the variation, if incorporated, will be compatible with other elements of work.

1.9.4 Review Schedule Extension

In addition to the normal submittal review period, a period of 14 calendar days will be allowed for the Government to consider submittals with variations.

1.10 SCHEDULING

Schedule and submit concurrently product data and shop drawings covering component items forming a system or items that are interrelated. Submit pertinent certifications at the same time. No delay damages or time extensions will be allowed for time lost in late submittals.

- a. Coordinate scheduling, sequencing, preparing, and processing of submittals with performance of work so that work will not be delayed by submittal processing. The Contractor is responsible for additional time required for Government reviews resulting from required resubmittals. The review period for each resubmittal is the same as for the initial submittal. Resubmit disapproved submittal items (codes C, D, E, G or X) within 10 calendar days after return.
- b. Submittals required by the contract documents are listed on the submittal register. If a submittal is listed in the submittal register but does not pertain to the contract work, the Contractor is to include the submittal in the register and annotate it "N/A" with a brief explanation. Approval by the Contracting Officer does not relieve the Contractor of supplying submittals required by the contract documents but that have been omitted from the register or marked "N/A."
- c. Resubmit the submittal register and annotate it monthly with actual submission and approval dates. When all items on the register have been fully approved, no further resubmittal is required.
- d. Unless otherwise indicated in specific paragraphs governing each submittal, the Government is afforded 10 calendar days to review and respond to an Information Only FIO submittal item, and 30 calendar days to review and respond to a Government Approved GA submittal item, from the date of receipt by the approval authority.

1.11 GOVERNMENT APPROVING AUTHORITY

When the approving authority is the Contracting Officer, the Government will:

- a. Note the date on which the submittal was received.
- b. Review submittals for approval within the scheduling period specified

and only for conformance with project design concepts and compliance with contract documents.

- c. Identify returned submittals with one of the actions defined in paragraph REVIEW NOTATIONS and with comments and markings appropriate for the action indicated.

Upon completion of review of submittals requiring Government approval, stamp and date submittals. Two copies of the submittal will be retained by the Contracting Officer and one copy of the submittal will be returned to the Contractor.

1.11.1 Review Notations

Submittals will be returned to the Contractor with the following notations:

- a. Submittals marked "approved" or "accepted" authorize proceeding with the work covered.
- b. Submittals marked "approved as noted" or "approved, except as noted, resubmittal not required," authorize proceeding with the work covered provided that the Contractor takes no exception to the corrections.
- c. Submittals marked "not approved," "disapproved," or "revise and resubmit" indicate incomplete submittal or noncompliance with the contract requirements or design concept. Resubmit with appropriate changes. Do not proceed with work for this item until the resubmittal is approved.
- d. Submittals marked "not reviewed" indicate that the submittal has been previously reviewed and approved, is not required, does not have evidence of being reviewed and approved by Contractor, or is not complete. A submittal marked "not reviewed" will be returned with an explanation of the reason it is not reviewed. Resubmit submittals returned for lack of review by Contractor or for being incomplete, with appropriate action, coordination, or change.
- e. Submittals marked "receipt acknowledged" indicate that submittals have been received by the Government. This applies only to "information-only submittals" as previously defined.

1.12 DISAPPROVED SUBMITTALS

Make corrections required by the Contracting Officer. If the Contractor considers any correction or notation on the returned submittals to constitute a change to the contract drawings or specifications, give notice to the Contracting Officer as required under the FAR clause titled CHANGES. The Contractor is responsible for the dimensions and design of connection details and the construction of work. Failure to point out variations may cause the Government to require rejection and removal of such work at the Contractor's expense.

If changes are necessary to submittals, make such revisions and resubmit in accordance with the procedures above. Provide with resubmission, a response to each Government review comment received on previous submittal a discussion of the corrective and revision made. No item of work requiring a submittal change is to be accomplished until the changed submittals are approved.

1.13 APPROVED SUBMITTALS

The Contracting Officer's approval of submittals is not to be construed as a complete check, and indicates only that

Approval or acceptance by the Government for a submittal does not relieve the Contractor of the responsibility for meeting the contract requirements or for any error that may exist, because under the Quality Control (QC) requirements of this contract, the Contractor is responsible for ensuring information contained within each submittal accurately conforms with the requirements of the contract documents.

After submittals have been approved or accepted by the Contracting Officer, no resubmittal for the purpose of substituting materials or equipment will be considered unless accompanied by an explanation of why a substitution is necessary.

1.14 APPROVED SAMPLES

Approval of a sample is only for the characteristics or use named in such approval and is not to be construed to change or modify any contract requirements. Before submitting samples, provide assurance that the materials or equipment will be available in quantities required in the project. No change or substitution will be permitted after a sample has been approved.

Match the approved samples for materials and equipment incorporated in the work. If requested, approved samples, including those that may be damaged in testing, will be returned to the Contractor, at its expense, upon completion of the contract. Unapproved samples will also be returned to the Contractor at its expense, if so requested.

Failure of any materials to pass the specified tests will be sufficient cause for refusal to consider, under this contract, any further samples of the same brand or make as that material. The Government reserves the right to disapprove any material or equipment that has previously proved unsatisfactory in service.

Samples of various materials or equipment delivered on the site or in place may be taken by the Contracting Officer for testing. Samples failing to meet contract requirements will automatically void previous approvals. Replace such materials or equipment to meet contract requirements.

1.15 WITHHOLDING OF PAYMENT

Payment for materials incorporated in the work will not be made if required approvals have not been obtained.

1.16 CERTIFICATION OF SUBMITTAL DATA

Certify the submittal data as follows on Form ENG 4025: "I certify that the above submitted items had been reviewed in detail and are correct and in strict conformance with the contract drawings and specifications except as otherwise stated.

____NAME OF CONTRACTOR _____ SIGNATURE OF CONTRACTOR

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PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Section --

SUBMITTAL REGISTER

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TITLE AND LOCATION

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ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASS SIF CATION REVIEW OR E W R	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION		DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 20 00	SD-01 Preconstruction Submittals														
			Schedule of Values	1.4	G RO												
			SD-07 Certificates														
			Request for Progress Payment	3.1.1	G RO												
		01 30 00	SD-01 Preconstruction Submittals														
			View Location Map	1.5	G RO												
			Progress And Completion	1.6													
			Pictures														
			SD-03 Product Data														
			Small Unmanned Aircraft	1.7													
			Systems (SUAS)														
			SD-07 Certificates														
			Partnering Meeting(s) Written	1.12.1.2	G RO												
			Synopsis														
			Signature Authority	3.3													
			Drug-Free Work Place Record	3.4													
			Insurance Requirements	3.5													
		01 32 01	SD-01 Preconstruction Submittals														
			Project Scheduler Qualifications	1.2	G RO												
			Initial Project Schedules	1.3.1	G RO												
			SD-07 Certificates														
			Periodic Schedule Update	1.4	G RO												
		01 33 00	SD-01 Preconstruction Submittals														
			Submittal Register	1.8	G												
		01 35 26	SD-01 Preconstruction Submittals														
			Dive Operations Plan	1.17	G DO												

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						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 35 26	Accident Prevention Plan	1.8.1	G DO												
			SD-06 Test Reports														
			Monthly Exposure Reports	1.5													
			Notifications and Reports	1.13													
			Reports	1.13.2	G DO												
			LHE Inspection Reports	1.13.3													
			SD-07 Certificates														
			Crane Operators, Signal Persons and Riggers	1.7.1.3	G DO												
			Standard Lift Plan	1.8.3.2	G DO												
			Critical Lift Plan	1.8.3.3	G DO												
			Naval Architecture Analysis	1.8.3.4	G DO												
			Activity Hazard Analysis (AHA)	1.9	G DO												
			Confined Space Entry Permit	1.10.1													
			Hot Work Permit	1.10.1													
			Certificate of Compliance	1.13.4	G DO												
			License Certificates	1.15	G DO												
			Radiography Operation Planning	1.15.1	G DO												
			Work Sheet														
			Portable Gauge Operations	1.15.1	G DO												
			Planning Worksheet														
		01 45 05	SD-01 Preconstruction Submittals														
			Contractor Quality Control (CQC) Plan	1.5.2	G RO												
			SD-06 Test Reports														
			Verification Statement	1.12.3													

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						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 50 02	SD-01 Preconstruction Submittals														
			Construction Site Plan	1.4.1	G RO												
			Mobilization/Demobilization Plan	1.4.2	G RO												
			Security Plan	1.4.3	G RO												
			Vegetation Protection and Restoration Plan	1.4.4	G DO												
			Documentation Of Existing Conditions	1.4.5	G RO												
			Haul Road Plan	2.2.1	G RO												
			SD-02 Shop Drawings														
			Temporary Electric Drawings	3.2.6													
			SD-03 Product Data														
			Project Identification And Safety Signs	2.1.2	G RO												
			SD-11 Closeout Submittals														
			Documentation Of Post Construction Restoration	3.10.1	G RO												
		01 57 20	SD-01 Preconstruction Submittals														
			Environmental Protection Plan	1.8	G DO												
			SD-02 Shop Drawings														
			Turtle Deflector Device	3.1.5.6	G DO												
			SD-06 Test Reports														
			Protected Listed Species Sighting Reports	3.1.5.2	G DO												
			Weekly Logs/Summary Of FWC Hopper Dredge Reports	3.1.5.7	G DO												

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						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION		DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 57 20	Weekly Protected Species Summary Report	3.1.5.3	G DO												
			SD-07 Certificates														
			Bird Monitoring Qualifications	1.9.1	G DO												
			Marine Animal Observer Qualifications	1.9.2	G DO												
			(Clamshell/Mechanical Only)														
			Endangered Species Observer and Trawl Supervisor Qualifications (For Hopper Dredges Only)	1.9.3	G DO												
			SD-11 Closeout Submittals														
			Protected Species Summary Report	3.1.5.3	G DO												
			Logs/Summary Of Bird Nesting Monitoring Report	3.1.5.8	G DO												
			Comprehensive Trawling Daily Logs	3.1.5.7	G DO												
			Invasive And Nuisance Species Report	3.4.2.1	G DO												
			Project Environmental Summary Sheet	1.11	G DO												
			Marine Animal Observers Final Logs	3.1.5.1	G DO												
		01 57 25	SD-01 Preconstruction Submittals														
			Turbidity Monitor Qualifications	1.3	G DO												

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						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 57 25	Turbidity Monitoring Plan	1.4	G DO												
			Draft Turbidity Sampling Map	1.4	G DO												
			Testing Equipment And Procedure	3.1.1.1	G DO												
			SD-03 Product Data														
			Calibration Standard	3.1.1.1	G												
			SD-06 Test Reports														
			Turbidity Monitoring Reports	3.2.2	G DO												
			Turbidity Exceedance Report	3.3.1	G DO												
			Updated Turbidity Exceedance Report	3.3.2	G DO												
		35 20 23	SD-01 Preconstruction Submittals														
			Notice of Intent to Dredge	1.4													
			Relocation of Navigation Aids	3.2.1													
			Marine Survey Inspection Certificates	3.14.2													
			SD-02 Shop Drawings														
			Drag Bar	3.10.2													
			Acceptance Sections Plan	3.6.1	G DO												
			SD-07 Certificates														
			Historical Period Shipwreck Sites	1.8													
			Surveys	3.7													
			Daily Report of Operations	3.1													
			Misplaced Material	3.5.8													
			Declaration of Inspection	1.11													
		35 20 26	SD-07 Certificates														

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GOVERNMENTAL SAFETY REQUIREMENTS

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B30.3	(2020) Tower Cranes
ASME B30.5	(2021) Mobile and Locomotive Cranes
ASME B30.7	(2021) Winches
ASME B30.8	(2020) Floating Cranes and Floating Derricks
ASME B30.9	(2021) Slings
ASME B30.20	(2021) Below-the-Hook Lifting Devices
ASME B30.22	(2023) Articulating Boom Cranes
ASME B30.23	(2022) Personnel Lifting Systems Safety Standard for Cableways, Cranes, Derricks, Hoists, Hooks, Jacks, and Slings
ASME B30.26	(2015; R 2020) Rigging Hardware

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ANSI/ASSP A10.22	(2007; R 2017) Safety Requirements for Rope-Guided and Non-Guided Workers' Hoists
ANSI/ASSP A10.34	(2021) Protection of the Public on or Adjacent to Construction Sites
ANSI/ASSP A10.44	(2020) Control of Energy Sources (Lockout/Tagout) for Construction and Demolition Operations
ANSI/ASSP Z244.1	(2016; R 2020) The Control of Hazardous Energy Lockout, Tagout and Alternative Methods
ANSI/ASSP Z359.0	(2018) Definitions and Nomenclature Used for Fall Protection and Fall Arrest
ANSI/ASSP Z359.1	(2024) The Fall Protection Code

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ANSI/ASSP Z359.2	(2023) Minimum Requirements for a Comprehensive Managed Fall Protection Program
ANSI/ASSP Z359.3	(2019) Safety Requirements for Lanyards and Positioning Lanyards
ANSI/ASSP Z359.4	(2013; R 2022) Safety Requirements for Assisted-Rescue and Self-Rescue Systems, Subsystems and Components
ANSI/ASSP Z359.6	(2016) Specifications and Design Requirements for Active Fall Protection Systems
ANSI/ASSP Z359.7	(2019) Qualification and Verification Testing of Fall Protection Products
ANSI/ASSP Z359.11	(2021) Safety Requirements for Full Body Harnesses
ANSI/ASSP Z359.12	(2019) Connecting Components for Personal Fall Arrest Systems
ANSI/ASSP Z359.13	(2013; R 2022) Personal Energy Absorbers and Energy Absorbing Lanyards
ANSI/ASSP Z359.14	(2021) Safety Requirements for Self-Retracting Devices for Personal Fall Arrest and Rescue Systems
ANSI/ASSP Z359.15	(2024) Safety Requirements for Single Anchor Lifelines and Fall Arresters for Personal Fall Arrest Systems
ANSI/ASSP Z359.16	(2016) Safety Requirements for Climbing Ladder Fall Arrest Systems
ANSI/ASSP Z359.18	(2017) Safety Requirements for Anchorage Connectors for Active Fall Protection Systems
ANSI/ASSP Z490.1	(2016) Criteria for Accepted Practices in Safety, Health, and Environmental Training

ASTM INTERNATIONAL (ASTM)

ASTM F855	(2023) Standard Specifications for Temporary Protective Grounds to Be Used on De-energized Electric Power Lines and Equipment
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INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 1048	(2016) Guide for Protective Grounding of Power Lines
IEEE C2	(2023) National Electrical Safety Code

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NATIONAL AERONAUTICS AND SPACE ADMINISTRATION (NASA)

- NASA NPR 8621.1 (2020d) NASA Procedural Requirements for Mishap and Close Call Reporting, Investigating, and Recordkeeping
- NASA NPR 8715.3 (2017d; Change 1) NASA General Safety Program Requirements
- NASA-STD 8719.12 (2021a; Change 2) Safety Standard for Explosives, Propellants, and Pyrotechnics

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

- NEMA Z535.2 (2011; R 2017) Environmental and Facility Safety Signs

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

- NFPA 10 (2022; ERTA 1 2021) Standard for Portable Fire Extinguishers
- NFPA 51B (2024) Standard for Fire Prevention During Welding, Cutting, and Other Hot Work
- NFPA 70 (2023) National Electrical Code
- NFPA 70E (2024) Standard for Electrical Safety in the Workplace
- NFPA 241 (2022) Standard for Safeguarding Construction, Alteration, and Demolition Operations
- NFPA 306 (2024) Standard for the Control of Gas Hazards on Vessels

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

- TIA-222 (2018H; Add 1 2019) Structural Standard for Antenna Supporting Structures and Antennas and Small Wind Turbine Support Structures
- TIA-1019 (2012; R 2016) Standard for Installation, Alteration and Maintenance of Antenna Supporting Structures and Antennas

U.S. ARMY CORPS OF ENGINEERS (USACE)

- EM 385-1-1 (2024) Safety and Occupational Health (SOH) Requirements

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

- 10 CFR 20 Standards for Protection Against Radiation
- 29 CFR 1910 Occupational Safety and Health Standards

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29 CFR 1910.146	Permit-required Confined Spaces
29 CFR 1910.147	The Control of Hazardous Energy (Lock Out/Tag Out)
29 CFR 1910.333	Selection and Use of Work Practices
29 CFR 1910.1000	Air Contaminants
29 CFR 1915	Confined and Enclosed Spaces and Other Dangerous Atmospheres in Shipyard Employment
29 CFR 1915.7	Competent person
29 CFR 1915.89	Control of Hazardous Energy (Lockout/Tags-Plus)
29 CFR 1919	Gear Certification
29 CFR 1926	Safety and Health Regulations for Construction
29 CFR 1926.16	Rules of Construction
29 CFR 1926.450	Scaffolds
29 CFR 1926.500	Fall Protection
29 CFR 1926.552	Material Hoists, Personal Hoists, and Elevators
29 CFR 1926.553	Base-Mounted Drum Hoists
29 CFR 1926.1400	Cranes and Derricks in Construction
29 CFR 1926.1431	Hoisting personnel
49 CFR 173	Shippers - General Requirements for Shipments and Packagings
CPL 02-01-056	(2014) Inspection Procedures for Accessing Communication Towers by Hoist
CPL 2.100	(1995) Application of the Permit-Required Confined Spaces (PRCS) Standards, 29 CFR 1910.146

1.2 DEFINITIONS

1.2.1 Competent Person (CP)

A Competent Person is one who is capable of identifying existing and predictable hazards in the surroundings or working conditions which are unsanitary, hazardous, or dangerous to employees, and who has the authorization to take prompt corrective measures to eliminate them.

1.2.2 Competent Person, Confined Spaces in Ships and Vessels (CPCSSV)

An individual, designated in writing by the employer, responsible for testing, inspecting, and evaluating spaces and meets the requirements of 29 CFR 1915.7.

1.2.3 Competent Person, Cranes and Rigging

A person designated in writing by the employer to be responsible for the immediate supervision, implementation, and monitoring of the Crane and Rigging Program. Must have sufficient training, knowledge, and experience in crane(s) and rigging and be capable of identifying, evaluating, and addressing existing and potential hazards and must have the authority to take prompt corrective measures with regard to such hazards.

1.2.4 Competent Person, Excavation/Trenching

A person designated in writing by the employer to be responsible for the immediate supervision, implementation, and monitoring of the Excavation and Trenching Plan. Must have training, knowledge, and experience in excavation and trenching to be capable of identifying, evaluating, and addressing existing and potential hazards. This individual must have the authority to take prompt corrective measures with regard to such hazards.

1.2.5 Competent Person, Fall Protection

A person designated in writing by the employer to be responsible for the immediate supervision, implementation, and monitoring of the Fall Protection Program. The CP for Fall Protection, who through training, knowledge, and experience in fall protection and rescue systems and equipment, is capable of identifying, evaluating, and addressing existing and potential fall hazards. The CP for Fall Protection has the authority to take prompt corrective measures with regard to such hazards.

1.2.6 Competent Person, Scaffolding

A person designated in writing by the employer to be responsible for immediate supervision, implementation, and monitoring of the Scaffold Work Plan. Must have training, knowledge, and experience in scaffolding to be capable of identifying, evaluating, and addressing existing and potential hazards. This individual must have the authority to take prompt corrective measures with regard to such hazards. CP qualifications must meet the requirements of EM 385-1-1.

1.2.7 High Hazard Activities

Example of High Hazard Activities are activities that involve work at heights, crane and rigging, excavations and trenching, scaffolding, electrical work, and confined space entry.

1.2.8 High Visibility Accident

A High Visibility Accident is any mishap which may generate publicity or high visibility.

1.2.9 Load Handling Equipment (LHE)

LHE is a term used to describe cranes, hoists and all other hoisting equipment (hoisting equipment means equipment, including crane, derricks,

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hoists and power operated equipment used with rigging to raise, lower or horizontally move a load).

1.2.10 Medical Treatment

Medical Treatment is treatment administered by a physician or by registered professional personnel under the standing orders of a physician. Medical treatment does not include first aid treatment even when provided by a physician or registered personnel.

1.2.11 Accident

Any unplanned event or series of events that results in death, injury, or illness to personnel, or damage to or loss of equipment or property. Within the context of this manual, accident is synonymous with mishap.

1.2.12 Mishap

A mishap is synonymous with accident.

1.2.13 Near Miss

A potentially serious accident that could have resulted in personal injury, death, or property damage, damage to the environment, or illness but did not occur due to one or more factors.

1.2.14 Operating Envelope

The Operating Envelope is the area surrounding any crane or load handling equipment. Inside this "envelope" is the crane, the operator, riggers and crane walkers, other personnel involved in the operation, rigging gear between the hook, the load, the crane's supporting structure (i.e. ground or rail), the load's rigging path, the lift and rigging procedure.

1.2.15 Qualified Person (QP)

The QP is a person designated in writing, who, by possession of a recognized degree, certificate, or professional standing, or extensive knowledge, training, and experience, has successfully demonstrated their ability to solve or resolve problems related to the subject matter, the work, or the project.

1.2.16 Qualified Person, Fall Protection (QP for FP)

A person with a recognized degree or professional certificate with extensive knowledge, training, and experience in the fall protection and rescue field. The QP for Fall Protection must be capable of designing, analyzing, evaluating, and specifying fall protection and rescue systems.

1.2.17 Recordable Injuries or Illnesses

Recordable Injuries or Illnesses are any work-related injury or illness that results in:

- a. Death, regardless of the time between the injury and death, or the length of the illness;
- b. Days away from work (any time lost after day of injury/illness onset);

- c. Restricted work;
- d. Transfer to another job;
- e. Medical treatment beyond first aid;
- f. Loss of consciousness; or
- g. A significant injury or illness diagnosed by a physician or other licensed health care professional, even if it did not result in (a) through (f) above

1.2.18 Government Property and Equipment

Interpret "USACE" property and equipment specified in USACE EM 385-1-1 as Government property and equipment.

1.2.19 Load Handling Equipment (LHE) Accident or Load Handling Equipment Mishap

A LHE accident occurs when any one or more of the eight elements in the operating envelope fails to perform correctly during operation, including operation during maintenance or testing resulting in personnel injury or death; material or equipment damage; dropped load; derailment; two-blocking; overload; or collision, including unplanned contact between the load, crane, or other objects. A dropped load, derailment, two-blocking, overload and collision are considered accidents, even though no material damage or injury occurs. A component failure (e.g., motor burnout, gear tooth failure, bearing failure) is not considered an accident solely due to material or equipment damage unless the component failure results in damage to other components (e.g., dropped boom, dropped load, or roll over). Document all LHE mishaps using the reporting form, ENG 3394, and a mandatory LHE Operator Drug Test.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" designation. Submittals not having a "G" or "S" designation are for information only. When used, a code following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Dive Operations Plan; G, DO

Accident Prevention Plan; G, DO

SD-06 Test Reports

Monthly Exposure Reports

Notifications and Reports

Reports; G, DO

LHE Inspection Reports

SD-07 Certificates

Crane Operators, Signal Persons and Riggers; G, D0

Standard Lift Plan; G, D0

Critical Lift Plan; G, D0

Naval Architecture Analysis; G, D0

Activity Hazard Analysis (AHA); G, D0

Confined Space Entry Permit

Hot Work Permit

Certificate of Compliance; G, D0

License Certificates; G, D0

Radiography Operation Planning Work Sheet; G, D0

Portable Gauge Operations Planning Worksheet; G, D0

1.4 PUBLIC HEALTH EMERGENCIES

In the event of a declared public health emergency, follow safety precautions as required by the Occupational Safety and Health Administration (OSHA) www.osha.gov, the Centers for Disease Control and Prevention (CDC) www.cdc.gov, and as required by federal, state and local requirements.

1.5 MONTHLY EXPOSURE REPORTS

Provide a Monthly Exposure Report by the fifth of each month. This report is a compilation of employee-hours worked each month for all site workers, both Prime and subcontractor. Failure to submit the report may result in retention of up to 10 percent of the voucher.

1.6 REGULATORY REQUIREMENTS

In addition to the detailed requirements included in the provisions of this Contract, comply with the most recent edition of USACE [EM 385-1-1](#), and the following federal, state, and local laws, ordinances, criteria, rules and regulations. Submit matters of interpretation of standards to the appropriate administrative agency for resolution before starting work. Where the requirements of this specification, applicable laws, criteria, ordinances, regulations, and referenced documents vary, the most stringent requirements govern.

1.7 SITE QUALIFICATIONS, DUTIES, AND MEETINGS

1.7.1 Personnel Qualifications

1.7.1.1 Site Safety and Health Officer (SSHO)

Provide a minimum of one Level 1 SSHO that meets the requirements of [EM 385-1-1](#). The SSHO must ensure that the requirements of [29 CFR 1926.16](#)

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are met for the project and must be present at the site of work and located so they have full mobility and reasonable access to all major work operations occurring during the shift. The SSHO or an equally-qualified Alternate SSHO must be at the work site at all times to implement and administer the Contractor's safety program and Government-accepted Accident Prevention Plan. The SSHO and Alternate SSHO must have the required training, experience, and qualifications in accordance with EM 385-1-1, and all associated sub-paragraphs.

The utilization of a Level 2 SSHO as identified and outlined in EM 385-1-1 does not apply in this contract. A Competent/Qualified Person (Level 3 SSHO) will be provided for all of the hazards identified in the Contractor's Safety and Health Program in accordance with the accepted Accident Prevention Plan, and will be on-site at all times when the work that presents the hazards associated with their professional expertise is being performed. The credentials of the Competent/Qualified Persons(s) will be approved by the Contracting Officer in consultation with the Safety Office. The Activity Hazard Analysis (AHA) is the document for this requirement.

1.7.1.1.1 Additional Site Safety and Health Officer (SSHO) Requirements and Duties

The Level 1 SSHO may not serve as the Quality Control Manager. The Level 1 SSHO may not serve as the Superintendent.

1.7.1.2 Competent Person Qualifications

Provide Competent Persons in accordance with EM 385-1-1 and herein. Competent Persons for high risk activities include confined space, cranes and rigging, excavation/trenching, fall protection, and electrical work. The CP for these activities must be designated in writing, and meet the requirements for the specific activity (i.e. competent person, fall protection).

The Competent Person identified in the Contractor's Safety and Health Program and accepted Accident Prevention Plan, must be on-site at all times when the work that presents the hazards associated with their professional expertise is being performed. Provide the credentials of the Competent Persons(s) to the Contracting Officer for information in consultation with the Safety Office.

1.7.1.2.1 Entry Supervisor for Confined Space Entry

Provide a Confined Space Entry Supervisor who meets the requirements of EM 385-1-1 and herein. The Entry Supervisor must supervise the entry into each confined space in accordance with EM 385-1-1.

Since this work involves operations that handle combustible or hazardous materials, this person must have the ability to understand and follow through on the air sampling, Personal Protective Equipment (PPE), and instructions of a Marine Chemist, Coast Guard authorized persons, or Certified Industrial Hygienist. Confined space and enclosed space work must comply with NFPA 306, OSHA 29 CFR 1915, Subpart B, "Confined and Enclosed Spaces and Other Dangerous Atmospheres in Shipyard Employment," or as applicable, 29 CFR 1910.146 for general industry.

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1.7.1.2.2 Competent Person for Scaffolding

Provide a Competent Person for Scaffolding who meets the requirements of EM 385-1-1 and herein.

1.7.1.2.3 Competent Person for Fall Protection

Provide a Competent Person for Fall Protection who meets the requirements of EM 385-1-1 and herein.

1.7.1.3 Crane Operators, Signal Persons and Riggers

Provide Operators, Signal Persons, and Riggers meeting the requirements in EM 385-1-1.

1.7.2 Personnel Duties

1.7.2.1 Duties of the Site Safety and Health Officer (SSHO)

The SSHO must:

- a. Conduct daily safety and health inspections and maintain a written log which includes area/operation inspected, date of inspection, identified hazards, recommended corrective actions, estimated and actual dates of corrections. Attach safety inspection logs to the Contractors' daily production report.
- b. Report and investigate all accidents and near misses and complete required accident report ENG Form 3394.
- c. Use and maintain OSHA's Form 300 to log work-related injuries and illnesses occurring on the project site for Prime Contractors and subcontractors, and make available to the Contracting Officer upon request. Post and maintain the Form 300A on the site Safety Bulletin Board.
- d. Maintain applicable safety reference material at the site of work including the EM 385-1-1, Safety Data Sheets (SDSs), APP, manufacturer instructions, etc.
- e. Attend all pre-construction conferences, tail-gate meetings, and pre-work meetings, to include preparatory meetings and periodic in-progress meetings.
- f. Review the APP and AHAs for compliance with EM 385-1-1, and approve, sign, implement and enforce them.
- g. Establish a Safety and Occupational Health (SOH) Deficiency Tracking System that lists and monitors outstanding deficiencies until resolution.
- h. Ensure all project personnel comply with SOH requirements (including subcontractors and visitors).
- i. Maintain a list of hazardous chemicals on site and their material Safety Data Sheets (SDS).
- j. Maintain a weekly list of high hazard activities involving energy, equipment, excavation, entry into confined space, and elevation, and

be prepared to discuss details during QC Meetings.

- k. Provide and keep a record of site safety orientation and indoctrination training for all project personnel, including subcontractors
- l. Conduct on-going evaluations of the APP throughout the life of the project to ensure it is current and site-specific. Submit changes, revisions, and updates to the APP for review and approval by the prime contractor.
- m. Ensure visitors are escorted, properly protected, are wearing or provided the appropriate PPE, and receive a safety brief on the hazards to be expected onsite and the SOH controls required.

Superintendent, QC Manager, and SSHO are subject to dismissal if the above or any other required duties are not being effectively carried out. If either the Superintendent, QC Manager, or SSHO are dismissed, project work will be stopped and will not be allowed to resume until a suitable replacement is approved and the above duties are again being effectively carried out.

1.7.3 Meetings

1.7.3.1 Preconstruction Conference

- a. Contractor representatives who have a responsibility or significant role in accident prevention on the project must attend the preconstruction conference. This includes the project superintendent, Site Safety and Occupational Health Officer, quality control manager, or any other assigned safety and health professionals who participated in the development of the APP (including the Activity Hazard Analyses (AHAs) and special plans, program and procedures associated with it).
- b. Discuss the details of the submitted APP to include incorporated plans, programs, procedures and a listing of anticipated AHAs that will be developed and implemented during the performance of the Contract. This list of proposed AHAs will be reviewed and an agreement will be reached between the Contractor and the Contracting Officer as to which phases will require an analysis. In addition, establish a schedule for the preparation, submittal, and Government review of AHAs to preclude project delays.
- c. Deficiencies in the submitted APP, identified during the Contracting Officer's review, must be corrected, and the APP re-submitted for review prior to the start of construction. Work is not permitted to begin until an APP is established that is acceptable to the Contracting Officer.

1.7.3.2 Safety Meetings

Conduct safety meetings on project sites to review past activities, plan for new or changed operations, review pertinent aspects of appropriate AHAs, establish safe working procedures for anticipated hazards, and provide pertinent SOH training and motivation). Meetings must be conducted at least once a month for all supervisors on the project location and at least once a week for all workers. Document the meetings. Include the date, persons in attendance, subjects discussed, and names of individual(s) who conducted the meeting. Include meeting

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minutes in the Daily QC report and furnish copies to the Contracting Officer. Notify the Contracting Officer of all scheduled meetings 7 calendar days in advance.

1.8 ACCIDENT PREVENTION PLAN (APP)

1.8.1 Accident Prevention Plan

A competent person must prepare the written site-specific APP. The APP must be job-specific and document all project, contract, and job-specific potential hazards in the workplace, and the company policies, controls, and work practices that will be used to minimize those hazards. They are an integral part of the planning and risk management process. Describe the methods to evaluate past safety performance of potential subcontractors in the selection process. Also, describe innovative methods used to ensure and monitor safe work practices of subcontractors. The Government considers the Prime Contractor to be the "controlling authority" for all work site safety and health of the subcontractors. Contractors are responsible for informing their subcontractors of the safety provisions under the terms of the Contract and the penalties for noncompliance, coordinating the work to prevent one craft from interfering with or creating hazardous working conditions for other crafts, and inspecting subcontractor operations to ensure that accident prevention responsibilities are being carried out. The APP must be signed by a corporate safety officer of the firm, the individual preparing the APP, the on-site superintendent, the designated SSHO, the Contractor Quality Control Manager, and any designated Certified Safety Professional (CSP) or Certified Health Physicist (CIH). The SSHO must provide and maintain the APP and a log of signatures by each subcontractor foreman, attesting that they have read and understand the APP, and make the APP and log available on-site to the Contracting Officer. If English is not the foreman's primary language, the Prime Contractor must provide an interpreter.

Submit the APP to the Contracting Officer within 30 calendar days of Contract award and not less than 10 calendar days prior to the date of the preconstruction conference for acceptance. Work cannot proceed without an accepted APP. Once reviewed and accepted by the Contracting Officer, the APP and attachments will be enforced as part of the Contract.

Disregarding the provisions of this Contract or the accepted APP is cause for stopping of work, at the discretion of the Contracting Officer, until the matter has been rectified. Continuously review and update the APP, as necessary, throughout the life of the Contract. Changes to the accepted APP must be made with the knowledge and concurrence of the Contracting Officer, project superintendent, SSHO and Quality Control Manager. Incorporate unusual or high-hazard activities not identified in the original APP as they are discovered. Should any severe hazard exposure (i.e. imminent danger) become evident, stop work in the area, secure the area, and develop a plan to remove the exposure and control the hazard. Notify the Contracting Officer within 24 hours of discovery. Eliminate and remove the hazard. In the interim, take all necessary action to restore and maintain safe working conditions in order to safeguard onsite personnel, visitors, the public (as defined by ANSI/ASSP A10.34), and the environment. At the time of APP submission, there may be unknown portions of the work. Include all known information in the APP prior to the start of work. Project-specific information that is yet to be known (for example, subcontractors to be used, type of fall protection to be used at a certain point in time, specific cranes or other load handling equipment (LHE) to be brought at the site of work, must be added and submitted prior to each preparatory phase/meeting, along with the appropriate AHAs.

1.8.2 Names and Qualifications

Provide plans in accordance with the requirements outlined in ENG Form 6293, including the following:

- a. Names and qualifications (resumes including education, training, experience and certifications) of site safety and health personnel designated to perform work on this project to include the designated Site Safety and Health Officer and other competent and qualified personnel to be used. Specify the duties of each position.
- b. Qualifications of competent and of qualified persons. As a minimum, designate and submit qualifications of competent persons for each of the following major areas: excavation; scaffolding; fall protection; hazardous energy; confined space; health hazard recognition, evaluation and control of chemical, physical and biological agents; and personal protective equipment and clothing to include selection, use and maintenance.

1.8.3 Plans

Provide plans in the APP in accordance with ENG Form 6293, Accident Prevention Plan (APP) Worksheet. Provide the completed ENG Form 6293 with the submission of the project Accident Prevention Plan., including the following:

1.8.3.1 Confined Space Entry Plan

Develop a confined or enclosed space entry plan in accordance with [EM 385-1-1](#), applicable OSHA standards [29 CFR 1910](#), [29 CFR 1915](#), and [29 CFR 1926](#), OSHA Directive [CPL 2.100](#), and any other federal, state and local regulatory requirements identified in this Contract. Identify the qualified person's name and qualifications, training, and experience. Delineate the qualified person's authority to direct work stoppage in the event of hazardous conditions. Include procedure for rescue by Contractor personnel and the coordination with emergency responders. (If there is no confined space work, include a statement that no confined space work exists and none will be created.)

1.8.3.2 [Standard Lift Plan](#) (SLP)

Plan all lifts to avoid situations where the operator cannot maintain safe control of the lift. Prepare a written SLP for every lift or series of lifts (if duty cycle or routine lifts are being performed). The SLP must be developed, reviewed, and accepted by all personnel involved in the lift. Keep a copy of the SLP on the LHE for the specific approved lift(s). Keep a copy of every SLP for a minimum of three months. Include in the plan the items identified in [EM 385-1-1](#) or use ENG Form 6203 (Crane Standard Lift Plan)

1.8.3.3 [Critical Lift Plan](#) - Crane or Load Handling Equipment

Provide a Critical Lift Plan prepared by a Qualified Person or Competent Person. The plan must include at a minimum, all requirements of the [EM 385-1-1](#). ENG Form 6213 (Load-Handling Equipment Crane Operation Critical Lift Plan) may be utilized. In addition, Critical Lift Plans are required for the following:

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- a. Lifts involving HAZMAT (for example, explosives, highly volatile substances).
- b. Hoisting personnel with LHE.
- c. Lifts made with more than one LHE.
- d. Lifts where the center of gravity (CG) could change.
- e. Lifts that exceed 75 percent of the load chart of the LHE during construction work.
- f. Lifts without the use of outriggers using rubber tire load charts.
- g. Lifts using more than one hoist on the same LHE.
- h. Lifts involving MLR assemblies or other non-routine or technically difficult rigging arrangements.
- i. Lifts involving submerged loads.
- j. Lifts out of the operator's view.
- k. Load Tests.
- l. When land based LHE mounted on barges, pontoons, or other means of flotation are required to travel while lifting the load.
- m. When two or more tower cranes are set up in the same operating envelope.
- n. Any lift the operator, lift director, supervisor, or employer believes should be considered critical.

1.8.3.3.1 Critical Lift Plan Planning and Schedule

Critical lifts require detailed planning and additional or unusual safety precautions. Develop and submit a critical lift plan to the Contracting Officer 30 calendar days prior to critical lift. Comply with load testing requirements in accordance with [EM 385-1-1](#).

1.8.3.3.2 Lifts of Personnel

In addition to the requirements of [EM 385-1-1](#) for lifts of personnel, demonstrate compliance with the requirements of [29 CFR 1926.1431](#).

1.8.3.4 Floating Cranes/Derricks, Crane Barges, and Auxiliary
Shipboard-Mounted Crane

Plan all lifts to avoid procedures that could result in configurations where the operator cannot maintain safe control of the lift. Prepare a written Standard Lift Plan (SLP) - Floating Plant meeting the requirements and criteria provided in [EM 385-1-1](#). Provide a [Naval Architecture Analysis](#) and include an LHE Manufacturer's Floating Service Load Chart in accordance with [EM 385-1-1](#), Chapter 16.

1.8.3.5 Multi-Purpose Machines, Material Handling Equipment, and Construction Equipment Lift Plan

Multi-purpose machines, material handling equipment, and construction equipment used to lift loads that are suspended by rigging gear, require proof of authorization from the machine OEM that the machine is capable of making lifts of loads suspended by rigging equipment. Written approval from a qualified registered professional engineer, after a safety analysis is performed, is allowed in lieu of the OEM's approval. Demonstrate that the operator is properly trained and that the equipment is properly configured to make such lifts and is equipped with a load chart.

1.8.3.6 Fall Protection and Prevention (FP&P) Plan

The plan must be in accordance with the requirements of [EM 385-1-1](#) and [ANSI/ASSP Z359.2](#), be site specific, and address all fall hazards in the work place and during different phases of construction. Address how to protect and prevent workers from falling to lower levels when they are at or above 6 feet or as exposed to hazards below. A competent person or qualified person for fall protection must prepare and sign the plan documentation. Include fall protection and prevention systems, equipment and methods employed for every phase of work, roles and responsibilities, assisted rescue, self-rescue and evacuation procedures, training requirements, and monitoring methods. Review and revise, as necessary, the Fall Protection and Prevention Plan documentation as conditions change, but at least annually, for lengthy projects, reflecting any changes during the course of construction due to changes in personnel, equipment, systems or work habits. Keep and maintain the accepted Fall Protection and Prevention Plan documentation at the job site for the duration of the project. Include the Fall Protection and Prevention Plan documentation in the Accident Prevention Plan (APP).

1.8.3.7 Rescue Plan

Provide a Rescue Plan in accordance with [EM 385-1-1](#) and [ANSI/ASSP Z359.2](#), and include in the FP&P Plan and as part of the APP. Include a detailed discussion of the following: methods of rescue; methods of self-rescue; equipment used; training requirement; specialized training for the rescuers; procedures for requesting rescue and medical assistance; and transportation routes to a medical facility.

1.8.3.8 Hazardous Energy Control Program (HECP)

Develop a HECP in accordance with [EM 385-1-1](#) Chapter 12, [29 CFR 1910.147](#), [29 CFR 1910.333](#), [29 CFR 1915.89](#), [ANSI/ASSP Z244.1](#), and [ANSI/ASSP A10.44](#). Submit this HECP as part of the Accident Prevention Plan (APP). Conduct a preparatory meeting and inspection with all effected personnel to coordinate all HECP activities. Document this meeting and inspection in accordance with [EM 385-1-1](#). Ensure that each employee is familiar with and complies with these procedures.

1.9 ACTIVITY HAZARD ANALYSIS (AHA)

Before beginning each activity, task or Definable Feature of Work (DFOW) involving a type of work presenting hazards not experienced in previous project operations, or where a new work crew or subcontractor is to perform the work, the Contractor(s) performing that work activity must prepare an AHA. AHAs must be developed by the Prime Contractor, subcontractor, or supplier performing the work, and provided for Prime

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Contractor review and approval before submitting to the Contracting Officer. AHAs must be signed by the SSHO, Superintendent, QC Manager and the subcontractor Foreman performing the work. Format the AHA in accordance with ENG Form 6206 (Activity Hazard Analysis (AHA)) or as directed by the Contracting Officer. Submit the AHA for review at least 30 calendar days prior to the start of each activity task, or DFO. The Government reserves the right to require the Contractor to revise and resubmit the AHA if it fails to effectively identify the work sequences, specific anticipated hazards, site conditions, equipment, materials, personnel and the control measures to be implemented.

AHAs must identify competent persons required for phases involving high risk activities, including confined entry, crane and rigging, excavations, trenching, electrical work, fall protection, and scaffolding.

1.9.1 AHA Management

Review the AHA list periodically (at least monthly) at the Contractor supervisory safety meeting, and update as necessary when procedures, scheduling, or hazards change. Use the AHA during daily inspections by the SSHO to ensure the implementation and effectiveness of the required safety and health controls for that work activity.

1.9.2 AHA Signature Log

Each employee performing work as part of an activity, task or DFO must review the AHA for that work and sign a signature log specifically maintained for that AHA prior to starting work on that activity. The SSHO must maintain a signature log on site for every AHA. Provide employees whose primary language is other than English, with an interpreter to ensure a clear understanding of the AHA and its contents.

1.10 DISPLAY OF SAFETY INFORMATION

1.10.1 Safety Bulletin Board

Within seven calendar days after commencement of contract work, the contractor must erect a safety bulletin board at the site of work. Where site, duration, or logistics of a project do not facilitate a bulletin board, an alternative Contracting Officer accepted method (for example, binder, digital) may be deemed as meeting the requirement for a bulletin board. Include and maintain information on safety bulletin board as required by [EM 385-1-1](#). Additional items required to be posted include:

- a. [Confined space entry permit](#).
- b. [Hot work permit](#).

1.10.2 Safety and Occupational Health (SOH) Deficiency Tracking System

Establish a SOH deficiency tracking system that lists and monitors the status of SOH deficiencies in chronological order. Use the tracking system to evaluate the effectiveness of the APP. A monthly evaluation of the data must be discussed in the QC or SOH meeting with everyone on the project. The list must be posted on the project bulletin board and updated daily, and provide the following information:

- a. Date deficiency identified;

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- b. Description of deficiency;
- c. Name of person responsible for correcting deficiency;
- d. Projected resolution date;
- e. Date actually resolved.

1.11 SITE SAFETY REFERENCE MATERIALS

Maintain safety-related references applicable to the project, including those listed in paragraph REFERENCES. Maintain applicable equipment manufacturer's manuals.

1.12 EMERGENCY MEDICAL TREATMENT

Contractors must arrange for their own emergency medical treatment in accordance with EM 385-1-1. Government has no responsibility to provide emergency medical treatment.

1.13 NOTIFICATIONS AND REPORTS

1.13.1 Accident and Near Miss Notification

Notify the Contracting Officer as soon as practical, but no later than the stated timelines included in EM 385-1-1, Table 2-1 Accident Reporting Required Timeline. The Contractor is responsible for obtaining appropriate medical and emergency assistance and for notifying fire, law enforcement, and regulatory agencies. Immediate reporting is required for electrical accidents, to include Arc Flash; shock; uncontrolled release of hazardous energy (includes electrical and non-electrical); load handling equipment or rigging; fall from height (any level other than same surface); and underwater diving. All accidents and near misses must be investigated, analyzed, and reported in accordance with EM 385-1-1.

Within notification include Contractor name; Contract title; type of Contract; name of activity, installation or location where accident occurred; date and time of accident; names of personnel injured; extent of property damage, if any; extent of injury, if known, and brief description of accident (for example, type of construction equipment used and PPE used). Preserve the conditions and evidence on the accident site until the Government investigation team arrives on-site and Government investigation is conducted. Assist and cooperate fully with the Government's investigation(s) of any accident or near miss.

1.13.2 Reports

The reporting and associated investigation of accidents and near misses is considered a leading indicator. Utilize the ENG Form 3394 (U.S. Army Corps of Engineers Mishap Notification and Investigation) to report accidents and near misses.

1.13.3 LHE Inspection Reports

Submit LHE inspection reports required in accordance with EM 385-1-1 and as specified herein with Daily Reports of Inspections.

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1.13.4 **Certificate of Compliance** and Pre-lift Plan/Checklist for LHE and Rigging

Provide a Certificate of Compliance (CoC) for Load Handling Equipment (LHE) and Rigging (ENG Form 6209) for LHE entering an activity under this Contract and in accordance with **EM 385-1-1**. Post certifications on the crane.

Develop a Standard Lift Plan (SLP) in accordance with **EM 385-1-1** and paragraph STANDARD LIFT PLAN (SLP) above for each lift planned. Submit SLP to the Contracting Officer for approval within 30 calendar days in advance of planned lift.

1.14 HOT WORK

1.14.1 Permit and Personnel Requirements

Submit and obtain a written permit prior to performing "Hot Work" (i.e. welding or cutting) or operating other flame-producing/spark producing devices, from the local fire department. A permit is required from the Explosives Safety Office for work in and around where explosives are processed, stored, or handled. CONTRACTORS ARE REQUIRED TO MEET ALL CRITERIA BEFORE A PERMIT IS ISSUED. Provide at least two **20 pound** 4A:20 BC rated extinguishers for normal "Hot Work". The extinguishers must be current inspection tagged, and contain an approved safety pin and tamper resistant seal. It is also mandatory to have a designated FIRE WATCH for any "Hot Work" done at this activity. The Fire Watch must be trained in accordance with **NFPA 51B** and remain on-site for a minimum of thirty (30) minutes after the exposure has ended or as determined by the Contractor and documented in the AHA.

When starting work in the facility, require personnel to familiarize themselves with the location of the nearest fire alarm boxes and place in memory the emergency fire department phone number. REPORT ANY FIRE, NO MATTER HOW SMALL, TO THE RESPONSIBLE FIRE DEPARTMENT IMMEDIATELY.

1.14.2 Work Around Flammable Materials

Obtain permit approval from a NFPA Certified Marine Chemist, or Certified Industrial Hygienist for "HOT WORK" within or around flammable materials (such as fuel systems or welding/cutting on fuel pipes) or confined spaces (such as sewer wet wells, manholes, or vaults) that have the potential for flammable or explosive atmospheres.

Whenever these materials, except beryllium and chromium (VI), are encountered in indoor operations, local mechanical exhaust ventilation systems that are sufficient to reduce and maintain personal exposures to within acceptable limits must be used and maintained in accordance with manufacturer's instruction and supplemented by exceptions noted in **EM 385-1-1**.

1.15 RADIATION SAFETY REQUIREMENTS

Submit **License Certificates**, employee training records, and Leak Test Reports for radiation materials and equipment to the Contracting Officer and Radiation Safety Office (RSO) for all specialized and licensed material and equipment proposed for use on the construction project (excludes portable machine sources of ionizing radiation including moisture density and X-Ray Fluorescence (XRF)). Maintain on-site records

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whenever licensed radiological materials or ionizing equipment are on Government property.

Protect workers from radiation exposure in accordance with 10 CFR 20, ensuring any personnel exposures are maintained As Low As Reasonably Achievable.

1.15.1 Radiography Operation Planning Work Sheet

Submit a Gamma and X-Ray Radiography Operation Planning Work Sheet to Contracting Officer 14 days prior to commencement of operations involving radioactive materials or radiation generating devices. For portable machine sources of ionizing radiation, including moisture density and XRF, use and submit the Portable Gauge Operations Planning Worksheet instead. The Contracting Officer will review the submitted worksheet and provide questions and comments.

Contractors must use primary dosimeters process by a National Voluntary Laboratory Accreditation Program (NVLAP) accredited laboratory.

1.15.2 Site Access and Security

Coordinate site access and security requirements with the Contracting Officer for all radiological materials and equipment containing ionizing radiation that are proposed for use on a government facility. For gamma radiography materials and equipment, a Government escort is required for any travels on the Installation. The Government authorized representative will meet the Contractor at a designated location outside the Installation, ensure safety of the materials being transported, and will escort the Contractor for gamma sources onto the Installation, to the job site, and off the Installation. For portable machine sources of ionizing radiation, including moisture density and XRF, the Government authorized representative will meet the Contractor at the job site.

Provide a copy of all calibration records, and utilization records for radiological operations performed on the site.

1.15.3 Loss or Release and Unplanned Personnel Exposure

Loss or release of radioactive materials, and unplanned personnel exposures must be reported immediately to the Contracting Officer, RSO, and Base Security Department Emergency Number.

1.15.4 Site Demarcation and Barricade

Properly demark and barricade an area surrounding radiological operations to preclude personnel entrance, in accordance with EM 385-1-1, Nuclear Regulatory Commission, and Applicable State regulations and license requirements, and in accordance with requirements established in the accepted Radiography Operation Planning Work Sheet.

Do not close or obstruct streets, walks, and other facilities occupied and used by the Government without written permission from the Contracting Officer.

1.15.5 Security of Material and Equipment

Properly secure the radiological material and ionizing radiation equipment at all times, including keeping the devices in a properly marked and

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locked container, and secondarily locking the container to a secure point in the Contractor's vehicle or other approved storage location during transportation and while not in use. While in use, maintain a continuous visual observation on the radiological material and ionizing radiation equipment. In instances where radiography is scheduled near or adjacent to buildings or areas having limited access or one-way doors, make no assumptions as to building occupancy. Where necessary, the Contracting Officer will direct the Contractor to conduct an actual building entry, search, and alert. Where removal of personnel from such a building cannot be accomplished and it is otherwise safe to proceed with the radiography, position a fully instructed employee inside the building or area to prevent exiting while external radiographic operations are in process.

1.15.6 Transportation of Material

Comply with 49 CFR 173 for Transportation of Regulated Amounts of Radioactive Material. Notify Local Fire authorities and the site Radiation Safety Officer (RSO) of any Radioactive Material use.

1.15.7 Schedule for Exposure or Unshielding

Actual exposure of the radiographic film or unshielding the source must not be initiated until after 5 p.m. on weekdays.

1.15.8 Transmitter Requirements

Adhere to the base policy concerning the use of transmitters, such as radios and cell phones. Obey Emissions control (EMCON) restrictions.

1.16 CONFINED SPACE ENTRY REQUIREMENTS

Confined space entry must comply with Chapter 34 of EM 385-1-1, OSHA 29 CFR 1926, OSHA 29 CFR 1910, OSHA 29 CFR 1910.146, and OSHA Directive CPL 2.100. Any potential for a hazard in the confined space requires a permit system to be used.

1.16.1 Entry Procedures

Prohibit entry into a permit required confined space by personnel for any purpose, including hot work, until the qualified person has conducted appropriate tests to ensure the confined or enclosed space is safe for the work intended and that all potential hazards are controlled or eliminated and documented. Comply with EM 385-1-1, Chapter 34 for entry procedures. Hazards pertaining to the space must be reviewed with each employee during review of the AHA.

1.16.2 Ventilation

Use natural or mechanical ventilation controls to provide continuous fresh air into the confined space based on the work that will be performed in accordance with EM 385-1-1.

1.16.3 Sewer Wet Wells

Sewer wet wells require continuous atmosphere monitoring with audible alarm for toxic gas detection.

1.16.4 Rescue Procedures and Coordination with Local Emergency Responders

Develop and implement an on-site rescue and recovery plan and procedures. The rescue plan must not rely on local emergency responders for rescue from a confined space.

1.17 DIVE SAFETY REQUIREMENTS

Develop a [Dive Operations Plan](#), AHA, emergency management plan, and personnel list that includes qualifications, for each separate diving operation. Submit these documents to the District Dive Coordinator (DDC) via the Contracting Officer for review and approval at least 30 working days prior to commencement of diving operations. These documents must be at the diving location at all times. Provide each of these documents as a part of the project file.

1.18 SEVERE STORM PLAN

In the event of a severe storm warning, the Contractor must comply with the applicable Storm Plan and:

- a. Secure outside equipment and materials and place materials that could be damaged in protected areas.
- b. Check surrounding area, including roof, for loose material, equipment, debris, and other objects that could be blown away or against existing facilities.
- c. Ensure that temporary erosion controls are adequate.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 CONSTRUCTION AND OTHER WORK

Comply with [EM 385-1-1](#), [NFPA 70](#), [NFPA 70E](#), [NFPA 241](#), the APP, the AHA, Federal and State OSHA regulations, and other related submittals and activity fire and safety regulations. The most stringent standard prevails.

PPE is governed in all areas by the nature of the work the employee is performing. Use personal hearing protection at all times in designated noise hazardous areas or when performing noise hazardous tasks. Safety glasses must be worn or carried/available on each person. Minimum requirements for PPE are provided in [EM 385-1-1](#).

3.1.1 Worksite Communication

Employees working alone in a remote location or away from other workers must be provided an effective means of emergency communications (i.e., cellular phone, two-way radios, land-line telephones or other acceptable means). The selected communication must be readily available (easily within the immediate reach) of the employee and must be tested prior to the start of work to verify that it effectively operates in the area/environment. Develop an employee check-in/check-out communication procedure to ensure employee safety.

3.1.2 Hazardous Material Exclusions

Notwithstanding any other hazardous material used in this Contract, radioactive materials or instruments capable of producing ionizing/non-ionizing radiation (with the exception of radioactive material and devices used in accordance with EM 385-1-1 such as nuclear density meters for compaction testing and laboratory equipment with radioactive sources) as well as materials which contain asbestos, mercury or polychlorinated biphenyls, di-isocyanates, lead-based paint, and hexavalent chromium, are prohibited. The Contracting Officer, upon written request by the Contractor, may consider exceptions to the use of any of the above excluded materials. Low mercury lamps used within fluorescent lighting fixtures are allowed as an exception without further Contracting Officer approval. Notify the Radiation Safety Officer (RSO) prior to excepted items of radioactive material and devices being brought on base.

3.1.3 Unforeseen Hazardous Material

Contract documents identify materials such as PCB, lead paint, and friable and non-friable asbestos and other OSHA regulated chemicals (i.e. 29 CFR 1910.1000). If material(s) that may be hazardous to human health upon disturbance are encountered during construction operations, stop that portion of work and notify the Contracting Officer immediately. Within 14 calendar days the Government will determine if the material is hazardous. If material is not hazardous or poses no danger, the Government will direct the Contractor to proceed without change. If material is hazardous and handling of the material is necessary to accomplish the work, the Government will issue a modification pursuant to FAR 52.243-4 Changes and FAR 52.236-2 Differing Site Conditions.

3.2 UTILITY OUTAGE REQUIREMENTS

Apply for utility outages in sufficient time as to not cause schedule delays or impacts. At a minimum, the written request must include the location of the outage, utilities being affected, duration of outage, any necessary sketches, and a description of the means to fulfill energy isolation requirements in accordance with EM 385-1-1, Chapter 11 (Isolation). Some examples of energy isolation devices and procedures are highlighted in EM 385-1-1, Chapter 12. Where outages involve Government or Utility personnel, coordinate with the Government on all activities involving the control of hazardous energy.

These activities include, but are not limited to, a review of HEC and HEC procedures, as well as applicable Activity Hazard Analyses (AHAs). In accordance with EM 385-1-1, Chapter 11 and NFPA 70E, work on energized electrical circuits must not be performed without prior Government authorization. Government permission is considered through the permit process and submission of a detailed AHA. Energized work permits are considered only when de-energizing introduces additional or increased hazard or when de-energizing is infeasible.

3.3 OUTAGE COORDINATION MEETING

After the utility outage request is approved and prior to beginning work on the utility system requiring shut-down, conduct a pre-outage coordination meeting in accordance with EM 385-1-1. This meeting must include the Prime Contractor, the Prime and subcontractors performing the work, the Contracting Officer, the Installation representative, and the

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Public Utilities representative. All parties must fully coordinate HEC activities with one another. During the coordination meeting, all parties must discuss and coordinate on the scope of work, HEC procedures (specifically, the lock-out/tag-out procedures for worker and utility protection), the AHA, assurance of trade personnel qualifications, identification of competent persons, and compliance with HEC training in accordance with EM 385-1-1. Clarify when personal protective equipment is required during switching operations, inspection, and verification.

3.4 CONTROL OF HAZARDOUS ENERGY (LOCKOUT/TAGOUT)

Provide and operate a Hazardous Energy Control Program (HECP) in accordance with EM 385-1-1 Chapter 12, 29 CFR 1910.147, 29 CFR 1915.89, ANSI/ASSP A10.44, NFPA 70E, and paragraph HAZARDOUS ENERGY CONTROL PROGRAM (HECP).

3.4.1 Safety Preparatory Inspection Coordination Meeting with the Government or Utility

For electrical distribution equipment that is to be operated by Government or Utility personnel, the Prime Contractor and the subcontractor performing the work must attend the safety preparatory inspection coordination meeting, which will also be attended by the Contracting Officer, and required by EM 385-1-1. The meeting will occur immediately preceding the start of work and following the completion of the outage coordination meeting. Both the safety preparatory inspection coordination meeting and the outage coordination meeting must occur prior to conducting the outage and commencing with lockout/tagout procedures.

3.4.2 Lockout/Tagout Isolation

Where the Government or Utility performs equipment isolation and lockout/tagout, the Contractor must place their own locks and tags on each energy-isolating device and proceed in accordance with the HECP. Before any work begins, both the Contractor and the Government or Utility must perform energy isolation verification testing while wearing required PPE detailed in the Contractor's AHA and required by EM 385-1-1. Install personal protective grounds, with tags, to eliminate the potential for induced voltage in accordance with EM 385-1-1, Chapter 12.

3.4.3 Lockout/Tagout Removal

Upon completion of work, conduct lockout/tagout removal procedure in accordance with the HECP. In accordance with EM 385-1-1, Chapter 12, each lock and tag must be removed from each energy isolating device by the authorized individual or systems operator who applied the device. Provide formal notification to the Government (by completing the Government form if provided by Contracting Officer), confirming that steps of de-energization and lockout/tagout removal procedure have been conducted and certified through inspection and verification. Government or Utility locks and tags used to support the Contractor's work will not be removed until the authorized Government employee receives the formal notification.

3.5 FALL PROTECTION PROGRAM

Establish a fall protection program, for the protection of all employees exposed to fall hazards. Within the program include company policy, identify roles and responsibilities, education and training requirements, fall hazard identification, prevention and control measures, inspection,

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storage, care and maintenance of fall protection equipment and rescue and evacuation procedures in accordance with ANSI/ASSP Z359.2 and EM 385-1-1, Chapter 21.

3.5.1 Training

Institute a fall protection training program. As part of the Fall Protection Program, provide training for each employee who might be exposed to fall hazards and using personal fall protection equipment. Provide training by a competent person for fall protection in accordance with EM 385-1-1. Document training and practical application of the competent person in accordance with EM 385-1-1 and ANSI/ASSP Z359.2 in the AHA.

3.5.2 Fall Protection Equipment and Systems

Enforce use of personal fall protection equipment and systems designated (to include fall arrest, restraint, and positioning) for each specific work activity in the Site Specific Fall Protection and Prevention Plan and AHA at all times when an employee is exposed to a fall hazard. Protect employees from fall hazards as specified in EM 385-1-1, Chapter 21.

Provide personal fall protection equipment, systems, subsystems, and components that comply with EM 385-1-1 Chapter 21, 29 CFR 1910, 29 CFR 1926.500 Subpart M, ANSI/ASSP Z359.0, ANSI/ASSP Z359.1, ANSI/ASSP Z359.2, ANSI/ASSP Z359.3, ANSI/ASSP Z359.4, ANSI/ASSP Z359.6, ANSI/ASSP Z359.7, ANSI/ASSP Z359.11, ANSI/ASSP Z359.12, ANSI/ASSP Z359.13, ANSI/ASSP Z359.14, ANSI/ASSP Z359.15, ANSI/ASSP Z359.16 and ANSI/ASSP Z359.18.

3.5.2.1 Additional Personal Fall Protection Measures

In addition to the required fall protection systems, other protective measures such as safety skiffs, personal floatation devices, and life rings, are required when working above or next to water in accordance with EM 385-1-1. Personal fall protection systems and equipment are required when working from an articulating or extendible boom, swing stages, or suspended platform. In addition, personal fall protection systems are required when operating other equipment such as scissor lifts. The need for tying-off in such equipment is to prevent ejection of the employee from the equipment during raising, lowering, travel, or while performing work.

3.5.2.2 Personal Fall Protection Equipment

Only a full-body harness with a shock-absorbing lanyard or self-retracting lanyard is an acceptable personal fall arrest body support device. The use of body belts is not acceptable. Harnesses must have a fall arrest attachment affixed to the body support (usually a Dorsal D-ring) and specifically designated for attachment to the rest of the system. Snap hooks and carabineers must be self-closing and self-locking, capable of being opened only by at least two consecutive deliberate actions and have a minimum gate strength of 3,600 lbs in all directions. Use webbing, straps, and ropes made of synthetic fiber. The maximum free fall distance when using fall arrest equipment must not exceed 6 feet, unless the proper energy absorbing lanyard is used. Always take into consideration the total fall distance and any swinging of the worker (pendulum-like motion), that can occur during a fall, when attaching a person to a fall arrest system. Equip all full body harnesses with Suspension Trauma Preventers

such as stirrups, relief steps, or similar in order to provide short-term relief from the effects of orthostatic intolerance in accordance with [EM 385-1-1](#).

3.5.3 Fall Protection for Roofing Work

Implement fall protection controls based on the type of roof being constructed and work being performed. Evaluate the roof area to be accessed for its structural integrity including weight-bearing capabilities for the projected loading.

- a. Low Sloped Roofs: Construction roofing activities on low-sloped roofs (less than or equal to 4:12 slope) with unprotected sides and edges 6 feet (1.8 m) or more above lower levels can utilize a Warning Line System (WLS). At minimum, the WLS must be used in combination with a safety monitor system in accordance with [EM 385-1-1](#).
- b. Steep-Sloped Roofs: Work on a roof having a slope greater than 4:12 (vertical to horizontal) requires use of conventional fall protection system (that is, guardrail system, safety net system, or personal fall protection system). This requirement also applies to residential or housing type construction.

3.5.4 Flexible Horizontal Lifelines (HLL)

Provide Flexible HLL in accordance with [EM 385-1-1](#). Commercially manufactured horizontal lifelines (HLL) must be designed, installed, certified and used, under the supervision of a qualified person, for fall protection as part of a complete fall arrest system which maintains a safety factor of 2 ([29 CFR 1926.500](#)). The competent person for fall protection may (if deemed appropriate by the qualified person) supervise the assembly, disassembly, use and inspection of the HLL system under the direction of the qualified person. Locally manufactured HLLs are not acceptable unless they are custom designed for limited or site specific applications by a Registered Professional Engineer who is qualified in designing HLL systems.

3.5.5 Guardrail and Safety Net Systems

Design, install and use guardrail and safety net systems in accordance with [EM 385-1-1](#) and [29 CFR 1926](#) Subpart M.

3.5.6 Rescue Plan and Procedures

When personal fall arrest systems are used, ensure that the mishap victim can self-rescue or can be rescued promptly should a fall occur. Prepare a Rescue Plan and include a detailed discussion of the following: methods of rescue; methods of self-rescue or assisted-rescue; equipment used; training requirement; specialized training for the rescuers; procedures for requesting rescue and medical assistance; and transportation routes to a medical facility. Include the Rescue Plan within the Activity Hazard Analysis (AHA) for the phase of work, in the Fall Protection and Prevention (FP&P) Plan, and the Accident Prevention Plan (APP). The plan must be in accordance with the requirements of [EM 385-1-1](#), [ANSI/ASSP Z359.2](#), and [ANSI/ASSP Z359.4](#).

3.6 WORK PLATFORMS

3.6.1 Scaffolding

Provide employees with a safe means of access to the work area on the scaffold. Climbing of any scaffold braces or supports not specifically designed for access is prohibited. Comply with the following requirements:

- a. Scaffold platforms greater than 20 feet in height must be accessed by use of a scaffold stair system.
- b. Ladders commonly provided by scaffold system manufacturers are prohibited for accessing scaffold platforms greater than 20 feet maximum in height.
- c. An adequate gate is required.
- d. Employees performing scaffold erection and dismantling must be qualified.
- e. Scaffold must be capable of supporting at least four times the maximum intended load, and provide appropriate fall protection as delineated in the accepted fall protection and prevention plan.
- f. Stationary scaffolds must be attached to structural building components to safeguard against tipping forward or backward.
- g. Special care must be given to ensure scaffold systems are not overloaded.
- h. Side brackets used to extend scaffold platforms on self-supported scaffold systems for the storage of material are prohibited. The first tie-in must be at the height equal to 4 times the width of the smallest dimension of the scaffold base.
- i. Scaffolding other than suspended types must bear on base plates upon wood mudsills (2 in x 10 in x 8 in minimum) or other adequate firm foundation.
- j. Scaffold or work platform erectors must have fall protection during the erection and dismantling of scaffolding or work platforms that are more than 6 feet.
- k. Delineate fall protection requirements when working above 6 feet or above dangerous operations in the Fall Protection and Prevention (FP&P) Plan and Activity Hazard Analysis (AHA) for the phase of work.

3.6.2 Elevated Aerial Work Platforms (AWPs)

Use restraint systems or PFAS on all AWPs. Workers must be anchored to the basket or bucket in accordance with manufacturer's specifications and instructions (anchoring to the boom may only be used when allowed by the manufacturer and permitted by the CP). Lanyards, to include adjustable energy absorbing lanyards, used in the restraint system must be sufficiently short to provide fall restraint and prohibit workers from climbing out of or being ejected from the platform. The climbing of rails is prohibited. Lanyards with built-in shock absorbers are acceptable. Self-retracting devices are not acceptable. Tying off to an adjacent pole or structure is not permitted unless a safe device for 100 percent tie-off

is used for the transfer.

Use of AWP's must be operated, inspected, and maintained as specified in the operating manual for the equipment and delineated in the AHA. Operators of AWP's must be designated as qualified operators by the Prime Contractor. Maintain proof of qualifications on site for review and include in the AHA.

3.7 EQUIPMENT

3.7.1 Material Handling Equipment (MHE)

- a. Material handling equipment such as forklifts must not be modified with work platform attachments for supporting employees unless specifically delineated in the manufacturer's printed operating instructions. Material handling equipment fitted with personnel work platform attachments are prohibited from traveling or positioning while personnel are working on the platform.
- b. The use of hooks on equipment for lifting of material must be in accordance with manufacturer's printed instructions. Material Handling Equipment Operators must be trained in accordance with OSHA 29 CFR 1910, Subpart N.
- c. Operators of forklifts or power industrial trucks must be licensed in accordance with OSHA.

3.7.2 Load Handling Equipment (LHE)

The following requirements apply. In exception, these requirements do not apply to commercial truck mounted and articulating boom cranes used solely to deliver material and supplies (not prefabricated components, structural steel, or components of a systems-engineered metal building) where the lift consists of moving materials and supplies from a truck or trailer to the ground; to cranes installed on mechanics trucks that are used solely in the repair of shore-based equipment; to crane that enter the activity but are not used for lifting; nor to other machines not used to lift loads suspended by rigging equipment. However, LHE accidents occurring during such operations must be reported.

- a. Equip cranes and derricks as specified in EM 385-1-1, Chapter 16.
- b. Notify the Contracting Officer 15 working days in advance of any LHE entering the activity so that necessary quality assurance spot checks can be coordinated. Contractor's operator must remain with the crane during the spot check. Rigging gear must be in accordance with OSHA, ASME B30.9 Standards safety standards.
- c. Comply with the LHE manufacturer's specifications and limitations for erection and operation of cranes and hoists used in support of the work. Perform erection under the supervision of a designated person (as defined in ASME B30.5). Perform all testing in accordance with the manufacturer's recommended procedures.
- d. Comply with ASME B30.5 for mobile and locomotive cranes, ASME B30.22 for articulating boom cranes, ASME B30.3 for construction tower cranes, ASME B30.8 for floating cranes and floating derricks, ASME B30.9 for slings, ASME B30.20 for below the hook lifting devices and ASME B30.26 for rigging hardware.

- e. When operating in the vicinity of overhead transmission lines, operators and riggers must be alert to this special hazard and follow the requirements of EM 385-1-1 Chapter 11, and ASME B30.5 or ASME B30.22 as applicable.
- f. Do not use crane suspended personnel work platforms (baskets) unless the Contractor proves that using any other access to the work location would provide a greater hazard to the workers or is impossible. Do not lift personnel with a line hoist or friction crane. Additionally, submit a specific AHA for this work to the Contracting Officer. Ensure the activity and AHA are thoroughly reviewed by all involved personnel.
- g. Inspect, maintain, and recharge portable fire extinguishers as specified in NFPA 10, Standard for Portable Fire Extinguishers.
- h. All employees must keep clear of loads about to be lifted and of suspended loads, except for employees required to handle the load.
- i. Use cribbing when performing lifts on outriggers.
- j. The crane hook/block must be positioned directly over the load. Side loading of the crane is prohibited.
- k. A physical barricade must be positioned to prevent personnel access where accessible areas of the LHE's rotating superstructure poses a risk of striking, pinching or crushing personnel.
- l. Maintain inspection records in accordance by EM 385-1-1, including shift, monthly, and annual inspections, the signature of the person performing the inspection, and the serial number or other identifier of the LHE that was inspected. Records must be available for review by the Contracting Officer.
- m. Maintain written reports of operational and load testing in accordance with EM 385-1-1, listing the load test procedures used along with any repairs or alterations performed on the LHE. Reports must be available for review by the Contracting Officer.
- n. Certify that all LHE operators have been trained in proper use of all safety devices (e.g. anti-two block devices).
- o. Take steps to ensure that wind speed does not contribute to loss of control of the load during lifting operations. At wind speeds greater than 20 mph, the operator, rigger and lift director must cease all crane operations, evaluate conditions and determine if the lift may proceed. Base the determination to proceed or not on wind calculations per the manufacturer and a reduction in LHE rated capacity if applicable. Include this maximum wind speed determination as part of the activity hazard analysis plan for that operation.
- p. Follow FAA guidelines when required based on project location.

3.7.3 Machinery and Mechanized Equipment

- a. Proof of qualifications for operator must be kept on the project site for review.

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- b. Manufacture specifications or owner's manual for the equipment must be on-site and reviewed for additional safety precautions or requirements that are sometimes not identified by OSHA or USACE EM 385-1-1. Incorporate such additional safety precautions or requirements into the AHAs.

3.7.4 Base Mounted Drum Hoists

- a. Operation of base mounted drum hoists must be in accordance with EM 385-1-1 and ANSI/ASSP A10.22.
- b. Rigging gear must be in accordance with applicable ASME/OSHA standards.
- c. When used on telecommunication towers, base mounted drum hoists must be in accordance with TIA-1019, TIA-222, ASME B30.7, 29 CFR 1926.552, and 29 CFR 1926.553.
- d. When used to hoist personnel, the AHA must include a written standard operating procedure. Operators must have a physical examination in accordance with EM 385-1-1 and trained, at a minimum, in accordance with EM 385-1-1. The base mounted drum hoist must also comply with OSHA Instruction CPL 02-01-056 and ASME B30.23.
- e. Material and personnel must not be hoisted simultaneously.
- f. Personnel cage must be marked with the capacity (in number of persons) and load limit in pounds.
- g. Construction equipment must not be used for hoisting material or personnel or with trolley/tag lines. Construction equipment may be used for towing and assisting with anchoring guy lines.

3.7.5 Use of Explosives

Explosives must not be used or brought to the project site without prior written approval from the Contracting Officer. Such approval does not relieve the Contractor of responsibility for injury to persons or for damage to property due to blasting operations.

Storage of explosives, when permitted on Government property, must be only where directed and in approved storage facilities. These facilities must be kept locked at all times except for inspection, delivery, and withdrawal of explosives.

3.8 EXCAVATIONS

Soil classification must be performed by a competent person in accordance with 29 CFR 1926 and EM 385-1-1.

3.8.1 Utility Locations

Provide a third party, independent, private utility locating company to positively identify underground utilities in the work area in addition to any station locating service and coordinated with the station utility department.

3.8.2 Utility Location Verification

Physically verify underground utility locations, including utility depth,

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by hand digging using wood or fiberglass handled tools when any adjacent construction work is expected to come within 3 feet of the underground system.

3.8.3 Utilities Within and Under Concrete, Bituminous Asphalt, and Other Impervious Surfaces

Utilities located within and under concrete slabs or pier structures, bridges, parking areas, and the like, are extremely difficult to identify. Whenever Contract work involves chipping, saw cutting, or core drilling through concrete, bituminous asphalt or other impervious surfaces, the existing utility location must be coordinated with station utility departments in addition to location and depth verification by a third party, independent, private locating company. The third party, independent, private locating company must locate utility depth by use of Ground Penetrating Radar (GPR), X-ray, bore scope, or ultrasound prior to the start of demolition and construction. Outages to isolate utility systems must be used in circumstances where utilities are unable to be positively identified. The use of historical drawings does not alleviate the Contractor from meeting this requirement.

3.9 ELECTRICAL

Perform electrical work in accordance with EM 385-1-1, Chapters 11 and 12.

3.9.1 Conduct of Electrical Work

As delineated in EM 385-1-1, electrical work is to be conducted in a de-energized state unless there is no alternative method for accomplishing the work. In those cases obtain an energized work permit from the Contracting Officer. The energized work permit application must be accompanied by the AHA and a summary of why the equipment/circuit needs to be worked energized. Underground electrical spaces must be certified safe for entry before entering to conduct work. Cables that will be cut must be positively identified and de-energized prior to performing each cut. Attach temporary grounds in accordance with ASTM F855 and IEEE 1048. Perform all high voltage cable cutting remotely using hydraulic cutting tool. When racking in or live switching of circuit breakers, no additional person other than the switch operator is allowed in the space during the actual operation. Plan so that work near energized parts is minimized to the fullest extent possible. Use of electrical outages clear of any energized electrical sources is the preferred method.

When working in energized substations, only qualified electrical workers are permitted to enter. When work requires work near energized circuits as defined by NFPA 70, high voltage personnel must use personal protective equipment that includes, as a minimum, electrical hard hat, safety shoes, insulating gloves and electrical arc flash protection for personnel as required by NFPA 70E. Insulating blankets, hearing protection, and switching suits may also be required, depending on the specific job and as delineated in the Contractor's AHA. Ensure that each employee is familiar with and complies with these procedures, 29 CFR 1910 Subpart S, and 29 CFR 1910.147.

3.9.2 Qualifications

Electrical work must be performed by QP with verifiable credentials who are familiar with applicable code requirements. Verifiable credentials consist of State, National and Local Certifications or Licenses that a

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Master or Journeyman Electrician may hold, depending on work being performed, and must be identified in the appropriate AHA. Journeyman/Apprentice ratio must be in accordance with State, Local requirements applicable to where work is being performed.

3.9.3 Arc Flash

Conduct a hazard analysis/arc flash hazard analysis whenever work on or near energized parts greater than 50 volts is necessary, in accordance with NFPA 70E.

All personnel entering the identified arc flash protection boundary must be QPs and properly trained in NFPA 70E requirements and procedures. Unless permitted by NFPA 70E, no Unqualified Person is permitted to approach nearer than the Limited Approach Boundary of energized conductors and circuit parts. Training must be administered by an electrically qualified source and documented.

3.9.4 Grounding

Ground electrical circuits, equipment and enclosures in accordance with NFPA 70 and IEEE C2 to provide a permanent, continuous and effective path to ground unless otherwise noted by EM 385-1-1.

Check grounding circuits to ensure that the circuit between the ground and a grounded power conductor has a resistance low enough to permit sufficient current flow to allow the fuse or circuit breaker to interrupt the current.

3.9.5 Testing

Temporary electrical distribution systems and devices must be inspected, tested and found acceptable for Ground-Fault Circuit Interrupter (GFCI) protection, polarity, ground continuity, and ground resistance before initial use, before use after modification and at least monthly. Monthly inspections and tests must be maintained for each temporary electrical distribution system, and signed by the electrical CP or QP.

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SECTION 01 45 00

RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE (RMS CM)

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this section to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety and Occupational Health
(SOH) Requirements

1.2 MEASUREMENT AND PAYMENT

The work of this section is not measured for payment. The Contractor is responsible for the work of this section, without any direct compensation other than the payment received for contract items.

1.3 CONTRACT ADMINISTRATION

The Government will use the Resident Management System (RMS) to assist in its monitoring and administration of this contract. The Government accesses the system using the Government Mode of RMS (RMS GM) and the Contractor accesses the system using the Contractor Mode (RMS CM). The term RMS will be used in the remainder of this section for both RMS GM and RMS CM. The joint Government-Contractor use of RMS facilitates electronic exchange of information and overall management of the contract. The Contractor accesses RMS to record, maintain, input, track, and electronically share information with the Government throughout the contract period in the following areas:

- Administration
- Finances
- Quality Control
- Submittal Monitoring
- Scheduling
- Closeout
- Import/Export of Data

1.3.1 Correspondence and Electronic Communications

For ease and speed of communications, exchange correspondence and other documents in electronic format to the maximum extent feasible. Some documents, such as correspondence, including pay requests and payrolls, may be required in paper format with original signatures. Paper documents will govern in the event of discrepancy with the electronic version. Contracting Officer will determine which documents require submission of paper copies.

1.3.2 Other Factors

Other portions of this document have a direct relationship to the reporting accomplished through RMS. Particular attention is directed to FAR 52.236-15 Schedules for Construction Contracts; FAR 52.232-27 Prompt Payment for Construction Contracts; FAR 52.232-5 Payments Under Fixed-Price Construction Contracts; Section 01 32 01 PROJECT SCHEDULE; Section 01 33 00 SUBMITTAL PROCEDURES; Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS; and Section 01 45 05 DREDGING/BEACH FILL PLACEMENT - QUALITY CONTROL.

1.4 RMS SOFTWARE

RMS is a web based application. Download, install and be able to utilize the latest version of RMS within 7 calendar days of receipt of the Notice to Proceed. RMS software, user manuals, access and installation instructions, program updates and training information are available from the RMS website (<https://rms.usace.army.mil>). The Government and the Contractor will have different access authorities to the same contract database through RMS. The common database will be updated automatically each time a user finalizes an entry or change.

1.5 CONTRACT DATABASE - GOVERNMENT

The Government will enter the basic contract award data in RMS prior to granting the Contractor access. The Government entries into RMS will generally be related to submittal reviews, correspondence status, and Quality Assurance(QA)comments, as well as other miscellaneous administrative information.

1.6 CONTRACT DATABASE - CONTRACTOR

Contractor entries into RMS establish, maintain, and update data throughout the duration of the contract. Contractor entries generally include prime and subcontractor information, daily reports, submittals, RFI's, schedule updates and payment requests. RMS includes the ability to import attachments and export reports in many of the modules, including submittals. The Contractor responsibilities for entries in RMS typically include the following items:

1.6.1 Administration

1.6.1.1 Contractor Information

Enter all current Contractor administrative data and information into RMS within 7 calendar days of receiving access to the contract in RMS. This includes, but is not limited to, Contractor's name, address, telephone numbers, management staff, and other required items.

1.6.1.2 Subcontractor Information

Enter all missing subcontractor administrative data and information into RMS CM within seven calendar days of receiving access to the Contract in RMS or within seven calendar days of the signing of the subcontractor agreement for agreements signed at a later date. This includes name, trade, address, phone numbers, and other required information for all subcontractors. A subcontractor is listed separately for each trade to be performed.

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1.6.1.3 Correspondence

Identify all Contractor correspondence to the Government with a serial number. Prefix correspondence initiated by the Contractor's site office with "S". Prefix letters initiated by the Contractor's home (main) office with "H". Letters are numbered starting from 0001. (e.g., H-0001 or S-0001). The Government's letters to the Contractor will be prefixed with "C" or "RFP".

1.6.1.4 Equipment

Enter and maintain a current list of equipment planned for use or being used on the jobsite, including the most recent and planned equipment inspection dates in accordance with Section 01 45 05 DREDGING/BEACH FILL PLACEMENT - QUALITY CONTROL.

1.6.1.5 Reports

Track the status of the project using the reports available in RMS. The value of these reports reflects the quality of the data input. These reports include but are not limited to the Progress Payment Request worksheet, Quality Control (QC) comments, Deficiency Tracking Log, Submittal Register Status, and Three-Phase Control worksheets in accordance with Sections 01 20 00 PRICE AND PAYMENT PROCEDURES, Section 01 32 01 PROJECT SCHEDULE, 01 33 00 SUBMITTAL PROCEDURES, and 01 45 05 DREDGING/BEACH FILL PLACEMENT - QUALITY CONTROL.

1.6.1.6 Request For Information (RFI)

Create and track all Requests For Information (RFI) in the RMS Administration Module for Government review and response. Provide status at weekly progress meeting in accordance with Section 01 32 01 PROJECT SCHEDULE.

1.6.1.7 Payroll Records

Submit and maintain all payroll records in the RMS Administration Module for Government review and acceptance. Payroll records must be submitted at a minimum monthly for the duration of the contract.

1.6.2 Finances

1.6.2.1 Pay Activity Data

Develop and enter a list of pay activities in conjunction with the project schedule in accordance with Section 01 32 01 PROJECT SCHEDULE. The sum of pay activities equals the total Contract amount, including modifications. Each pay activity must be assigned to a Contract Line Item Number (CLIN). The sum of the activities assigned to a CLIN equals the amount of each CLIN.

1.6.2.2 Payment Requests

Prepare all progress payment requests using RMS. Update the work completed under the Contract at least monthly, measured as percent or as specific quantities. After the update, generate a payment request and prompt payment certification using RMS. Submit the signed prompt payment certification and payment request as well as supporting data, either electronically or by hard copy.

1.6.3 Quality Control (QC)

Enter and track implementation of the 3-phase QC Control System, QC testing, transferred and installed property and warranties in RMS. Prepare daily reports, identify and track deficiencies, document progress of work, and support other Contractor QC requirements in RMS. Maintain all data on a daily basis. Ensure that RMS reflects all quality control methods, tests and actions contained within the Contractor Quality Control (CQC) Plan and Government review comments. Provide within seven calendar days of Government acceptance of the CQC Plan.

1.6.3.1 Quality Control (QC) Reports

The Contractor's Quality Control (QC) Daily Report in RMS is the official report. The Contractor can use other supplemental formats to record QC data, but information from any supplemental formats are to be consolidated and entered into the RMS QC Daily Report. Any supplemental information may be entered into RMS as an attachment to the report. QC Daily Reports must be finalized and signed in RMS within 24 hours after the date covered by the report. Provide the Government a printed signed copy of the QC Daily Report, unless waived by the Contracting Officer.

1.6.3.2 Deficiency Tracking

Use the QC Daily Report Module to enter and track deficiencies. Deficiencies identified and entered into RMS by the Contractor or the Government will be sequentially numbered with a QC or QA prefix for tracking purposes. Enter each deficiency into RMS the same day that the deficiency is identified. Monitor, track and resolve all QC and QA entered deficiencies. A deficiency is not considered to be corrected until the Government indicates concurrence in RMS.

1.6.3.3 Three-Phase Control Meetings

Maintain scheduled and actual dates and times of preparatory and initial control meetings in RMS, and upload meeting minutes of the meetings. Worksheets for the three-phase control meetings are generated within RMS.

1.6.3.4 Labor and Equipment Hours

Enter labor and equipment exposure hours on a daily basis. Roll up the labor and equipment exposure data into a monthly exposure report.

1.6.3.5 Accident/Safety Reporting

Both the Contractor and the Government enter safety related comments in RMS as a deficiency. The Contractor must monitor, track and show resolution for safety issues in the QC Daily Report area of the RMS QC Module. In addition, follow all reporting requirements for accidents and incidents as required in [EM 385-1-1](#), Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS and as required by any other applicable Federal, State or local agencies.

1.6.3.6 Definable Features of Work

Enter each feature of work, as defined in the approved CQC Plan, into the RMS QC Module. A feature of work may be associated with a single or multiple pay activities, however a pay activity is only to be linked to a

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single feature of work.

1.6.3.7 Activity Hazard Analysis

Import activity hazard analysis electronic document files into the RMS QC Module utilizing the document package manager. Prepare activity hazard analysis in accordance with Section 01 35 26 GOVERNMENT SAFETY REQUIREMENTS.

1.6.4 Submittal Management

Enter all current submittal register data and information into RMS within seven calendar days of receiving access to the Contract in RMS. Contractor is responsible for entering Activity Numbers for each submittal item corresponding to their Schedule. The Contractor will import the information shown in the submittal register following the specification Section 01 33 00 SUBMITTAL PROCEDURES unless otherwise directed by the Contracting Officer. Group electronic submittal documents into transmittal packages to send to the Government, except very large electronic files, samples, spare parts, mock ups, color boards, or where hard copies are specifically required. Track transmittals and update the submittal register in RMS on a daily basis throughout the duration of the Contract.

Contractor may request submittals to be added to RMS throughout the lifecycle of the project. Approval/disapproval of Contractor requests is at the discretion of the Contracting Officer. Submittals entry into RMS must be updated daily or as practical for performance of the Contract, as approved by the Contracting Officer.

1.6.5 Schedule

Enter and update the contract project schedule in RMS by either manually entering all schedule data or by importing the Standard Data Exchange Format (SDEF) file, based on the requirements in Section 01 32 01 PROJECT SCHEDULE.

1.6.6 Closeout

Closeout documents, processes and forms are managed and tracked in RMS by both the Contractor and the Government. Ensure that all closeout documents are entered, completed and documented within RMS.

1.7 IMPLEMENTATION

Use of RMS as described in the preceding paragraphs is mandatory. Ensure that sufficient resources are available to maintain contract data within the RMS system. RMS is an integral part of the Contractor's required management of quality control.

1.8 NOTIFICATION OF NONCOMPLIANCE

Take corrective action within seven calendar days after receipt of notice of RMS non-compliance by the Contracting Officer.

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PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

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SECTION 01 45 05

DREDGING/BEACH FILL PLACEMENT - QUALITY CONTROL

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM C1077	(2017) Standard Practice for Agencies Testing Concrete and Concrete Aggregates for Use in Construction and Criteria for Testing Agency Evaluation
ASTM D3666	(2016) Standard Specification for Minimum Requirements for Agencies Testing and Inspecting Road and Paving Materials
ASTM D3740	(2023) Minimum Requirements for Agencies Engaged in the Testing and/or Inspection of Soil and Rock as Used in Engineering Design and Construction
ASTM E90	(2009; R2016) Standard Test Method for Laboratory Measurement of Airborne Sound Transmission Loss of Building Partitions and Elements
ASTM E329	(2023) Standard Specification for Agencies Engaged in Construction Inspection, Testing, or Special Inspection
ASTM E543	(2021) Standard Specification for Agencies Performing Non-Destructive Testing

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2024) Safety and Occupational Health (SOH) Requirements
ER 1110-3-12	(2021) Military Engineering and Design Quality Management

1.2 PAYMENT

Separate payment will not be made for providing and maintaining an effective Quality Control program. Include all associated costs in the applicable Line Items and Pricing Schedule item.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.

SD-01 Preconstruction Submittals

Contractor Quality Control (CQC) Plan; G, R0

SD-06 Test Reports

Verification Statement

1.4 GENERAL REQUIREMENTS

Establish and maintain an effective quality control (QC) system that complies with FAR 52.246-12 Inspection of Construction. QC is comprised of plans, procedures, and organization necessary to produce an end product that complies with the Contract requirements. The QC system covers all construction operations, both onsite and offsite, and must be keyed to the proposed construction sequence. The Quality Control Manager, Superintendent, Site Safety and Health Officer (SSHO), and all on-site supervisors are responsible for the quality of work and are subject to removal by the Contracting Officer for non-compliance with the quality requirements specified in the Contract. The Quality Control Manager must maintain a physical presence at the work site at all times and is the primary individual responsible for all quality control.

1.5 QUALITY CONTROL (QC) PROGRAM REQUIREMENTS

Establish and maintain a QC program as described in this section. The QC program consists of a QC Organization, QC Plan, QC Plan Meeting(s), a Coordination and Mutual Understanding Meeting, QC meetings, three phases of control, submittal review and approval, testing, completion inspections, QC certifications, and documentation necessary to provide materials, equipment, workmanship, fabrication, construction and operations that comply with the requirements of this Contract. The QC program must cover on-site and off-site work and be keyed to the work sequence. No construction work or testing may be performed unless the QC Manager is on the work site. The QC Manager must report to an officer of the firm and not be subordinate to the Project Superintendent or the Project Manager. The QC Manager, Project Superintendent and Project Manager must work together effectively. Although the QC Manager is the primary individual responsible for quality control, all individuals will be held responsible for the quality of work on the job.

1.5.1 Meetings

1.5.1.1 Quality Control Plan Meeting

Prior to submission of the QC Plan, the Contractor may request a meeting with the Contracting Officer to discuss the QC Plan requirements of this Contract.

The purpose of this meeting is to develop a mutual understanding of the QC Plan requirements prior to plan development and submission and to agree on

the Contractor's list of Definable Feature of Work (DFOW).

1.5.1.2 Coordination and Mutual Understanding Meeting

After the Preconstruction Conference, before start of construction, and prior to acceptance by the Government of the CQC Plan, meet with the Contracting Officer and discuss the Contractor's quality control system. During the meeting, a mutual understanding of the system details must be developed, including the forms for recording the CQC operations, control activities, testing, administration of the system for both onsite and offsite work, and the interrelationship of Contractor's Management and control with the Government's Quality Assurance. Minutes of the meeting will be prepared by the Government and signed by the Contractor and the Government. Provide a copy of the signed minutes to all attendees and include in the QC Plan. At a minimum the Coordination and Mutual Understanding Meeting must be repeated when a new QC Manager is appointed. There can be other occasions when subsequent conferences will be called by either party to reconfirm mutual understandings or address deficiencies in the CQC system or procedures which can require corrective action by the Contractor.

1.5.1.2.1 Purpose

The purpose of this meeting is to develop a mutual understanding of the QC details, including documentation, administration for on-site and off-site work, design intent, environmental requirements and procedures, coordination of activities to be performed, and the coordination of the Contractor's management, production, and QC personnel. At the meeting, the Contractor must explain in detail how three phases of control will be implemented for each DFOW, as well as how each DFOW will be affected by each management plan or requirement as listed below:

- a. QC reporting and documentation.
- b. Procedures for noise and acoustics management.
- c. Environmental Protection Plan.
- d. Environmental regulatory requirements.
- e. Safety.
- f. Turbidity monitoring.
- g. Plant and equipment location and positioning.
- h. Endangered species monitoring.

1.5.1.2.2 Coordination of Activities

Coordinate activities included in various sections to assure efficient and orderly installation of each component. Coordinate operations included under different sections that are dependent on each other for proper installation and operation.

1.5.1.2.3 Attendees

As a minimum, the Contractor's personnel required to attend include an officer of the firm, the Project Manager, Project Superintendent, QC

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Manager, Alternate QC Manager, QC Specialists, Environmental Manager, and subcontractor representatives. Each subcontractor who will be assigned QC responsibilities must have a principal of the firm at the meeting.

1.5.1.3 Quality Control (QC) Meetings

After the start of construction, conduct weekly QC meetings led by the QC Manager at the work site with the Project Superintendent, the QC Specialists, and the other personnel as necessary. The QC Manager is to prepare the minutes of the meeting and provide a copy to the Contracting Officer within 2 working days after the meeting. The Contracting Officer may attend these meetings. As a minimum, accomplish the following at each meeting:

- a. Review the minutes of the previous meeting.
- b. Review the schedule and the status of work and deficiencies/rework. Review the most current approved schedule (in accordance with schedule specification) and the status of work and deficiencies/rework.
- c. Review the status of submittals and Request For Information (RFIs).
- d. Review the work to be accomplished in the next 3 weeks as defined by the schedule section paragraph WEEKLY PROGRESS MEETINGS in Section 01 32 01 PROJECT SCHEDULE and all documentation required for that work.
- e. Review Testing Plan and Log including status of tests performed since last QC Meeting.
- f. Resolve QC and production problems. Discuss status of pending change orders.
- g. Address items that may require revising the QC Plan.
- h. Review Accident Prevention Plan (APP) and effectiveness of the safety program.
- i. Review environmental requirements and procedures.
- j. Review Environmental Management Plan.
- k. Review Waste Management Plan.
- l. Review the status of training completion.

1.5.2 Contractor Quality Control (CQC) Plan

Submit no later than 30 days after receipt of notice to proceed, the CQC Plan proposed to implement the requirements FAR 52.246-12 Inspection of Construction. The Government will consider an interim plan for the first 30 days of operation. Construction may be permitted to begin only after acceptance of the CQC Plan and other Contract requirements or acceptance of an interim plan applicable to the particular feature of work to be started. Work outside of the accepted interim plan will not be permitted to begin until acceptance of a CQC Plan or another interim plan containing the additional work.

1.5.2.1 Content of Contractor Quality Control (CQC) Plan

Provide a CQC Plan, prior to start of construction that includes a table of contents, with major sections identified, pages numbered sequentially, and that documents the proposed methods and responsibilities for accomplishing quality control during the construction of the project. The CQC Plan must at a minimum include the following sections:

- a. A description of the quality control organization and acknowledgment that the CQC staff will implement the three phase control system for all aspects of the work specified.
- b. An organizational chart showing the quality control organization with individual names and job titles and lines of authority up to an executive of the company at the home office.
- c. NAMES AND QUALIFICATIONS: Names and qualifications, in resume format, (including position titles and durations for qualifying experiences) for each person in the QC organization. Include the Construction Quality Management (CQM) for Contractors course certifications for the QC personnel as required by the paragraph CONSTRUCTION QUALITY MANAGEMENT TRAINING.
- d. DUTIES, RESPONSIBILITY AND AUTHORITY OF QC PERSONNEL: Duties, responsibilities, and authorities of each person in the QC organization.
- e. OUTSIDE ORGANIZATIONS: A listing of outside organizations, such as architectural and consulting engineering firms, that will be employed by the Contractor and a description of the services these firms will provide.
- f. APPOINTMENT LETTERS: Letters signed by an officer of the firm appointing the QC Manager and Alternate QC Manager, and stating that they are responsible for implementing and managing the QC program as described in this Contract. Include in this letter the responsibility of the QC Manager and Alternate QC Manager to implement and manage the three phases of control, and their authority to stop work that is not in compliance with the Contract. Letters of direction are to be issued by the QC Manager to all other QC Specialists or quality control representatives outlining their duties, authorities, and responsibilities. Include copies of the letters in the QC Plan.
- g. SUBMITTAL PROCEDURES AND INITIAL SUBMITTAL REGISTER: Procedures for reviewing, approving, scheduling, and managing submittals, including those of subcontractors, offsite fabricators, suppliers, and purchasing agents. Provide the name(s) of the person(s) in the QC organization authorized to review and certify submittals prior to approval. Provide the initial submittal of the Submittal Register as specified in Section 01 33 00 SUBMITTAL PROCEDURES.
- h. TESTING LABORATORY INFORMATION: Testing laboratory information required by the paragraph ACCREDITATION REQUIREMENTS, as applicable.
- i. TESTING PLAN AND LOG: A Testing Plan and Log that includes the tests required, associated feature of work required, referenced by the specification paragraph number requiring the test, the frequency, and the person responsible for each test.

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- j. Procedures to complete construction deficiencies from identification through acceptable corrective action. Establish verification procedures that identified deficiencies have been corrected. This phase is performed prior to beginning work on each definable feature of work, after all required plans, documents, materials are approved, and after copies are at the work site.
- k. Reporting procedures, including proposed reporting formats.
- l. Procedures for submitting and reviewing design changes/variations prior to submission to the Contracting Officer.
- m. LIST OF DEFINABLE FEATURES: A Definable Feature of Work (DFOW) is a task that is separate and distinct from other tasks and has control requirements and work crews unique to that task. A DFOW is identified by different trades or disciplines, or it is work by the same trade in a different environment. A DFOW is by definition any item or activity on the construction schedule, and the schedule specification provides direction regarding how the DFOWs are to be structured. Include in the list of DFOWs for all activities on the Construction Schedule. Although each section of the specifications can generally be considered as a definable feature of work, there are frequently more than one definable features under a particular section. Identify the specification section number and schedule activity ID for each DFOW listed. The DFOW list will be reviewed in coordination with the construction schedule and agreed upon during the Coordination of Mutual Understanding Meeting.
- n. PROCEDURES FOR PERFORMING AND TRACKING THE THREE PHASES OF CONTROL: Identify procedures used to ensure the three phases of control to manage the quality on this project. For each Definable Feature of Work (DFOW), a Preparatory and Initial phase checklist will be filled out during the Preparatory and Initial phase meetings. Conduct the Preparatory and Initial Phases and meetings with a view towards obtaining quality construction by planning ahead and identifying potential problems for each DFOW.
- o. PROCEDURES FOR COMPLETION INSPECTION: Procedures for identifying and documenting the completion inspection process. Include in these procedures the responsible party for punch out inspection, pre-final inspection, and final acceptance inspection.
- p. TRAINING PROCEDURES AND TRAINING LOG: Procedures for coordinating and documenting the training of personnel required by the Contract.
- q. ORGANIZATION AND PERSONNEL CERTIFICATIONS LOG: Procedures for coordinating, tracking and documenting all certifications required for entities such as subcontractors, testing laboratories, suppliers, and personnel. The QC Manager will ensure that certifications are current, appropriate for the work being performed, and will not lapse during any period of the Contract that the work is being performed.

1.5.3 Acceptance of the Quality Control (QC) Plan

The Contracting Officer's acceptance of the Contractor QC Plan, or interim plan applicable to the particular feature of work to be started, is required prior to the start of construction. The Government may consider an interim plan for the first 30 days of operation. Work outside of the accepted interim plan will not be permitted to begin until acceptance of a

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CQC Plan or another interim plan containing the additional work. The Contracting Officer reserves the right to require changes in the QC Plan and operations as necessary, including removal or addition of personnel, to ensure the specified quality of work. The Contracting Officer reserves the right to interview any member of the QC organization at any time to verify the submitted qualifications. All QC organization personnel are subject to acceptance by the Contracting Officer. The Contracting Officer may require the removal of any individual for non-compliance with quality requirements specified in the Contract.

1.5.4 Notification of Changes

Notify the Contracting Officer, in writing, of any proposed changes in the QC Plan or changes to the QC organization personnel. Proposed changes are subject to acceptance by the Contracting Officer.

1.6 QUALITY CONTROL (QC) ORGANIZATION

1.6.1 Personnel Requirements

The requirements for the CQC organization are a Site Safety and Health Officer (SSHO), QC Manager, and enough qualified personnel to ensure safety and contract quality control for workmanship and materials, including environmental compliance. The SSHO reports directly to a senior project (or corporate) official independent from the QC Manager. The SSHO will also serve as a member of the CQC Staff Personnel identified in the technical provisions as requiring specialized skills to assure the required work is being performed properly. The CQC staff always maintains a presence at the site during progress of the work and have complete authority and responsibility to take any action necessary to ensure Contract compliance. The CQC staff will be subject to acceptance by the Contracting Officer. Provide adequate office space, filing systems and other resources as necessary to maintain an effective and fully functional CQC organization. Promptly complete and furnish all letters, material submittals, shop drawing submittals, schedules, and all other project documentation to the CQC organization. The CQC organization is responsible for always maintaining these documents and records at the site, except as otherwise acceptable to the Contracting Officer.

1.6.2 Quality Control (QC) Manager

1.6.2.1 Duties

Provide a QC Manager at the work site to implement and manage the QC program. The QC Manager must be employed by the Prime Contractor. The only duties and responsibilities of the QC Manager are to manage and implement the QC program on this Contract. The QC Manager must attend the Coordination and Mutual Understanding Meeting, perform the three phases of control, perform submittal review and approval, ensure testing is performed and provide QC certifications and documentation required in this Contract. The QC Manager is responsible for managing and coordinating the three phases of control and documentation performed by the QC Specialists, testing laboratory personnel and any other inspection and testing personnel required by this Contract. The QC Manager is the manager of all QC activities.

1.6.2.2 Qualifications

The QC Manager must be an individual with a minimum of 3 years combined

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experience in the following positions: Project Superintendent, QC Manager, Project Manager, Project Engineer or Construction Manager on similar size and type construction Contracts which included the major trades that are part of this Contract. The individual must be familiar with the requirements of EM 385-1-1 and have experience in the areas of hazard identification, safety compliance, and sustainability.

The QC Manager and all members of the QC organization must be capable of reading, writing, and conversing fluently in the English language.

1.6.2.3 Construction Quality Management Training

In addition to the above experience and education requirements, the QC Manager must have completed the CQM for Contractors course. If the QC Manager does not have a current certification, obtain the CQM for Contractors course certification within 90 days of award. This course is periodically offered by the Naval Facilities Engineering Systems Command and the Army Corps of Engineers. Contact the Contracting Officer for information on the next scheduled class.

The Construction Quality Management Training certificate expires after 5 years. If the QC Manager's certificate has expired, retake the course to remain current.

1.6.3 Organizational Changes

Maintain the QC staff with personnel as required by the specification section at all times. When it is necessary to make changes to the QC staff, revise the CQC Plan to reflect the changes and submit the changes to the Contracting Officer for acceptance.

1.6.4 Alternate Quality Control (QC) Manager Duties and Qualifications

Designate an alternate for the QC Manager at the work site to serve in the event of the designated QC Manager's absence. The qualification requirements for the Alternate QC Manager must be the same as for the QC Manager.

1.6.5 Assistant Quality Control (QC) Manager Duties and Qualifications

Provide a full-time assistant to the QC Manager at the work site to perform the three phases of control, perform submittal review, ensure testing is performed, and prepare QC certifications and documentation required by this Contract. The Assistant QC Manager must be on the work site during supplemental work shifts and perform the duties of the QC Manager during such supplemental shift work. The qualification requirements for the Assistant QC Manager must be the same as the QC Manager, unless otherwise approved by the Contracting Officer. The individual must be familiar with the requirements of EM 385-1-1 and have experience in the areas of hazard identification and safety compliance. The Assistant QC Manager must be capable of reading, writing, and conversing fluently in the English language.

1.6.6 Quality Control (QC) Specialists

Provide a separate QC Specialist at the work site for each of the areas as listed in the Matrix listed below, who must assist and report to the QC Manager and who may perform production related duties but must be allowed sufficient time to perform their assigned quality control duties. These

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individuals or specialized technical companies are employees of the Prime or subcontractor. QC Specialists must be physically present at the work site with frequency as indicated in the Experience Matrix below, to participate in the QC Meetings, perform the three phases of control, including participation in Preparatory and Initial Phase meetings, and to perform and document Follow-up inspections as an extension of the QC Manager for each definable feature of work in their area of responsibility. QC Specialist must assist and be present for training events, and Critical System Acceptance inspections by the Government. Qualification, experience, Area of Responsibility, and frequency of QC surveillance are provided in Matrix listed herein.

Experience Matrix		
1. Area	2-1. Qualification 2-2. Experience	3-1. Area of Responsibility 3-2. Frequency
Surveyor	Persons currently working under direct supervision of Professional Surveyor and Mapper (PSM).	Note: See paragraph Professional Surveyor and Mapper (PSM).

1.6.6.1 Professional Surveyor and Mapper (PSM)

Contract requires survey quality control, quantity surveys for progress payments, and as-built surveys to be certified by a PSM registered in the State the work is being performed. This person must directly supervise contractor surveys in accordance with FAR Clause 52.236-16, QUANTITY SURVEYS. PSM must certify field notes, computations, and other records relating to Quantity or As-Built surveys and layout of work.

1.7 SUBMITTAL AND DELIVERABLES REVIEW AND APPROVAL

Procedures for submission, review and approval of submittals are described in Section 01 33 00 SUBMITTAL PROCEDURES. Procedures must include field verification of relevant dimensions and component characteristics by the QC organization prior to submittal being sent to the Contracting Officer. The CQC organization is responsible for certifying that all submittals and deliverables are in compliance with the Contract.

1.8 THREE PHASES OF CONTROL

CQC enables the Contractor to ensure that the construction, including that of subcontractors and suppliers, complies with the requirements of the Contract. At least three phases of control must be conducted by the QC Manager to adequately cover both on-site and off-site work for each definable feature of the construction work as follows:

1.8.1 Preparatory Phase

Document the results of the preparatory phase actions by separate minutes prepared by the QC Manager and attach to the daily CQC report. Instruct applicable workers as to the acceptable level of workmanship required to meet Contract specifications.

Notify the Contracting Officer at least 2 business days in advance of each preparatory phase meeting. The meeting will be conducted by the QC

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Manager and attended by the QC Specialists, the Project Superintendent, SSHO, and the foreman responsible for the DFOV. When the DFOV will be accomplished by a subcontractor, that subcontractor's foreman must attend the preparatory phase meeting. This phase is performed prior to beginning work on each definable feature of work, after all required plans/documents/materials are approved/accepted, and after copies are at the work site. Perform the following prior to beginning work on each DFOV:

- a. Review each paragraph of the applicable specification sections, reference codes, and standards. Make available during the preparatory inspection a copy of those sections of referenced codes and standards applicable to that portion of the work to be accomplished in the field. Maintain and make available in the field for use by Government personnel until final acceptance of the work.
- b. Review the Contract drawings.
- c. Verify that field measurements are as indicated on construction or shop drawings or both before confirming product orders, to minimize waste due to excessive materials.
- d. Verify that appropriate shop drawings and submittals for materials and equipment have been submitted and approved.
- e. Review the testing plan and ensure that provisions have been made to provide the required QC testing.
- f. Examine the work area to ensure that the required preliminary work has been completed and complies with the Contract and ensure any deficiencies/rework items in the preliminary work have been corrected and confirmed by the Contracting Officer.
- g. Review coordination of equipment/material delivery to designated prepared areas to execute the work.
- h. Examine the required materials, equipment and sample work to ensure that they are on hand and conform to the approved shop drawings and submitted data and are properly stored.
- i. Check to assure that all materials and equipment have been inspected, tested, submitted, and approved.
- j. Discuss specific controls to be used, construction methods, construction tolerances, workmanship standards, and the approach that will be used to provide quality construction by planning ahead and identifying potential problems for each DFOV. Ensure any portion of the plan requiring separate Contracting Officer acceptance has been approved.
- k. Review the APP and appropriate Activity Hazard Analysis (AHA) to ensure that applicable safety requirements are met, and that required Safety Data Sheets (SDS) are submitted.
- l. Discussion of procedures for controlling quality of the work including repetitive deficiencies. Document construction tolerances and workmanship standards for that feature of work.
- m. Discuss schedule and execution of the initial control phase and confirmation or construction quality compliance.

n. Before and After Dredge Survey Requirements and Procedures.

1.8.2 Initial Phase

Notify the Contracting Officer at least 2 business days in advance of each initial phase. When construction crews are ready to start work on a DFOW, conduct the initial phase with the QC Specialists, the Project Superintendent, SSHO, and the foreman responsible for that DFOW. Observe the initial segment of the DFOW to ensure that the work complies with Contract requirements. Document the results of the initial phase in the daily CQC Report and in the Initial Phase Checklist. Repeat the initial phase for each new crew to work on-site when acceptable levels of specified quality are not being met. Indicate the exact location of initial phase for definable feature of work for future reference and comparison with follow-up phases. Perform the following for each DFOW:

- a. Check work to ensure that it is in full compliance with Contract requirements. Review minutes of the preparatory meeting.
- b. Verify adequacy of controls to ensure full Contract compliance. Verify required control inspection and testing comply with the Contract.
- c. Establish level of workmanship and verify that it meets the minimum acceptable workmanship standards. Compare with required sample panels as appropriate.
- d. Resolve any issues with environmental compliance observations and testing, quantity survey, and/or control.
- e. Ensure that testing is performed by the approved laboratory.
- f. Check work procedures for compliance with the APP and the appropriate AHA to ensure that applicable safety requirements are met.
- g. Review project specific work plans to ensure all preparatory work items have been completed and documented.

1.8.3 Follow-Up Phase

Perform the following for on-going DFOW daily, or more frequently as necessary, until the completion of each DFOW. The Final Follow-Up for any DFOW will clearly note in the daily report the DFOW is completed, and all deficiencies/rework items have been completed in accordance with the paragraph DEFICIENCY/REWORK ITEMS LIST. Each DFOW that has completed the Initial Phase and has not completed the Final Follow-up must be included on each daily report. If no work was performed on that DFOW for the period of that daily report, it must be so noted. Document all Follow-Up activities for DFOWs in the daily CQC Report:

- a. Ensure the work including control testing complies with Contract requirements until completion of that particular work feature. Record checks in the CQC documentation.
- b. Maintain the quality of workmanship required.
- c. Ensure that testing is performed by the approved laboratory.

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- d. Ensure that deficiencies/rework items are being corrected. Conduct final follow-up checks and correct all deficiencies prior to the start of additional features of work which may be affected by the deficient work.
- e. Do not build upon nor conceal non-conforming work.
- f. Assure manufacturers' representatives have performed necessary inspections if required and perform safety inspections.

1.8.4 Additional Preparatory and Initial Phases

Conduct additional preparatory and initial phases on the same DFOV if the quality of on-going work is unacceptable, if there are changes in the applicable QC organization, if there are changes in the on-site production supervision or work crew, if work on a DFOV has not started within 45 days of the initial preparatory meeting or has resumed after 45 days of inactivity, or if other problems develop.

1.8.5 Deficiency/Rework Items List

The QC Manager must maintain a list of work that does not comply with the Contract, identifying what items need to be corrected, the activity ID number associated with the item, the date the item was originally discovered, the date the item will be corrected by, and the date the item was corrected.

The list shall be reviewed at each weekly QC Meeting:

- a. There is no requirement to report a deficiency/rework item that is corrected the same day it is discovered.
- b. No successor task may be advanced beyond the preparatory phase meeting until all deficiencies/rework items have been cleared by the QC Manager and concurred with by the Contracting Officer. This must be confirmed as part of the Preparatory Phase activities.
- c. Attach a copy of the "Deficiency/Rework Items List" to the last daily CQC Report of each month.
- d. The Contractor is responsible for including those items identified by the Contracting Officer.
- e. All deficiencies/rework items must be confirmed as corrected by the QC Manager, and concurred by the Contracting Officer, prior to commencement of any completion inspections per paragraph COMPLETION INSPECTIONS unless specifically exempted by the Contracting Officer.
- f. Non-Compliance with these requirements shall be grounds for removal in accordance with paragraph ACCEPTANCE OF THE QUALITY CONTROL (QC) PLAN.
- g. All delays, concurrent or related to failure to manage, monitor, control, and correct deficiencies/rework items are entirely the responsibility of the Contractor and shall not be made the subject, or any component of any request for additional time or compensation.

1.9 TESTING

Perform specified or required tests to verify that control measures are

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adequate to provide a product which conforms to Contract requirements. Upon request, furnish to the Government duplicate samples of test specimens for possible testing by the Government. Testing includes operation and acceptance tests when specified. Procure the services of an U.S. Army Corps of Engineers approved testing laboratory or establish an approved testing laboratory. Perform the following activities and record and provide the following data:

- a. Verify that testing procedures comply with Contract requirements.
- b. Verify that facilities and testing equipment are available and comply with testing standards.
- c. Check test instrument calibration data against certified standards.
- d. Verify that recording forms and test identification control number system, including all test documentation requirements, have been prepared.
- e. Record results of all tests taken, both passing and failing on the CQC report for the date taken. Specification paragraph reference, location where tests were taken, and the sequential control number identifying the test. If approved by the Contracting Officer, actual test reports are submitted later with a reference to the test number and date taken. Provide an information copy of tests performed by an offsite or commercial test facility directly to the Contracting Officer. Failure to submit timely test reports as stated results in nonpayment for related work performed and disapproval of the test facility for this Contract.

1.9.1 Laboratory Accreditation Authorities

All testing laboratories must be validated by the USACE Material Testing Center (MTC) for the tests to be performed. Information on the USACE MTC with web-links to both a list of validated testing laboratories and for the laboratory inspection request for can be found at:

<https://mtc.erdcdren.mil>

1.9.2 Capability Check

The Contracting Officer retains the right to check laboratory equipment in the proposed laboratory and the laboratory technician's testing procedures, techniques, and other items pertinent to testing, for compliance with the standards set forth in this Contract. Laboratories utilized for testing soils, concrete, asphalt, and steel must meet criteria detailed in [ASTM D3740](#) and [ASTM E329](#).

1.9.2.1 Capability Recheck

If the selected laboratory fails the capability check, the Contractor will bear all costs associated with each succeeding recheck of the laboratory or the checking of a subsequent laboratory.

1.9.3 Test Results

Cite applicable Contract requirements, tests or analytical procedures used. Provide actual results and include a statement that the item tested or analyzed conforms or fails to conform to specified requirements. If the item fails to conform, notify the Contracting Officer immediately.

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Conspicuously stamp the cover sheet for each report in large red letters "CONFORMS" or "DOES NOT CONFORM" to the specification requirements, whichever is applicable. Test results must be signed by a testing laboratory representative authorized to sign certified test reports. Furnish the signed reports, certifications, and other documentation to the Contracting Officer via the QC Manager.

1.10 COMPLETION INSPECTIONS

1.10.1 Post-Construction Measurement and Inspection

Near end of work on each acceptance section, notify Contracting Officer to conduct post-construction survey and inspection to ensure contract conformance. Contracting Officer reserves right to witness Contractor performed survey to conduct quality assurance or pay on Contractor conducted survey. Submit certified survey data and correct contract drawings to show as-built information. Spot check grading for berm for beach placement and confirm beach fill is within allowable tolerances. Notify Contracting Officer to schedule a joint inspection of each acceptance section once deficiencies have been corrected.

1.10.2 Pre-Final Inspection

Near completion of project Contracting Officer and Contractor will conduct a Pre-Final Inspection. In addition to completion of quantity survey, Contracting Officer will inspect for demobilization of temporary facilities and cleanup of staging areas used to ensure contract conformance. Contractor and Contracting Officer will inspect for final grading, tilling, and removal of escarpments. Contracting Officer will perform Pre-Final Inspection to verify work is complete and ready for Owner acceptance. Contracting Officer Pre-Final Inspection may result in additional work such as beach tilling to be done. Contractor's CQC System Manager must ensure items are corrected before notifying Contracting Officer so that a Final Inspection with the Customer can be scheduled. Correct any items noted on Pre-Final Inspection in a timely manner. Pre-Final Inspection and deficiency corrections must be accomplished within project completion period including time for establishment of dune vegetation.

1.10.3 Final Acceptance Inspection

Final Inspection will be scheduled by Contracting Officer based upon results of Pre-Final Inspection. Notify the Contracting Officer in writing when ready for Final Inspection and propose a date for the Final Inspection. Include a statement in the notification assuring the Contracting Officer that deficiencies will be corrected and the work will be acceptable by the date proposed for Final Inspection. The Contracting Officer will schedule the Final Inspection within 14 working days of Contractor's notification. The Contracting Officer will notify Owner or Sponsor to attend a Final Inspection. Contractor, Project Manager/Superintendent, and CQC System Manager must attend the Final Inspection. In addition to Contracting Officer and Owner or Sponsor, persons from local government or other agencies may attend. Contractor's failure to have contract work completed for this inspection will be cause for Contracting Officer to bill the Contractor for Contracting Officer's additional inspection cost in accordance with the Clause INSPECTION OF CONSTRUCTION.

1.11 QUALITY CONTROL (QC) CERTIFICATIONS

1.11.1 Contractor Quality Control (CQC) Report Certification

Contain the following statement within the CQC Report: "On behalf of the Contractor, I certify that this report is complete and correct and equipment and material used, and work performed during this reporting period is in compliance with the Contract drawings and specifications to the best of my knowledge, except as noted in this report."

1.12 DOCUMENTATION AND INFORMATION FOR THE CONTRACTING OFFICER

Maintain current and complete records of on-site and off-site QC program operations and activities.

Contact the Contracting Officer for sample forms or print from RMS-QCS as needed. Prior to commencing work on construction, the Contractor must obtain a copy set of the current report forms. The report forms will consist of the Contractor Quality Control (CQC) Report, CQC Report (Continuation Sheet), Contractor Production Report, Contractor Production Report (Continuation Sheet), Preparatory Phase Checklist, Initial Phase Checklist, Testing Plan and Log, and Rework Items List. Unless otherwise provided by the Contracting Officer, Contractor may use the forms provided as related material located at <https://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/ufgs-01-45-00>.

1.12.1 Construction Documentation

Reports are required for each day that work is performed and must be attached to the Contractor Quality Control Report prepared for the same day. Maintain current and complete records of on-site and off-site QC program operations and activities. Reports are required for each day work is performed. Account for each calendar day throughout the life of the Contract.

The Project Superintendent and the QC Manager must prepare and sign the Contractor Production and CQC Reports, respectively. Every space on the forms must be filled in. Use N/A if nothing can be reported in one of the spaces. The reporting of work must be identified by terminology consistent with the construction schedule. In the "Remarks" sections of the reports, enter pertinent information including directions received, problems encountered during construction, work progress and delays, conflicts or errors in the drawings or specifications, field changes, safety hazards encountered, instructions given and corrective actions taken, delays encountered, a record of visitors to the work site, quality control problem areas, deviations from the QC Plan, construction deficiencies encountered, and meetings held. For each entry in the report(s), identify the Schedule Activity No. that is associated with the entered remark.

1.12.2 Quality Control Activities

CQC and Contractor Production reports will be prepared daily to maintain current records providing factual evidence that required quality control activities and tests have been performed. Include in these records the work of subcontractors and suppliers on an acceptable form that includes, as a minimum, the following information:

- a. The name and area of responsibility of the Contractors and any

subcontractors.

- b. Operating plant/equipment with hours worked, idle, or down for repair.
- c. Work performed each day, giving location, description, and by whom. When a Network Analysis Schedule (NAS) is used, identify each item of work performed each day by NAS activity number.
- d. Control phase activities performed. Preparatory, and Initial phase Checklists associated with the DFOV referenced to the construction schedule. Follow-up phase activities identified to the DFOV. If testing or specific QC Specialist activities are associated with the Follow-up phase activities for a specific DFOV note this and include those reports.
- e. Test and control activities performed with results and references to specifications and drawings requirements. Identify the control phase (Preparatory, Initial, Follow-up). List of deficiencies noted, along with corrective action in accordance with the paragraph DEFICIENCY/REWORK ITEMS LIST.
- f. Quantity of materials received at the site with statement as to acceptability, storage, and reference to specifications and drawings requirements.
- g. Submittals and deliverables reviewed, with Contract reference, by whom, and action taken.
- h. Offsite surveillance activities, including actions taken.
- i. Job safety evaluations stating what was checked, results, and instructions or corrective actions.
- j. Instructions given/received and conflicts in plans and specifications.

1.12.3 Verification Statement

Indicate a description of trades working on the project; the number of personnel working; weather conditions encountered; and any delays encountered. Cover both conforming and deficient features and include a statement that equipment and materials incorporated in the work and workmanship comply with the Contract.

Furnish the original and one copy of these records in report form to the Government by 10:00 AM the next working day after the date covered by the report. As a minimum, prepare and submit one report for every 7 days of no work and on the last day of a no work period. All calendar days need to be accounted for throughout the life of the Contract. The first report following a day of no work will be for that day only. Reports need to be signed and dated by the QC Manager. Include copies of test reports and copies of reports prepared by all subordinate quality control personnel within the QC Manager Report.

1.12.4 Reports from the Quality Control (QC) Specialist(s)

Document inspection results on a QC specialist report prepared each day work is performed in their area of responsibility. The report must include a description of the visual inspection or observation performed, a written summary of findings, a conclusion on compliance with the Contract

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documents, and signature of the QC Specialist. In person inspections must be documented with Video/photographs. Video/photographic documentation of deficiencies must include before and after conditions and physical measurements, as necessary. Forward the QC daily report to the QC Manager who must include the report with the submission of their daily QC Report to the Government each day. Every site visit by the QC Specialist must be documented on a QC Specialist daily report.

1.12.5 Quality Control Validation

Establish and maintain the following in an electronic folder. Divide folder into a series of tabbed sections as shown below. Ensure folder is updated at each required progress meeting.

- a. CQC Meeting minutes in accordance with paragraph QUALITY CONTROL (QC) MEETINGS.
- b. All completed Preparatory and Initial Phase Checklists, arranged by specification section, further sorted by DFOW referenced to the construction schedule. Submit each individual Phase Checklist the day the phase event occurs as part of the CQC daily report.
- c. All milestone inspections, arranged by Activity Number referenced to the construction schedule.
- d. An up-to-date copy of the Testing Plan and Log with supporting field test reports, arranged by specification section referenced to the DFOW to which individual reports results are associated. Individual field test reports will be submitted within 2 working days after the test is performed in accordance with the paragraph QUALITY CONTROL ACTIVITIES.
- e. Copies of all Contract modifications, arranged in numerical order. Also include documentation that modified work was accomplished.
- f. An up-to-date copy of the paragraph DEFICIENCY/REWORK ITEMS LIST.
- g. Upon commencement of Completion Inspections of the entire project or any defined portion, maintain up-to-date copies of all punch lists issued by the QC staff to the Contractor and subcontractors and all punch lists issued by the Government in accordance with the paragraph COMPLETION INSPECTIONS.

1.12.6 Testing Plan and Log

As tests are performed, the QC Manager will record on the "Testing Plan and Log" the date the test was performed and the date the test results were forwarded to the Contracting Officer. Attach a copy of the updated "Testing Plan and Log" to the last daily CQC Report of each month. Provide a copy of the final "Testing Plan and Log" to the preparer of the Operation & Maintenance (O&M) documentation.

1.13 NOTIFICATION ON NON-COMPLIANCE

The Contracting Officer will notify the Contractor of any detected non-compliance with the Contract. Take immediate corrective action after receipt of such notice. Such notice, when delivered to the Contractor at the work site, is deemed sufficient for the purpose of notification. If the Contractor fails or refuses to comply promptly, the Contracting Officer may issue an order stopping all or part of the work until

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satisfactory corrective action has been taken. No part of the time lost due to such stop orders will be made the subject of a claim for extension of time for excess costs or damages by the Contractor.

1.14 DELIVERY, STORAGE, AND HANDLING

Designate receiving/storage areas for incoming material to be delivered according to installation schedule and to be placed convenient to work area in order to minimize waste due to excessive materials handling and misapplication. Store and handle materials in a manner as to prevent loss from weather and other damage. Keep materials, products, and accessories covered and off the ground, and store in a dry, secure area. Prevent contact with material that may cause corrosion, discoloration, or staining. Protect all materials and installations from damage by the activities of other trades.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

-- End of Section --

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SECTION 01 50 02

TEMPORARY CONSTRUCTION FACILITIES

PART 1 GENERAL

1.1 SUMMARY

In addition to Construction Facilities this Section covers:

Temporary Utilities
Construction Aids
Vehicular Access and Parking
Project Identification

See Section 01 57 20 ENVIRONMENTAL PROTECTION for requirements including silt control, trailer placement, fueling restrictions, dust control, solid waste, and cleanup. Upon completion of project, clean up and restore area in accordance with Clause CLEANING UP.

a. Construction Facilities include, but are not limited to, the following:

- (1) Contractor Offices
- (2) Information Bulletin Board
- (3) Material and Equipment Storage Area
- (4) Fueling Area
- (5) Secured Storage Area
- (6) Employee Parking Area
- (7) Debris Container (dumpster)
- (8) Construction Signage to include Project Sign; Safety Sign; and, Construction Warning Signs

b. Temporary Utilities include, but are not limited to, the following:

- (1) Water
- (2) Electric
- (3) Sewage
- (4) Communications
 - (i) Telephone
 - (ii) Internet (10 Mbps or higher speed)
- (5) Lighting

1.2 REFERENCES

The publications listed below form a part of this specification to extent referenced. The publications are referred to in text by basic designation only.

FLORIDA DEPARTMENT OF TRANSPORTATION (FDOT)

FDOT Specifications

(2024) Standard Specifications for Road
and Bridge Construction

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INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C2 (2023) National Electrical Safety Code

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023) National Electrical Code

NFPA 241 (2022) Standard for Safeguarding
Construction, Alteration, and Demolition
Operations

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety and Occupational Health
(SOH) Requirements

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.

SD-01 Preconstruction Submittals

Construction Site Plan; G, R0

Mobilization/Demobilization Plan; G, R0

Security Plan; G, R0

Vegetation Protection and Restoration Plan; G, D0

Documentation Of Existing Conditions; G, R0

Haul Road Plan; G, R0

SD-02 Shop Drawings

Temporary Electric Drawings

SD-03 Product Data

Project Identification And Safety Signs; G, R0

SD-11 Closeout Submittals

Documentation Of Post Construction Restoration; G, R0

1.4 PRECONSTRUCTION REQUIREMENTS

Submit the following preconstruction submittal items no later than 30 calendar days after Notice to Proceed.

1.4.1 Construction Site Plan

Prior to the start of work, submit for Government approval a site plan showing the locations and dimensions of temporary facilities (including layouts and details, equipment and material storage area (onsite and

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offsite), and access and haul routes, avenues of ingress/egress to the fenced area and details of the fence installation. Identify any areas which may have to be graveled to prevent the tracking of mud. Indicate if the use of a supplemental or other staging area is desired. Show locations of safety and construction fences, site trailers, construction entrances, trash dumpsters, temporary sanitary facilities, and worker parking areas.

1.4.2 Mobilization/Demobilization Plan

Plan must include, but not be limited to, the following:

a. Mobilization Requirements:

- (1) Methods, equipment and materials
- (2) Connection of utilities
- (3) Placement of site facilities and temporary controls
- (4) Construction of facilities

b. Demobilization Requirements (methods, equipment and materials required to clean-up and restore site at project conclusion):

- (1) Collection, recycle and disposal of solid waste
- (2) Contract-generated material
- (3) Utility disconnection
- (4) Removal of Contractor facilities
- (5) Repair and restoration of site (i.e., fences, roads, or permanent facilities)

1.4.3 Security Plan

Prepare a Security Plan for Contracting Officer describing site security as follows:

- a. Day and night security
- b. Weekend and holiday security
- c. General security duties

1.4.4 Vegetation Protection and Restoration Plan

Prior to clearing for access, submit a Vegetation Protection and Restoration Plan describing the methods to be used for protection of vegetation that is not to be disturbed or otherwise impacted. Include restoration measures for vegetation cleared from access areas and in the event of the accidental damage to, or destruction of, said vegetation.

1.4.5 Documentation of Existing Conditions

Before beginning any mobilization or work, survey the site and photo-document the work area including all staging areas, haul routes, parking areas, and vegetation to show existing limits that could potentially be impacted. Include in the record a description of each picture taken as well as the conditions of roadways, utilities, and other site features, and the location and extent of existing damage prior to starting work. It is the Contractor's responsibility to document all pertinent existing conditions that may affect the subject property.

1.5 EXISTING UTILITIES

Investigate the availability of utilities at the project site. Provide potable water, sanitary sewer, electricity, communications and optional utilities require at the project site. Provide internet services with 10 Mbps minimum speed and wireless capabilities to the Contractor's Field Office complex to include Government office space.

1.5.1 Water

In addition to the above, provide and maintain an adequate supply of water for construction use at no additional cost to the Government. Install and maintain necessary supply connections and piping for same, but only at such locations and in such manner as may be approved by the Contracting Officer. Provide and maintain temporary toilet and washing facilities. Install and maintain toilet and washing facilities in a location approved by the Contracting Officer. Refer to paragraph AVAILABILITY AND USE OF UTILITY SERVICES below.

1.5.2 Electricity

In addition to the above, furnish all electric current at no additional cost to the Government. Install, connect, and maintain all temporary lines in a workmanlike manner satisfactory to the Contracting Officer. Remove temporary lines in like manner at no additional cost to the Government prior to completion of the construction. Refer to paragraph AVAILABILITY AND USE OF UTILITY SERVICES below.

1.6 WORK AND ACCESS DATA

1.6.1 Staging and Access Areas

Obtain and coordinate staging areas and submit for Contracting Officer's approval. Access areas are shown on the contract drawings for the Contractor's use. The Contracting Officer will field-determine the final limits of the access areas indicated on the drawings in coordination with the Local Sponsor and the Contractor. Investigate and obtain any additional areas which may be necessary for construction operations at the Contractor's expense. The additional areas will be subject to the approval of the Contracting Officer.

1.6.2 Contractor Responsibilities

Exclude the public from the work area in the immediate vicinity of his operations including but not limited to fill placement, grading, and transport. Minimize areas closed to the public. Install warning signs to warn the public and all commercial recreational boats of all construction activities. Provide and maintain all water and land access routes necessary for his equipment and plant to and from the work sites. Ascertain the environmental conditions which can affect access, such as climate, terrain, and winds.

1.6.3 Transport of Government Personnel

Provide safe access for the Contracting Officer's personnel and dedicated government personnel at the project site, to all points ashore and afloat, as required by the EM 385-1-1. Provide a dedicated four passenger UTV with an operator dedicated for Government personnel project access which meets the requirements of the EM 385-1-1.

1.7 ADJACENT PROPERTY AND STRUCTURES

Immediately repair any damage to private or public property within the project boundaries, including staging site(s) and work and access areas/roads. Repair any damage as a result of the Contractor's operations at no cost to the Government.

PART 2 PRODUCTS

2.1 TEMPORARY SIGNAGE

2.1.1 Bulletin Board

Prior to commencement of work activities, provide a clear weatherproof covered bulletin board not less than 915 by 1220 mm (36 by 48 inches) in size for displaying the Equal Employment Opportunity poster, a copy of the wage decision contained in the contract, Wage Rate Information poster, WH1090 Paid Sick Leave for Contractors poster (<https://www.dol.gov/whd/regs/compliance/wh1090.pdf>), Safety and Health Information as required by EM 385-1-1 and other information approved by the Contracting Officer. Display legible copies of the aforementioned data until work is completed. Remove the bulletin board upon completion of work. The bulletin board will remain the property of the Contractor. Coordinate requirements herein with Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS.

2.1.2 Project Identification and Safety Signs

Provide signs, their content, and location as indicated on the web site provided in paragraph "Construction Forms and Details" below. Erect the signs within 15 days after receipt of the Notice to Proceed. Correct the data required by the safety sign, with light colored metallic or non-metallic numerals. Upon completion of the project, remove the signs from the site.

2.1.3 Construction Forms and Details

See the CONSTRUCTION FORMS & DETAILS module at the following web address: <https://www.saj.usace.army.mil/About/Divisions-Offices/Engineering/Design-Branch/Specifications-Section/>

2.1.4 Warning Signs

Post temporary signs, tags, and labels to give workers and the public adequate warning and caution of construction hazards according to the EM 385-1-1 Chapter 4. Attach signs to the perimeter fencing every 150 feet or as determined by the Contracting Officer warning the public of the presence of construction hazards. Install signs requiring unauthorized persons to keep out of the construction site. Update the data required by the safety signs daily.

2.1.5 Haul Road Traffic and Construction Warning Signs

Use 1/2 inch HDO plywood, or aluminum sheet, conforming to FDOT Specifications Section 700 Uniform Traffic Manual Standard Signs, non-standard signage high visibility orange with black lettering. Orange fabric mesh is acceptable for temporary survey crew use.

2.2 TEMPORARY TRAFFIC CONTROL

During construction, provide access and temporary relocated roads as necessary to maintain traffic. Maintain and protect traffic on all affected roads during the construction period except as otherwise approved by the Contracting Officer. Install measures for the protection and diversion of traffic, including the provision of watchmen and flagmen, erection of barricades, placing of lights around and in front of equipment and the work, and the erection and maintenance of adequate warning, danger, and direction signs, as required by the State and local authorities having jurisdiction. Protect the traveling public from damage to person and property. Limit the interface between the Contractor's traffic on roads selected for hauling material to and from the site with public traffic. Investigate the adequacy of existing roads and the allowable load limit on these roads. Repair any damage to roads caused by construction operations.

2.2.1 Haul Roads

Construct access and haul roads necessary for proper prosecution of the work under this contract in accordance with [EM 385-1-1](#) Chapter 4. Construct with suitable grades and widths; sharp curves, blind corners, and dangerous cross traffic are to be avoided. Submit [haul road plan](#) for approval. Provide necessary lighting, signs, barricades, and distinctive markings for the safe movement of traffic. Provide a method of dust control to ensure safe operation at all times. Location, grade, width, and alignment of construction, and hauling roads area subject to approval by the Contracting Officer. Provide adequate lighting to assure full and clear visibility for full width of haul road and work areas during any night work operations.

2.2.2 Barricades

Erect and maintain temporary barricades to limit public access to hazardous areas. Whenever safe public access to paved areas such as roads, parking areas or sidewalks is prevented by construction activities or as otherwise necessary to ensure the safety of both pedestrian and vehicular traffic barricades will be required. Securely place barricades clearly visible with adequate illumination to provide sufficient visual warning of the hazard during both day and night.

2.3 TEMPORARY FENCING

Provide fencing along the construction site and at all open excavations and tunnels as indicated in the Contractor's Temporary Facilities shop drawings to control access by unauthorized personnel. Provide highly visible safety fencing to be seen by pedestrians and vehicular traffic. Specific fencing requirements are as described herein. All fencing will meet the requirements of [EM 385-1-1](#).

2.3.1 Polyethylene Mesh Safety Fencing

Temporary safety fencing must be a high visibility orange colored, high density polyethylene grid, a minimum of 48 inches high and maximum mesh size of 2 inches. Extend fencing from the grade to a minimum of 48 inches above the grade and be tightly secured to T-posts spaced as necessary to maintain a rigid and taut fence.

2.3.2 Chain Link Panel Fencing

Temporary panel fencing must be galvanized steel chain link panels 6 feet high. Multiple fencing panels may be linked together at the bases to form long spans as needed. Weight down each panel base using sand bags or other suitable materials in order for the fencing to withstand anticipated winds while remaining upright. Fencing must remain rigid and taut with a minimum of 200 pounds of force exerted on it from any direction with less than 4 inches of deflection.

2.3.3 Post-Driven Chain Link Fencing

Temporary post-driven fencing must be galvanized chain link fencing 6 feet high supported by a tightly secured to galvanized steel posts driven below grade. Locate fence posts on minimum 10 foot centers. Posts may be set in various surfaces such as sand, soil, asphalt or concrete as necessary. Chain link fencing must remain rigid and taut with a minimum of 200 pounds of force exerted on it from any direction with less than 4 inches of deflection. Completely remove fencing and posts at the completion of construction. Restore any surfaces disturbed or damaged to its original condition. Locate and identify underground utilities prior to setting fence posts. Equip fence with a lockable gate. Gate must remain locked when construction personnel are not present.

2.4 TEMPORARY WIRING

Provide temporary wiring in accordance with EM 385-1-1 Chapter 11, NFPA 241 and NFPA 70. Include monthly inspection and testing of all equipment and apparatus.

2.5 STORAGE CONTAINERS

Welded steel construction, locking, shipping containers or equal.

PART 3 EXECUTION

3.1 GENERAL REQUIREMENTS

3.1.1 Identification of Employees

Furnish identification as approved by the Contracting Officer to each employee and require each employee engaged on the work to display them. Immediately deliver prescribed identification to the Contracting Officer for cancellation upon release of any employee. When required, obtain and provide fingerprints of persons employed on the project. Wear identifying markings on hard hats clearly identifying the company for whom the employee works.

3.1.2 Employee Parking

- a. Park employee's vehicles in areas designated by Contractor, away from construction traffic, within reasonable walking distance of site. Maintain area free of ruts, mud holes and puddles. Place gravel where required by deteriorated conditions.
- b. Protect unattended equipment as it may be subject to vandalism.
- c. Storage trailers and storage area with Government material will be locking type with lighting.

3.1.3 Onsite Information

Keep copy of contract drawings, specifications, and other contract documents at Contractor's Office onsite, available for use at all times.

3.1.4 Janitorial and Other Services

- a. Daily -- Sweep and mop floors; empty wastebaskets; service and sanitize bathrooms and other common areas; restock cleaning supplies and toiletries such as, but not limited to, hand soap, hand sanitizer, paper towels, and toilet paper.
- b. Weekly -- Wash floors, windows (inside and out).
- c. Cut grass as needed to keep below 3 inches high.
- d. Pest control for all field offices.

3.2 AVAILABILITY AND USE OF UTILITY SERVICES

Install temporary facilities and utilities in accordance with [IEEE C2](#), [EM 385-1-1](#), [NFPA 70](#). Obtain necessary construction, building, zoning, or soil erosion and sediment control approvals required by local authorities and utility companies. Equip trailer(s) with wind tie downs in accordance with local wind and building code requirements.

3.2.1 Lighting

Electric light, non-glare type luminaries to provide a minimum illumination, level of 30-foot candles at desk height level. Comply with temporary lighting, wiring and Ground Fault Circuit Interrupter (GFCI) requirements found in [EM 385-1-1](#).

3.2.2 Heating and Cooling

Adequate equipment to maintain internal ambient air temperature of 75 degrees F in 105 degrees F and 70 degrees F in 30 degrees F. Equip trailer(s) with smoke alarms. If bottled gas is used to heat, equip trailer(s) with carbon monoxide alarms.

3.2.3 Water

Non-potable well water may be used to flush toilets, and for dish and hand washing. Provide bulk potable water in storage tanks or bottled water for drinking.

3.2.4 Fire Extinguisher

Refer to [EM 385-1-1](#). Non-toxic, dry chemical, fire extinguisher meeting Underwriters Laboratories, Inc., approval for Class A, Class B, and Class C fires with a minimum rating of 2A; 10B; and 10C.

3.2.5 Utility Lines

Install, connect and modify temporary lines as coordinated with owning utility. Conform requirements in accordance with [IEEE C2](#) and [NFPA 70](#) for Temporary Electric Lines. Remove temporary line at completion of project.

3.2.6 Electric Power

Submit [temporary electric drawings](#) including electricity supply and

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lighting signed by licensed electrician- from electrical source to layout locations, fixtures, and materials, to include outdoor lighting. Provide uninterrupted electrical service sufficient to operate office systems. Directly purchase electrical power.

3.3 PROTECTION AND MAINTENANCE OF TRAFFIC

During construction, provide access and temporary relocated roads as necessary to maintain traffic. Maintain and protect traffic on all affected roads during the construction period except as otherwise approved by the Contracting Officer. Provide measures for the protection and diversion of traffic, including the provision of watchmen and flagmen, erection of barricades, placing of lights around and in front of equipment and the work, and the erection and maintenance of adequate warning, danger, and direction signs, as required by the State and local authorities having jurisdiction. Protect the traveling public from damage to person and property. Ensure limited interference between the Contractor's traffic on roads selected for hauling material to and from the site with public traffic. Investigate the adequacy of existing roads and the allowable load limit on these roads. Repair any damage to roads caused by construction operations.

3.4 MAINTENANCE OF HAUL AND ACCESS ROADS

Providing and maintain safe and passable means of ingress and egress to the site and to all project features on the site for Contractor, Government, and third party entities. Requirements herein are applicable to both new and existing haul and access roads. Repair and re-grade haul and access roads as necessary. Maintain roadway drainage, including maintaining roadway crown and profile and keeping drainage ditches clear. Repair potholes, mud holes, ruts, and/or depressions of 6 inches deep or more within 24 hours of discovery. Check condition of haul and access roads at least weekly and after any severe weather event. All requirements of [EM 385-1-1](#) concerning access and haul roads also apply.

3.5 CONTRACTOR'S TEMPORARY FACILITIES

If suitable office space cannot be furnished on board the dredge or other water craft, furnish accommodations ashore as indicated in paragraph ACCOMMODATIONS AND MEALS FOR INSPECTORS below. Staging and storage may also be either on board the dredge or other water craft, or on shore at the Contractor's discretion. For accommodations or staging ashore, Place the field offices, staging areas, stockpile storage, and temporary buildings in areas as approved by the Contracting Officer and in accordance with Section [01 57 20](#) ENVIRONMENTAL PROTECTION. Make temporary movement or relocation of Contractor facilities when approved by the Contracting Officer. Provide a general layout sketch of temporary site facilities including, but not limited to, the following:

- a. Trailer locations
- b. Parking areas
- c. Material storage
- d. Equipment lay down area
- e. Areas for gravel
- f. Fuel areas
- g. Supplemental or other staging area
- h. Temporary well, water supply
- i. Septic field or holding tanks, port-a-lets
- j. Contaminated water handling

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- k. Concrete cleaning area and methods of disposal
- l. Fences -- location and dimensions, entrance and exit points, and details of installation
- m. Utilities (poles, connections and services)
- n. Explosives storage

3.5.1 Contractor Field Office

Provide on-site field office of sufficient size and staff capability to manage project activities. Provide communications and computer capabilities to manage the work, including implementing RMS, facsimile, electronic mail, reproduction, to track schedule and cost. Include a conference area with table and chairs to accommodate a minimum of 8 people. Provide handicap access with a 1 on 12 ramp and gravel parking spot marked with handicap parking sign. Provide high-speed internet services (10 Mbps or higher speed rate) and wireless capabilities to the field office facilities/compound to include Government office space to facilitate RMS, SharePoint (file transfer) and visitor usage.

3.5.1.1 Office Space for Government Personnel

Provide a minimum of 80 square feet air conditioned office space (with locking door) for exclusive use by Government personnel with the following:

- a. Two office desks with lockable drawers (60 inches by 30 inches, laminated top) and two swivel office chairs.
- b. Five drawer, legal size, lockable filing cabinet.
- c. A 1.5 cf microwave oven and 7 cf refrigerator.
- d. A minimum of two designated Government parking spaces per each active staging/access area.
- e. Temporary toilet and washing facilities, for Government personnel, either in or outside of the Government office space. Toilet and washing facilities are not considered part of the minimum 80 square feet office space.

3.5.2 Appearance of Trailer(s)

Maintain trailer(s) used for both office and material storage purposes in clean, neat exterior appearance and in good repair. Trailer(s) requiring exterior painting or maintenance will not be allowed on site until determined satisfactory by Contracting Officer.

3.5.3 Storage Area

Construct a temporary 6 foot high chain link fence around trailers and materials. Fence posts may be driven, in lieu of concrete bases, where soil conditions permit. Do not place or store trailers, materials, or equipment outside the fenced area unless such trailers, materials, or equipment are assigned a separate and distinct storage area by the Contracting Officer away from the vicinity of the construction site but within the installation boundaries. Do not stockpile materials outside the fence in preparation for the next day's work. Park mobile equipment, such as tractors, wheeled lifting equipment, cranes, trucks, and like equipment within the fenced area at the end of each work day.

3.5.3.1 Maintenance of Storage Area

Keep fencing in a state of good repair and proper alignment. Should the Contractor elect to traverse with construction equipment or other vehicles, cover grassed or unpaved areas which are not established roadways, with a layer of gravel as necessary to prevent rutting and the tracking of mud onto paved or established roadways; gravel gradation will be at the Contractor's discretion. Mow grass located within the boundaries of the construction site for the duration of the project. Edge or trim neatly, grass and vegetation along fences, buildings, under trailers, and in areas not accessible to mowers.

3.5.4 Waste Storage

Provide dumpsters or suitable debris containers. Prevent windblown trash; cover as needed. Dispose of offsite when needed. Refer to Section 01 57 20 ENVIRONMENTAL PROTECTION.

3.5.5 Fuel Storage and Fueling Operations

Refer to Section 01 57 20 ENVIRONMENTAL PROTECTION. Provide light when fueling at night.

3.6 SITE SECURITY

Site Security is the Contractor's responsibility. Provide site security person(s) at all times, even when there is not on-site activities. Maintain 24-hour security. Site security must include, but not be limited, to:

- a. Limit vehicular access to authorized vehicles and personnel only.
- b. Limit public access to travel along access road only. Keep access road closed at all times except for passage of authorized personnel and vehicles.
- c. Maintain a list of authorized personnel and subcontractors available at site offices.
- d. Maintain a sign-in log documenting visitors, deliveries, and security incidents. Include date, name, address, company, time in and time out for each employee and visitor.
- e. Check fenced areas, equipment, trailers on a daily basis. If damage is observed or vandalism is found report to Contracting Officer.

3.6.1 Visitors

No visitors will be allowed on site without knowledge of Contractor and permission of Contracting Officer. Direct visitors to report upon arrival to Contractor's Field Office for site safety and accident prevention briefing. Provide visitors appropriate protective equipment (i.e., ear plugs, safety glasses, etc.).

3.7 ACCOMMODATIONS AND MEALS FOR INSPECTORS

- a. Furnish regularly to inspectors, for office purposes, a suitable separate room on board the dredge or other craft upon which they are employed or, if not available thereon. Furnish suitable alternate

accommodations ashore at a location approved by the Contracting Officer, suitable transportation between the alternate accommodations and the dredge or other craft upon which they are employed.

- b. If the Contractor maintains on this work an establishment for the subsistence of the Contractor's own employees, furnish to inspectors employed on the work, and to all Government agents who may visit the work on official business, meals of a quality satisfactory to the Contracting Officer. The meals furnished will be paid for by the Government employees at a rate no greater than the JTR meal rate per person for the county in which work is performed.

3.8 CLEANUP

Remove construction debris, temporary fencing, waste materials, packaging material and the like from the work site daily. Clean away any dirt or mud tracked onto paved or surfaced roadways. Store materials resulting from demolition activities which are salvageable within the fenced area described above or at the supplemental storage area. Neatly stack stored material not in trailers, whether new or salvaged, when stored. Refer to Section 01 57 20 ENVIRONMENTAL PROTECTION for solid waste and post construction cleanup.

3.9 RESTORATION OF STORAGE AREA

Remove the fence upon completion of the project. After removal of trailers, materials, and equipment from within the fenced area, the area will become the property of the Contractor. Restore areas used by the Contractor for the storage of equipment or material, or other use, to the original or better condition. Remove gravel used to traverse grassed areas and restore the area to its original condition, including top soil and seeding as necessary.

3.10 RESTORATION OF STAGING AND ACCESS AREAS

Restore areas used for staging or access to the original or better condition. Replace any fencing, signage, curbing, or any other structural features disturbed by the Contractor's activities. Provide revegetation of access and staging areas with sod or seeding. Include a survival warranty of 90 percent of the plant material for 90 days for revegetation. Install vegetation with fertilization and irrigation, at no additional expense to the Government. Replace shrubs and trees to preexisting conditions in accordance with the requirements of Section 01 57 20 ENVIRONMENTAL PROTECTION.

3.10.1 Documentation of Post Construction Restoration

Following the end of work and demobilization, survey the site and photo-document the post-construction condition of all areas surveyed for the Documentation of Existing Conditions. Include in the record conditions of roadways, utilities, and other site features, and the location and extent of any changes observed since documentation of pre-construction conditions. It is the Contractor's responsibility to document all pertinent post-construction conditions that may affect the subject property.

-- End of Section --

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SECTION 01 57 20

ENVIRONMENTAL PROTECTION

PART 1 GENERAL

1.1 SCOPE

This Section covers prevention of environmental damage as the result of construction operations under this contract and for those measures set forth in other Technical Requirements of these specifications. For the purpose of this specification, environmental damage is defined as the presence of hazardous, physical, chemical, or biological elements or agents which adversely affect human health or welfare; unfavorably alter ecological balances; affect other species, biological communities, or ecosystems; or degrade the quality of the environment for aesthetic, cultural, and/or historical purposes. The control of environmental damage requires consideration of land, water, and air, and includes management of visual aesthetics, noise, solid waste, radiant energy and radioactive materials, as well as other pollutants.

1.2 ENVIRONMENTAL LAWS, REGULATIONS AND DOCUMENTS

1.2.1 Miscellaneous Environmental Laws And Regulations

There are numerous environmental laws and regulations. At the Federal level, the applicable laws and regulations include compliance with the Clean Water Act (CWA); Clean Air Act (CAA); Coastal Zone Management Act (CZMA); Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA); Endangered Species Act (ESA); Fish and Wildlife Coordination Act (FWCA); Marine Protection, Research, and Sanctuaries Act (MPRSA); Magnuson-Stevens Fishery Conservation and Management Act (MSFCMA); National Environmental Policy Act (NEPA); National Historic Preservation Act (NHPA); National Pollutant Discharge Elimination System (NPDES); Research and Sanctuaries Act (RSA); Native American Graves Protection and Repatriation Act (NAGPRA); Resource Conservation and Recovery Act (RCRA); Rivers and Harbors Act (R&H); Safe Drinking Water Act (SDWA); Toxic Substance Control Act (TSCA); Wild and Scenic Rivers Act (WSRA); Federal Insecticide, Fungicide, and Rodenticide Act (FIFRA); Marine Mammal Protection Act (MMPA); Migratory Bird Treaty Act (MBTA); Code of Federal Regulations (CFRs); Executive Orders (EO); and, Environmental Protection Agency (EPA) requirements.

1.2.2 NEPA Reference

Commitments for the Contractor made in the NEPA document have been incorporated into these specifications. For additional background information, the NEPA document can be found at the following website: <https://www.saj.usace.army.mil/About/Divisions-Offices/Planning/Environmental-Branch/Environmental-Documents/>

1.3 REFERENCES

Any publication listed below forms a part of this specification to the extent referenced. Publications are referred to in the text by basic

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designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2024) Safety and Occupational Health (SOH) Requirements
EM 1110-1-1003	(2013) NAVSTAR Global Positioning System Surveying
ER 1110-1-5	(1984) Plant Pest Quarantined Areas and Foreign Soil Samples

Code of Federal Regulations (CFR) indicated herein may be found at
<https://www.govinfo.gov/app/collection/cfr>.

1.4 QUALITY CONTROL

Establish and maintain quality control for environmental protection of all items set forth herein. Record on daily quality control reports or attachments thereto, any problems in complying with laws, regulations and ordinances, and corrective action taken.

1.5 PERSONNEL TRAINING

The Contractor is to train their personnel in all phases of environmental protection.

1.6 PERMITS AND AUTHORIZATIONS

Comply with all requirements under the terms and conditions set out in the following permit(s) and authorization(s) obtained by the Corps of Engineers listed below.

a. Florida Department of Environmental Protection Permit No. 0157891-024-JC; Effective Date: February 15, 2022; Expiration Date: February 15, 2037 and Modifications No. 0157891-025-JM issued on August 23, 2024.

The above permits indicate issuing agency approval of work required by this contract. Permits are available at the following web address:
<https://www.saj.usace.army.mil/About/Divisions-Offices/Planning/Environmental-Branch/Environmental-Compliance/>

Obtain and comply with the requirements of all other permits or licenses required for construction of this project in accordance with the Clause PERMITS AND RESPONSIBILITIES.

1.7 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.

SD-01 Preconstruction Submittals

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Environmental Protection Plan; G, D0

SD-02 Shop Drawings

Turtle Deflector Device; G, D0

SD-06 Test Reports

Protected Listed Species Sighting Reports; G, D0

Weekly Logs/Summary Of FWC Hopper Dredge Reports; G, D0

Weekly Protected Species Summary Report; G, D0

SD-07 Certificates

Bird Monitoring Qualifications; G, D0

Marine Animal Observer Qualifications (Clamshell/Mechanical Only);
G, D0

Endangered Species Observer and Trawl Supervisor Qualifications
(For Hopper Dredges Only); G, D0

SD-11 Closeout Submittals

Protected Species Summary Report; G, D0

Logs/Summary Of Bird Nesting Monitoring Report; G, D0

Comprehensive Trawling Daily Logs; G, D0

Invasive And Nuisance Species Report; G, D0

Project Environmental Summary Sheet; G, D0

Marine Animal Observers Final Logs; G, D0

1.8 ENVIRONMENTAL PROTECTION PLAN

Submit a draft [Environmental Protection Plan](#) no later than 5 calendar days after Notice to Proceed for review and acceptance by the Contracting Officer and cc the Environmental Compliance POC. The Government will consider an interim plan for the first 30 days of operations. However, furnish an acceptable final plan no later than 30 calendar days after receipt of Notice to Proceed. Acceptance of the plan will not relieve the Contractor of their responsibility for adequate and continuing control of pollutants and other environmental protection measures. Acceptance of the plan is conditional and predicated on satisfactory performance during construction. The Government reserves the right to require the Contractor to make changes to the Environmental Protection Plan or operations if the Contracting Officer determines that environmental protection requirements are not being met. No physical work at the site is to begin prior to acceptance of the plan or an interim plan covering the work to be performed.

The EPP is to have a Table of Contents. Pages of the EPP will be numbered, including appendices. Sections of the plan and all appendices are to be indexed in the table of contents. The Environmental Protection Plan is to

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include but not be limited to the following:

- a. A list of Federal, State, and local laws, regulations, and permits concerning environmental protection, pollution control, and abatement that are applicable to the proposed operations and the requirements imposed by those laws, regulations, and permits.
- b. Methods for protection of features to be preserved within authorized work areas. Prepare a listing of methods to protect resources needing protection, i.e., trees, shrubs, vines, grasses and ground cover, landscape features, air and water quality, fish and wildlife, soil, historical, archeological, and cultural resources.
- c. Procedures to be implemented to provide the required environmental protection and to comply with the applicable laws and regulations. Provide written assurance that immediate corrective action will be taken to correct pollution of the environment due to accident, natural causes, or failure to follow the procedures set out in accordance with the environmental protection plan.
- d. A permit or license for and the location of the solid waste disposal area.
- e. Drawings showing locations of any proposed temporary excavations or embankments for haul roads, stream crossing, material storage areas, structures, sanitary facilities, and stockpiles of excess or spoil materials.
- f. Environmental monitoring plans for the job site, including land, water, air, and noise monitoring.
- g. Traffic control plan.
- h. Methods of protecting surface and ground water during construction activities.
- i. Spill prevention. Specify all potentially hazardous substances to be used on the job site and intended actions to prevent accidental or intentional introduction of such materials into the air, ground, water, wetlands, or drainage areas. Specify the provisions to be taken to meet Federal, State, and local laws and regulations regarding labeling, storage, removal, transport, and disposal of potentially hazardous substances. List the Corps environmental POC's with phone numbers and email addresses. Specify that the contractor will notify the Corps Environmental Branch POC's by phone the same day any spill if possible or no later than the morning of the next day (to include weekends).
- j. Spill contingency plan for hazardous, toxic, or petroleum material. There must be a secondary containment (110% of full tank capacity) for any fuel storage tanks (including pumps/generators). This is to include temporary fuel storage above 550 gallons. Any fuel storage container with more than 550 gallon capacity, trailer mounted or otherwise, left on the job site for more than 24 hours must comply with the secondary containment with over head cover unless specifically authorized by the Contracting Officer. Fuel storage tanks must have over head cover to prevent rainwater from entering the secondary containment. The spill response material quantity must be sufficient to absorb/contain the maximum expected fuel storage.

- k. Work area plan showing the proposed activity in each portion of the area and identifying the areas of limited use or nonuse. Include measures for marking the limits of use areas.
- l. Include a statement as to the person who will be responsible for implementation of the Environmental Protection Plan. The Contractor personnel responsible is to report directly to the Contractor's top management and is to have the authority to act for the Contractor in all environmental protection matters.
- m. Recycling and Waste Management Plan. Executive Order 12873 of 20 October 1993 requires a number of considerations in planning a project. Fallen trees should not be burned or buried to the maximum extent practical. Mulching, composting, and other uses for trees should be considered. Also, recovery of metals at the job site, including aluminum cans, should be considered with proceeds to be retained by the Contractor. Non-Federal recycling and waste minimization efforts are to also be incorporated into this plan.
- n. Operational plan to achieve protection of sea turtles during hopper dredge(s) operation.
- o. Migratory Bird Protection Plan. Construction activities are to be conducted in a manner as not to impact migratory birds or induce their nesting.
- p. Steps to be taken to construct the project in such a manner as not to impact gopher tortoises.
- q. Invasive and Nuisance Species Transfer Prevention Plan. The invasive and nuisance species transfer prevention plan is to identify specific transfer prevention procedures and designated cleaning sites/locations. The Contractor is to provide a report with photographs verifying that equipment brought on the site was cleaned. The prevention plan and report are to be submitted to the Invasive Species Management Branch, Operations Division.

1.9 QUALIFICATIONS

The individual(s) are to have a combination of education and experience that demonstrates they have the knowledge and background to identify the species of concern and be able to fulfill the requirements of the environmental specification. Resume/qualification submittals must specifically address the knowledge/experience with the species of concern. Any specific requirements for an environmental observer/monitor called out in environmental coordination or authorization documents (such as agency coordination letters, biological opinions, or permits) regarding the project area must be also addressed in the submittals.

1.9.1 Bird Monitoring Qualifications

No later than 5 calendar days after Notice to Proceed, the Contractor will furnish to the Contracting Officer and cc the Environmental Compliance POC for approval, the qualifications of the bird monitor/observer. The qualifications of the bird monitor will have prior experience and demonstrate the ability to identify bird species, general and nesting behavior characteristics, nests and eggs, and a knowledge of habitat requirements. In addition, references must be provided to verify

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non-educational experience. The Contractor will consult with and coordinate all monitoring plans and activities with the Contracting Officer and the Project Biologist.

1.9.2 Marine Animal Observer Qualifications (Clamshell/Mechanical Only)

Submit to the Contracting Officer and cc the Environmental Compliance POC for approval, the qualifications of the marine animal observer no later than 5 calendar days after Notice to Proceed. A resume of demonstrated experience monitoring manatees and marine turtles and their behaviors in association with in-water construction projects is required. Specifically, include a table with the names and locations of specific in-water construction projects that observers have worked on. Also, the table is to include for each in-water construction project the permit number(s), the estimated amount of time (hours) spent observing, the estimated number of manatees and marine turtles seen, and whether the work was ever halted due to manatees or marine turtles in the area. Complete the FWC Protected Marine Species Observer Information Form (see CONSTRUCTION FORMS AND DETAILS) for each marine animal observer to document their experience.

Additional Requirements:

- 1) Dedicated Observers are to be outfitted with equipment needed for observation of and for reaction to observed protected marine species (such as binoculars, polarized sunglasses, lights, two-way radios, contact information for dredge operator and contracting officer, etc.). Specify the equipment used to aid in observation and communication;
- 2) A description (including a map) of where the dedicated observers will be located during work (including when tugboats and crew vessels are mooring to the dredge) and how they will cover the areas required. Please note: To maximize visibility, the observers are to be in elevated positions, when possible;
- 3) The length of work shifts for the dedicated observers; and

1.9.3 Endangered Species Observer and Trawl Supervisor Qualifications (For Hopper Dredges Only)

Submit to the Contracting Officer and cc the Environmental Compliance POC the names of the endangered species observer-biologists (ESOs) to perform the monitoring for this contract no later than 5 calendar days after Notice to Proceed. Only NMFS pre-approved ESOs are authorized to monitor threatened/endangered species takes by hopper dredges. A pre-approved list of ESOs can be obtained by contacting NOAA Fisheries' Southeast Region, Protected Resources Division at (727) 824-5312.

1.10 SUBCONTRACTORS

Assurance of compliance with this section by subcontractors will be the responsibility of the Contractor.

1.11 NOTIFICATION

The Contracting Officer will notify the Contractor in writing of any observed noncompliance with the aforementioned Federal, State, or local laws or regulations, permits and other elements of the environmental

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protection specifications. After receipt of such notice, inform the Contracting Officer of proposed corrective action and take such action as may be approved. If the Contractor fails to comply promptly, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No time extensions are to be granted or costs or damages allowed to the Contractor for any such suspension.

Within 30 calendar days following project completion, the Contractor is to submit to the Contracting Officer and cc the Environmental Compliance POC the [Project Environmental Summary Sheet](#). The Project Environmental Summary Sheet is to provide absence or occurrence of environmental incidents, as indicated in paragraph CONSTRUCTION FORMS AND DETAILS below.

1.12 CONTRACTOR PERSONNEL QUALIFICATIONS IN POLLUTION CONTROL

Provide qualified personnel to perform all phases of environmental protection, including methods of detecting and avoiding pollution, familiarization with pollution standards, both statutory and contractual, and careful installation and monitoring of the project to ensure adequate and continuous environmental pollution control. Quality Control and supervisory personnel are to be thoroughly knowledgeable of Federal, State, and local laws, regulations, and permits as listed in the Environmental Protection Plan submitted by the Contractor. Identify the Quality Control personnel in the Quality Control Plan submitted in accordance with Section 01 45 05 DREDGING/BEACH FILL PLACEMENT - CONTRACTOR QUALITY CONTROL.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 PROTECTION OF ENVIRONMENTAL RESOURCES

For contract work, comply with all applicable Federal, State, or local laws and regulations. The environmental resources within the project boundaries and those affected outside the limits of permanent work under this Contract will be protected at least during the entire period of this Contract. Confine activities to areas defined by the drawings and specifications. Environmental protection is to be as stated in the following subparagraphs.

3.1.1 General Project Environmental Design and Installation Criteria

Some project sites have features that must not be impacted in any way, including cultural, historic, or archeological features. At all sites, minimize disturbance to existing features at the site to the extent possible, including vegetative, topographic, and drainage pattern features. Wetland, seagrass or hardbottoms impacts (temporary access, detours, staging areas, and other work area impacts) to project sites should be avoided and may require separate permitting action. Any wetlands temporarily impacted are to have its soil restored upon project completion. Expansion of previously permitted project footprints may likewise require separate permitting action.

In all cases, the installation of temporary systems to aid construction will provide for protection of the environment during handling,

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installing, storing, utilizing, transporting, servicing, testing, refilling, transferring, pumping, processing, removing waste products, repairing and maintaining systems and their components. Necessary protection is to also be considered that would prevent contamination of the environment from impacts to the systems caused by storm water runoff and flooding. Retrofit of temporary systems on project sites to modern environmental protection design standards will also be considered.

In the event environmental protection measures fail, implement procedures to control and correct environmental damage. Coordinate with the Contracting Officer if environmental protection measures fail.

3.1.1.1 Sewage-Based Systems Environmental Design and Installation Criteria

In general, there will be no waste or debris discharges of any kind for a project unless authorized by the Contracting Officer. This is to include the Contractor's providing sufficient temporary sanitary equipment and facilities for the project. The design and/or installation of temporary or permanent sewage systems are to ensure that waters will be free of effects of sewage discharges. Adhere to Federal, State, or local codes and requirements regarding sewage in the design, such as those of the EPA and, in the case of the State, Chapter 62-620 (Wastewater Facilities) of the FAC. Adhere to Best Management Practices from the applicable agencies in the design.

3.1.2 Protection of Land Resources

Prior to the beginning of any construction, identify all land resources to be preserved or avoided within the work area. Remove materials displaced into not cleared areas. Do not remove, cut, deface, injure, or destroy land resources including trees, shrubs, vines, grasses, topsoil, and land forms without special permission from the Contracting Officer. Engage a qualified tree surgeon to perform all tree surgery. Repair injuries to bark, trunk, branches, and roots of protected trees by dressing, cutting, and painting as specified for Class I Fine Pruning, of the National Arborist Association Pruning Standards for Shade Tree or as per State's Agricultural Extension Agency Guidelines, immediately as occurrences arise. No ropes, cables, or guys are to be fastened to or attached to any trees for anchorage unless specifically authorized. Where such special emergency use is permitted, provide effective protection for land and vegetation resources at all times as defined in the following subparagraphs.

3.1.2.1 Work Area Limits

Prior to any construction, mark the areas that are not required to accomplish all work to be performed under this Contract. Mark or fence isolated areas within the general work area to be saved and protected. Protect existing trees designated to remain. Provide protection of tree roots against noxious materials in solution caused by run-off or spillage. Fires are to be located outside the canopy of protected trees. Do not store materials, trailers, or equipment within the drip line of any protected tree. Protect monuments and markers before construction operations commence. Where construction operations are to be conducted during darkness, the markers will be visible. Convey to personnel the purpose of marking and/or protection of all necessary objects.

3.1.2.2 Contractor Facilities and Other Work Areas

Locate facilities, staging and other work areas as indicated in Section 01 50 02 TEMPORARY CONSTRUCTION FACILITIES, and in compliance with permits listed in the paragraph PERMITS AND AUTHORIZATIONS above and obtained for performance of work. Manage borrow or dredge material management (DMMA) areas to minimize erosion and to prevent sediment from entering nearby watercourses, wetlands, or lakes. Manage and control dredged material areas to limit spoil intrusion into areas designated on the drawings and to prevent erosion of soil or sediment from entering nearby watercourses, wetlands, or lakes. Develop dredged material areas in accordance with the grading plan indicated on the drawings. Control temporary excavation and embankments for plant and/or work areas to protect adjacent areas from despoilment. If there is suspicion that sediment may be unsuitable for disposal at a specified location, immediately take measures to contain the suspect sediment and notify the Contracting Officer.

3.1.2.3 Solid Wastes

The Contracting Officer will approve the type of solid waste containers, staging areas on land for these containers, the roads to haul the solid waste, and the final solid waste disposal site. In addition, comply with Federal, State and local regulations pertaining to the use of the solid waste disposal site.

Place solid wastes (excluding clearing debris) in containers which are emptied on a regular schedule. Conduct handling and disposal to prevent contamination.

3.1.2.4 Fuel, Oil, and Lubricants

Manage fuel, oil, and lubricants in order to prevent spills and evaporation. Provide a 4-foot square, 16-gauge metal pan with borders banded up and welded at corners right below the bibb for fuel dispensers to prevent spills. Edges of the pans are to be 8-inch minimum in depth to ascertain that no contamination of the ground takes place. Clean pans by an approved method immediately after every dispensing of fuel and wastes disposed of offsite in an approved area. If any spilling of fuel occurs, immediately recover the contaminated ground and dispose of it offsite in an approved area. Store petroleum waste generated in marked corrosion-resistant containers and recycled or disposed of in accordance with 40 CFR 279, State, and local regulations. Any fueling operation is to be conducted such that any accidental spills will be prevented from entering surface waters or wetlands.

For any fuel stored onsite overnight at the project work site in excess of 550 gallons, whether attached to a tractor trailer or not, the fuel storage area/fuel dispensing area is to be located within a bermed and if practical impervious lined area. If directed by the Contracting Officer, at no additional cost to the government, a fuel storage system with less than 550 gallons may be required to be within a bermed area. The bermed area must be sized to be able to contain the maximum amount of fuel stored on site and be sloped to allow access/egress to the vehicles being fueled at this location. Fueling operations, to the maximum extent practical, are to be conducted within this bermed/lined fuel storage area. These preventive measures to contain potential spills shall be taken by the contractor at no additional cost to the government. If the construction activity is close to an OFW, the fuel storage bermed area must be lined unless this is not practical. The Contracting Officer may determine if

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other measures are necessary to prevent any fuel spills from entering nearby surface water/wetlands, and these additional actions will be required of the contractor at no additional cost to the government. Any fueling operation shall be manned 100% of the time to ensure a rapid response to any potential fuel spillage. The fuel storage container is to be protected from vandalism by appropriate security measures which could include a fenced area with a lockable gate with high security locks on the gate and/or high security locks on the fuel dispensing equipment. These security measures are to be provided by the contractor at no additional cost to the government. The appropriate level of security measures are to be determined by the Contracting Officer as appropriate to the individual job location. For fuel stored at the job site in tanks less than 550 gallons, the Contracting Officer may determine a secondary containment/and or double walled tank be required at no additional cost to the government.

3.1.2.5 Hazardous Waste

Hazardous wastes are defined in 40 CFR 261. Ensure that hazardous wastes are stored and disposed of in accordance with 40 CFR 261 and State and local regulations. Ensure that hazardous wastes are packed, labeled, and transported in accordance with 49 CFR 173 and State and local regulations.

3.1.2.6 Hazardous Materials

Ensure that hazardous materials are labeled, stored, and transported in accordance with 49 CFR 173, State and local regulations.

3.1.2.7 Disposal of Other Materials

Handle other materials than previously discussed (Construction and Demolition, vegetative waste, etc.) as directed by the Contracting Officer.

3.1.3 Preservation and Recovery of Historic, Archeological, and Cultural Resources

3.1.3.1 Applicable Law

A number of Federal laws require protection of cultural resources. Two laws, in particular, can be potentially involved with construction activities: (1) the National Historic Preservation Act, as amended; and, (2) the Abandoned Shipwreck Act.

3.1.3.2 Known Resources

If known historic, archeological, and cultural resources within the Contractor's work area(s) are present, it will be designated as an avoidance area on the contract drawings or other documents. If so designated, install protection for these resources and preserve during the Contract's duration. Do not dredge, spud, or anchor within the designated avoidance area. Do not distribute maps or other information on these resource locations except for distribution among the Contractor's staff with a need to know technical responsibility for protecting the resources.

3.1.3.3 Inadvertent Discoveries

If, during construction activities, items that may have historic or archaeological origin are observed, such observations are to be reported immediately to the Contracting Officer so that the USACE Project Archaeologist may be notified. Cease all activities adjacent to the

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discovery that may result in the destruction of these resources and prevent employees from further removing, or otherwise damaging, such resources. Once reported, USACE Project Archaeologist will initiate coordination with the appropriate federal, tribal and state agencies to determine if archaeological investigation is required. Additional work in the area of the discovery will be suspended at the site until all federal and state regulations have been successfully completed and the USACE staff members provide further directive. Project activities in the vicinity of the discovery may not resume until the Contracting Officer approves work to proceed.

The possibility of encountering submerged cultural resources is inherent in dredging and snagging operations. Such findings could include shipwrecks, shipwreck debris fields (such as steam engine parts, or wood planks and beams), anchors, ballast rock, concreted iron objects, concentrations of coal, prehistoric watercraft (such as log dugouts), and other structural features intact or displaced. The materials may be deeply buried in sediment, resting in shallow sediments or above them, or protruding into water. Suspected cultural materials inadvertently gathered from a water-saturated context should be kept moist by re-immersion, spraying, or some other expedient means of wetting until the Contracting Officer provide further directives in coordination with the USACE Project Archaeologist. No interviews or other contact with the media or federal, tribal or state agencies are to occur without clear authorization from the Contracting Officer.

In the unlikely event that unmarked human remains are identified, they will be treated in accordance with Section 872.05 Florida Statutes. Immediately cease all work and ground disturbing activities within a 50-meter diameter of the unmarked human remains and notify the Contracting Officer is to be immediately notified. The USACE Project Archaeologist will notify the local medical examiner, and the appropriate State Historic Preservation Office (SHPO) and Tribal Historic Preservation Office(s) (THPO). Based on the circumstances of the discovery, equity to all parties, and considerations of the public interest, the Contracting Officer may modify, suspend, or terminate work activities. Such activity is not to resume without written authorization from the State Archeologist and from the Contracting Officer.

3.1.3.4 Claims for Downtime due to Inadvertent Discoveries

Upon discovery and subsequent reporting of a possible inadvertent discovery of cultural resources, continue work well away from, or otherwise protectively avoiding, the area of interest, or in some other manner that strives to continue productive activities in keeping with the Contract. If an inadvertent discovery be of the nature that substantial impact(s) to the work schedule are evident, such delays are to be coordinated with the Contracting Officer. Contract adjustments resulting from compliance with this paragraph are to be determined in accordance with Clause DIFFERING SITE CONDITIONS.

3.1.4 Protection of Water Resources

Keep construction activities under surveillance, management, and control to avoid pollution of surface, ground waters, and wetlands. Implement special management techniques as set out below to control water pollution by the listed construction activities which are included in this Contract. Use construction methods that protect wetland and surface water areas from damage due to mechanical grading, erosion, sedimentation and

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turbid discharges. Storageing or stockpiling of equipment, tools, or materials within wetlands or along the shoreline within the littoral zone is not allowed unless specifically authorized.

3.1.4.1 Turbidity

Conduct operations in a manner to minimize turbidity. Refer to Section 01 57 25 TURBIDITY AND DISPOSAL MONITORING for further instructions.

3.1.4.2 Oil, Fuel, and Hazardous Substance Spill Prevention and Mitigation

Prevent oil, fuel, or other hazardous substances from entering the air, ground, drainage, local bodies of water, or wetlands. This is to be accomplished by design and procedural controls. In the event that a spill occurs despite the design and procedural controls, the following is to occur:

- (1) Take immediate action to contain and cleanup any spill of oil, fuel or other hazardous substance.
 - (2) Immediately report spills to the Contracting Officer, the National Response Center (NRC) at 1-800-424-8802. For spills into wetlands or waterways, regardless of quantity, and terrestrial spills over 25 gallons, report to the Florida Department of Environmental Protection's (FDEP) State Warning Point at 1-800-320-0519 within 24 hours. This immediate response must include the suspension of any pumping/dewatering activities that could transport the spilt material to another location. Pumping must be suspended until the spill site is cleaned up and has no visible chemical sheen on the water surface. The Contractor is fully responsible, at no addition cost to the government for the remediation the spill cleanup to include the spill site and any location the spilt material was transferred to.
 - (3) Spill contingency planning is to be strictly in accordance with the criteria of 40 CFR, Part 109.
 - (4) To control the spread of any potential spill, absorbent materials are to be readily available and capable of absorbing the contents of the single largest tank.
 - (5) To control the spread of any potential spill, provide a written certification of commitment of manpower, equipment, and materials required to expeditiously cleanup and dispose of spill materials.
- a. Spill Preventive Systems: System design and installation requirements have been discussed at the beginning of this Section. Temporary or portable tanks will conform to applicable Federal, State and local codes and requirements and is not to be placed where they may be affected by storm, flooding, or washout. Diversionary structures for spills are to be put in place in advance where practical. Fuel storage locations to include pumps and generators will be placed and have features such as earthen berms, to prevent any fuel spills to surface water or wetlands. Both spill preventive systems and any deviations from associated requirements will be approved by the Contracting Officer prior to implementation.
 - b. Liabilities: Liable in the amounts established in 40 CFR, Part 113 when it can be shown that oil was discharged as a result of willful

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negligence or willful misconduct. The penalty for failure to report the discharge of oil will be in accordance with the provision of 33 CFR, Part 153.

3.1.5 Protection of Fish and Wildlife Resources

Keep construction activities under surveillance, management, and control to minimize interference with, disturbance to, and damage of fish and wildlife. List species that require specific attention along with measures for their protection in the Environmental Protection Plan prior to the beginning of construction operation.

In the event that a threatened, endangered, or Marine Mammal Protection Act (MMPA) species is harmed as a result of construction activities, cease all work and notify the Contracting Officer. The order of contact within the Corps of Engineers will be as follows:

Order of Contact of Corps Personnel

<u>Title</u>	<u>Telephone Number</u>
Corps, Inspector	Onsite/After hours to be provided
Corps, Project Engineer	Onsite/After hours to be provided
Contracting Officer	To be provided
Administrative Contracting Officer	To be provided
Area Engineer for Area Office	To be provided

Information provided above will be updated during the pre-construction meeting.

3.1.5.1 Listed Species Protection

Endangered and Threatened Species and MMPA species that may be in the project area

Common Name	Scientific Name	ESA Status
Florida manatee	Trichechus manatus latirostris	Threatened
Sperm whale	Physeter macrocephalus	Endangered
Humpback whale	Megaptera novaeangliae	MMPA protected
Common Bottlenose Dolphin	Tursiops truncatus	Marine Mammal Protection Act Protected
Shortnose sturgeon	Acipenser brevirostrum	Endangered
Gulf sturgeon	Acipenser oxyrinchus desotoi	Threatened
Green sea turtle	Chelonia mydas	Threatened
Loggerhead sea turtle	Caretta caretta	Threatened
Kemp's ridley sea turtle	Lepidochelys kempii	Endangered

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Common Name	Scientific Name	ESA Status
Leatherback sea turtle	<i>Dermochelys coriacea</i>	Endangered
Hawksbill sea turtle	<i>Eretmochelys imbricata</i>	Endangered
Smalltooth sawfish	<i>Pristis pectinata</i>	Endangered
Giant Manta Ray	<i>Manta birostris</i>	Threatened
Eastern Indigo snake	<i>Drymarchon couperi</i>	Threatened
Nassau Grouper	<i>Epinephelus striatus</i>	Threatened
Scalloped hammerhead	<i>Sphyrna lewini</i>	Threatened
Oceanic Whitetip	<i>Carcharhinus longimanus</i>	Threatened
Piping plover	<i>Charadrius melodus</i>	Threatened
Red knot	<i>Calidris canutus rufa</i>	Threatened
Bald eagle	<i>Haliaeetus leucocephalus</i>	Bald and Golden Eagle Act protected
Wood Stork	<i>Mycteria americana</i>	Threatened

Instruct personnel associated with the project of the potential presence of listed species as noted in the Table 3.1.5.1 in the area and the need to avoid collisions with and/or harming these animals. All construction personnel are to be advised that there are civil and criminal penalties for harming, harassing, or killing, or attempting to harm, harass or kill or listed species as noted in the Table 3.1.5.1 in the area which are protected under the Marine Turtle Protection Act, Marine Mammal Protection Act of 1972, the Endangered Species Act of 1973, and/or the Florida Manatee Sanctuary Act. The Contractor will be held responsible for any listed species as noted in the Table 3.1.5.1 in the area harmed, harassed, or killed as a result of construction activities.

- a. Siltation Barriers: Made of material in which manatees cannot become entangled, properly secured, and regularly monitored to avoid manatee entrapment. Barriers must not block manatee entry to or exit from essential habitat.
- b. Special Operating Conditions:
 - (1) All vessels associated with the project will operate at no wake/idle speeds at all times while in waters where the draft of the vessel provides less than a 4-foot clearance from the bottom, and vessels will follow routes of deep water whenever possible. Boats used to transport personnel must be shallow-draft vessels, preferably of the light-displacement category, where navigational safety permits. Mooring bumpers must be placed on all barges, tugs, and similar large vessels wherever and whenever there is a potential for manatees to be crushed between two moored vessels. The bumpers are to provide a minimum stand-off distance of four

feet under maximum designed compression.

- (2) Manatee or Marine Turtle Sighting: If a manatee(s) or marine turtle(s) are sighted within 100 yards of the project area, all appropriate precautions will be implemented by the Contractor to ensure protection of the manatee or marine turtle. These precautions will include the operation of all moving equipment, including vessels, no closer than 50 feet of a manatee or marine turtle. If a manatee or marine turtle are closer than 50 feet to moving equipment or the project area, the equipment will be shut down and all construction activities are to cease within the waterway to ensure protection of the manatee or marine turtle. For unanchored vessels, operators are to disengage the propeller and drift out of the potential impact zone. If drifting would jeopardize the safety of the vessel then idle speed may be used to leave the potential impact zone. Construction activities will not resume until animal(s) has moved beyond the 50-foot radius of the project operation, or until 30 minutes elapses if the animal(s) has not reappeared within 50 feet of the operation. Animals will not be herded away or harassed into leaving. If construction activity will cease, notify the Contracting Officer.
 - (3) Report interactions with threatened or endangered (T&E) species to the Contracting Officer. Cease dredging operations each time an T&E species is potentially taken until the Contracting Officer provide notification to resume dredging. Promptly following the potential take of an T&E species, the Government will perform a risk assessment to include a review of all potential circumstances which contributed to the take, a review of the DQM data, and a physical inspection of the dredge and its operating procedures in accordance with the hopper dredge inspection checklist, project specifications and applicable biological opinions and permits. The Government, in coordination with the Contractor, will develop a risk management plan containing specific actions to minimize risk of additional potential take. The Government may terminate dredging or redirect dredging to another acceptance section if it is determined that take numbers are at or near the incidental take statement limits or additional take jeopardizes other Government approved regional dredging projects authorized by the South Atlantic/Gulf of America Regional Biological Opinion and the risk of additional take by the project is determined to be high.
 - (4) If a sea turtle or sturgeon is taken by hopper dredge (dead or alive), a submittal of incident information is to be made into the Operations and Dredging Endangered Species System (ODESS).
- c. Marine Animal Monitoring (Clamshell/Mechanical Only): During clamshell dredging operations, a qualified dedicated observer is to monitor for the presence of manatees and marine turtles. The dedicated observer must be in a location with direct line of sight for all in-water construction activity (i.e. direct line of sight of clamshell bucket or excavator arm entering water). The dedicated observer is to have experience in manatee and marine turtle observation and be equipped with polarized sunglasses to aid in observing. Nighttime lighting of waters within and adjacent to the work area is to be illuminated, using shielded or low-pressure sodium-type lights, to a degree that allows the dedicated observer to sight any manatee or marine turtle on the surface within 200 feet of the operation. The dredge operator must gravity-release the clamshell

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bucket only at the water surface, and only after confirmation that there are no manatees or marine turtles within the safety distance identified in the standard construction conditions.

- d. Port Sutton Channel and Turning Basin Only: No clamshell dredging and disposal activities are authorized to occur between dusk and dawn. Movement of a work barge, other associated vessels, or any in- water work is not to be performed after sunset, when the possibility of spotting manatees is negligible.
- e. Port Sutton Channel and Turning Basin Only: Between November 1 and March 31, at least two dedicated marine animal observers are to be present when all in-water work is being performed. The dedicated observers are to perform no other duties that may interfere with their ability to observe for protected marine species (including manatees, marine turtles, dolphins, etc.) during dredging activities.
- f. **Marine Animal Observers Final Logs:** A final report summarizing the dedicated observer logs are to be submitted to the Contracting Officer and FWC at ImperiledSpecies@MyFWC.com no later than 30 days following demobilization of each event. This final report is to include the following:
 - 1) the project name;
 - 2) state permit number;
 - 3) names of all dedicated observers and their contact information;
 - 4) reporting period;
 - 5) a summary of all sightings, collisions, and injuries (in spreadsheet form); and
 - 6) actions taken in response to each sighting, collision, or injury.
- g. If an injured or dead marine mammal or marine turtle is discovered in the vicinity of the project location while mobilized, the following shall occur:
 - 1) All work in the active work zone of the incident, shall cease immediately.
 - (2) Contractor shall maintain contact with the injured or dead animal to the greatest extent practicable until the appropriate authorities (i.e., FWC or NMFS) arrive to retrieve the animal.
 - (3) In the case of a marine turtle take, work may resume while the submitted details and other reports (such as internal FWC rescue or necropsy reports) are assessed by FWC. In the case of a marine mammal take, the Contracting Officer in coordination with the agencies (i.e., FWC, NMFS, and DEP) will determine when work may resume.
 - (4) Details of the incident shall be sent to FWC at ImperiledSpecies@myfwc.com within 24 hours. Information shall include, but not be limited to: a narrative of the incident; photographs/videos of the incident and surrounding environment (if possible); a GPS point where the animal was discovered; names,

titles and contact numbers of all personnel onsite at the time of the incident; and the name, title and contact number for the designated representative. Additional information may be requested by FWC, and telephone interviews may be required.

- (5) Once the circumstances have been reviewed, Contracting Officer will provide their written determination to the Contractor on how to proceed: whether operations can proceed normally; whether modifications to the protective measures are required in order to proceed; or if operations need to cease temporarily due to the need for further investigation. If modifications to the protective measures are required, construction shall not resume until the protective measures are made by the Contractor.

- h. Manatee Signs: Post temporary signs concerning manatees prior to and during all in-water project activities at sufficient locations to be regularly and easily viewed by all personnel engaged in water-related activities. Remove all signs upon completion of the project. Two temporary signs that have already been approved for this use by the FF&WCC must be used at each location. One sign which reads CAUTION BOATERS - WATCH FOR MANATEES must be posted. A second sign measuring at least 8 ½ inch by 11 inch must explain the requirements for Idle Speed/No Wake and the shutdown of in-water operations. These signs can be viewed at MyFWC.com/manatee.

3.1.5.2 Protected Listed Species Sighting Reports

Report any take concerning any Listed Species in the Table 3.1.5.1; or sighting of any injured or incapacitated of Listed Species to the Contracting Officer by notifying the personnel indicated in the table ORDER OF CONTACT OF CORPS PERSONNEL above.

Provide a copy of the incidental take report in PDF format with photographs to the Contracting Officer, Project Biologist, and Environmental Compliance POC within 24 hours of the incident. Report any collision with and/or injury to a manatee to the Florida Fish and Wildlife Conservation Commission Manatee Hotline 1-888-404-FWCC (3922) and imperiledspecies@myFWC.com, as well as the U.S. Fish and Wildlife Service, Jacksonville Field Office 904-731-3336 for North Florida. Report any collision with (and/or injury to) a marine turtle to the Sea Turtle Stranding and Salvage Network (STSSN) at SeaTurtleStranding@myfwc.com. Report any take of Smalltooth sawfish to 1-844-SAW-FISH or email Sawfish@MyFWC.com. Report any take of a Giant manta ray to manta.ray@noaa.gov.

3.1.5.3 Protected Species Summary Report

If a hopper dredge is used, submit all details regarding potential takes in ODESS and the Quality Control Report. Any incidents, including sightings, collisions with, injuries, or killing of manatees, sea turtles (not associated with a take via the dredge) or whales occurring during the contract period are to be recorded in a log that will be turned into the Contracting Officer at the end of the project. Maintain paper copies of all forms entered into the ODESS until the project is deemed complete by the Contracting Officer. An electronic copy of all incident reports/photos associated with take via the dredge is to be included in the summary report. [Weekly Protected Species Summary Report](#) is to be submitted weekly in electronic spreadsheet format to the Contracting Officer and the Project Biologist. An example spreadsheet is available

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upon request.

Maintain a log detailing all incidents, including sightings, collisions with, injuries, or killing of Listed Species occurring during the contract period. The data is to be recorded on forms provided by the Contracting Officer (sample forms are on the web site indicated in paragraph CONSTRUCTION FORMS AND DETAILS below). Within 30 calendar days following project completion, a [Protected Species Summary Report](#) summarizing the above incidents and sightings is to be submitted to the Contracting Officer and cc the Environmental Compliance POC.

3.1.5.4 Endangered Species Observers (Hopper Dredge Only)

During dredging operations, observers approved by the National Oceanic and Atmospheric Administration - Fisheries (NOAA-Fisheries/NMFS) must be aboard to monitor for the presence of the listed species. Observer coverage must be 100 percent (24hr/day) and is to be conducted year round. During transit to and from the disposal area, the observer is to monitor from the bridge during daylight hours for the presence of T&E species, especially the North Atlantic right whale, during the period November 1 through April 30. The ESO(s) are responsible for personally monitoring, handling, and reporting all captured listed species at all times when the hopper dredge is operating. The ESO on-duty will stand watch to detect listed species in the area and to alert the captain of their presence to avoid vessel collision whenever the vessel is moving. The on-duty ESO will only be responsible for standing watch and not performing other tasks such as inspecting or handling captures when the vessel is in motion. During dredging operations, while dragheads are submerged, the observer is to continuously monitor the inflow and/or overflow screening for turtles and/or turtle parts and sturgeon and/or sturgeon parts. Upon completion of each load cycle, dragheads should be monitored as the draghead is lifted from the sea surface and is placed on the saddle in order to assure that listed species that may be impinged within draghead are not lost and un-accounted for. Observers are to physically inspect dragheads and inflow and overflow screening/boxes for threatened and endangered species take. Hopper dredge operators will not open the hydraulic doors on the inflow boxes prior to inspection by the ESO for evidence of listed species take. The screened area must be accessible to the ESO to ensure 100% observer coverage. The ESO must inspect the contents of all inflow screening boxes after every load, including opening the box (where applicable and safely accessible) and looking inside at all contents for evidence of listed species entrainment. If the contents are not clearly visible and identifiable from a location outside of the box, then in limited instances, the ESO may be required to enter the inflow box to identify contents. If the inflow box cannot be observed due to clogging, the box contents cannot be dumped or flushed unless overflow screening that captures contents for observation by the ESO is operational and monitored for evidence of take. ESOs are to visually monitor box contents as they are dumped or flushed into the hopper.

- a. Monitoring Reports: The results of the monitoring are to be recorded on the appropriate observation sheets and entered into the ODESS system (See Appendix A attached after the end of this Section) and Quality Control Daily Report. There is a form for each load and a form for an incident involving a listed species, specific to the species (turtles, sturgeon, etc.). Complete load forms regardless of whether any incidents with sturgeon (Gulf, Shortnose or Atlantic), whales, or sea turtles occur. In the event of any sea turtle or sturgeon (Gulf,

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Atlantic or Shortnose) is potentially taken by the dredge, appropriate incident reporting forms are to be completed and entered into the ODESS system. Additionally, photograph all specimens with the digital camera built into the ODESS tablet. Upload these photographs to the respective incident reports on the ODESS within 4-hours from observing the incident. Do not commence dredging of subsequent loads until all appropriate reports are completed from the previous dredging load to ensure completeness and thoroughness of documentation associated with the incidental take exports. Submit reports to the Contracting Officer within 4-hours of the take. Observer forms can be accessed on the web site indicated in paragraph CONSTRUCTION FORMS AND DETAILS below.

- b. Digital Photographs: Use a digital camera embedded in the ODESS tablet, with an image resolution capability of at least 300 dpi, in order to photographically report all incidental takes, without regard to species, during dredging operations. Upload photos taken associated with the take to the ODESS with the incident information.
- c. A copy of all incident forms, load forms, and photographs from incidents with listed species is to be included in the Protected Species Summary Report.

3.1.5.5 Disposition of Endangered or Threatened Species Carcasses

a. Listed Species Taken by Hopper Dredge:

- (1) Dead ESA-listed species including Turtles or sturgeon (regardless of species): Upon removal of carcass and/or parts from the draghead or screening, observers are to take photographs as to sufficiently document major characteristics of the carcass or parts including but not limited to dorsal, ventral, anterior, and posterior views. For all photographs taken, a backdrop is to be prepared to document the dredge name, observer company name, contract title, time, date, species, load number, location of dredging, and specific location taken (draghead, screening, etc.). Carcass/parts are to be scanned for flipper and Passive Integrated Transponder (PIT) tags. Any identified tags are to be recorded on the ODESS Turtle Incident Form or ODESS Sturgeon Incident Form (as appropriate) that is included in the ODESS Program Forms available through the ODESS tablet (or on the ODESS website if the tablet is not available). Also, fill out the ODESS Dredge Load Form and take photos for submission. Animal parts which cannot be positively identified to species, on board the dredge or barge(s) are to be preserved by the observer(s) for later identification. A tissue sample must be collected from any lethally taken ESA species submitted under the process stated in the Protocol for Collecting Tissue Samples from Turtles for Genetic Analysis found in the CONSTRUCTION FORMS AND DETAILS below. All genetic samples collected are to be submitted to NMFS within 30 days of collection, and verification of submittal to NMFS is to be provided to the Contracting Officer and USACE Project Biologist. After all data collection is complete, the carcass/parts should be marked (spray paint works well), weighted down and disposed of, or provided to a state agency staff member, in accordance with the direction of the Contracting Officer.
- (2) Live ESA-listed species including Turtles: Observer(s) is to measure, weigh, scan for PIT tags, tag (Iconnel flipper and PIT

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tags (if PIT tag not located during scan, and only if observer is qualified to tag using PIT tags)), and photograph any live listed species incidentally taken by the dredge. Observer(s) (or their authorized representative) is to coordinate with the Contracting Officer and Environmental branch staff prior to transport. Handling of any marine turtles captured during hopper dredging is to be conducted only by persons with prior experience and training in these activities, such as a NMFS-approved marine turtle observer, or by persons who have submitted documentation to the Contracting Officer of meeting the FWC Marine Turtle Conservation Guidelines specific to stranding activities. The Contracting Officer or their designee who transport live or dead marine turtles or marine turtle parts into, out of, or within, the state of Florida will notify FWC in writing specifying the number, species of turtle, type of specimen, and the destination after transport is complete. Before transport, the Contracting Officer or their designee must coordinate with FWC at SeaTurtleStranding@myfwc.com to determine the appropriate facility to receive live marine turtles for rehabilitation. The Contracting Officer or their designee will abide by the State of Florida's FWC Marine Turtle Conservation Guidelines (<https://myfwc.com/media/3133/fwc-mtconservationhandbook.pdf>) specific to transport of live stranded turtles.

b. Turtles Taken by Relocation Trawler:

- (1) Live Turtles: At least one (1) crewmember, who possesses the required permits for handling endangered species, experienced in sea turtle capture or is a NMFS-approved observer, is to be on board the trawler during the trawl and act as the sea turtle trawling and relocation supervisor. Only a NMFS-approved observer or observer candidate in training under the direct supervision of a NMFS-approved observer is to conduct the sampling operations of tagging, measuring, weighing and collecting tissue samples from turtles. Each turtle that is captured is to be identified to species and age class (juvenile, sub-adult, adult), digitally photographed, scanned for Passive Integrated Transponder (PIT) tags, measured, tagged (Inconel flipper tag and PIT tag, if one not located by scan, and only if observer is qualified to tag using PIT tags), and released and the data recorded on the Sea Turtle Tagging and Relocation Report found in the CONSTRUCTION FORMS AND DETAILS below. External tags are to also be noted and data recorded into the observer's log. Turtles are to be released at locations at least 3 nautical miles (nmi) away from the project area, at time intervals designated by the Sea Turtle Trawling and Relocation Supervisor who is proficient in the viable handling and relocating of sea turtles.

Weight/Size Measurements and Tissue Collection. Turtles must be measured, tagged, and weighed when safely possible, prior to release. Turtle measurements is to be recorded and must include, at a minimum, weight when possible, straight-line length, straight-line width, body depth, and tail length. Turtles are to be tagged with NMFS No. 681 Inconel tags in each of the front flippers and PIT tags according to NMFS protocol (found at https://www.sefsc.noaa.gov/turtles/TM_579_SEFSC_STRTM.pdf and on the ODESS website under the Information tab). Aseptic conditions are maintained for tags and tag attachment. A tissue sample is to be collected from any trawl captured turtle and submitted under

the process stated in the Protocol for Collecting Tissue Samples from Turtles for Genetic Analysis found in the CONSTRUCTION FORMS AND DETAILS below. All genetic samples collected are to be submitted to NMFS within 30 days of collection, and verification of submittal to NMFS is to be provided to the Contracting Officer. Obtain permits related to trawling from the appropriate State agencies.

- (2) Dead Turtles: Any turtle found dead in the nets during relocation trawling efforts is to be digitally photographed, measured, weighed (if possible), scanned for tags. The carcass should then be placed on ice and kept cold (but not frozen) as quickly as possible and the Contracting Officer is to be contacted immediately. The Contracting Officer will contact the Florida State Sea Turtle Salvage and Stranding network (SeaTurtleStranding@myFWC.com) to arrange for the carcass to be transferred for necropsy. The turtle is to be kept on ice (but not frozen) until the appropriate state agency can be contacted. If the state agency declines to recover the carcass for necropsy, the carcass is to be marked (spray paint works well), weighted down and disposed of in accordance with the direction of the Contracting Officer.

Weight/Size Measurements and Tissue Collection. Turtles are to be measured, and weighed. Measurements are to be recorded and must include, at a minimum, weight when possible, straight-line length, straight-line width, body depth, and tail length. The Carcass is to also be scanned for flipper and Passive Integrated Transponder (PIT) tags. Any identified tags are to be recorded on the Sea Turtle Incidental Take Form that is included in the Endangered Species Observer Program Forms located on the web site indicated in paragraph CONSTRUCTION FORMS AND DETAILS below. A tissue sample is to be collected from any lethally taken sea turtle and submitted under the process stated in the Protocol for Collecting Tissue Samples from Turtles for Genetic Analysis found in the CONSTRUCTION FORMS AND DETAILS below. All genetic samples collected are to be submitted to NMFS within 30 days of collection, and verification of submittal to NMFS is to be provided to the Contracting Officer. Obtain permits related to trawling from the appropriate State agencies.

3.1.5.6 Hopper Dredge Equipment

Hopper dredge dragheads are to be equipped with rigid sea turtle deflectors which are rigidly attached. Deflectors are to be solid with no openings in the face. Alternative designs will be considered provided sufficient information is included indicating a particular modification is effective in minimizing potential turtle takes. The Government technical staff will coordinate with NMFS on the effectiveness of this alternate design. A recommendation from the Government technical staff and NMFS will be provided to the Contracting Officer for final approval or disapproval of the alternate design. Do not presume that a decision on an alternative design will be provided during the contracting period. The Contractor's unit price is to be based on the original, solid faced deflector design. No adjustment in unit price will be made for the approval or denial of an alternate deflector design. No dredging is to be performed by a hopper dredge without an installed turtle deflector device approved by the Contracting Officer. Sample Turtle Deflector Design Details are on the first web site indicated in paragraph CONSTRUCTION

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FORMS AND DETAILS below. For each dredge operating on the project, dredging operations must not begin until the USACE Sea Turtle Deflector Checklist for Hopper Dredges for USACE and USACE/Army-permitted projects has been completed by a USACE-approved inspector. The completed checklist for each dredge is to be submitted by the USACE-approved inspector to the Contracting Officer and forwarded to the Project Biologist. The checklist can be downloaded from the ODESS website - <https://odess.usace.army.mil>

If a hopper dredge is used for this work, submit detail drawings showing the proposed [Turtle Deflector Device](#) and its attachment to the equipment no later than 30 calendar days after notice to proceed. Include the approach angle for any depths to be dredged during this Contract on the submitted drawing. A copy of the approved drawings and calculations are to be available on the vessel during the life of this Contract. Do not commence any dredging work until approval of the installed turtle deflector device.

a. Deflector Design:

- (1) The leading vee-shaped portion of the deflector is to have an included angle of less than 90 degrees. Internal reinforcement is to be installed in the deflector to prevent structural failure of the device. The leading edge of the deflector is to be designed to have a plowing effect of at least 6 inch depth when the drag head is being operated. Appropriate instrumentation or indicator is to be used and kept in proper calibration to insure the critical approach angle. (Information Only Note: The design approach angle or the angle of lower drag head pipe relative to the average sediment plane is very important to the proper operation of a deflector. If the lower drag head pipe angle in actual dredging conditions varies tremendously from the design angle of approach used in the development of the deflector, the 6 inch plowing effect does not occur. Therefore, every effort should be made to ensure this design approach angle is maintained with the lower drag pipe.)
- (2) If adjustable depth deflectors are installed, they must be rigidly attached to the drag head using either a hinged aft attachment point or an aft trunnion attachment point in association with an adjustable pin front attachment point or cable front attachment point with a stop set to obtain the 6 inch plowing effect. This arrangement allows fine-tuning the 6 inch plowing effect for varying depths. After the deflector is properly adjusted there is to be no openings between the deflector and the drag head that are more than 4 inches by 4 inches.
- (3) Draghead Screening: If the Contractor has the draghead(s) screened, the screening for the draghead production opening must be 8 inches by 8 inches or larger. Pumps employed must be sufficient to operate efficiently with this screening as a minimum.

b. InFlow/Overflow Basket Design:

- (1) Install baskets or screening over the hopper inflow/overflow(s) with no greater than 4 inch by 4 inch openings. The method selected is to depend on the construction of the dredge used and is to be approved by the Contracting Officer prior to commencement of dredging. The screening is to provide 100 percent coverage of the hopper inflow and overflow(s). The screens and/or baskets are

to remain in place throughout the performance of the work. 100 percent inflow and overflow screening is to be maintained throughout the project. If 100 percent inflow and overflow screening cannot be maintained due to clogging of the baskets by dredged material, or other material including reeds, shell, etc, coordinate with the Contracting Officer to determine an appropriate path forward.

- (2) Install and maintain floodlights suitable for illumination of the baskets or screening to allow the observer to safely monitor the hopper basket(s) during non-daylight hours or other periods of poor visibility. Safe access is to be provided to the inflow baskets or screens to allow the observer to inspect for turtles, turtle parts or damage.

c. Hopper Dredge Operation:

- (1) Operate the hopper dredge to minimize the possibility of taking sea turtles or sturgeon and to comply with the requirements stated in the Incidental Take Statement provided by the NMFS in the appropriate Biological Opinion.
- (2) The turtle deflector device and inflow and overflow screens are to be maintained in operational condition for the entire dredging operation.

When initiating dredging, suction through the drag heads is to be allowed just long enough to prime the pumps, then the drag heads are to be placed firmly on the bottom. When lifting the drag heads from the bottom, suction through the drag heads must be allowed just long enough to clear the lines, and then must cease. Pumping water through the drag heads is to cease while maneuvering or during travel to/from the disposal area. (Information Only Note: Optimal suction pipe densities and velocities occur when the deflector is operated properly. If the required dredging section includes compacted fine sands or stiff clays, a properly configured arrangement of teeth may enhance dredge efficiency which reduces total dredging hours and turtle takes. The operation of a drag head with teeth is to be monitored for each dredged section to ensure that excessive material is not forced into the suction line.

- (3) When excess high-density material enters the suction line, suction velocities drop to extremely low levels causing conditions for plugging of the suction pipe. Dredge operators should configure and operate their equipment to eliminate all low level suction velocities. Pipe plugging cannot be corrected by raising the drag head off the bottom. Arrangements of teeth and/or the reconfiguration of teeth should be made during the dredging process to optimize the suction velocities.)
- (4) Raising the drag head off the bottom to increase suction velocities is not acceptable. The primary adjustment for providing additional mixing water to the suction line should be through water ports. To ensure that suction velocities do not drop below appropriate levels, monitor production meters throughout the job and adjust primarily the number and opening sizes of water ports. Water port openings on top of the drag head or on raised stand pipes above the drag head is to be screened

before they are utilized on the dredging project. If a dredge section includes sandy shoals on one end of a tract line and mud sediments on the other end of the tract line, adjust the equipment to eliminate drag head pick-ups to clear the suction line.

- (5) Near the completion of each payment section, perform sufficient surveys to accurately depict those portions of the acceptance section requiring cleanup. Keep the drag head buried a minimum of 6 inches in the sediment at all times. Although the over depth prism is not the required dredging prism, achieve the required prism by removing the material from the allowable over depth prism.
 - (6) During turning operations the pumps are to either be shut off or reduced in speed to the point where no suction velocity or vacuum exists.
 - (7) These operational procedures are intended to stress the importance of balancing the suction pipe densities and velocities in order to keep from taking sea turtles. Develop a written operational plan to minimize turtle takes and submit it as part of the Environmental Protection Plan.
 - (8) Comply with all requirements of this specification and the accepted Environmental Protection Plan. The contents of this specification and the Environmental Protection Plan is to be shared with all applicable crew members of the hopper dredge.
 - (9) To prevent impingement or entrainment of listed species within the water column, dredging pumps will be disengaged by the operator when the dragheads are not actively dredging and therefore working to keep the draghead firmly on the bottom. Pumps will be disengaged when lowering dragheads to the bottom to start dredging, turning, or lifting dragheads off the bottom at the completion of dredging. Hopper dredges may utilize a bypass or other system that would allow pumps to remain engaged, but result in no suction passing through the draghead. Pumping water through the dragheads is not allowed while maneuvering or during travel to/from the disposal or pumpout area. The dredge operator will ensure the draghead is embedded in sediment when pumps are operational, to the maximum extent practicable.
 - (10) All waterport or other openings on the hopper dredge are required to be screened to prevent listed species from entering the dredge.
- d. Hopper Dredge Lighting Requirements: From May 1st through October 31st, sea turtle nesting and emergence season, all lighting aboard hopper dredges and hopper dredge pumpout barges operating within three nautical miles of sea turtle nesting beaches is to be limited to the minimal lighting necessary to comply with U.S. Coast Guard and Occupational Safety and Health Administration (OSHA) requirements. All non-essential lighting on the dredge and pumpout barge is to be minimized through reduction, shielding, lowering and appropriate placement of lights to reduce potential disorientation effects on female sea turtles approaching the nesting beaches and sea turtle hatchlings heading seaward.

3.1.5.7 Sea Turtle Trawling and Relocation (For Hopper Dredges Only)

- a. Sea Turtle Trawling and Relocation: Initiation of sea turtle

relocation via trawling will be determined by the Contracting Officer following the take of two sea turtles, without regard to species, and continue until as directed by the Contracting Officer. The results of each trawl is to be recorded on Sea Turtle Trawling Report available from the ODESS website (<https://odess.usace.army.mil/>). An interim trawling report is to be submitted within 24 hours after the third turtle take by the dredge. A final report is to be prepared after the completion of all trawling efforts. Both reports are to be submitted in hard copy and electronic format to the Contracting Officer, and must summarize the results of the trawling (with all forms and including total trawling times, number of trawls and number of captures). Any turtles captured during the trawling effort are to be measured and tagged in accordance with standard biological sampling procedures with sampling data recorded on Sea Turtle Tagging and Relocation Report available from the ODESS website (<https://odess.usace.army.mil/>). Any captured sea turtles are to be relocated from the work area no less than 3 miles away from the location recorded on the Sea Turtle Tagging and Relocation Report form.

- b. Sea Turtle Trawling Procedures: A NMFS approved sea turtle trawling and relocation supervisor is to provide sea turtle trawl observers and nets to capture and relocate sea turtles and is to conduct any initiated sea turtle trawling. Turtles are to be captured with trawl nets to relocate them away from the project area to reduce the potential of sea turtle take. Methods and equipment is to be standardized including nets, trawling direction to tide, length of station, length of tow, and number of tows per station. Data on each tow is to be recorded using Sea Turtle Trawling Report available from the ODESS website (<https://odess.usace.army.mil/>). The trawler is to be equipped with two 60-foot nets constructed from 8-inch mesh (stretch) fitted with mud rollers and flats as specified in Turtle Trawl Nets Specifications on the first web site indicated in paragraph CONSTRUCTION FORMS AND DETAILS below. Paired net tows are to be made for 10 to 12 hours per day or night. Trawling is to be conducted with the tidal flow using repetitive 15-30 minute (total time) tows in the channel or borrow area. Tows are to be made as close to the operating dredge as the trawler captain determines is safe. Positions at the beginning and end of each tow are to be determined from GPS Positioning equipment. Tow speed must be recorded at the approximate midpoint of each tow. Refer to [EM 1110-1-1003](#), paragraph 5.3 and Table 5-1, for acceptable GPS criteria.
- c. Water Quality and Physical Measurements: Water temperature measurements are to be taken at the water surface each day using a laboratory thermometer or a thermometer built into the trawling vessel. Weather conditions are to be recorded from visual observations and instruments on the trawler. Weather conditions, air temperature, wind velocity and direction, sea state-wave height, and precipitation is to be recorded on the Sea Turtle Trawling Report on the second web site indicated in paragraph CONSTRUCTION FORMS AND DETAILS below. High and low tides are to be recorded.
- d. Initiation of Trawling: Trawling will be initiated at the discretion of the Contracting Officer. Initiate trawling and relocation activity in the dredging area as quickly as possible (within 48-72 hours) from the notification of the Contracting Officer. Continue trawling until suspended by the Contracting Officer.
- e. Approved Trawling Supervisor: Trawling is to be conducted under the

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supervision of a biologist approved by the NMFS. A letter of approval from NMFS is to be provided to the Contracting Officer prior to commencement of trawling. Approved trawl supervisors can be verified through NMFS-NERO (978)-281-9291.

- f. Turtle Excluder Devices: Approval for trawling for sea turtles without Turtle Excluder Devices (TEDs) must be obtained from NMFS. Approval for capture and relocation of sea turtles must be obtained from the Florida Fish and Wildlife Conservation Commission (FF&WCC). Approvals must be submitted to the Contracting Officer prior to trawling.
- g. Daily Reporting: A daily log is to be kept for each capture trawling operation. The results of each trawl are to be recorded on the Sea Turtle Trawling Report posted on the second web site indicated in paragraph CONSTRUCTION FORMS AND DETAILS below. Positions at the beginning and end of each tow are to be determined from GPS positioning equipment. Tow speed is to be recorded at the approximate midpoint of each tow. See [EM 1110-1-1003](#), paragraph 5.3 and Table 5-1, for acceptable GPS criteria. Data to be recorded in the comments field must also include:
- h. [Weekly Logs/Summary Of FWC Hopper Dredge Reports](#): The Contractor is to e-mail the Contracting Officer, (MTP@MyFWC.com), and cc the Environmental Compliance POC weekly reports to the Imperiled Species Management section on Friday each week that trawling is conducted in Florida waters. These weekly reports are to include the species and number of turtles captured in Florida waters, general health, and release information. The Contractor is to provide a summary (on FWC provided Excel spreadsheet) of the following: all trawling activity, all turtles captured in Florida waters, all measurements, the latitude and longitude (in decimal degrees) of captures and tow start-stop points, and times for the start-stop points of the tows (including those tows on which no turtles are captured).
- i. [Comprehensive Trawling Daily Logs](#): Within 30 calendar days of completion of the project, submit a copy of the logs regarding capture trawling to the Contracting Officer, (MTP@MyFWC.com), and cc the Environmental Compliance POC.

3.1.5.8 Protection of Migratory Bird Species

Keep construction activities under surveillance, management, and control to prevent impacts to migratory birds and their nests. All construction personnel are to be advised that migratory birds are protected by the Florida Endangered and Threatened Species Act of 1977, Title XXVIII, Chapter 372.072, and the U.S. Fish and Wildlife Service pursuant to the Migratory Bird Treaty Act of 1918 and the Endangered Species Act of 1973, as amended. The Contractor may be held responsible for harming or harassing the birds, their eggs or their nests as a result of the construction.

- a. Monitoring of Construction Area: Conduct monitoring of the construction area beginning 15 February or 10 days prior to commencement of construction whichever is later through 15 September, if construction activities occur during that period. Daily monitoring using the Daily Bird Monitoring Report is to be conducted during the dawn or dusk time frames by a bird monitor approved by the Contracting Officer. (Caution is to be taken by the monitor to avoid disturbance

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to the nesting birds.) Maintain a daily log detailing monitoring and nesting activity (not all bird species are listed). Sample monitoring report are on CONSTRUCTION FORMS AND DETAILS below. Within 30 days after completion of construction, submit a [Logs/Summary Of Bird Nesting Monitoring Report](#) to the Contracting Officer and cc the Environmental POC detailing nesting and nesting success/failure including species, number of nests created, location, number of eggs, number of offspring generated during the project and reasons for nesting success or failure, if known.

- b. Nesting Activity Notification: Report any nesting activity observed to the Contracting Officer who will have sole authority for any work stoppages, creation of the buffer area, or restart of construction activities. In addition, notify the personnel indicated in the table ORDER OF CONTACT OF CORPS PERSONNEL above.
- c. Nesting Within Construction Area:
 - (1) If nesting begin within the construction area, a temporary, 300-foot buffer is to be created around the nests and marked with signs to avoid entry. The area is to be left undisturbed until nesting is completed or terminated, and the chicks fledge. The decision to allow construction in a former nesting site will be determined by the Contracting Officer in consultation with the U.S. Fish and Wildlife Service and the FF&WCC. Access to the nesting sites by humans (except limited access when accompanied by the bird monitor or Contracting Officer), equipment or pets under control of the Contractor is prohibited.
 - (2) If nesting occurs within the construction area, place and maintain a bulletin board and maintained in the contracting shed with the location map of the construction site showing the bird nesting areas and a warning, clearly visible, stating that BIRD NESTING AREAS ARE PROTECTED BY THE FLORIDA THREATENED AND ENDANGERED SPECIES ACT AND THE FEDERAL MIGRATORY BIRD TREATY ACT.
 - (3) Birds will find the top of the dike or the flat interior desirable nesting habitat. Any stoppage of construction activity could induce nesting, subsequently, construction could be altered or stopped to avoid impacting the birds. Areas which are potentially suitable for nesting can be altered to make the area undesirable. One approved method is the placement of stakes at 10- to 15-foot intervals and tie flagging between the stakes in a web fashion. This may dissuade bird nesting until construction can be resumed. In addition, the disposal area basin can be flooded prior to the beginning of nesting season to the elevation required for displacement from the disposal of dredged material in order to make the basin undesirable for bird nesting.
- d. Work Delay: Delays in work due to the fault of negligence of the Contractor or the Contractor's failure to comply with this specification is not compensable. Any adjustments to the contract performance period or price that are required as a result of compliance with this section is to be made in accordance with the Clause SUSPENSION OF WORK.

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3.1.5.9 Protection of Gopher Tortoise (GT) Populations (*Gopherus polyphemus*)

The Contractor shall keep construction activities under surveillance, management, and control to prevent impacts to GTs and their burrows. The Contractor shall stay at least 25 feet from the entrance of individual burrows. All construction personnel shall be advised that GTs are listed by the State of Florida as a Threatened Species and protected by the FAC, Chapter 68A-27.004. The Contractor shall be held responsible for taking, harming, or harassing the tortoises, their eggs or their burrows as a result of the construction. The destruction of GT burrows constitutes taking under this law. If a burrow cannot be avoided, the Contractor shall contact the Contracting Officer and Project Biologist prior to any construction activity within 25 feet of the borrow. In addition, if a burrow cannot be avoided, the Contractor shall obtain a relocation permit from FWCC and abide by the trapping and relocation permitting conditions listed in the State of Florida's "GOPHER TORTOISE PERMITTING GUIDELINES" located on the web at: <http://www.myfwc.com/license/wildlife>.

Taking: If the construction work kills tortoises, it will be the Contractor's responsibility to obtain an emergency take permit from the State and the FFWCC and pay the fine associated with the permit. Taking includes the entombment or killing of gopher tortoises as a result of bulldozing, grading, paving, or building construction. If such occurs, the Contractor shall notify the Project Biologist.

3.1.6 Seagrass Protection Measures

- a. Instruct all personnel associated with the project of the presence of seagrasses and the need to avoid contact with seagrasses.
- b. All construction personnel are to be advised that there are civil and criminal penalties for harming or destroying seagrasses. The Contractor may be held responsible for any seagrasses harmed or destroyed due to construction activities.
- c. Do not anchor, place pipeline, or stage equipment in a manner that will cause any damage to seagrasses or hardbottoms. Anchoring, placing pipeline, or staging equipment is to avoid these sensitive areas. If such activities cannot be done without affecting these sensitive areas, the activities are to cease and the Contracting Officer and Project Biologist are to be immediately notified (no later than the morning following the next working day if the incident occurs after normal working hours). Any actual or potential incident involving damage to, or disturbance of, seagrasses or hardbottoms is to be reported.

3.1.7 Protection of Air Resources

Keep construction activities under surveillance, management, and control to minimize pollution of air resources. All activities, equipment, processes and work operated or performed by the Contractor in accomplishing the specified construction is to be in strict accordance with the applicable air pollution standards of the State of Florida (Florida Statute, Chapter 403 and others and Chapters 200 series of the FAC) and all Federal emission and performance laws and standards, including the U.S. Environmental Protection Agency's Ambient Air Quality Standards.

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3.1.7.1 Particulates

Particulates, such as dust, is to be controlled at all times, including weekends, holidays, and hours when work is not in progress. Maintain excavations, stockpiles, haul roads, permanent and temporary access roads, plant sites, spoil areas, borrow areas, and work areas within or outside the project boundaries free from particulates that would cause air pollution standards to be exceeded or that would cause a hazard or nuisance. Provide equipment and approved methods to control particulates as the work proceeds and before a problem develops.

3.1.7.2 Odors

Odors are to be controlled at all times for all construction activities.

3.1.8 Protection of Sound Intrusions

Keep construction activities under surveillance and control to minimize damage to the environment by noise.

3.2 POSTCONSTRUCTION CLEANUP

Clean up any area(s) used for construction.

3.3 PRESERVATION AND RESTORATION OF LANDSCAPE AND AQUATIC VEGETATION DAMAGES

Restore all landscape features and aquatic vegetation damaged or destroyed during construction operations outside the limits of the approved work areas. Such restoration is to be a part of the Environmental Protection Plan as defined in subparagraph ENVIRONMENTAL PROTECTION PLAN of paragraph SUBMITTALS above. This work is to be accomplished at the Contractor's expense.

3.4 INVASIVE AND NUISANCE SPECIES

3.4.1 Prevention of Invasive and Nuisance Species Transfer

Thoroughly clean equipment prior to and following work on the project site to ensure that items/materials including, but not limited to, soil, vegetative debris, eggs, mollusk larvae, seeds, and vegetative propagules are not transported from a previous work location to this project site, nor transported from this project site to another location. Prevention protocols require cleaning all equipment surfaces, including but not limited to, undercarriages, tires, and sheet metal. All equipment, including but not limited to, heavy equipment, vehicles, trailers, ATV's, and chippers must be cleaned. Smaller equipment, including, but not limited to, chainsaws, loppers, shovels, and backpack sprayers, must be cleaned and inspected to ensure they are free of eggs, vegetative debris, vegetative propagules, etc. Utilize any method accepted by the Government; common accepted methods include pressure washing and steam cleaning/washing equipment. Prevention protocols are to also address clothing and personal protective equipment.

Prior to the commencement of work, complete and provide an invasive and nuisance species transfer prevention plan to the Corps for approval. This plan is to be part of the Environmental Protection Plan as defined in subparagraph ENVIRONMENTAL PROTECTION PLAN above. The invasive and nuisance species transfer prevention plan is to identify specific transfer

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prevention procedures and designated cleaning sites/locations. Prevention protocols may vary depending upon the nature of the project site. Ensure all equipment coming onto and leaving the project site is inspected and not harboring materials that would spread, or potentially spread, invasive and nuisance species onto or off the project site. Provide a report with photographs, verifying equipment brought on site was cleaned and provide a report verifying equipment was cleaned prior to removal from the project site.

Work delays due to the fault and/or negligence of the Contractor or the Contractor's failure to comply with this specification is not to be compensable.

3.4.2 Invasive and Nuisance Species Reporting

Report sightings of invasive and nuisance species not identified and documented in the pre-construction condition within 24 hours. The reporting is to include the date, time, location (latitude and longitude preferred), environmental conditions, circumstances surrounding the sighting, disposition/behavior of the species, and any other notable observations. Pictures of the reported species are desirable. Reporting is to be made to the Jacksonville District Corps of Engineers Planning Division, Environmental Branch and Operations Division, Invasive Species Management Branch, InvasivePreventionPlan@usace.army.mil. Points of contact will be provided in the Preconstruction Conference.

3.4.2.1 Invasive And Nuisance Species Report

Within 30 calendar days following completion of work, submit a report with photographs verifying equipment brought on site was cleaned to the Contracting Office, Project Biologist, and Environmental Compliance POC. Provide a report with photographs verifying equipment was cleaned prior to removal from the project site.

3.5 MAINTENANCE OF POLLUTION CONTROL FACILITIES

Maintain all constructed facilities and pollution control facilities and devices for the duration of the contract or for that length of time construction activities create the particular pollutant.

3.6 CONSTRUCTION FORMS AND DETAILS

See the Construction Forms & Details module at the following web address: <https://www.saj.usace.army.mil/About/Divisions-Offices/Engineering/Design-Branch/Specifications-Section/>

Also, see web site address <https://odess.usace.army.mil> to obtain the Sea Turtle/Post-Hopper Dredging Project Checklist, Endangered Species Observer Program Forms, Sea Turtle Tagging and Relocation Report, and Sea Turtle Trawling Report. Links to these forms are under the heading Turtle Information.

-- End of Section --

SECTION 01 57 20

Appendix A

ODESS GUIDANCE



Operations and Dredging Endangered Species System (ODESS) Guidance

MONITORING ENDANGERED SPECIES

Maintenance of US waterways for navigation is essential for national and international trade, job creation, and national security as well as for hydropower, flood protection, municipal water supply, agricultural irrigation, recreation, and regional development.

Typically, cutterhead pipeline, hopper, and mechanical dredges are used to maintain navigation depths in these channels and to construct new waterways. However, hopper dredging in the southeastern US potentially impacts five species of threatened or endangered sea turtles (Kemp's ridley, leatherback, loggerhead, green, hawksbill) and three species of sturgeon (shortnose, gulf, and atlantic).

To monitor these impacts and assist in the evaluation of these data, the US Army Corps of Engineers (USACE) Engineer Research and Development Center (ERDC) created the Sea Turtle Data Warehouse (STDW) in 1992. The database was designed as a central repository for current and historical data on impacted sea turtles.

In 2014 the US Army Corps of Engineers, National Dredging Quality Management Program (DQM), at the request of USACE Headquarters (HQUSACE), partnered with ERDC; the US Army Corps of Engineers, Mobile District, Spatial Data Branch (CESAM-OP-J) and Bowhead to develop a new and enhanced STDW database and website. This new Program is referred to as the Operations & Dredging Endangered Species System (ODESS) Program. The use of ODESS will facilitate enhanced monitoring and data collection, enable faster transmittal of information, and meet threatened and endangered species reporting requirements to the National Marine Fisheries Service (NMFS).

In order to monitor dredging impacts on threatened and endangered aquatic species, the dredge shall be equipped with a dedicated tablet computer running ODESS software to track and document the presence of sea turtle, sturgeon, and marine mammal species during dredging operations.

The ODESS system, which consists of a tablet computer with an Internet connection, shall be a standalone system, exclusive to other systems, and shall have USACE ODESS data



collection and reporting software, referred to as the ODESS Field Collector (FC) tool, installed by USACE ODESS support personnel. In the event hardware or software problems prevent the storage or transmission of the collected data, paper copies of the latest ODESS forms and information shall be maintained and submitted to ODESS Support and the USACE Inspector or Contracting Officer according to the schedule outlined in the contract specifications.

OBSERVER QUALIFICATIONS AND TRAINING

Prior to the initiation of the project, Observers shall be familiar with the operation of the ODESS FC tool and proficient in its use so as to be able to prepare and transmit the results of their observations. ODESS system webinar training can be requested by contacting ODESS Support at ODESS@usace.army.mil or 1.877.840.8024.

Depending on the target audience (Observer, dredging Contractor, USACE District personnel, or other Federal agencies), ODESS training could, in addition to the webinar training, consist of demonstrating the steps involved in setting up the FC tool on the dredge, loading Observer-collected data and attachments into the FC tool, submitting these data and attachments to the ODESS database, and/or navigating around the ODESS public website to view and pull down data and/or decision-making information for later analysis.

OBSERVER DATA COLLECTION AND REPORTING

Observers shall record the results of the threatened and endangered species monitoring (described in Section 3.1.5.1, "Endangered Species Protection" of Section 01 57 20, "Environmental Protection," of this specification) in the ODESS system by filling in the appropriate electronic forms on the ODESS FC tool and transmitting the data to the ODESS database. If there is an issue with recording data straight to the FC tool due the logistical nature of how the Observer is collecting this data, paper copies of these forms can be downloaded from the ODESS public website (<https://dqm.usace.army.mil/odess/>) and later entered into the FC tool when the Observer has the best opportunity.

Start of the Project

Prior to the start of dredging, Observers shall verify that the ODESS FC tool is installed and operational on a dredge's dedicated tablet computer and that a viable Internet connection is available. In addition, before a project is initiated, on the ODESS FC tool homepage Observers shall retrieve (or "pull down") project-specific information from the ODESS database and perform a one-time setup of the dredging project by establishing the dredge name and time zone.

During the Project

The following forms shall be used in the FC tool and submitted to the ODESS database at the indicated reporting frequency.

Load Data Form

Observers shall complete the Load Data Form, including a description of screen contents and sea conditions, based on their observations. This form shall be completed and

transmitted to the ODESS database for each load. At the end of each Observer shift, or when an Internet signal is available (not to exceed 24 hours from the start of the shift), the Observer shall submit all of his/her Load Data Forms. If this is not possible due to hardware or software problems, the Observer shall revert to email submission of the forms to ODESS@usace.army.mil.

Sea Turtle Incidental Data Form

If a sea turtle or its remains are identified during a load inspection, after the appropriate parties are notified (according to the requirements identified in the endangered species compliance section of the contract specification), a Sea Turtle Incidental Data Form shall be completed and submitted to the ODESS database as soon as possible (not to exceed 12 hours after the incident). Any applicable documentation (scanned copies of the paper Observer load and incident forms, species photos, etc.) shall be included as electronic attachments (.JPG or .PDF) and submitted using the FC tool.

Sturgeon Incidental Data Form

If a sturgeon or sturgeon parts are identified during a load, after the appropriate parties are notified (according to the requirements identified in the endangered species compliance section of the contract specification), a Sturgeon Incidental Data Form shall be completed and submitted to the ODESS database as soon as possible (not to exceed 12 hours after the incident). Any applicable documentation (scanned copies of the paper Observer load and incident forms, species photos, etc.) shall be included as electronic attachments (.JPG or .PDF) and submitted using the FC tool.

Marine Mammal Observation Data Form

If a large whale is observed, both the Dredge Load and the Marine Mammal Observation Data Forms shall be completed and submitted (not to exceed 12 hours after the observation) to ODESS Support at ODESS@usace.army.mil consistent with the endangered species compliance section of the contract specification (Section 3.1.5.1, "Endangered Species Protection" of Section 01 57 20, "Environmental Protection").

Observers are required to use the FC tool to send all incident attachments and any necessary documentation (i.e., pictures, etc). Do not send attachments via personal email unless the FC tool is unavailable. Also, the paper forms, if needed, should be used as either a "scratch pad" for data collection notes or used any time the FC tool becomes unavailable. The FC tool is the **primary** means of observer data collection and reporting **not** the paper forms. It is not required to scan, attach and submit a copy of the load and incident paper forms as part of the electronic incident record unless it is needed to support the electronic incident record (e.g., species diagram markups).

End of the Project

At the completion of project, the dredging Contractor shall coordinate with a designated USACE district QA/QC point of contact (POC) to determine whether electronic or paper copies of all applicable Observer paper forms will be submitted for the project. Information previously entered on the Post Hopper Dredging Checklist will be available on the ODESS public website (<https://dqm.usace.army.mil/odess>) for the dredging project.

HARDWARE REQUIREMENTS

The dredge shall be equipped and the Contractor is responsible for an ODESS hardware system consisting of a tablet computer, wireless keyboard, wireless mouse and data modem (or equivalent onboard internet connection) along with a proper tote bag and setup location for the afore mentioned hardware components. If a hardware problem occurs, or if a part of the system is physically damaged, the Contractor shall be responsible for repairing it within 48 hours of determination of the condition. The Contractor shall also keep ODESS personnel updated on the status of the onboard ODESS system and the progress of any repairs.

Computer

The Contractor shall provide a dedicated onboard tablet computer for use by the observers and shall have ODESS software installed on it prior to project initiation. This computer shall be located and oriented to allow data entry and data viewing. It must meet or exceed the following specifications:

Tablet Hardware Component	Specification
CPU	Intel or AMD processor with a (non-overclocked) clock speed of at least 2.4 gigahertz (GHz)
Hard Disk	128 gigabytes (GB); solid state internal storage
RAM	4 gigabytes (GB)
Network Adapter	Internal wired or wireless network hardware to match internet connection
Video Adapter	Support for 1024x768 resolution at 16-bit color depth
Display	>= 10.8 in.
Integrated Camera	2MP HD webcam (front); 8MP (back)
Ports	1 free USB port

Internet Access

The Contractor shall maintain an Internet connection capable of transmitting data to the ODESS database. The telemetry system shall always be available and have connectivity in the contract area. If connectivity is lost, unsent data shall be stored locally within the FC tool and transmitted upon restoration of connectivity. The Contractor shall acquire and install all necessary hardware and software to make the Internet connection available for data transmission to the ODESS database. The hardware and software must be configured to allow remote access to the computer by USACE ODESS personnel. Coordination between the dredging company's IT and ODESS Support may be required in order to configure remote access through any security, firewall, router, and telemetry systems. Telemetry systems must be capable of meeting these minimum reporting requirements in all operating conditions.

SOFTWARE REQUIREMENTS

ODESS personnel shall be responsible for installing and testing all ODESS software tools on the dedicated onboard ODESS tablet computer. No other software which conflicts with the ODESS function of recording and transmitting data shall be installed on the tablet computer. The Contractor shall be responsible for installing and/or maintaining any necessary manufacturer-provided software for the installed hardware. If any software problem occurs, the Contractor shall contact ODESS Support at ODESS@usace.army.mil or 1.877.840.8024.

The ODESS tablet computer shall have the following minimum software installed in support of the ODESS system.

Software Component	Specification
Operating System	Windows 10, Contractor-installed
Browser**	Chrome, Internet Explorer, Contractor-installed
ODESS Software	Field Collector (FC) tool, USACE ODESS Support-installed
Remote Access Software	Team Viewer, USACE ODESS Support-installed
**Latest version recommended, Chrome is preferred.	

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SECTION 01 57 25

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SECTION 01 57 25

TURBIDITY AND DISPOSAL MONITORING

PART 1 GENERAL

1.1 SCOPE

The work covered by this section consists of furnishing all labor, materials, and equipment, and performing all work required to obtain, analyze, and report the results of turbidity and disposal monitoring.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. The following is to be submitted in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Turbidity Monitor Qualifications; G, D0

Turbidity Monitoring Plan; G, D0

Draft Turbidity Sampling Map; G, D0

Testing Equipment And Procedure; G, D0

SD-03 Product Data

Calibration Standard; G

SD-06 Test Reports

Turbidity Monitoring Reports; G, D0

Turbidity Exceedance Report; G, D0

Updated Turbidity Exceedance Report; G, D0

1.3 TURBIDITY MONITOR QUALIFICATIONS

No later than 10 calendar days after NTP, and prior to hiring, the Contractor is to furnish to the Contracting Officer for approval, the names and qualifications of those individuals performing turbidity monitoring, along with 24-hour contact information. Appropriate qualifications for a turbidity monitor must be familiar with the dredging techniques being used and professional experience monitoring water quality without a record of permit violations. Each monitor is to submit a resume which includes the names and locations of dredging projects worked on, permit number (if known), and dates spent monitoring. In addition, monitor needs to certify they have read and are familiar with the FDEP Standard Operating Procedures (SOPs) for sampling and analysis.

1.4 TURBIDITY MONITORING PLAN

The Contractor is to submit, no later than 10 calendar days after NTP, the following:

- a. A statement as to the person who is to be responsible for turbidity monitoring and reporting to be performed.
- b. Method of turbidity monitoring and reporting that will be performed.
- c. A [Draft Turbidity Sampling Map](#) is to have a georeferenced map, overlaid on an aerial photograph, indicating the location of the current construction activity, the sampling locations (background and compliance), the visible plume pattern of the applicable mixing zone, and the direction of flow. Also include on the map the boundaries of any benthic resources or location of nearby Outstanding Florida Waters, if applicable.
- d. The turbidity monitoring requirements of any special areas such as outstanding Florida waters.
- e. Documentation that the person(s) who will be conducting the turbidity monitoring meets the following requirements:
 - i. is independent of the construction contractor;
 - ii. has formal training in water quality monitoring; and
 - iii. has professional experience in monitoring turbidity for coastal construction projects; and submit a scope of work for the turbidity monitoring to ensure that the appropriate equipment is available to conduct the monitoring correctly at any location, and under any safe conditions.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 MONITORING REQUIREMENTS

3.1.1 General

Conducted sampling in accordance with techniques described in the latest edition of Standard Methods published by the American Public Health Association (APHA), American Water Works Association (AWWA), and Water Pollution Control Federation (WPCF), and other current techniques recognized by the scientific community and approved by the Jacksonville District, Corps of Engineers. Analyze samples obtained for turbidity analysis within 30 minutes of collection. Take samples with a sampler obtaining samples uncontaminated by water from any other depth. Additionally, any additional requirement for turbidity monitoring called out in the FDEP permit must be complied with. If there is any conflict with the afore mentioned references with the FDEP permit, the FDEP requirements have precedence.

3.1.1.1 Turbidity Monitoring Equipment

Monitoring required for turbidity is to be measured in Nephelometric

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Turbidity Units (NTU) using a Nephelometer. Global Positioning System (GPS) is also required to record sampling stations. Provide the Government with a certification, attesting to the accuracy of [Testing Equipment And Procedure](#) no later than 10 calendar days after NTP. Provide a duplicate of the [Calibration Standard](#) used to calibrate the testing instrument as well as a complete set of operating instructions for the turbidity testing equipment no later than 10 calendar days after NTP. This standard will be used throughout the project to maintain the calibration of the equipment.

Provide instruments used to measure turbidity that are fully calibrated with primary standards within one month of the commencement of the project, and at least once a month throughout the project. Verify calibration with secondary standards each morning prior to use, after each time the instrument is turned on, and after field sampling using two secondary turbidity standards that bracket the anticipated turbidity samples. If the post-sampling calibration value deviates more than 8 percent from the previous calibration value, report results as estimated and include a description of the problem in the field notes.

Perform analysis of turbidity samples in accordance with DEP-SOP-001/01 FT 1600 Field Measurement of Turbidity:
<https://floridadep.gov/dear/quality-assurance/content/dep-sops>

3.1.2 Dredging and Disposal Locations

Obtain maps and electronic files (Shapefiles, KML, or similar) for all Outstanding Florida Waters (OFW) within the project area. Maps and electronic files can be downloaded from the DEP GIS website at the following link: <https://ca.dep.state.fl.us/mapdirect/?focus-standard>.

Conduct monitoring at dredging and disposal locations provided sea, wind, visibility, marine traffic and other conditions allow for safe performance of the activity. Document all instances where turbidity monitoring within the bounds of the above parameters could not be performed. If weather related, provide National Weather Service (NWS) advisories for the day and time turbidity monitoring was not conducted.

3.1.2.1 Frequency

Both compliance sites Type 1 and 2 apply to the dredging and disposal location.

a. Compliance Site Type 1 (where the mixing zone does not abut seagrass): Twice daily at least 4 hours apart during all dredging, filling or disposal operations. If overflow occurs from filling a hopper dredge, disposal barge or scow barge, then monitoring shall occur at least once during each loading event when the turbidity reaches the edge of the mixing zone.

b. Compliance Site Type 2 (where a shortened mixing zone abuts seagrass): Hourly, when visual observation indicates a turbidity plume extends into areas containing seagrasses. If overflow occurs from filling a hopper dredge, disposal barge or scow barge, then monitoring frequency shall be increased to every 30 minutes during loading and up to 30 minutes after overflow has ceased.

Monitoring for a hopper dredge shall be conducted for each hopper dredge load during daylight hours. At the dredge site, sampling shall

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be conducted after overflow from the hopper begins and the associated turbidity plume has reached the edge of the mixing zone. At the fill placement site, sampling shall be conducted after discharge from the hopper begins and the associated turbidity plume has reached the edge of the mixing zone.

Sampling is to be conducted while the highest project-related turbidity levels are crossing the edge of the mixing zone. Since turbidity levels can be related to pumping rates, the dredge pumping rates shall be recorded, and provided to the Department upon request. The compliance samples and the corresponding background samples shall be collected at approximately the same time, i.e., background sample shall immediately follow the compliance sample.

3.1.2.2 Location

- a. Background: Sampling is to occur at surface (approximately one foot below the surface), mid-depth (for sites with depths greater than 6 feet), and bottom (approximately 6 feet above the bottom for sites with depths greater than 25 feet). All background sampling is to occur clearly outside the influence of any artificially generated turbidity plume or the influence of an outgoing inlet plume.

Dredge site: Samples are to be collected at least 300 meters up-current from the source of turbidity at the dredge site.

Dredged Material Management Area 2D: At least 300 meters upcurrent from the point of discharge and outside of any turbidity generated by the project.

- b. Compliance: Sampling shall occur at surface (approximately one foot below the surface), mid-depth (for sites with depths greater than 6 feet), and bottom (approximately 6 feet above the bottom for sites with depths greater than 25 feet).

Dredge Location: Samples shall be collected 150 meters down-current from the clamshell, cutterhead or the hopper dredge overflow point, or 10 meters from the edge of the nearest SAV bed in the downcurrent direction, whichever is closest to the clamshell, cutterhead, or overflow point and from any other source of turbidity generated by the dredge, in the densest portion of any visible turbidity plume. If no plume is visible, follow the likely direction of flow.

Dredged Material Management Area 2D: Samples shall be collected 150 meters downcurrent from the point of discharge or 10 meters from the edge of the nearest SAV bed in the downcurrent direction, whichever is closest to the discharge point and within the densest portion of any project-related turbidity plume.

3.2 TURBIDITY TESTS

3.2.1 Testing

Whenever there is doubt as to the adequacy of the testing or validity of the results, the Contracting Officer may direct that additional tests be performed at no additional cost to the Government.

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3.2.2 Reporting

Record the monitoring data on summary forms, and as a daily QC test in the RMS system, that contain the pertinent information in the following paragraphs. If there is no construction activity requiring turbidity on a given day, the contractor is to fill out the daily turbidity monitoring report and explain why no monitoring occurred on that day. A standardized form meeting the criteria below, such as DEP's 'Turbidity Monitoring Form' and/or 'Turbidity Monitoring Form - Additional Monitoring Stations' is to be utilized. If one of the DEP forms is used, follow the associated instructions for guidance on naming, filing and distributing the reports. DEP's standardized forms and instructions can be found at <https://floridadep.gov/rcp/beaches-inlets-ports/content/tools-applicants>. Other example forms are on the web site indicated [.../content/tools-applicants](#). Other example forms are on the web site indicated in paragraph CONSTRUCTION FORMS AND DETAILS below. Submit other data in the form supplied by the laboratory chosen to do the analysis. Electronically submit all [Turbidity Monitoring Reports](#) to the Contracting Officer within 24 hours after completion of each test and submit weekly to the Environmental Compliance POC and FDEP within 7 days of collection. Electronic mail addresses of the Corps and FDEP personnel to receive these reports are provided below. Reports are to be provided in Adobe (.pdf) or Spreadsheets (.xls) files.

NAME	ORGANIZATION	E-MAIL ADDRESS
To be determined	USACE ACO	To be determined
To be determined	CESAJ-PD-E POC	To be determined
JCP Compliance Officer	FDEP	jcpcompliance@dep.state.fl.us

Information to be determined will be provided in the Preconstruction Conference. Names and contact information for the Corps personnel to receive these reports will be provided at the Preconstruction Conference. See Section [01 57 20](#) ENVIRONMENTAL PROTECTION.

3.2.2.1 Report Contents

Monitoring reports are to include the following information for each day during the construction period:

- a. Permit number.
- b. A statement describing the methods used in collection, handling, storage, and quality control methods used in the analysis of the samples.
- c. Turbidity meter calibration documentation.
- d. A statement by the individual responsible for implementation of the sampling program concerning the authenticity, precision, limits of detection, and accuracy of the data.
- e. Specific Parameters:
 - (1) Time of day and date samples were taken.
 - (2) Depth of water body.
 - (3) Depth of each sample.
 - (4) Antecedent weather conditions including wind direction and velocity.

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- (5) Tidal stage and direction of flow.
 - (6) Water temperature, C degrees.
 - (7) Water sample location.
 - (8) Sampling results, the net difference between compliance and background results, and whether the turbidity level is in compliance.
 - (9) Provide GPS coordinates (x and y and/or Latitude/Longitude) for all sampling stations along with the coordinates of the dredge and discharge pipe and the distance between the sampling station and dredge/discharge pipe for each sampling event to demonstrate compliance with the stated sampling distances.
- f. A description of any factors influencing the dredging or disposal operation or the sampling program. Furnish reports daily even when no sampling is conducted. When sampling is not conducted, give a brief statement in the report explaining the reason for not conducting the sampling, such as dredge not working due to mechanical problems or no sampling taken due to high seas. If unable to collect samples due to severe weather conditions, include a copy of a current weather report from a reliable and independent source, such as an online weather service.
- g. A georeferenced map, overlaid on an aerial photograph, indicating the location of the current construction activity, the sampling locations (background and compliance), the visible plume pattern of the applicable mixing zone, and the direction of flow. Also include on the map the boundaries of any benthic resources or location of nearby Outstanding Florida Waters, if applicable. When possible, coordinates should be provided in decimal degrees with a 5-decimal level of precision (i.e., 0.000001). Please also indicate the datum.

3.3 NOTIFICATION

3.3.1 Turbidity Exceedance

If turbidity exceeds background levels by more than 29 NTU, immediately notify via email the Contracting Officer, FDEP, and Corps Compliance staff member. In addition, cease all dredging or disposal activity immediately and take all measures to reduce turbidity. Do not resume dredging or disposal until corrective measures have been taken and turbidity has returned to acceptable levels as determined by proper testing described in subparagraph DREDGING AND DISPOSAL LOCATIONS above. Provide a [Turbidity Exceedance Report](#) to include the following to the Contracting Officer and FDEP within 24 hours of the time when the Contractor first becomes aware of the discharge (email acceptable).

- a. the Project Name;
- b. the Permit Number;
- c. location of the turbidity exceedance (shown on map with GPS coordinates);
- d. Level (NTUs above background) of the turbidity exceedance;
- e. the time and date that the exceedance occurred; and
- f. the time and date that construction ceased.

3.3.2 Re-Commencing Construction

Prior to re-commencing construction, an [Updated Turbidity Exceedance Report](#) is to be emailed to the Government Quality Assurance Representative (QAR), Contracting Officer, Administrative Contracting Officer (ACO), and the

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Environmental Compliance POC in addition to FDEP. with the same information that was included in the Exceedance Report, plus the following information:

- a. turbidity monitoring data collected during the shutdown documenting the decline in turbidity levels and achievement of acceptable levels;
- b. corrective measures that were taken; and
- c. cause of the exceedance.

3.4 WORK DELAY

Delays in work due to the fault or negligence of the Contractor or the Contractor's failure to comply with this specification are not compensable. Any adjustments to the contract performance period or price that are required as a result of compliance with this section are to be made in accordance with the provisions of the Clause SUSPENSION OF WORK.

3.5 CONSTRUCTION FORMS AND DETAILS

See the Construction Forms & Details module at the following web address:
<https://www.saj.usace.army.mil/About/Divisions-Offices/Engineering/Design-Branch/Specifications-Section/>

-- End of Section --

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SECTION 35 20 23

DREDGING

PART 1 GENERAL

1.1 SCOPE

The work covered by this section consists of furnishing all labor, materials, and equipment, and performing all excavation and disposal of all material as specified herein and indicated on the drawings. This scope also includes all necessary measures for protection of the environment. Environmental protection requirements under this Contract are as important to overall completion of the work as other technical aspects. Failure to meet the requirements of these specifications for environmental protection may result in work stoppages or termination for default. No part of the time lost due to any such work stoppages can be made the subject of claims for extensions of time or for excess costs or damages by the Contractor. If the Contractor fails or refuses to promptly repair any damage caused by violation of the provisions of these specifications, the Contracting Officer may have the necessary work performed and charge the cost thereof to the Contractor.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM D1586/D1586M	(2018) Standard Test Method for Standard Penetration Test (SPT) and Split-Barrel Sampling of Soils
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U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2024) Safety and Occupational Health (SOH) Requirements
EM 1110-1-1000	(2015) Photogrammetric and LiDAR Mapping
EM 1110-1-1002	(2012) Survey Markers and Monumentations
EM 1110-1-1003	(2013) NAVSTAR Global Positioning System Surveying
EM 1110-1-1005	(2007) Control and Topographic Surveying
EM 1110-1-2909	(2012) Geospatial Data and Systems
EM 1110-2-1003	(2013) Hydrographic Surveying
EM 1110-2-1009	(2018) Structural Deformation Surveying

ERDC/ITL TR-19-7

(2021) A/E/C CAD Standard - Release 6.1

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

33 CFR 81

72 COLREGS: Implementing Rules

33 CFR 84

Annex I: Positioning and Technical Details
of Lights and Shapes

33 CFR 89

Inland Navigation Rules: Implementing Rules

33 CFR 156.120

Requirements for transfer

29 CFR 1926.21

Safety Training and Education

FLORIDA BOARD OF PROFESSIONAL SURVEYORS AND MAPPERS (FBPSM)

FBPSM

Minimum Technical Standards, Chapters 177,
472, 61G17

Code of Federal Regulations (CFR) indicated herein may be found at
<https://www.govinfo.gov/app/collection/cfr>.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Notice of Intent to Dredge

Relocation of Navigation Aids

Marine Survey Inspection Certificates

SD-02 Shop Drawings

Drag Bar

Acceptance Sections Plan; G, D0

SD-07 Certificates

Historical Period Shipwreck Sites

Surveys

Daily Report of Operations

Misplaced Material

Declaration of Inspection

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1.4 NOTICE OF INTENT TO DREDGE

Prior to commencement of work on this contract, notify the Commander, Seventh Coast Guard District of intended operations to dredge and request that it be published in the Local Notice to Mariners. This notification must be given in enough time so that it appears in the Notice to Mariners at least two weeks prior to the commencement of this dredging operation. Provide a copy of the notification to the Contracting Officer.

1.5 DREDGING RESTRICTIONS

1.5.1 Order of Work

There is no specific order of work for this project. A reach is defined as a single acceptance section unless otherwise approved by the Contracting Officer. Complete a reach in its entirety before dredging begins on any subsequent reach. If multiple dredges are utilized, then multiple reaches may be worked simultaneously. Prior to the start of dredging, coordinate the dredging sequence and needed dredging access timing in Ybor Channel with the Contracting Officer and local dry dock operators. The coordination is to occur at least 30 days in advance of desired timing of dredging access.

1.5.2 Airport Restrictions

Due to close proximity of the Peter O. Knight Airport, all construction cranes and other temporary facilities and equipment must comply with the Federal Aviation Administration requirements. Coordinate and obtain FAA approvals from Peter O. Knight Airport in advance of construction activity which will affect the flight envelope for the airport. The Contractor is to investigate and plan for potential significant lead time necessary for obtaining FAA permit. FAA requirements are applicable for Cut-D and the Lower Sparkman Cut.

1.5.3 Transportation of Material

Water and dredge material are not permitted to overflow or spill out of barges or hopper dredges during transport to the disposal site.

1.6 SPECIAL INSTRUCTIONS

Dredging of insitu rock is not required; but if encountered, report the location and extent of any insitu rock above grade or required depth to the Contracting Officer.

1.7 PUMPING OF BILGES

Contractors are warned that pumping oil or bilge water containing oil into navigable waters, or into areas which would permit the oil to flow into such waters, is prohibited by Section 13 of the River and Harbor Act of 1899, approved 3 March 1899 (30 Stat. 1152; 33 U.S.C. 407). Violation of this prohibition is subject to the penalties under the referenced Acts.

1.8 HISTORICAL PERIOD SHIPWRECK SITES

If any shipwreck, artifact, or other objects of antiquity that have scientific or historical value, or are of interest to the public, are discovered, located, and/or recovered, the Contractor acknowledges:

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- a. The site(s), articles, or other materials are the property of the State of Florida, with title vested in the Department of State, Division of Historical Resource.
- b. To immediately notify the Contracting Officer.

1.9 UTILITY CROSSINGS

1.9.1 General

Investigate the location of all utility crossings. Take precautions against damages which might result from his operations in the vicinity of the utility crossings. Suspend dredging until the damage is repaired and approved by the Contracting Officer if any damage occurs as a result of dredging operations. Costs of such repairs and downtime of the dredge and attendant plant will be at the Contractor's expense.

1.9.2 Known Utility Crossings

A natural gas pipeline is known to cross Cut-C (TB) and Cut-G (PT). A reclaimed water and force main utility is known to cross Port Sutton Channel. A water main and sewer pipeline are known to cross Cut-D (HB). A sewer pipeline is known to cross the Lower Sparkman Channel. An electrical transmission line and force main line is known to cross the Upper Sparkman Channel. The approximate areas of these utilities are shown in the drawings. Upon award of this contract, call 811 OR 1-800-432-4770 (Sunshine State One Call) to determine the exact location.

1.10 PERMITS

Refer to the Clause PERMITS AND RESPONSIBILITIES and the paragraph PERMITS AND AUTHORIZATIONS of Section 01 57 20 ENVIRONMENTAL PROTECTION.

1.11 FUEL OIL TRANSFER OPERATIONS

In accordance with U.S. Coast Guard regulations ([33 CFR 156.120](#)), couplings used in fuel oil transfer operations on any vessel with a capacity of 250 or more barrels of oil must be either a bolted or full-threaded connection; or a quick-connect coupling approved by the Commandant; or an automatic back-pressure shutoff nozzle used to fuel the vessel. Submit an executed fuel oil transfer (Declaration) form signed by the tanker operator to the Contracting Officer for each refueling operation. Notify the U.S. Coast Guard prior to any refueling. A copy of the [Declaration of Inspection](#) for Refueling is on the web site indicated in the paragraph CONSTRUCTION FORMS AND DETAILS below.

1.12 SIGNAL LIGHTS

Display signal lights and conduct operations in accordance with the General Regulations of the Department of the Army and of the Coast Guard governing lights and day signals to be displayed by towing vessels with tows on which no signals can be displayed, vessels working on wrecks, dredges, and vessels engaged in laying cables or pipe or in submarine or bank protection operations, lights to be displayed on dredge pipe lines, and day signals to be displayed by vessels of more than 65 feet in length moored or anchored in a fairway or channel, and the passing by other vessels of floating plant working in navigable channels, as set forth in Commandant U.S. Coast Guard Instruction M16672.2, Navigation Rules: International-Inland (COMDTINST M16672.2), or [33 CFR 81](#) Appendix A

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(International) and 33 CFR 84 through 33 CFR 89 (Inland) as applicable.

1.13 NOTICE TO MARINERS -- DREDGING CONTRACTS

If any objects are encountered on the channel bottom which could be a hazard to navigation during dredging operations, immediately notify the Contracting Officer of the location of said object and provide any other pertinent information necessary for the Contracting Officer to prepare and issue a Notice to Mariners.

1.14 FINAL CLEANUP

Final cleanup, as stated in the paragraph COMMENCEMENT, PROSECUTION, AND COMPLETION OF WORK, includes the removal of all the Contractor's plant and equipment either for disposal or reuse. Dispose of plant and/or equipment and/or materials in a manner and at locations approved by the Contracting Officer. Unless otherwise approved by the Contracting Officer, the Contractor will not be permitted to abandon any equipment in the disposal area or other areas adjacent to the worksite. Failure to promptly remove all plant, pipeline, equipment, and materials upon completion of the dredging will be considered a delay in the completion of the final cleanup and demobilization work. In such case, the Government will exercise its right as stated in Clause DEFAULT (FIXED-PRICE CONSTRUCTION) to remove any plant and/or equipment and/or materials at the Contractor's expense.

1.15 WORK VIOLATIONS

Work done in violation of these specifications or a verbal or written stop order of the Contracting Officer will be considered as unsatisfactory progress for purposes of progress payments in accordance with Clause PAYMENTS UNDER FIXED-PRICE CONSTRUCTION CONTRACTS.

1.16 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM SYSTEM (DREDGES AND DUMP SCOWS)

1.16.1 System Requirements

See Section 35 20 26 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM - SCOW ULLAGE PROFILE and Section 35 20 27 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM - PIPELINE HYDRAULIC DREDGE and/or 35 20 28 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM - HOPPER DREDGE. However, in the event of NDQMP System failure (not fully operational), notify the Contracting Officer. If the Contractor cannot use the NDQMP System within 48 hours of failure, cease all dredging operations for the vessel until the NDQMP System is fully operational. Any delays resulting from NDQMP System failure will be at the Contractor's expense.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 DAILY REPORT OF OPERATIONS

Prepare in RMS and submit the Daily Report of Operations, using either ENG Form No. 27A or ENG Form No. 4267, for each dredge and/or unloader working. Submit report daily and not in groups (groups = multi-days reports packaged together at one time).

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3.2 NOTIFICATION OF COAST GUARD

3.2.1 Relocation of Navigation Aids

Navigation aids located within or near the areas required to be dredged will be removed, if necessary, by the U.S. Coast Guard in advance of dredging operations. Within 7 calendar days following receipt of Notice to Proceed, notify the United States Coast Guard Sector St. Pete Waterways Management Division ATON Officer, of any plans to dredge adjacent to any aids which require relocation to facilitate dredging. Send the notification via phone to (727) 502-1528, ATTN: BOSON4 Joseph Holt. Submit a formal written request after notification and provide a copy to the Contracting Officer. Contact the U.S. Coast Guard, Joseph.I.Holt@uscg.mil, for information concerning the position to which the aids will be relocated. Do not remove, change the location of, obstruct, willfully damage, make fast to, or interfere with any aid of navigation.

3.2.2 Dredging Aids

Obtain approval from the U.S. Coast Guard for all buoys, dredging aid markers to be placed in the water, and dredging aid markers affixed with a light prior to the installation. Do not color or place dredging aid markers and lights in a manner that they will obstruct or be confused with navigation aids.

3.3 WORK AREA

The Contractor will be permitted to exclude the public from the work areas in the immediate vicinity of his dredging, transporting, and disposal operations including dredging, transporting, and disposal operations. Prevent public access to the discharge end of the pipeline. Erect, maintain, and move as necessary, a restrictive barrier around the discharge of the hydraulic pipeline. Construct the barrier to prevent the public from approaching the discharge from any direction closer than 40 feet. Post signs in a conspicuous location with the wording "DANGER - HIGH PRESSURE DISCHARGE FROM DREDGE". Enforcement is the Contractor's responsibility at no additional cost to the Government. Coordinate enforcement with local enforcement agencies and will be subject to approval of the Contracting Officer. Additionally, place a safety person at the discharge end of the disposal pipeline. The safety person must always be present during discharge operations and will maintain radio communication between the dredge and the disposal operation.

3.3.1 Access

Provide and maintain access necessary for all equipment and plant to and from the work site, mooring area, and disposal area. Ascertain the environmental conditions which can affect the access such as climate, winds, currents, waves, depths, shoaling, and scouring tendencies.

3.3.2 Protection of Existing Waterways

Conduct operations in such a manner that material or other debris are not pushed outside of dredging limits or otherwise deposited in existing side channels, basins, docking areas, or other areas being utilized by vessels. Change the method of operations as may be required to comply with the above requirements. Should any bottom material or other debris be pushed into areas described above, as a result of the Contractor's

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operations, the same must be promptly removed by and at the expense of the Contractor to the satisfaction of the Contracting Officer.

3.3.3 Adjacent Property and Structures

No dredging will be permitted within 25 feet of any structure. Any damage to private or public property or structures resulting from the Contractor's negligence will result in suspension of dredging and require prompt repair at the Contractor's expense as a prerequisite to the resumption of dredging. Verify the location of all structures prior to dredging, and adjust the dredging limits as necessary to ensure the offset distance.

3.3.4 Subaqueous Cable Crossings

Verify the locations and depths of all utility crossings and take precautions against damages which might result from the operations, especially the sinking of dredge spuds and/or anchors into the channel bottom, in the vicinity of utility crossings. If any damage occurs as a result of the operations, suspend dredging until the damage is repaired and approved by the Contracting Officer. Costs of such repairs and downtime of the dredge and attendant plant will be at the Contractor's expense.

3.4 CHARACTER OF MATERIALS

- a. Materials to be removed under this Contract are described in Section 00 31 32 GEOTECHNICAL DATA REPORT as referred to in Section 01 11 00 SUMMARY OF WORK.
- b. Due to typical harbor activities, debris such as trash, rope, chain, cable, tires, miscellaneous scrap metal, woody debris, etc., may be encountered in the channel. Bring all such materials to the surface and dispose of in a disposal area obtained by the Contractor in a manner approved by the Contracting Officer.

3.5 DISPOSAL OF EXCAVATED MATERIAL

3.5.1 General

Transport and dispose excavated material to the disposal area DMPF (Dredge Material Placement Facility) 2D, as designated on the drawings. The average distance from Tampa Harbor to which the material will have to be transported is approximately 7 nautical miles and the maximum distance is approximately 12 nautical miles. All dredged material is to be disposed of within the north cell of DMPF-2D.

3.5.2 Upland Disposal Area

3.5.2.1 Maximum Height of Dredged Materials

Do not place dredged materials and/or water higher than 2.5 feet below the lowest point of the dike crest. Constantly monitor the water height while pumping to insure that this clearance is maintained. Materials deposited in the disposal area above the maximum height or outside the disposal area will result in suspension of dredging operations and require the removal of such materials as a prerequisite to the resumption of dredging.

Do not block or impair drainage of areas adjacent to the diked disposal

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area. Excavate and maintain ditches necessary to prevent blocking or impairing drainage. Provide an adequate number and size of ditches to eliminate all blockage or impairment of drainage adjacent to the diked disposal area.

3.5.2.2 Disposal Area Drainage

If necessary, excavate and maintain ditches to drain all low areas in the dredged material and disposal area to the weirs. Provide an adequate number and size of ditches to eliminate all ponding of water within the limits of the disposal area.

3.5.3 Spillage/Leakage During Transport to Disposal Site

Water and excavated material are not permitted to overflow, leak out, or spill out of the barges, dump scows, or hopper dredges while in transit to the disposal site. Transit to the disposal site begins as soon as dredged material loading into the disposal vessel is completed and the vessel begins moving to the disposal site. Failure to repair leaks or change the method of operation which is resulting in the overflow, leakage, or spillage will result in suspension of dredging operations and require prompt repair or change of operation to prevent overflow, leakage, or spillage as a prerequisite to the resumption of dredging. Appropriate measures may include, but are not limited to, up-to-date U.S. Coast Guard and/or American Bureau of Shipping certification of all disposal-related vessels, maintenance (inspection and/or replacement) of gaskets on barge doors, minimization of excess free liquids in barge loads, pre-transit testing of barge door hydraulics, and pre-transport verification of appropriate weather and sea state conditions. Notify the Contracting Officer within 24 hours if any apparent leaking or spilling of dredged material occurs as indicated by an average loss of draft during transit from the dredging area to the disposal site (forward draft loss plus aft draft loss divided by 2) in excess of 2.0 feet. Excessive leakage may be classified as a mis-dump and be subject to the actions described below in paragraph "Misplaced Material" below.

3.5.4 Placing of Dredged Material

During placement of dredged material in the disposal areas, provide an attendant at each disposal area to maintain constant radio contact with the dredge. This will enable the Contractor's personnel at the disposal areas to immediately notify the dredge in the event of dike or pipeline failure. In the event of dike or pipeline failure, suspend the dredging operations and promptly repair of the dike or pipeline as a prerequisite to the resumption of dredging.

3.5.5 Barges

Water and dredged materials are not be permitted to overflow or spill out of barges while transporting to the disposal site(s). Failure to repair leaks or change the method of operation which is resulting in overflow or spillage will result in suspension of dredging operations and require prompt repair or change of operation to prevent overflow or spillage as a prerequisite to the resumption of dredging.

3.5.6 Dredge Pipelines

3.5.6.1 Dredge Discharge Pipeline

Plainly mark the pipeline access routes with conspicuous stakes, targets and/or buoys to be maintained throughout the contract operations. Maintain a tight dredge discharge pipeline to prevent spilling of dredged material or dredge water outside of the disposal area. Provide and maintain radio communication between the dredge and the disposal areas and the dredge and the Contracting Officer. Inspect the pipeline at least twice daily for leaks. Failure to immediately repair leaks in the discharge pipeline will result in suspension of dredging operations and require prompt repair of pipeline as a prerequisite to the resumption of dredging. Repair any damage to private or public property resulting from the Contractor's operations at the Contractor's expense.

3.5.6.2 Submerged Pipeline

In the event the Contractor elects to submerge the pipeline, rest the pipeline on the bottom, and the top of the submerged pipeline and any anchor securing the submerged pipeline cannot be higher than the required project depth for the channel in which the submerged pipeline is placed. If a pipeline material which is buoyant or semi-buoyant, such as PVC pipe or similar low-density materials is used, securely anchor the pipeline to prevent the pipeline from lifting off the bottom under any conditions. Conduct daily inspections of the submerged pipeline to ensure buoyancy has not loosened the anchors. Remove all anchors when the submerged pipeline is removed. Mark the location of the entire length of submerged pipeline with signs, buoys, lights, and flags conforming to U.S. Coast Guard regulations.

3.5.6.3 Floating Pipeline

Should the Contractor's pipeline not rest on the bottom, it will be considered a floating pipeline and will be visible on the surface and clearly marked. In no case will the Contractor's pipeline be allowed to fluctuate between the surface and the bottom or lie partly submerged. Install lights on the floating pipeline as required in paragraph SIGNAL LIGHTS above. Support the lights either by buoys or by temporary piling, provided by the Contractor and approved by the Contracting Officer. Where the pipeline does not cross a navigable channel, space the flashing yellow all-around lights not over 200 feet apart, unless closer spacing is required by U.S. Coast Guard personnel, in which case the requirements of the U.S. Coast Guard will govern, at no additional cost to the Government.

3.5.7 Booster Pumps

Install booster pumps at least 300 feet from any residential-type building or house. Provide booster pumps, their prime movers, and any auxiliary equipment fitted or equipped with mufflers, noise control enclosures, or other engineering noise control methods, measures, and features such that steady noise emanating from this equipment does not exceed 85 decibels on the A scale at slow response, and impulsive noise does not exceed 140 decibels. Maintain such items throughout the course of the work.

3.5.8 [Misplaced Material](#)

Materials deposited above the maximum indicated elevation or outside of the disposal area template shown will require the degrading or removal of

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such materials at the Contractor's expense. Notify the U.S. Coast Guard Marine Safety Office of any misplaced material as stated in the Clause OBSTRUCTION OF NAVIGABLE WATERWAYS.

3.5.9 Cumulative Summary Disposal Log

Maintain and provide a cumulative summary disposal log which contains the following information:

- a. Load Number.
- b. Disposal Vessel Name and Type (e.g. scow).
- c. Estimated Volume of Load.
- d. Description of Material Disposed or Placed.
- e. Source of Dredged Material.
- f. Disposal/Placement Area (i.e. beach, nearshore, ODMDS, etc.)
- g. Vessel Transit Draft Change.

Furnish a pdf of the cumulative summary disposal log to the Contracting Officer within 24 hours of collection as part of the Contractor's Quality Control (CQC) daily report indicated in paragraph DAILY REPORT OF OPERATIONS above.

3.6 REQUIRED DEPTH, ALLOWABLE OVERDEPTH, AND SIDE SLOPES

3.6.1 Acceptance Sections

For purposes of payment, the work to be done will be divided into acceptance sections. Prior to any dredging, prepare and provide a proposed [Acceptance Sections Plan](#) for approval by the Contracting Officer. Generally, the work should be divided into areas that equate to approximately 30 days of dredging or that constitute separable or definable features of work.

3.6.2 Required Depth

The material actually removed from the designated areas to be dredged, to a depth of not more than the required depth shown on the drawings, will be estimated and paid for in accordance with the provisions contained in the subparagraphs EXCAVATION, UNCLASSIFIED of Section 01 20 00 PRICE AND PAYMENT PROCEDURES.

3.6.3 Allowable Overdepth

To cover the inaccuracies of the dredging process, material actually removed from the designated areas to be dredged, to a depth below the required depth of not more than the allowable overdepth shown on the drawings, will be measured and paid for in accordance with the provisions contained in the subparagraph EXCAVATION UNCLASSIFIED of Section 01 20 00 PRICE AND PAYMENT PROCEDURES.

3.6.4 Side Slopes

Although dredging of side slope material may be necessary to provide the required project channel dimensions (depth and width), the side slopes shown on the drawings are provided for payment purposes only. Side slopes may be formed by box cutting, step cutting, or dredging along the side slope. Material actually removed, within the limits approved by the Contracting Officer, to provide for final side slopes not flatter than that shown on the contract drawings, but not in excess of the amount

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originally lying above this limiting side slope, will be measured and paid for in accordance with the provisions contained in subparagraphs EXCAVATION UNCLASSIFIED of Section 01 20 00 PRICE AND PAYMENT PROCEDURES. Such amount will be estimated and paid for whether dredged in original position or by box cut dredging whereby a space is dredged below the allowable side slope plane on the bottom of the slope for upslope material capable of falling into the cut. End slopes and transition slopes will not be estimated or paid for under this contract. In such cases, a 0 horizontal on 1 vertical will be used with no upslope allowance provision applied outside the required prism.

3.6.5 Excessive Dredging

It is recognized that the limits of dredging need to extend beyond the pay prism in order to remove material from within the pay prism. Therefore, the limits of active dredging may exceed the limits of the pay prism in vertical dimension as well as horizontal dimension to the extent necessary to remove material from the pay prism. Indirect impacts (sideslope sloughing, etc.) may extend even beyond the limits of active dredging. Nevertheless, material taken from beyond the limits of the pay prism as described in subparagraphs ALLOWABLE OVERDEPTH and SIDE SLOPES above will be deducted from the total amount dredged as excessive overdepth dredging, or excessive sideslope dredging, for which payment will not be made. Nothing herein will be construed to prevent payment for the removal of shoals performed in accordance with the applicable provisions of the paragraphs FINAL EXAMINATION AND ACCEPTANCE or SHOALING of this Section.

3.6.6 Areas to be Dredged

Based on information currently available to the Government, areas known to require dredging are depicted on the drawings as shaded areas. The actual areas to be dredged may vary from the shaded areas shown in the drawings. In order to provide the required project dimensions within and throughout the project limits shown on the drawings, remove material located within the project limits as directed by the Contracting Officer, regardless of whether the material is in a shaded area or not. Payment for all dredged material dredged from a shaded area will be made at the applicable contract unit price. Payment for all material dredged from a non-shaded area may be made at the applicable contract unit price; however, the Contractor may request an adjustment to the contract unit price in accordance with DFAR Clause 252.243-7002 REQUESTS FOR EQUITABLE ADJUSTMENT and/or FAR Clause 52.243-4 CHANGES.

3.7 SURVEYS

3.7.1 General

Provide 10 days advance notice, in writing, to the Contracting Officer of the need for a pre-dredging survey or after-dredging survey for final acceptance for each acceptance section. Surveys will be performed in accordance with the paragraph QUANTITY SURVEYS; paragraph LAYOUT OF WORK of Section 01 11 00 SUMMARY OF WORK; Section 01 45 05 DREDGING/BEACH FILL PLACEMENT - QUALITY CONTROL; EM 1110-1-1000, EM 1110-1-1002, EM 1110-1-1003, EM 1110-1-1005, EM 1110-1-2909, and EM 1110-2-1003; FBPSM; and, ERDC/ITL TR-19-7. A copy of the EM's can be downloaded from the following web site:
<http://www.publications.usace.army.mil/USACEPublications/EngineerManuals.aspx>.
A copy of the ERDC/ITL TR-19-7 can be downloaded from the following web site: <https://caddimcenter.erdcdren.mil/default.aspx?p=a&t=1&i=2>.

3.7.2 Contractor Representative

The Contractor or his authorized representative will be notified when soundings and/or sweepings are to be made and will be offered the opportunity to be on board the Government survey vessel, but may decline the offer. All in-place measurement surveys and final acceptance sweep surveys will be performed by the Government. The Contractor's representative must be fully knowledgeable in offshore construction subsurface surveying procedures, techniques, equipment, and horizontal and vertical calibration methods, and state-of-the-art horizontal and vertical accuracy limitations. The Contractor's representative may observe and review, in progress, the adequacy and accuracy of the survey for in-place payment purposes, and for the potential existence of collusion, fraud, or obvious error in the data.

3.7.3 Survey Certification

- a. Immediately upon completion of any survey, the Contractor's representative, based on their on-site review of the survey execution, determine that the survey contains no evidence of collusion, fraud, obvious error, and that subsequent horizontal and vertical corrections are accurately annotated on the subsurface record.
- b. The Contractor's authorized representative must bring aboard the survey vessel a blank copy of the Certification Statement and attest to an acceptable survey by signing the Certification Statement before leaving the vessel. Sample copy of the Certification Statement is on the web site indicated in paragraph CONSTRUCTION FORMS AND DETAILS below.
- c. If the Contractor's authorized representative observes (and quantifies) specific documentary evidence of either fraud, collusion, or obvious error, the survey will be immediately rerun. Resurveys will totally supersede any previously run survey and will be run over the full reach of any Acceptance Section.
- d. If acceptability is not acquired after performing one resurvey of an Acceptance Section, a meeting will be held between the Contractor and the Contracting Officer to expeditiously resolve the issue causing rejection of the survey. Contractor equipment and personnel standby time to resolve acceptability of the survey will be at the Contractor's expense.
- e. In no case will a previously unacceptable survey be later judged acceptable by the Contractor; unless such a reassessment/reevaluation is performed within 24 hours after the original survey, and prior to initiating any resurvey action based upon identifiable collusion, fraud, or obvious error.
- f. Should the Contractor or their authorized representative refuse to certify to the acceptability of a survey for contract payment without identifiable collusion, fraud, or obvious error, then the following actions will follow:
 - (1) Preconstruction (pre-dredging) Survey: Do not commence excavation until representatives of the Contractor and Contracting Officer have met and resolved the basis for refusal of certification. Should the Contractor commence excavation prior to obtaining an

acceptable survey, the Contractor will be liable for any excavation performed. If a resurvey is performed, and accepted, prior excavation will not be measured, estimated, or paid for.

(2) Post-Construction (after-dredging) Survey: The 2-week survey window allowed under subparagraph EXCAVATION, UNCLASSIFIED of Section 01 20 00 PRICE AND PAYMENT PROCEDURES will be indefinitely extended until a final survey is accepted. Any material accretion which might occur due to such a time extension will neither be measured, estimated, or paid for.

(3) Refusal to Certify: Contractor equipment and personnel standby time to resolve refusal to certify to the acceptability of a survey when there is no identifiable collusion, fraud, or obvious error will be at the Contractor's expense and resultant delays will not be the basis for time extensions of the contract.

g. Intermediate surveys taken between the pre-dredging and post-dredging surveys will not be considered for the purposes of determining quantities for final payment and acceptance of the area dredged.

3.7.4 Positioning – Horizontal Location and Vertical/Tidal Determination

3.7.4.1 Real Time Kinematic (RTK) GPS

The government will use RTK GPS for horizontal positioning and the determination of Real Time water levels (tide corrections) for measurement and payment purposes. All data will be collected relative to the mapping datums and horizontal coordinate system(s) as noted within the survey notes on the accompanying plan sheets. All surveying and mapping will be exclusively tied, either by occupation or calibration, to the Project Survey Control Points as identified by tabulation and called for by individual survey notes on the plan sheets. The Kinematic Tidal Datum (KTD) model file (next section) employed for the design surveys/site investigations contributing to the design of this project will be the sole method to relate GPS surveyed heights (NAVD88) to the chart datum, be it Mean Lower Low Water (MLLW) or pool datum. The Contractor is responsible for providing all applicable RTK equipment for their surveying and dredging operations.

3.7.4.2 Kinematic Tidal Datum (KTD)

A file listing the separations between the North American Vertical Datum of 1988 (NAVD88) and the Chart Datum (MLLW or Pool) will be provided to the Contractor for entry into the hydrographic survey software. Obtain the Kinematic Tidal Datum file from the Construction Representative prior to commencing surveying and dredging operations.

3.8 ENCOUNTER OF MUNITIONS AND EXPLOSIVES OF CONCERN (MEC) IN PROJECT AREA

See the "Tampa Harbor O&M North of Sta. 0+00 Cut-E (TB), Probability Assessment For Munitions and Explosives of Concern, Hillborough County, FL" in Appendix A after the end of this section.

3.8.1 Munitions and Explosives of Concern (MEC)

Munition and Explosives of Concern (MEC) are military munitions that may pose unique explosives safety risks and include Unexploded Ordnance (UXO), Discarded Military Munitions (DMM), and Munitions Constituents (MC)

present in high enough concentrations to pose an explosive hazard.

3.8.2 Procedure Upon MEC Encounter

Use due caution when working in coastal areas. In the event a suspect MEC is encountered, follow the 3Rs of explosive safety: Recognize, Retreat, Report. Recognize that munitions are dangerous; Retreat - do not approach, touch, move or disturb it, but carefully leave the area; and Report immediately what you saw and where you saw it to local law enforcement - call 911 and then inform the Contracting Officer. Cease all work efforts until the suspect MEC has been evaluated by emergency response personnel. In the event that the suspect MEC is positively identified as MEC, cease further work efforts until a new MEC probability assessment has been conducted.

3.8.3 Munition Use and Disposal Throughout Florida

The State of Florida was used extensively for war preparations in the 1940s and is home to a number of Formerly Used Defense Site (FUDS). More information on the FUDS program and sites throughout Florida can be found at the below web address.

<https://www.usace.army.mil/Missions/Environmental/Formerly-Used-Defense-Sites/FUDS-GIS>
FUDS are not the only source of MEC along the coastline of Florida. MEC may be found along the entire coastline of Florida, both on land and in State and Federal waters. The militaries of the U.S. and other countries have conducted live-fire training and combat operations at sea for centuries, and commonly disposed of excess, obsolete, and unserviceable munitions in coastal waters.

3.9 INSPECTION

3.9.1 Quality Assurance Representative (QAR)

The QAR will be notified prior to the establishment of horizontal control work (baseline layout, ranges, station flags, shore-based control for EPS/RPS, etc.) and vertical control work (tide staff(s), upland cross sections, construction elevations top/invert, maximum/minimum elevations of dredged materials within disposal area(s), etc.), but the presence or absence of the QAR will not relieve the Contractor of their responsibility for proper execution of the work in accordance with the specifications. The Contractor will be required:

- a. To furnish, on the request of the Contracting Officer or any QAR, the use of such boats, boatmen, laborers, and material forming a part of the ordinary and usual equipment and crew of the dredging plant as may be reasonably necessary in inspecting and supervising the work. However, the Contractor will not be required to furnish such facilities for the surveys prescribed in the paragraph FINAL EXAMINATION AND ACCEPTANCE of this Section.
- b. To furnish, on the request of the Contracting Officer or any QAR, suitable transportation from all points on shore designated by the Contracting Officer to and from the various pieces of plant, and to and from the disposal area.

3.9.2 Failure to Comply

In conjunction with the Clause INSPECTION OF CONSTRUCTION, should the Contractor refuse, neglect, or delay compliance with these requirements,

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the specific facilities may be furnished and maintained by the Contracting Officer and the cost thereof will be deducted from any amounts due or to become due the Contractor.

3.10 FINAL EXAMINATION AND ACCEPTANCE

3.10.1 Final Examination of Work

As soon as practicable and no later than two (2) weeks after the completion of the entire work or any section thereof (if the work is divided into sections) as in the opinion of the Contracting Officer will not be subject to damage by further operations under the contract, such work will be thoroughly examined at the cost and expense of the Government by sounding or by sweeping, or both, as determined by the Contracting Officer. Should any shoals, lumps, or other lack of contract depth be disclosed by this examination, the Contractor will be required to remove same by dragging the bottom in accordance with the subparagraph BED LEVELING below or by dredging at the contract rate of dredging.

The Contractor or authorized representative will be notified when soundings and/or sweepings are to be made and will be permitted to accompany the survey party, but may decline. When the area is found to be in a satisfactory condition, it will be accepted finally. Should more than two sounding or sweeping operations by the Government over an area be necessary by reason of work for the removal of shoals disclosed at a prior sounding or sweeping, the cost of such third and any subsequent soundings or sweeping operations will be charged against the Contractor at the rate of \$11,600 for each day in which the Government plant is engaged in sounding or sweeping and/or is in route to or from the site or held at or near the said site for such operation.

3.10.2 Bed Leveling

Bed leveling by dragging the bottom with a drag bar or other apparatus will be allowed only in the designated dredging areas shown on the drawings. Dragging in areas outside of the designated dredging areas shown on the drawings is specifically prohibited without written approval of the Contracting Officer. Submit drawings and one photograph showing [drag bar](#) equipment used for final leveling work.

3.10.3 Bed Leveling - Reporting and Documenting

Fully document all bed leveling activity including date and time for initiation and completion of bed leveling in the Contractor's Quality Control Report (QCR).

3.10.4 Final Acceptance

Final acceptance of the whole or a part of the work and the deductions or corrections of deductions made thereon will not be reopened after having once been made, except on evidence of collusion, fraud or obvious error, and the acceptance of a completed section will not change the time of payment of the retained percentages of the whole or any part of the work.

3.11 SHOALING

If, before the Contract is completed, shoaling occurs in any section previously accepted, including shoaling in the finished channel because of the natural lowering of the side slopes, redredging at contract price,

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within the limits of available funds may be done if agreeable to both the Contractor and the Contracting Officer.

3.12 CONTINUITY OF WORK

Full payment for, and formal acceptance of acceptance sections, will not be rendered until the acceptance section is excavated to the full depth and extent required by this contract, except by decision of the Contracting Officer. The Contracting Officer reserves the right to withhold the invoiced amounts necessary to protect the Government's interests until acceptance of an acceptance section is achieved.

3.13 NOISE CONTROL

Equip all hauling and excavating equipment and dredge/barges, boats, and tugs used on this work with satisfactory mufflers or other noise abatement devices. Conduct the operations to comply with all Federal, State and local laws pertaining to noise. Limit the use of horns and whistle signals to the minimum necessary in order to ensure as quiet an operation as possible.

3.14 GENERAL DREDGING SAFETY REQUIREMENTS

3.14.1 Dredge Plant Restriction Definitions

To increase the safety of dredges on Federal contracts, the contract plans, and specifications may identify a location within the Federal channel (usually indicated by a channel station or navigation aid) requiring minimum equipment standards. This is a Certification Required Line (CRL). The following are specific designations that may apply to this Contract.

3.14.2 Certification of Attendant and Auxiliary Plant

In addition to the requirements of [EM 385-1-1](#), all supporting floating plant for dredging operations (to include, but not limited to, anchor barges, deck barges, deck/work barges, tug boats, tenders, workboats, pump-out barges, derrick barges, and spider barges) must have a current marine survey inspection certificate. The only exceptions to this requirement include pipeline support pontoons or floats and outboard skiffs under 16 feet in length. The certification must be issued by a licensed and accredited marine surveyor within the previous 12 months. The marine surveyor must be accredited by either the National Association of Marine Surveyors or the Society of Accredited Marine Surveyors. No attendant plant that does not have a valid certification will be used on this Contract. The certifications and inspection must be appropriate for the intended use of the plant in all locations specified in this Contract. Prior to commencement of work, submit [marine survey inspection certificates](#) of all floating plant and equipment.

3.14.3 Minimum Tow-Boat Requirements

In addition to the requirements of [EM 385-1-1](#), when dredging with non-self-propelled dredges seaward of the CRL-L or CRL-C, provide the following.

- a. One primary towboat of adequate size and horsepower to move all attendant and support plant against prevailing tides, currents, and winds during normal and severe weather conditions. Always make the

Tampa Harbor, Maintenance Dredging
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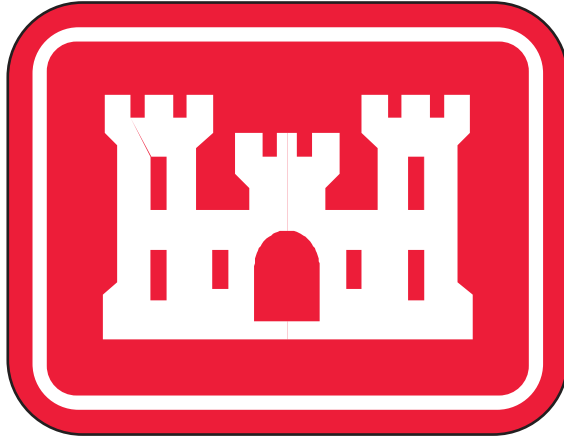
towboat available within the project work area.

- b. A towboat(s) of adequate size and horsepower to move the dredge(s) against prevailing tides, currents, and winds during normal and severe weather conditions must be able to mobilize to the work site, and move the dredge(s) to safe harbor with 4 to 6 hours of first notification.

3.15 CONSTRUCTION FORMS AND DETAILS

See the CONSTRUCTION FORMS & DETAILS module at the following web address:
<https://www.saj.usace.army.mil/About/Divisions-Offices/Engineering/Design-Branch/Specifications-Section/>

-- End of Section --



Tampa Harbor O & M
North of Sta. 0+00 Cut-E (TB)
Probability Assessment
For
Munitions and Explosives of Concern

Hillsborough County, FL

January 23, 2025

Prepared by
Environmental and Munitions Design Center
Baltimore District
U.S. Army Corps of Engineers

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1.0 Introductions

This Munitions and Explosives of Concern (MEC) probability assessment has been prepared for the U.S. Army Corps of Engineers (USACE) Jacksonville District (CESAJ) Navigation, Operations & Maintenance Branch in support of Tampa Harbor dredging contracts.

This document was prepared by USACE Baltimore District (CENAB) Environmental and Munitions Design Center (EMDC), Engineering Division. The document summarizes historical activities involving military munitions and assesses the probability of encountering MEC. This assessment documents the findings and recommendations for Tampa Harbor dredging contracts.

2.0 Assessment Area Background

The assessment area for this document includes all cuts north of Sta. 0+00 Cut-E (TB). See Figure 1. A site-history and past use review of the assessment area was completed. Records and reports involving military munitions within the scope of the assessment area were reviewed to determine the probability of encountering MEC. Only ranges and historical battles that took place over water were evaluated.

The following historical records were reviewed.

- Online Search of Civil War battle areas
- FUDS Portal
- USACE Archives
- Joint Explosive Ordnance Disposal (JEOD) Portal – EOD Incident reports.

3.0 Evaluation for Potential Munitions.

3.1 Online Search of Civil war battle areas.

The **Battle of Tampa**, also known as the "Yankee Outrage at Tampa", was a minor engagement of the American Civil War fought June 30 – July 1, 1862, between the United States Navy and a Confederate artillery company charged with "protecting" the village of Tampa, Florida. Although small, Tampa's port was a key hub of trade for Central Florida, and several blockade runners from Tampa regularly slipped past the Union naval blockade that extended down the Atlantic coast around to Florida's west coast.

On June 30, 1862, the Union gunboat USS Sagamore steamed into Tampa Bay to demand the surrender of the small Confederate (CSA) garrison at Fort Brooke. This demand was rebuffed by Captain John William Pearson, to which Captain A.J. Drake of the Sagamore responded with a warning that his ship would open fire at 6:00 p.m. that same day. The gunboat began a

long-range bombardment of the fort and town that evening as scheduled, to which the fort responded by returning fire with its three larger cannons. Neither side inflicted real damage or suffered any casualties, and after firing several more rounds the next morning from beyond the range of the fort's guns, the Sagamore returned to its blockade station near the mouth of Tampa Bay.

3.2 FUDS Portal

Mullet Key Bomb & Gun Range-

The site was utilized by the Army for a coastal defense site, a sub-post of MacDill Field, and a bombing and gunnery range.

Fort Dade-

The Army used the site as a prison, a training area, and as a bombing and gunnery range. During the period of ownership by the War Department, other Federal agencies were granted use of portions of the island: Department of Commerce, Treasury Department, YMCA. The project was authorized as an Ordnance and Explosive Waste (OEW) project in March 1994 (for investigation of the site as a former bombing and gunnery range).

Passage Key Air to Ground Gun-

During World War II, aircrews assigned to the Sarasota Army Airfield used Passage Key in Tampa Bay for a strafing and skip bombing range. There were two banks of targets facing north and south about 500 feet apart, and each bank had six targets. Crews trained with machine guns and practice and possibly live bombs. The targets are now completely submerged.

Terra Ceia Skip Bombing Target-

Skip bombing was a low-level bombing technique independently developed by several of the combatant nations in World War II, notably Italy, Australia, Britain, Soviet Union and the United States. It allowed an aircraft to attack shipping by skipping the bomb across the water like a stone. Dropped at very low altitudes, the bomb never rises more than about 5 meters (16 ft) above the surface of the water, ensuring that it will hit the side of the ship if it is aimed correctly.

3.3 USACE Archives.

There are several FUDS sites around the Tampa Harbor area three have the potential to produce dud or practice ordnance:

- Pinellas Practice Bomb Range - land based and will not be part of this assessment.

- Mullet Key Bomb & Gun Range.
- Fort Dade.

Mullet Key Bomb & Gun Range

Consists of two Military Munitions Response Program (MMRP) sites and is located approximately 2.5 miles West of the I-275/Sunshine Skyway Bridge.

Project 01 (Mortar and Artillery ranges) covering 29,719 underwater acres west of the Key. Project 01 was used from 1900 to 1920 and destroyed by a hurricane in 1921, the extent of their uses is unknown. A Remedial Investigation/Feasibility Study (RI/FS) was conducted with 1 MEC round, 37mm/2 lb. Hotchkiss round, being located.

Project 02 Bombing and Gunnery Ranges, comprised of 2,466 acres, consists of four overlapping ranges: two High Explosive Bomb Targets, one Practice Bomb Target, and one Air-to-ground Gunnery Range. This Munitions Response Site (MRS) was used between 1942 and 1947.

Fort Dade

Consists of three project sites.

Project 2, Bombing and Gunnery Range is the only MRS of the three that had the potential to produce MEC. An RI/FS completed on 30 Sep 2004 determined that “no clean-up” was required.

3.4 Joint Explosive Ordnance Disposal (JEOD) Portal – EOD Incident reports

26 September 2016 Explosive Ordnance Disposal (EOD) responded to St. Petersburg Beach regarding an ordnance item being discovered on the beach. The item was a 3 pounder/ Hotchkiss 37MM Projectile.

4.0 Summary and Conclusion

After reviewing historical documents, it was determined that all live munition training ranges are located at a distance that would not impact dredging operations within the assessment area. Although the area has a documented historical battle that took place, operations and maintenance (O&M) dredging within the assessment area has been ongoing for decades without MEC discoveries. In addition, all material is to be disposed of into an upland site with no public access.

5.0 Recommendations

The CENAB Military Munitions Design Center recommends classifying this section of Tampa Harbor as having a '**Low Probability**' of encountering MEC, in accordance with DESR 6055-09 Edition 1.

It is recommended that 3R's awareness be provided to all site workers. The following website is a good resource of information for the 3R's briefing, <https://www.denix.osd.mil/uxo/>.

If MEC is encountered, it should be reported by calling 911 and all work must stop until a MEC re-assessment is completed.

Point of contact for this MEC Probability Assessment is David Holland, Ordnance & Explosives Safety Specialist, U.S. Army Corps of Engineers, Baltimore District.

HOLLAND.DAVID.L.10964502
 Digitally signed by
 HOLLAND.DAVID.L.1096450296
 Date: 2025.01.23 11:39:33 -05'00'

Signature: 96

David Holland, Ordnance and Explosive Safety Specialist

WHITTINGTON.ALBERT.J.116
 Digitally signed by
 WHITTINGTON.ALBERT.J.1163137425
 Date: 2025.01.23 12:24:50 -05'00'

Signature: 3137425

Albert Whittington, Ordnance and Explosive Safety Technical Advisor

USACE CESAJ Commander or Designee

Name: Milan A. Mora

MORA.MILLAN.A.1367741470
 Digitally signed by MORA.MILLAN.A.1367741470
 Date: 2025.01.28 08:39:19 -05'00'

Signature:

APPENDIX A FIGURES

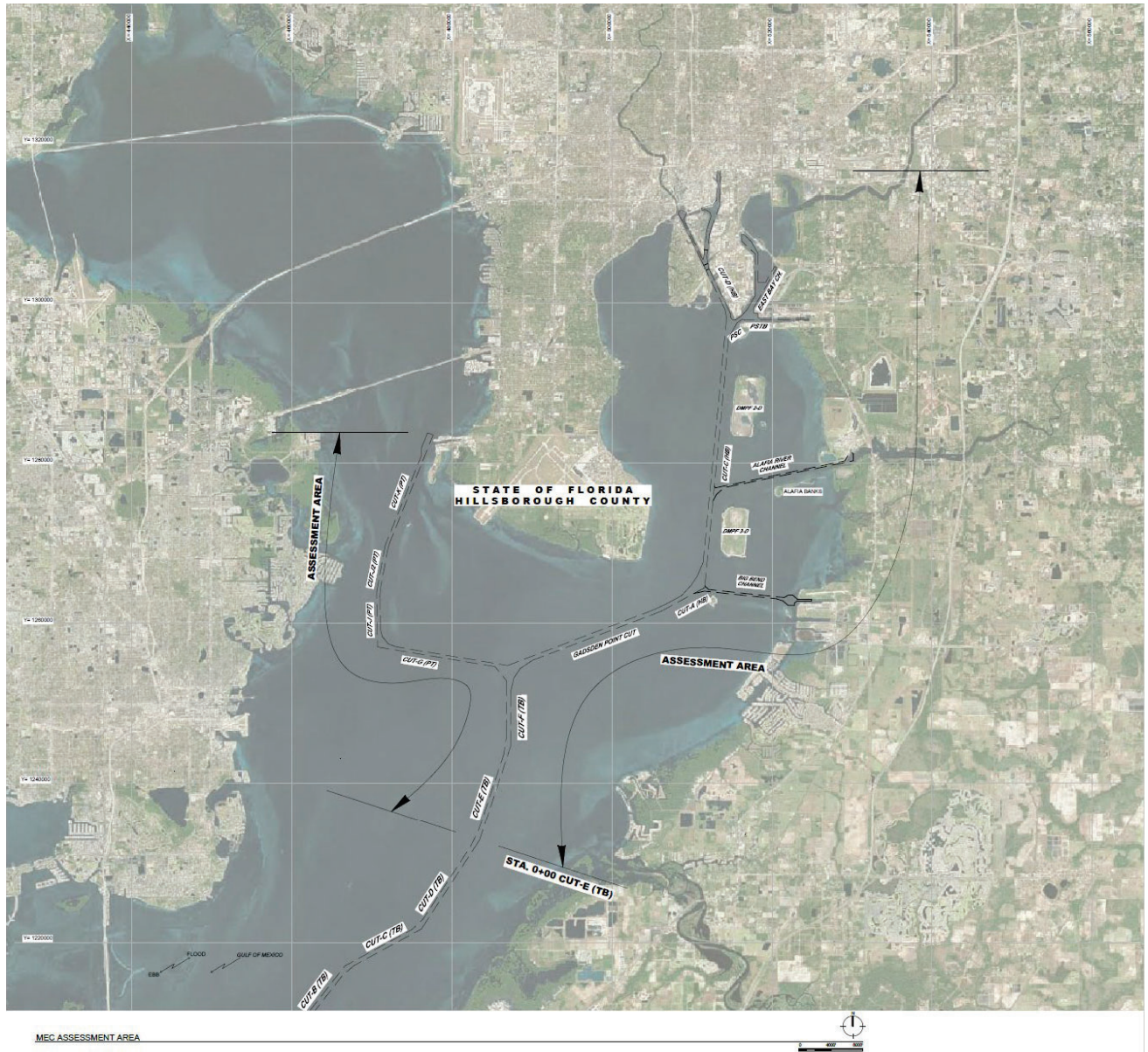


Figure – 1 (Assessment Area)

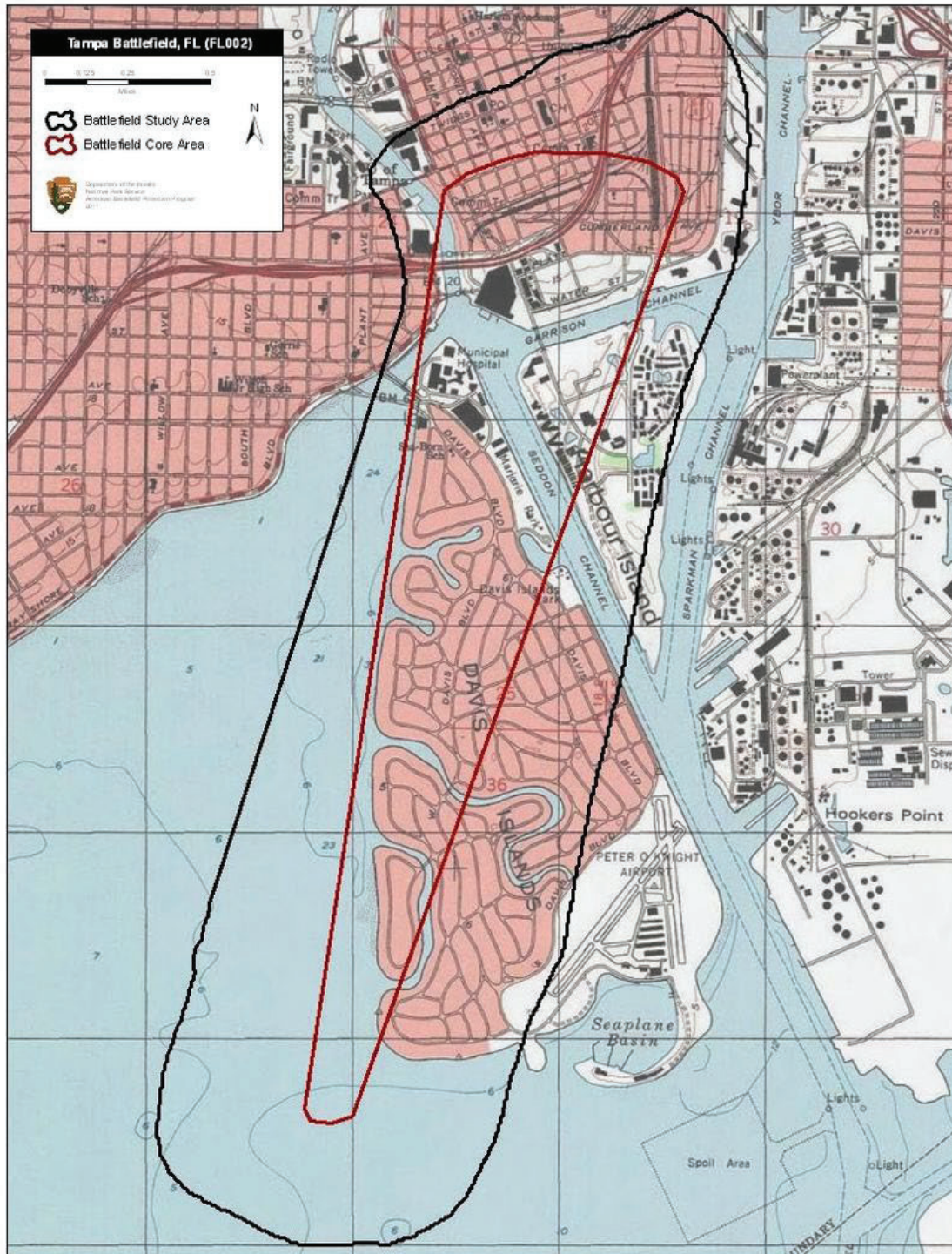


Figure – 2 (Battle of Tampa Area)

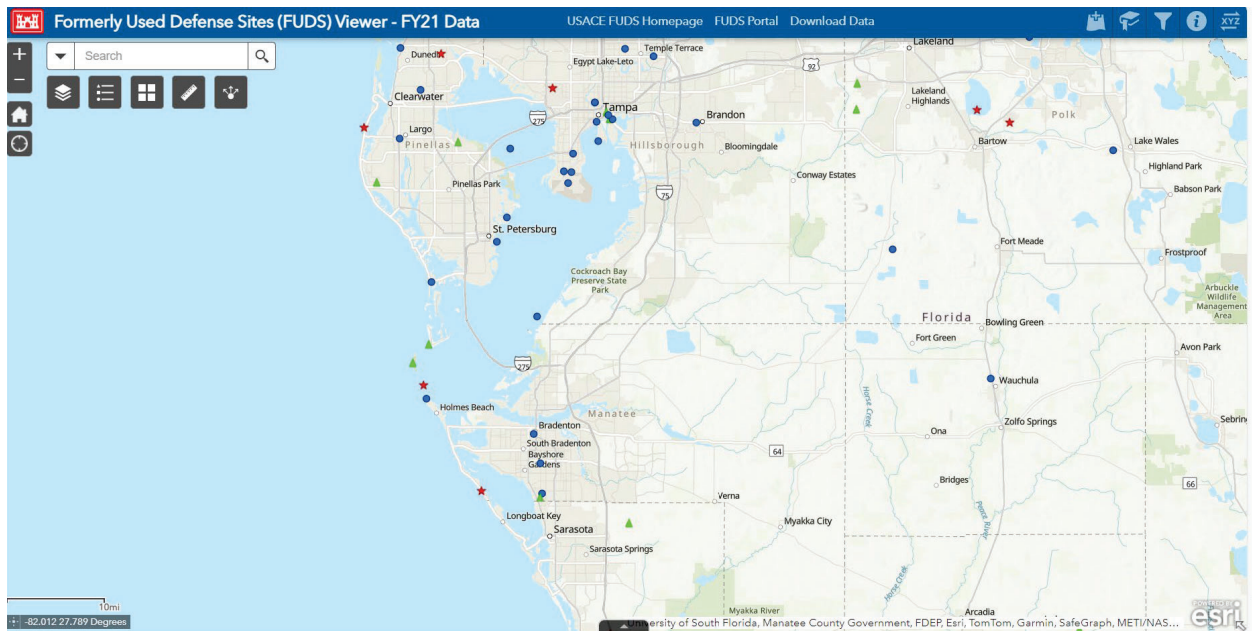


Figure – 3 (FUDS)

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-- End of Section Table of Contents --

SECTION 35 20 26

NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM - SCOW ULLAGE PROFILE

PART 1 GENERAL

1.1 DESCRIPTION

The work under this contract requires use of the National Dredging Quality Management Program (DQM) to monitor the dredge's status at all times during the contract and to manage data history. For the purpose of these specifications, a scow is defined as any non-self-propelled vessel used to transport dredged material. This includes, but is not limited to, split-hull scows, pocket scows, hopper barges, and deck barges.

This performance-based specification section identifies the minimum required output and the precision and instrumentation requirements. The requirements may be satisfied using equipment and technical procedures selected by the Contractor.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-07 Certificates

National Dredging Quality Management Program Certification,
Jacksonville District; G, R0

1.3 PAYMENT

Separate payment for installation, operation, and maintenance of the DQM-certified system as specified herein for the duration of the dredging operations is not allowed; all costs in connection therewith are considered a subsidiary obligation of the Contractor and are covered under the contract unit price for dredging in the bidding schedule.

1.4 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM CERTIFICATION

1.4.1 Certification

The Contractor is required to have a current certification from DQM for the scow instrumentation system to be used under this contract. Criteria for certification is based on the most recent specification posted on the DQM website <https://dqm.usace.army.mil/Specifications/Index.aspx>, Verify compliance with these criteria by onsite quality assurance (QA) checks conducted by the DQM Support Center Data Acquisition and Analysis Team and by periodic review of the transmitted data. If a system is installed specifically for this contract, in order to ensure that it is capable of transmitting quality data to the DQM database, the QA checks will take place either prior to the start of the contract or, with prior approval of

the local USACE District, as soon as practical after dredging commences. DQM Certification is valid for one year from the date of certification and is contingent upon the system's ability to meet the performance requirements as outlined in paragraph PERFORMANCE REQUIREMENTS. If issues with data quality are not corrected within 48 hours, the system certification will be revoked and additional QA checks by the Data Acquisition and Analysis Team may be necessary. Annual DQM Certification must be based on the following:

A series of QA checks as outlined on the DQM website
<https://dqm.usace.army.mil/Certifications/Index.aspx>

Verification of data acquisition and transfer as described in paragraph PERFORMANCE REQUIREMENTS

Review of the Dredge Plant Instrumentation Plan (DPIP) as described in paragraph DREDGE PLANT INSTRUMENTATION PLAN (DPIP)

1.4.2 Quality Assurance (QA)

The Dredging Contractor must have personnel who are familiar with the system instrumentation and who have the ability to recalibrate the sensors on site during the QA process. The Dredging Contractor must coordinate pickup times and locations and provide transportation to and from any platform with a DQM system to team personnel in a timely manner. The Dredging Contractor must also have on site for the QA checks a tug capable of towing the scow. As a general rule, DQM Data Acquisition and Analysis Team personnel will come with personal protective equipment (PPE) consisting of hardhats, steel toe boots, and life jackets. If additional safety equipment is needed, such as eye protection, safety harnesses, work gloves, or personal location beacons, provide these items to the team while on site. Submit a test data package to the DQM database from the system on each scow and have it accepted by the DQM Support Center prior to scow compliance checks. Also submit data collected during the QA Checks from the scow monitoring system to the DQM database and the Data Acquisition and Analysis Team personnel while on site. It is the Dredging Contractor's obligation to inform the QA team if the location designated for the QA checks has any site-specific safety concerns prior to their arrival on site.

The owner or operator of the scow must contact DQM at DQM-AnnualQA@rpsgroup.com on an annual basis, or at least three weeks prior to the proposed beginning of dredging, to schedule QA checks. This notification is meant to make the Data Acquisition and Analysis Team aware of a target date and the contract on which the plant will be used. At least one week prior to the target date, the Dredging Contractor must contact the Data Acquisition and Analysis Team and verbally coordinate a specific date and location. The Contractor must then follow up this conversation with a written email confirmation. The owner/operator must coordinate the QA checks with all local authorities including, but not limited to, the local USACE Contracting Officer.

1.4.3 Recertification

Recertification is required for any yard work which produces modification to displacement (for example, a change in scow lines, or repositioning or repainting hull marks), modification to bin volume (change in bin dimensions or addition or subtraction of structure), or changes in sensor type or location; report these changes in the sensor log section of the

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DPIP. A system does not have to be transmitting data between jobs; however, in order to retain certification during this period, the system sensors or hardware will not be disconnected or removed from the scow. If the system is powered down, retain calibration coefficients.

1.5 DREDGE PLANT INSTRUMENTATION PLAN (DPIP)

The Contractor must have a digital copy of the DPIP on file with the DQM Support Center. While working on site, the Contractor must also maintain on the dredge a copy of the DPIP which is easily accessible to Government personnel at all times. This document must describe the sensors used, configuration of the system, how sensor data will be collected, how quality control on the data will be performed, and how sensors/data reporting equipment will be calibrated and repaired if they fail. A description of computed scow-specific data and how the sensor data will be transmitted to the DQM database must also be included. The Contractor must submit to the DQM Support Center any addendum or modifications made to the plan, subsequent to its original submission, prior to start of work.

A complete list of the required DPIP contents is provided on the DQM website <https://dqm.usace.army.mil/Certifications/Index.aspx>. Submit to the DQM Support Center any addendum or modifications made to the plan, subsequent to its original submission, prior to the start of work. Any changes to the computation methods must be approved by the DQM Support Center prior to their implementation.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 REQUIREMENTS FOR REPORTED DATA

Provide, operate, and maintain all hardware and software to meet these specifications. The Contractor is responsible for replacement, repair, and calibration of sensors and other necessary data acquisition equipment needed to supply the required data. Complete repairs within 48 hours of any sensor failure. Notify the Contracting Officer upon completion of a repair, replacement, installation, modification, or calibration. The Contracting Officer may request recalibration of sensors or other hardware components at any time during the contract as deemed necessary.

Keep a log of sensor repair, replacement, installation, modification, and calibration in the onsite copy of the DPIP. The log must contain a three-year history of sensor maintenance, including the time of sensor failures (and subsequent repairs), the time and results of sensor calibrations, the time of sensor replacements, and the time that backup sensor systems were initiated to provide the required data. It must also contain the name of the person responsible for the sensor work. Install sensors that are capable of collecting parameters within specified accuracies and resolutions indicated in the following subparagraphs.

With the exception of position and any value calculated, reported sensor values will represent a weighted average with the highest and lowest values not included in the calculated average for the given interval. The averaging routine used will be consistent across all event triggers. This information will be documented in the DPIP sections that say "Calculations done external to the instrumentation." These data-reporting requirements cover the collection of electronic data on a scow through the entire dredging cycle. Disposal events can consist of both open-water disposal

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and offloading. Open-water disposal is the placement of material via bottom doors or split hull. Offloading is the placement of material via either hydraulic or mechanical means.

3.1.1 Scow Name

Assign a unique name for each scow that will remain constant from one dredging operation to the next.

3.1.2 Contract Number

The USACE-assigned contract number for the project will be reported.

3.1.3 Load Number

A DQM load number must document the end of a disposal event for a given SCOW.

3.1.4 Horizontal Positioning

Record horizontal positioning as the geographic coordinates of the vessel as indicated by the location of the Global Positioning System (GPS) antenna. Obtain all locations using a positioning system operating with a minimum accuracy level of **3 to 10 feet** horizontal Circular Error Probable (CEP). Report positions as Latitude/Longitude WGS 84 in decimal degrees. West Longitude and South Latitude values are reported as negative.

3.1.5 Date and Time

Report the date and time to the nearest second and referenced to Universal Time Coordinated (UTC) based on a 24-hour format: yyyy-mm-dd hh:mm:ss.

3.1.6 Hull Status

Hull status is meant to reflect a condition when material could be removed or released from the scow. For this contract, hull status must register closed prior to leaving the disposal area.

3.1.6.1 Open-Water Disposal

Indicate an open split hull or open bottom door of a scow by reporting an "OPEN" value. Indicate a closed split hull or closed bottom door of a scow by reporting a "CLOSED" value. An open status must be indicated as the bin starts to open, and a closed status must be indicated only once the bin is fully closed. For pocket scows, the open/closed status must correspond to the compartment which is first to open and last to close.

3.1.7 Offloading

Offloading occurs when the scow is being unloaded, either by hydraulic or mechanical means. When offloading is occurring, a value of "true" must be reported; otherwise, a "false" value must be reported. The only permissible values are "true" and "false".

3.1.8 Course

Provide scow course-over-ground (COG) using industry-standard equipment. Provide scow course-over-ground (to the nearest whole degree) with values from 000 (true north) to 359 degrees referenced to a clockwise positive

direction convention.

3.1.9 Speed

Provide scow speed-over-ground in knots using industry-standard equipment with a minimum accuracy of 1.0 knot and resolution to the nearest 0.1 knot.

3.1.10 Heading

Provide scow heading using industry-standard equipment. The scow heading must be accurate to within 5 degrees and reported to the nearest whole degree with values from 000 (true north) to 359 degrees referenced to a clockwise positive direction convention.

3.1.11 Draft

Report all draft measurements in feet, tenths, and hundredths with an accuracy of plus or minus 0.1 foot relative to observed physical draft readings. Report the measurements at a resolution of two decimal places (hundredths of a foot). The reported forward draft value must be equal to the sum of the visual forward port and starboard draft mark readings divided by two. The reported aft draft value must be equal to the sum of the visual aft port and starboard draft mark readings divided by two. Forward draft, aft draft, and average draft will be reported. Place sensors at an optimum location on the scow to be reflective of observed physical draft mark readings at any trim or list. Minimum accuracies are conditional to relatively calm water. The reported sensor value is an average of at least ten samples per event, with at least one maximum value and one minimum value removed, and the minimum eight remaining values averaged. When the average draft is calculated for the purpose of determining displacement, maintain significant digits for average draft such that if forward draft were 0.15 and aft draft were 0.1, then the average draft would be 0.125.

3.1.12 Displacement

Report scow displacement in long tons, based on the most accurate method available for the scow. The minimum standard of accuracy for displacement is interpolation from the displacement table, based on the average draft. For this contract the density of water used to calculate displacement is 64.2* [lbs/cubic foot](#) and is used for an additional interpolation between the fresh and salt water tables.

*The water density used is project-/location-specific. Enter the appropriate water density in the blank:

Fresh Water: [62.43 lbs/cubic foot](#)
Salt Water: [64.11 - 64.30 lbs/cubic foot](#)

3.1.13 Bin Ullage Sounding

Report all ullage soundings in [feet](#), tenths, and hundredths with an accuracy of plus or minus 0.1 foot with respect to the combing and be representative of the forward and aft extents of the hopper as close to the centerline as is possible. Report the measurements at a resolution of two decimal places (hundredths of a foot). If sensors must be offset from the centerline of the bin, they will be offset to opposite sides of the vessel. Forward ullage, aft ullage, and average ullage soundings will be reported. The reported sensor value is an average of at least ten samples

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per event, with at least one maximum value and one minimum value removed, and the minimum eight remaining values averaged. When the average ullage is calculated for the purpose of determining the hopper volume, maintain significant digits for the average ullage such that if the forward ullage were 0.15 and aft ullage were 0.1, then the average ullage would be 0.125. Special arrangements for pocket scows may be made in consultation with the DQM Support Center.

3.1.14 Bin Volume

Report scow bin volume in cubic yards based on the most accurate method available for the scow. The minimum standard of accuracy for bin volume is interpolation from the bin volume table based on the average ullage soundings.

3.2 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM SYSTEM REQUIREMENTS

The Contractor's DQM system must be capable of collecting, displaying, and transmitting information to the DQM database. The parameters which must be reported to the DQM database include trip number, date and time, hull status, offloading status, scow course, scow speed, scow heading, draft, displacement, ullage, and bin volume. Provide an easily accessible, permanent visual display on the scow to show in real time the parameters collected by the system in the same units as the data submitted to the DQM database. In the event a reported parameter is calculated based on multiple sensors, the sensor values as used in the equation must be able to be viewed in addition to the required parameter. If a hardware problem occurs, or if a part of the system is physically damaged, the Contractor is responsible for repairing it within 48 hours of determination of the condition.

3.2.1 Telemetry

The Contractor may select any commercial satellite, cellular phone, or other data communications systems available, as long as it is capable of transmitting real-time data as well as enough additional bandwidth to clear historically queued data when a connection is reobtained. If connectivity is lost, queue and transmit unsent data upon restoration of connectivity. Delays in pushing real-time data to the DQM database will not exceed four hours. Exceptions to these requirements may be granted by the DQM Center on a case-by-case basis with consideration for contract-specific requirements, site-specific conditions, and extreme weather events.

The data transmission process from the scow to the DQM database must be automated. The data may be sent from the scow directly to the DQM database or to a shore-based system. Data transmitted to the DQM database will be raw data; use repeatable automated software or programming routine to process any shoreside data. Include a description of this process in the DPIIP.

3.2.2 Data Reporting Frequency

Log disposal activities with high temporal and spatial resolution. Log data as a series of events. Each set of measurements (time, position, etc.) will be considered an event. Collect any required information in paragraph REQUIREMENTS FOR REPORTED DATA, that is not an averaged variable (that is, draft and ullage) within 1 second of the reported time. Measure data with sufficient frequency by the scow system to resolve the events to

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the accuracy specified in the following table. Any averaged variable must be collected and computed within this sampling interval. Reporting intervals must be consistent and not change for the data collected on a given scow. This interval will be documented by the Contractor in the DPIP.

Event Type	Event Trigger Descriptions	Event Time Resolution	Event Position Resolution
Loading	No change in position with hull status closed An elapsed time of 1 hour since the last event. NONCLOSURE In the event a scow has completed an open water disposal and transited back to a holding station without closing the hull, the sampling must be changed to once per hour.	1 minute	N/A
Sailing	Change in position with hull status closed Time from the last sample equals 1 minute.	1 second	plus or minus 10 ft
Offloading	Offloading material, hull status reported as open A position must be recorded within 1 minute arrival at the offload location and within one second of the material starting to be removed from scow. The time from the last sample equals 1 minute. STANDBY OFFLOADING In the even a scow is not being actively offloaded at the offload location for a time equal to one hour, the sampling interval must be equal to once an hour.	1 second 1 minute	plus or minus 10 ft

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Event Type	Event Trigger Descriptions	Event Time Resolution	Event Position Resolution
Open Water Disposal	Hull status open A data point must be recorded within 1 second of the hull status going from closed to open and again within 1 second of the hull status going from open to closed. Between these events, report the data at equal interval from 6 to 12 seconds. This interval must always remain consistent for the dredge plant.	1 second	plus or minus 10 ft

Example: The scow is stationary for 1 hour and 15 minutes, and then it sails to the disposal area. You will have a "Loading/Stationary" event at time 0, time 1 hour, and time 1 hour and 15 minutes. Then, for "Sailing," within 1 second of an elapsed time of 1 minute from the 1 hour and 15 minutes event, another event occurs.

3.2.3 Data Transmission to the Web Service

Use a Simple Object Access Protocol (SOAP) web service to report sensor data to the DQM database. Transmit data as it is collected in real time and pushed to the DQM web service. If the web service is not available or returns an error message, store the data in a queue and transmit upon re-establishment of the connection, starting with the oldest data in the queue and continuing until real-time transmission is restored. Delays in pushing real-time data to the DQM database will not exceed four hours. Exceptions to these requirements may be granted by the DQM Support Center on a case-by-case basis with consideration for contract-specific requirements, site-specific conditions, and extreme weather events.

Contact dqm-support@usace.army.mil to obtain the web service URL and the appropriate key credentials and communication protocol.

The data transmission method call takes two arguments: a string containing the plant identifier assigned by the DQM Support Center and a second string containing the XML-formatted sensor data. The method returns the string "OK" if the data is received. If the data is not received, either the web service or the client application throws an error.

3.2.4 XML-Formatted Sensor Data String

Pass each scow event as a string on one continuous line of data. The example below is broken up by variable for ease of reading:

```
<?xml version="1.0"?>
```

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```
<SCOW_DREDGING_DATA version="2.5">
  <SCOW_NAME>AU1994</SCOW_NAME>
  <PLANT_IDENTIFIER>1999</PLANT_IDENTIFIER>
  <CONTRACT>W123BA-09-D-0087_RL01</CONTRACT>
  <TRIP_NUMBER>34</TRIP_NUMBER>
  <X_POSITION>-81.670632</X_POSITION>
  <Y_POSITION>41.528987</Y_POSITION>
  <DATE_TIME>2010-08-14 10:50:15</DATE_TIME>
  <SCOW_SPEED>0.0</SCOW_SPEED>
  <SCOW_COURSE>0.0</SCOW_COURSE>
  <HULL_STATUS>OPEN</HULL_STATUS>
  <OFFLOADING>FALSE</OFFLOADING>
  <SCOW_HEADING></SCOW_HEADING>
  <SCOW_FWD_DRAFT></SCOW_FWD_DRAFT>
  <SCOW_AFT_DRAFT></SCOW_AFT_DRAFT>
  <SCOW_AVG_DRAFT></SCOW_AVG_DRAFT>
  <ULLAGE_FWD></ULLAGE_FWD>
  <ULLAGE_AFT></ULLAGE_AFT>
  <ULLAGE_AVG></ULLAGE_AVG>
  <SCOW_BIN_VOLUME></SCOW_BIN_VOLUME>
  <SCOW_DISPLACEMENT></SCOW_DISPLACEMENT>
  <ADDITIONAL_DATA>Some more scow info, if needed</ADDITIONAL_DATA>
</SCOW_DREDGING_DATA>
```

Format DATE_TIME values as YYYY-MM-DD HH:MM:SS, as shown above. If, for any reason, a field has no value, the enclosing XML tags will be sent with nothing between them (for example, <DRAFT_AFT></DRAFT_AFT>). The web service cannot handle a "null" value or any other indicators of no value collected.

3.2.5 Contractor Data Backup

Maintain an archive of all data sent to the DQM database during the dredging contract. The Contracting Officer may require, at no increase in the contract price, that the Contractor provide a copy of these data covering specified time periods. Provide the data in the HTML format which would have been transmitted to the DQM database. Submit data via storage medium acceptable to the Contracting Officer.

At the end of the dredging contract, contact the DQM Support Center prior to discarding the data. The DQM Support Center will verify that all data has been received and appropriately archived before giving the Contractor discard permission. Record in a separate section at the end of the scow's onsite copy of the DPIP the following information:

- Person who made the call
- Date of the call
- DQM representative who gave permission to discard

3.3 PERFORMANCE REQUIREMENTS

The Contractor's DQM system must be fully operational at the start of dredging operations and fully certified prior to moving dredge material on the contract (see paragraph NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM CERTIFICATION). To meet contract requirements for operability, in addition to certification, the Contractor's system must provide a data string with values for all parameters while operating, as described within the specifications. Additionally, all hardware must be compliant with DPIP requirements (see paragraph DREDGE PLANT INSTRUMENTATION PLAN

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(DPIP)). Quality data strings are considered to be those providing values for all parameters reported when operating according to the specification. Make repairs necessary to restore data return compliance within 48 hours. Failure by the Contractor to report the required data within the specified time window for scow measurements (see paragraph DATA REPORTING FREQUENCY, and paragraph DATA TRANSMISSION TO THE WEB SERVICE) and failure to receive DQM certification prior to dredging will result in withholding of up to 10 percent of the contract progress payment.

3.4 LIST OF ITEMS TO BE PROVIDED BY THE CONTRACTOR

DPIP - <https://dqm.usace.army.mil>

Paragraph DREDGE PLANT INSTRUMENTATION PLAN (DPIP)

DQM SYSTEM

Sensor instrumentation - Paragraph REQUIREMENTS FOR REPORTED DATA

SCOW DATA

Event documentation - Paragraph DATA REPORTING FREQUENCY

Data reports - Paragraph DATA TRANSMISSION TO THE WEB SERVICE

QA EQUIPMENT ON THE DREDGE

Clear and accurate draft marks

Ullage tape

-- End of Section --

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SECTION 35 20 27

NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM - PIPELINE HYDRAULIC DREDGE

PART 1 GENERAL

1.1 DESCRIPTION

The work under this contract requires use of the National Dredging Quality Management Program (DQM) to monitor the dredge's status at all times during the contract and to manage data history.

This performance-based specification section identifies the minimum required output and the precision and instrumentation requirements. The requirements may be satisfied using equipment and technical procedures selected by the Contractor.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-07 Certificates

National Dredging Quality Management Program Certification,
Jacksonville District; G, R0

1.3 PAYMENT

Separate payment for installation, operation, and maintenance of the DQM-certified system as specified herein for the duration of the dredging operations is not allowed; all costs in connection therewith are considered a subsidiary obligation of the Contractor and are covered under the contract unit price for dredging in the bidding schedule.

1.4 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM CERTIFICATION

The Contractor is required to have a current certification from the DQM Program for the cutter/suction head hydraulic dredge instrumentation system to be used under this contract. Standard Operating Procedures (SOP) and criteria for certification are presented on the DQM website at <https://dqm.usace.army.mil>.

1.5 DREDGE PLANT INSTRUMENTATION PLAN (DPIP)

The Contractor must have a digital copy of the Dredge Plant Instrumentation Plan (DPIP) on file with the DQM Support Center. While working on site, the Contractor must also maintain on the dredge a copy of the DPIP, which is easily accessible to Government personnel at all times. This document must accurately describe the sensors used, the configuration of the system, how sensor data will be collected, how quality control on the data will be performed, and how the

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sensors/data-reporting equipment will be calibrated and repaired if it fails. A description of the computed dredge-specific data and how the sensor data will be transmitted to the DQM database will also be included. Prior to the start of work, the Contractor must submit to the DQM Support Center any addendum or modifications made to the plan subsequent to its original submission. Requirements and a template for the DPIP are available on the DQM website at <https://dqm.usace.army.mil>.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 REQUIREMENTS FOR REPORTED DATA

Provide, operate, and maintain all hardware and software to meet these specifications. The Contractor is also responsible for the replacement, repair, and calibration of the sensors and other necessary data acquisition equipment needed to supply the required data. Document and complete the procedure to complete a repair as soon as practical. If repair is not possible within two business days of any sensor failure, submit a plan and timeline to complete the repair. Upon completion of a repair, replacement, installation, modification, or calibration, notify the Contracting Officer. The Contracting Officer may request recalibration of the sensors or other hardware components at any time during the contract as deemed necessary.

Keep a log of sensor repair, replacement, installation, modification, and calibration in the dredge's onboard copy of the DPIP. The log must contain a three-year history of sensor maintenance, including the time of the sensor failures (and subsequent repairs), the time and results of sensor calibrations, the time of sensor replacements, and the time that backup sensor systems were initiated to provide the required data. It must also contain the name of the person responsible for the sensor work. Install sensors that are capable of collecting parameters within the specified accuracies and resolutions indicated in the following subparagraphs and transmit these parameters to the DQM database. Transmit all data in JSON message bundles. Each bundle can contain multiple message types. Transmit sensor data as work event messages, and transmit data which relates to the operational state of the dredge or its sensors as state event messages. (See paragraph PARAMETER TRANSMISSION TO THE WEB SERVICE.)

3.1.1 Message Bundle Data

Every message bundle must contain descriptive data that relates the message to a given dredge plant and date/time. Identify the start of a message bundle by the tag "DQM_data".

3.1.1.1 Messages

Messages contain operational data that populates the DQM database for a dredge plant. A message must consist of an event type and its associated data (as defined in paragraph DREDGE EVENTS - WORK EVENT and paragraph DREDGE EVENTS - STATE EVENT), a date/time stamp indicating when the event occurred or started, and a comment providing clarification or metadata about the situation. There are multiple event types, but they all fall into one of two categories - work events and state events.

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3.1.1.1.1 Message Time

In a work event message, message time is the date and time that the data is collected from the sensors; in a state event message, message time is the date and time that the state event begins. Report and reference the message time to the nearest second and to Coordinated Universal Time (UTC) time based on a 24-hour format (YYYY-MM-DD HH:MM:SS). In order to ensure accuracy and reliability, synchronize the time stamp to UTC format from an accurate, unchangeable source (for example, a GPS National Marine Electronics Association (NMEA) datastring). Identify message time by the tag "msg_time".

3.1.1.1.2 Comment

Comments concerning the work event or state event messages being transmitted provide descriptive information that relates to the data. An example of a comment for work event data is information about a sensor issue; an example of a comment for state event data is a description of operations. Identify a comment by the introductory tag "comment", and the comment must not consist of more than 250 characters.ext

3.1.1.2 Dredge Events - Work Event

There are two types of dredge event messages - work event messages and state event messages. Work event messages contain data that are instantaneously collected or calculated from sensors and are logged as a series of events. Work events are triggered by a time interval change (as described in paragraph WORK EVENT MESSAGES). Initiate all work event messages by the header tag "work_event".

3.1.1.2.1 Vertical Correction

Obtain the variation of the water level from the vertical datum for the river stage or tidal gage described in the state events using appropriate equipment to give the water level with an accuracy of plus or minus 0.1 ft. Enter vertical correction values above project datum described in the dredging specification with a positive sign and those below with a negative sign. The tag for vertical correction is "vert_correction".

3.1.1.2.2 Cutter/Suction Head Location and Movement

Monitor the X, Y, and Z components of the cutter/suction head location. Additional calculations made from the observed values determine the rates of movement to track the progress of the dredge.

3.1.1.2.2.1 Cutter/Suction Head Horizontal Position

Obtain the forwardmost point of the cutter/suction head using a positioning system operating with a minimum accuracy level of 3-10 feet horizontal Circular Error Probable (CEP). Report it as Latitude/Longitude WGS 84 in decimal degrees with West Longitude and South Latitude values reported as negative. Identify position values by the tags "ch_latitude" and "ch_longitude".

3.1.1.2.2.2 Cutter/Suction Invert Depth

Cutter/suction invert depth is the depth of the invert of the suction mouth relative to the surface of the water. Instrumentation must be capable of reporting to an accuracy of plus or minus 0.5 foot and a

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resolution to the nearest 0.1 foot with no tidal adjustments. Minimum accuracies are conditional to relatively calm water. Use the tag "ch_depth" to identify the cutter/suction head depth.

3.1.1.2.2.3 Cutter/Suction Head Heading

The cutter/suction head heading is the angle of the centerline of the cutter/suction head and dredge ladder measured relative to true north. Provide all headings using industry-standard equipment. The heading must be accurate to within 5 degrees and reported to the nearest whole degree with values from 000 (true north) to 359 degrees referenced to a clockwise positive direction convention. Use the tag "ch_heading" to identify the cutter/suction head heading.

3.1.1.2.3 Dredge Activity

Monitor dredge activity using a combination of the following parameters.

3.1.1.2.3.1 Slurry Velocity

Use a flow-metering device, calibrated according to the manufacturer's specifications, to record the slurry velocity to the nearest 0.01 fps with an accuracy of plus 0.5 fps. If the manufacturer does not specify a frequency of recalibration, conduct calibration prior to the commencement of work. Measure the slurry velocity for the same pipeline inside diameter as that used for the slurry density measurement. Associate the tag "slurry_velocity" with this value.

3.1.1.2.3.2 Slurry Density

Use a density-metering device, calibrated according to the manufacturer's specifications, to record the slurry density to the nearest 0.01 g/cc. It is understood that the accuracy of this sensor can vary based on several factors, including the type of material, the magnitude of the cut, and the length of time since calibration. If the manufacturer does not specify a frequency of recalibration, conduct calibration prior to the commencement of work. Continuous monitoring of this sensor ensures that drift and other factors inherent in the dredging process can be accounted for in monitoring dredge activity. Associate the tag "slurry_density" with this value.

3.1.1.2.3.3 Pump RPM

The pump rpm is the number of revolutions per minute measured for the slurry pump shaft. Measure the shaft revolution rate (rev/min) with the highest level of accuracy that is standard on the vessel's operational displays either at the bridge or in the engine room. Identify this value by the tag "rpm".

3.1.1.2.3.4 Pump Vacuum

Measure the vacuum pressure of the dredge pump(s) (inches of mercury) as near to the eye as practicable in the pump's suction pipe with the highest level of accuracy that is standard on the vessel's operational displays either at the leverman's controls or in the engine room. Identify vacuum pressure by the tag "vacuum".

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3.1.1.2.3.5 Pump Outlet Pressure

Measure the pump outlet pressure in the discharge line on the pump side of the flap valve in terms of pounds per square inch (psi) on a gauge. Identify pump outlet pressure by the tag "outlet_psi".

3.1.1.2.4 Outfall Information (Open Water/Spill Barge Disposal)

Monitor the X and Y position of the terminal end of the outfall pipe continuously and report the position as part of the work event string.

3.1.1.2.4.1 Discharge Horizontal Position

Obtain the horizontal position of the outfall end of the discharge pipe using a positioning system operating with a minimum accuracy level of 3-10 feet horizontal Circular Error Probable (CEP). Report it as Latitude/Longitude WGS 84 in decimal degrees with West Longitude and South Latitude values being reported as negative. Identify position values by the tags "outfall_latitude" and "outfall_longitude".

3.1.1.3 Dredge Events - State Event

There are two types of dredge event messages - work event messages and state event messages. State event messages provide information about the current state of the dredge equipment or operations. They are created and sent only when a state changes. Since state events often cannot be collected in real time, state events are tagged with a date time stamp (referenced to Coordinated Universal Time (UTC)) that indicates when the state change happened relative to the work event message tag. This data is considered to be "true" until another state event tag of the same type is received. Indicate each type of state event message by a specific header tag as enumerated in the following subparagraphs. State events can be transmitted along with work event message bundles directly by the contractor using the indicated format, or they can be entered on the "State" tab in the DQM-provided software. However, they will be sent only if the state value changes.

3.1.1.3.1 Message Time

The state event time is the date and time that the event starts. Enter the leverman's time to the nearest second as local time and automatically convert to and report in UTC based on a 24-hour format (YYYY-MM-DD HH:MM:SS). Identify message time by the tag "msg_time".

3.1.1.3.2 Contract Event

Report information concerning the contract under which dredging is being performed at the start and completion of each contract using the header tag "contract_event".

3.1.1.3.2.1 Contract Number

Report the USACE-assigned contract number for the project using the tag "contract_number".

3.1.1.3.2.2 Contract Start and End

Report the start and end of a contract using the tag "event_type" with the appropriate value of "start" or "end".

3.1.1.3.3 Tide Station/River Stage Gage Event

Group together properties associated with the vertical correction (see paragraph VERTICAL CORRECTION) for the tide station/river stage gage under the header tag "station_event". This information must be sent at the start of the contract and each time the dredge has moved enough to change the station being used.

3.1.1.3.3.1 Station Name

The station name is a concise name defining the tide station/river stage gage being referred to. It must be introduced by the tag "station_name", and it must not consist of a descriptor of more than 25 characters.

3.1.1.3.4 Length of Pipe Event

Report the leverman's estimate of the length of pipe downflow from the dredge pump, measured to the nearest whole foot, under the header tag "pipe_length_event". This information must be sent at the start of the contract and at the completion of each 24-hour period ending at midnight local time.

3.1.1.3.4.1 Floating Pipe

Report the total length of floating pipe with the tag "length_floating".

3.1.1.3.4.2 Submerged Pipe

Report the total length of floating pipe with the tag "length_submerged".

3.1.1.3.4.3 Shore Pipe

Report the total length of shore pipe with the tag "length_land".

3.1.1.3.5 Booster Pump

Include information concerning the booster pumps being used under the header tag "booster_pump_event". A message must be sent to indicate any change in the status of the booster pumps being used.

3.1.1.3.5.1 Number of Booster Pumps

Upon the addition or removal of a booster pump, report the total number of booster pumps being used with the tag "booster_total".

3.1.1.3.6 Dredge Advance

Measure the dredge advance, the total forward progress of the dredge relative to the centerline of the cut, to the nearest whole foot and cumulatively calculate over a 24-hour period from midnight to midnight local time. Identify it by the tag "advance_daily". Report in Greenwich Mean Time (GMT) the msg_time associated with this tag as the first timestamp of the following 24-hour period (based on the local time) rather than as midnight of the day for which the value was calculated.

3.1.1.3.7 Outfall Information

Monitor and send the X and Y position of the terminal end of the outfall

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pipe at the start of the contract and thereafter according to the following table. Discharge Heading and Pipe Elevation may be omitted if the dredge is not discharging into an upland disposal site. For beach nourishment, the horizontal X and Y position of the outfall must be sent at the start of the contract and at the completion of each 24-hour period ending at midnight local time.

Discharge Location	Horizontal Position	Discharge Pipe Elevation	Discharge Outfall Heading
Open Water	Continuous Work Event	N/A	N/A
Scow	Upon Change	N/A	N/A
Beach	Every 24 Hours	N/A	N/A
Upland	Upon Change	Upon Change	Upon Change

3.1.1.3.7.1 Discharge Location

Report information on where the slurry is being discharged with the tag "outfall_location". Acceptable values include "upland", "open water", "beach", and "scow".

3.1.1.3.7.2 Discharge Horizontal Position

Obtain the horizontal position of the outfall end of the discharge pipe using a positioning system operating with a minimum accuracy level of 3-10 feet horizontal Circular Error Probable (CEP). Report it as Latitude/Longitude WGS 84 in decimal degrees with West Longitude and South Latitude values being reported as negative. Identify position values by the tags "outfall_latitude" and "outfall_longitude".

3.1.1.3.7.3 Discharge Outfall Heading

The discharge outfall heading is the angle relative to true north measured from the centerline of the pipe in the direction of discharge. Provide all headings using industry-standard equipment. They must be accurate to within 5 degrees and report to the nearest whole degree with values from 000 (true north) to 359 degrees referenced to a clockwise positive direction convention. Identify the discharge heading by the tag "outfall_heading".

3.1.1.3.7.4 Discharge Pipe Elevation

The discharge pipe elevation is the height of the outfall measured in feet and tenths of a foot relative to the project datum. The required accuracy is contingent upon contract requirements. Use the tag "outfall_elevation" to identify this elevation.

3.1.1.3.8 Non-effective Work Event

Report delays and dredge downtime at the conclusion of the event. Submit the reason for the non-effective work time under the header tag "non_eff_event" within 24 hours of the event.

3.1.1.3.8.1 Non-effective Work Interval

Report the start and end times for the non-effective work event using the

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tags "msg_start_time" and "msg_end_time".

3.1.1.3.8.2 Dredge Function Code

Transmit the dredge operator indication of production delays, as listed on Form 4267, at the end of the non-effective interval. Identify dredge function event messages by the tag "function_code" consisting of one of the following standardized entries to indicate the operation:

AGV	Assisting Grounded Vessels
CCH	Change Cutterhead
CCSH	Clear Cutter Suction
CLPJ	Change Location Bar
COLL	Collision
CPPL	Clear Pump Pipeline
CPR	Change Impeller
DR	Dike Repair
FBD	Fire Boat Drills
HPL	Handling Pipe Line
HSL	Handling Swing Line
HSP	Handling Shore Pipe
LDNE	Loss Due to Natural Elements
LDPV	Loss Due to Passing Vessel
LNL	Transfer to New Location
MISC	Miscellaneous
MOB	Mobilization & Demobilization
MSC	Miscellaneous/Non-pay
OC	Out of Commission
OR	Operating Repairs
P	Preparation
PREP	Preparation & Making Up Tow
RPL	Repair Pipeline
SB	Sounding & Buoying
SBT	Stand-By Time as Directed
SH	Sundays-Holidays
TFS	Taking on Fuel & Supplies
TOW	Time on Tow
WAP	Waiting Attendant Plant

3.1.1.3.8.3 Additional Comments

Use the "comment" tag to provide additional explanation for the noted delays or downtimes. For example, when the code "LDPV" (Loss Due to Passing Vessel) is indicated, list the name of the vessel and the number of tows with the "comment" tag.

3.2 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM SYSTEM REQUIREMENTS

The Contractor's DQM system must be capable of collecting and transmitting information to the DQM system. Record the applicable parameters from paragraph REQUIREMENTS FOR REPORTED DATA as local events and transmit continuously to the DQM database anytime an Internet connection is available. To accomplish this transmission, there are two options. The first option is to equip the vessel with a DQM computer system consisting of a computer, monitor, keyboard, mouse, data modem, Universal Power Supply (UPS), and network hub. Provide a standalone computer system, exclusive to the DQM monitoring system, with USACE DQM software installed on it. If a hardware problem occurs, or if a part of the system is physically damaged, the Contractor is responsible for repairing it within

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two business days of the determination of the condition or submitting a plan and timeline for repair if the repair will take more than two business days.

The second option is to send data directly to DQM's web endpoint as outlined in paragraph DIRECT DATA TRANSMISSION.

3.2.1 Computer Requirements

Provide a dedicated onboard computer for use by the Dredging Quality Management system. This computer must run the USACE DQM software and receive data from the Contractor's data-reporting interface. This computer must meet or exceed the following performance specifications:

CPU:	Intel or AMD processor with a (non-overclocked) clock speed of at least 1.6 gigahertz (GHz)
Hard drive:	250 gigabytes (GB); internal
RAM:	4 gigabytes (GB)
Ethernet adapter:	Internal network card with an RJ-45 connector
Ports:	1 free serial port with standard 9-pin connectors; 1 free USB port
Other hardware:	Keyboard, mouse, monitor

Install a fully licensed copy of Windows 10 Professional Operating System or higher on the computer specified above. Also install any necessary manufacturer-provided drivers for the installed hardware.

Locate and orient this computer to allow data entry and data viewing as well as to provide access to data ports for connection of external hardware.

3.2.2 Software

The DQM computer's primary function is to transmit data to the DQM shoreside database. Do not install software which conflicts with this function on this computer. The DQM computer must have the USACE-provided Dredging Quality Management Onboard Software (DQMOBS) installed on it by DQM personnel.

3.2.3 UPS

Supply an Uninterruptible Power Supply (UPS) for the computer and networking equipment. It must interface with the DQM computer to communicate UPS status, and it must provide backup power at 1 kVA for a minimum of 10 minutes. Ensure that sufficient power outlets are available to run all specified equipment.

3.2.4 Direct Data Transmissions

Option 2: Direct Data Transmissions to the DQM Web Service - Use the web service to report sensor data to the DQM database. Data must be transmitted as it is collected in real time and pushed to the DQM web service. If the web service is not available or returns an error message, the data must be stored in a queue and transmitted upon re-establishment

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of the connection, starting with the oldest data in the queue and continuing until real-time transmission is restored.

Contact dqm-support@usace.army.mil to obtain the web service URL and the appropriate key credentials and communication protocol. If this option is chosen, a display of the raw data being transmitted to DQM must be easily accessible and visible onboard the dredge

3.2.5 Internet Access

Maintain an Internet connection capable of transmitting real-time data to the DQM server as well as enough additional bandwidth to clear historically queued data when a connection is re-established. If connectivity is lost, queue and transmit unsent data upon restoration of connectivity. Delays in pushing real-time data to the DQM database will not exceed four hours. Exceptions to these requirements may be granted by the DQM Support Center on a case-by-case basis with consideration for contract-specific requirements, site-specific conditions, and extreme weather events.

Acquire and install all necessary hardware and software to make the Internet connection available for data transmission to the DQM web service. Configure the hardware and software to allow the DQM Support Center remote access to this computer, and the telemetry system must be capable of meeting these minimum reporting requirements in all operating conditions.

In areas with poor cellular service and at the local District's discretion, it may be required to manually download the data on a daily basis using the protocol for retrieving and submitting backup files provided by the DQM Support Center. This method of data transmission will be used only if Internet connectivity is unavailable at the dredging site, and it will be considered a temporary measure.

3.2.6 Data Routing Requirements

Onboard sensors continually monitor dredge conditions, operations, and efficiency and route this information to the shipboard dredge-specific system (DSS) computer to assist in guiding dredge operations. Portions of this Contractor-collected information, as described in this specification, must be routed to the DQM computer on a real-time basis utilizing one of the two options outlined in this section. Use an RS-232 serial interface with a baud rate of 9600 or 19200 bps to send standard sensor data to the DQM computer. Configure the serial interface as 8 bits, no parity, and no flow control.

Digitally log and transmit information regarding changes in the state of the dredge as close to the time of the occurrence as possible. These events can either be included in a separate message bundle going to the DQM onboard computer, or they can be entered on the "State" tab in the DQM Pipeline Software, or entered online in the DQM State Data Tool.

3.3 DREDGE MONITORING DATA

3.3.1 General

Onboard sensors continuously collect dredging data in support of the dredge Contractor's operations. Portions of this Contractor-collected information, as described in this specification, and store and transmit

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calculations based on them to the DQM database on a near real-time basis. Additionally, digitally log and transmit information regarding the state of the dredge.

3.3.2 Data Measurement Frequency

The frequency of data transmission is dependent on the type of message being sent. Work Event messages contain data that are instantaneously collected or calculated from sensors and are logged as a series of events. State event messages are activated by a change in the dredge state.

3.3.2.1 Work Event Messages

Log data as a series of events. Each event must consist of a dataset containing dredge information (as defined in paragraph REQUIREMENTS FOR REPORTED DATA). Consider each set of measurements (for example, time and position) an event with a 6-12 second interval between work events. This interval must remain consistent across event types for the dredge plant.

Record a standard data string within one second of an event trigger with the time stamp and all parameters reflecting when the event happened.

3.3.2.2 State Event Messages

Consider a set of descriptive information (event name, time, description, comment) a state event. Record these events within 24 hours of a change in state with the time stamp reflecting when the event happened.

3.3.3 Parameter Transmission to the Web Service

Format the data as JSON (JavaScript Object Notation, as defined at <http://www.json.org>) strings of arbitrary length. These JSON strings represent a hierarchical data structure consisting of a message bundle which may contain 0-3 automatic data messages and any number of manual data messages.

A tag/parameter is reported only when it contains a value. Do not include "Null" value strings in a message bundle.

Message bundle

```
{
  "DQM_Data": {
    "plant_identifier": <integer value 0000-9999>,
    "transmission_time": <24-hour UTC time YYYY-MM-DD HH:MM:SS>,
    "messages": {
      {
        "work_event": {
          "msg_time": <24-hour UTC time YYYY-MM-DD HH:MM:SS>,
          "vert_correction": <floating point 100th decimal place>,
          "ch_latitude": <decimal to 6 decimal places>,
          "ch_longitude": <decimal to 6 decimal places>,
          "ch_depth": <floating point 100th decimal
place>,
          "ch_heading": <integer value 000-359>,
          "slurry_velocity": <floating point 100th decimal place>,

```

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```

        "slurry_density":    <floating point 100th decimal place>,
        "pump_rpm":          <integer>,
        "vacuum":            <floating point 100th decimal place>,
        "outlet_psi":        <floating point 100th decimal place>,
        "comment":           <string>},
    },
    {
        "contract_event": {
            "msg_time":      <24-hour UTC time YYYY-MM-DD HH:MM:SS>,
            "contract_number": <string>,
            "event_type":    <string - "start" or "end">,
            "comment":       <string>
        }
    },
    {
        "station_event": {
            "msg_time":      <24-hour UTC time YYYY-MM-DD HH:MM:SS>,
            "station_name":  <string>,
            "comment":       <string>
        }
    },
    {
        "pipe_length_event": {
            "msg_time":      <24-hour UTC time YYYY-MM-DD HH:MM:SS>,
            "length_floating": <integer>,
            "length_submerged": <integer>,
            "length_land":    <integer>,
            "comment":       <string>
        }
    },
    {
        "booster_pump_event": {
            "msg_time":      <24-hour UTC time YYYY-MM-DDHH:MM:SS>,
            "booster_total": <integer>,
            "comment":       <string>
        }
    },
    {
        "advance_Event": {
            "msg_time":      <24-hour UTC time YYYY-MM-DD HH:MM:SS>,
            "advance_daily": <integer>,
            "comment":       <string>
        }
    },
    {
        "outfall_position": {
            "msg_time":      <24-hour UTC time YYYY-MM-DD HH:MM:SS>,
            "outfall_location": <string-"upland", "beach", "scow",
"open water">
            "outfall_latitude": <decimal to 6 decimal places>,
            "outfall_longitude": <decimal to 6 decimal places>,
            "outfall_heading": <integer value 000-359>,
            "outfall_elevation": <floating point 10th decimal place>,
            "comment":       <string>
        }
    },
    {
        "non_eff_event": {

```

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```
"msg_start_time":    <24-hour UTC time YYYY-MM-DD HH:MM:SS>,  
"msg_end_time":      <24-hour UTC time YYYY-MM-DD HH:MM:SS>,  
"function_code":     <string - 1 to 4 characters>,  
"comment":           <string>  
  
    }  
  }  
}
```

3.3.4 Contractor Data Backup

Maintain an archive of all data sent to the DQM computer during the dredging contract. The Contracting Officer may require, at no increase in the contract price, that the Contractor provide a copy of these data covering specified time periods. Provide the data in the same JSON format as would have been transmitted to the DQM computer. There must be no line breaks between the parameters, and each record string must be on separate line. The naming convention for the files must be *<dredgename>_<StartYYYYMMddhhmmss>_<EndYYYYMMddhhmmss>.txt*.

Data submission must be via a storage medium acceptable to the Contracting Officer.

At the end of the dredging contact, call the National DQM Support Center prior to discarding the data. The DQM Support Center will verify that all data has been received and appropriately archived before giving the Contractor discard permission. Record the following information in a separate section at the end of the dredge's onboard copy of the DPIP:

Person who called the National DQM Support Center
Date of the call
DQM representative who gave permission to discard the data

3.4 PERFORMANCE REQUIREMENTS

The Contractor's National Dredging Quality Management Program's data transmission must be fully operational at the start of dredging operations. To meet contract requirements for operability, the Contractor's system must provide an accurate data string return and be compliant with hardware requirements. Data string return is defined as the number of quality records within an event or state tag sent by the contractor's system to the DQM database. Quality data strings are considered to be those providing accurate values for all parameters reported when operating according to the specification. Make repairs necessary to restore data return compliance within two business days, or submit a plan and timeline for repair if the repair will take more than two business days. Failure by the Contractor to report quality data within the specified time window for dredge measurements as stated in the specifications (see paragraph INTERNET ACCESS, paragraph DATA MEASUREMENT FREQUENCY, and paragraph PARAMETER TRANSMISSION TO THE WEB SERVICE, will result in withholding of up to 10 percent of the contract progress payment.

3.5 LIST OF ITEMS TO BE PROVIDED BY THE CONTRACTOR

DPIP
<https://dqm.usace.army.mil>

DQM System

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Paragraph 3.2 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM SYSTEM
REQUIREMENTS, including all subparagraphs

Dredge Data
Paragraph DREDGING MONITORING DATA

-- End of Section --

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SECTION 35 20 28

NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM - HOPPER DREDGE

PART 1 GENERAL

1.1 DESCRIPTION

The work under this contract requires use of the National Dredging Quality Management Program (DQM) to monitor the dredge's status at all times during the contract and to manage data history.

This performance-based specification section identifies the minimum required output and the precision and instrumentation requirements. The requirements may be satisfied using equipment and technical procedures selected by the Contractor.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" or classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-07 Certificates

National Dredging Quality Management Program Certification,
Jacksonville District; G, RO

1.3 PAYMENT

Separate payment for installation, operation, and maintenance of the DQM-certified system as specified herein for the duration of the dredging operations is not allowed; all costs in connection therewith are considered a subsidiary obligation of the Contractor and are covered under the contract unit price for dredging in the bidding schedule.

1.4 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM CERTIFICATION

1.4.1 Certification

The Contractor is required to have a current certification from DQM for the hopper dredge instrumentation system to be used under this contract. Criteria for certification is based on the most recent specification posted on the DQM website:
<https://dqm.usace.army.mil/Specifications/Index.aspx>. Verify compliance with these criteria by annual onsite quality assurance (QA) checks conducted by the DQM Support Center Data Acquisition and Analysis Team and by periodic review of the transmitted data. DQM Certification is valid for one year from the date of the annual QA checks. Certification is contingent upon the system's ability to continuously meet the performance requirements as outlined in paragraph PERFORMANCE REQUIREMENTS. If issues with data quality are not corrected within 48 hours, the system certification will be revoked and additional QA checks by the Data

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Acquisition and Analysis Team may be necessary.

Annual DQM Certification must be based on the following:

- a. A series of quality assurance checks as outlined on the DQM website: <https://dqm.usace.army.mil/Certifications/Index.aspx>.
- b. Verification of data acquisition and transfer as described in paragraph Performance Requirements.
- c. Review of the Dredge Plant Instrumentation Plan (DPIP) as described in paragraph DREDGE PLANT INSTRUMENTATION PLAN (DPIP).

1.4.2 Recertification

The owner or operator of the dredge must contact DQM at DQM-AnnualQA@rpsgroup.com on an annual basis, or at least three weeks prior to certification expiration, to schedule QA checks for renewal. This notification is meant to make the Data Acquisition and Analysis Team aware of a target date for the annual QA checks for the dredge. At least one week prior to the target date, the Contractor must contact the Data Acquisition and Analysis Team and verbally coordinate a specific date and location. The Contractor must then follow up this conversation with a written email confirmation. The owner/operator must coordinate the QA checks with all local authorities, including but not limited to, the local USACE Contracting Officer.

Recertification is required for any yard work which produces modification to displacement (change in dredge lines, or repositioning or repainting hull marks), modification to bin volume (change in bin dimensions, or addition or subtraction of structure), or changes in sensor type or location; report these changes in the sensor log section of the DPIP. A system does not have to be transmitting data between jobs; however, in order to retain its certification during this period, the system sensors or hardware will not be disconnected or removed from the dredge. If the system is powered down, retain calibration coefficients.

1.5 DREDGE PLANT INSTRUMENTATION PLAN (DPIP)

The Contractor must have a digital copy of the DPIP on file with the DQM Support Center. While working on site, the Contractor must also maintain on the dredge a copy of the DPIP which is easily accessible to Government personnel at all times. This document must describe the sensors used, configuration of the system, how sensor data will be collected, how quality control on the data will be performed, and how sensors/data reporting equipment will be calibrated and repaired if they fail. A description of the computed dredge-specific data and how the sensor data will be transmitted to the DQM database will also be included. A complete list of the required DPIP contents is provided on the DQM website <https://dqm.usace.army.mil/Certifications/Index.aspx>

Submit to the DQM Support Center any addendum or modifications made to the plan, subsequent to its original submission, prior to the start of work. Any changes to the computation methods must be approved by the DQM Support Center prior to their implementation.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 REQUIREMENTS FOR REPORTED DATA

Provide, operate, and maintain all hardware and software to meet these specifications. The Contractor is responsible for replacement, repair, and calibration of sensors and other necessary data acquisition equipment needed to supply the required data. Complete repairs within 48 hours of any sensor failure. Notify the Contracting Officer upon completion of a repair, replacement, installation, modification, or calibration. The Contracting Officer may request recalibration of sensors or other hardware components at any time during the contract as deemed necessary.

Keep a log of sensor repair, replacement, installation, modification, and calibration in the dredge's onboard copy of the DPIP. The log must contain a three-year history of sensor maintenance, including the time of the sensor failures (and subsequent repairs), the time and results of sensor calibrations, the time of sensor replacements, and the time that backup sensor systems were initiated to provide required data. It must also contain the name of the person responsible for the sensor work. Install sensors that are capable of collecting parameters within specified accuracies and resolutions indicated in the following subparagraphs. Reported sensor values for ullage, draft, and draghead depth will represent a weighted average with the highest and lowest values not included in the calculated average for the given interval. This information will be documented in the DPIP sections that say "Calculations done external to the instrumentation."

3.1.1 Date and Time

Report the date and time to the nearest second and referenced to UTC time based on a 24-hour format: mm/dd/yyyy hh:mm:ss. The reported time must be the time reported by the GPS in the NMEA string.

3.1.2 Load Number

A load number must document the end of a disposal event. Load numbering will begin at number 1 at the start of the contract and will be incremented by 1 at the completion of each disposal event or emptying of the hopper. Whenever possible, calculate the load number off of the sensors aboard the dredge, which must be a mathematically repeatable routine. Make efforts to include logic that avoids false load number increments. Do not allow the routine to miss any disposal event. If manual incrementing of the load number is in place, pay extra attention to this value in the quality control process.

3.1.3 Horizontal Positioning

Obtain all locations using a positioning system operating with a minimum accuracy level of 3 to 10 feet horizontal Circular Error Probable (CEP). Report positions as Latitude/Longitude WGS 84 in decimal degrees. West Longitude and South Latitude values are reported as negative.

3.1.3.1 Vessel Horizontal Positioning

Record vessel horizontal positioning as geographic coordinates of the vessel as indicated by the location of the GPS antenna.

3.1.3.2 Draghead Horizontal Positioning

Record draghead horizontal positioning as geographic coordinates of the heel on the centerline of the draghead(s). Describe any offset calculations from the GPS antenna in the DPIIP.

3.1.4 Hull Status

Monitor open/closed status of the hopper dredge, corresponding to the split/non-split condition of a split-hull hopper dredge. For dredges with hopper doors, the status of a single door that is the first opened during normal disposal operations may be monitored. An "open" value indicates that the hopper door is open or, in the case of split-hull dredges, that the hull is split. A "closed" value indicates that the hopper doors are closed or, in the case of split-hull dredges, that the hull is not split.

For this contract, hull status must register closed prior to leaving the disposal area.

3.1.5 Dredge Course

Provide dredge course-over-ground (COG) using industry-standard equipment. Provide dredge course-over-ground to the nearest whole degree with values from 000 (true north) to 359 degrees referenced to a clockwise positive direction convention.

3.1.6 Dredge Speed

Provide dredge speed-over-ground in knots using industry-standard equipment with a minimum accuracy of 1 knot and resolution to the nearest 0.1 knot.

3.1.7 Dredge Heading

Provide dredge heading using industry-standard equipment. The dredge heading must be accurate to within 5 degrees and reported to the nearest whole degree with values from 000 (true north) to 359 degrees referenced to a clockwise positive direction convention.

3.1.8 Tide

Obtain tide data using appropriate equipment to give the water level with an accuracy of plus or minus 0.1 foot and a resolution of 0.01 foot. Enter tide values above project datum described in the dredging specification with a positive sign and those below with a negative sign.

3.1.9 Draft

Report all draft measurements in feet, tenths, and hundredths with an accuracy of plus or minus 0.1 foot relative to observed physical draft readings. Report the measurements at a resolution of two decimal places (hundredths of a foot). The reported forward draft value is equal to the sum of the visual forward port and starboard draft mark readings divided by two. The reported aft draft value is equal to the sum of the visual aft port and starboard draft mark readings divided by two. Forward draft, aft draft, and average draft will be reported. Place sensors at an optimum location on the vessel to be reflective of observed physical draft mark readings at any trim or list. Minimum accuracies are conditional to

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relatively calm water. The sensor value reported must be an average of at least ten samples per event, with at least one maximum value and one minimum value removed, and the minimum eight remaining values averaged. When the average draft is calculated for the purpose of determining displacement, maintain significant digits for average draft such that if forward draft was 0.15 and aft draft was 0.1, the average draft would be 0.125.

3.1.10 Hopper Ullage Sounding

Report all reported ullage soundings in feet, tenths, and hundredths with an accuracy of plus or minus 0.1 foot with respect to the combing and be representative of the forward and aft extents of the hopper as close to the centerline as is possible. Report the measurements at a resolution of two decimal places (hundredths of a foot). Report forward ullage and aft ullage soundings. Mount sensors to avoid discharge flume turbulence, foam, and any structure that could produce sidelobe errors. If sensors must be offset from the centerline of the hopper, they will be offset to opposite sides of the vessel. If more than one fore or one aft sensor are used, place the sensors near the corners of the hopper, and report the average value of the fore sensors and the average value of the aft sensors. The sensor value reported must be an average of at least ten samples per event, with at least one maximum value and one minimum value removed, and the minimum eight remaining values averaged. When the average ullage is calculated for the purpose of determining hopper volume, maintain significant digits for average ullage such that if forward ullage was 0.15 and aft ullage was 0.1, then the average ullage would be 0.125.

3.1.11 Hopper Volume

Report hopper volume in cubic yards, based on the most accurate method available for the dredge. The minimum standard of accuracy for hopper volume is interpolation from the certified hopper volume table, based on the average fore and aft ullage soundings.

3.1.12 Displacement

Report dredge displacement in long tons, based on the most accurate method available for the dredge. The minimum standard of accuracy for displacement is interpolation from the displacement table, based on the average draft. For this contract the density of water used to calculate displacement is 64.2* lbs/cubic foot, and it will be used for an additional interpolation between the fresh and salt water tables.

*The water density used is project-/location-specific. Enter the appropriate water density in the blank:

Fresh Water: 62.43 lbs/cubic foot
Salt Water: 64.11 - 64.30 lbs/cubic foot

3.1.13 Empty Displacement

Report empty displacement in long tons, which is the lightship value of the dredge, or the weight of the dredge with no material in the hopper, adjusted for fuel and water consumption.

3.1.14 Draghead Depths

Report draghead depths with an accuracy of plus or minus 0.5 foot and a

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resolution to the nearest 0.1 foot as measured from the surface of the water with no tidal adjustments. Minimum accuracies are conditional to relatively calm water. The sensor value reported must be an average of at least ten samples per event, with at least one maximum value and one minimum value removed, and the minimum eight remaining values averaged.

3.1.15 Slurry Densities

Use a density metering device, calibrated according to the manufacturer's specifications, to record the slurry density of each dragarm to the nearest 0.001 g/cc with an accuracy of plus or minus 0.01 g/cc. If the manufacturer does not specify a frequency of recalibration, conduct calibration prior to commencement of work.

3.1.16 Slurry Velocities

Use a flow metering device, calibrated according to the manufacturer's specifications, to record the slurry velocity of each dragarm to the nearest 0.01 fps with an accuracy of plus or minus 0.5 fps. If the manufacturer does not specify a frequency of recalibration, conduct calibration prior to commencement of work. Measure the slurry velocity in the same pipeline inside diameter as that used for the slurry density measurement.

3.1.17 Pump RPM

Measure the RPM of any pump being used to move material with the highest level of accuracy that is standard on the vessel operational displays, either at the bridge, at the drag tender's controls, or in the engine room. Dredges with multiple pumps per side must report RPM for the pump that best describes the dredging process (typically the outboard pump).

3.1.18 Sea Suction Valve for Dragarm

If sea suction can be taken to bypass suction through the draghead, the sea suction location and valve status will be reported. The status of the valve will change from "closed" to "open" when the valve starts to open and will register "closed" when the valve is fully closed. When applicable, the state of the latch will be reported as "true" or "false." Report the sea suction location in a standard non-changing name string of no more than 20 characters. These field values will always occur in the XML string as a set. The DQM system can accommodate only up to four unique sea suction locations. Suggested options for the naming convention can be found in the example dataset in paragraph DATA FORMAT.

3.1.19 Pumpout

When the hopper dredge is being pumped out, report a "true" value; when it is not, report a "false" value. The only permissible values are "true" and "false."

3.2 NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM SYSTEM REQUIREMENTS

The Contractor's DQM system must be capable of collecting, displaying, and transmitting information to the DQM database. Record the applicable parameters from paragraph REQUIREMENTS FOR REPORTED DATA as events locally and continually transmitted to the DQM database anytime an Internet connection is available. Equip the Dredge with a DQM computer system, consisting of a computer, monitor, keyboard, mouse, data modem, UPS, and

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network hub. Provide a standalone computer system, exclusive to the DQM monitoring system, and with USACE DQM software installed on it. If a hardware problem occurs, or if a part of the system is physically damaged, the Contractor is responsible for repairing it within 48 hours of determination of the condition.

3.2.1 Computer Requirements

Provide a dedicated onboard computer for use by the DQM system. This computer must run USACE software and receive data from the Contractor's data-reporting interface. This computer must meet or exceed the following performance specifications:

CPU:	Intel or AMD processor with a (non-overclocked) clock speed of at least 1.6 gigahertz (GHz)
Hard drive:	250 gigabytes (GB); internal
RAM:	4 gigabytes (GB)
Ethernet adapter:	Internal network card with an RJ-45 connector
Ports:	1 free serial port with standard 9-pin connectors; 1 free USB port
Other hardware:	Keyboard, mouse, monitor

Install a fully licensed copy of Windows 10 Professional Operating System or higher on the computer specified above. Also install any necessary manufacturer-provided drivers for the installed hardware. Locate and orient the computer to allow data entry and data viewing, as well as to provide access to data ports for the connection of external hardware.

3.2.2 Software

The DQM computer's primary function is to transmit data to the DQM shoreside database. Do not install software which conflicts with this function on this computer. The DQM computer must have the USACE-provided Dredging Quality Management Onboard Software (DQMOBS) installed on it by DQM personnel along with USACE-selected software for remote support and management.

3.2.3 UPS

Supply an Uninterruptible Power Supply (UPS) for the computer and networking equipment. The UPS must provide backup power at 1 kVA for a minimum of ten minutes. The UPS must interface with the DQM computer to communicate UPS status. Ensure that sufficient power outlets are available to run all specified equipment.

3.2.4 Internet Access

Maintain an Internet connection capable of transmitting real-time data to the DQM server and supporting remote access, as well as enough additional bandwidth to clear historically queued data when a connection is re-obtained. If connectivity is lost, queue and transmit unsent data upon restoration of connectivity. Delays in pushing real-time data to the DQM database will not exceed four hours. Exceptions to these requirements may be granted by the DQM Support Center on a case-by-case basis with

consideration for contract-specific requirements, site-specific conditions, and extreme weather events.

Acquire and install all necessary hardware and software to make the Internet connection available for data transmission to the DQM web service. Configure the hardware and software to allow the DQM Support Center remote access to this computer. Coordination between the dredging company's IT and the DQM Support Center may be required in order to configure remote access through any security, firewall, router, and telemetry systems. Telemetry systems must be capable of meeting these minimum reporting requirements in all operating conditions.

3.2.5 Data Routing Requirements

Onboard sensors must continually monitor dredge conditions, operations and efficiency and route this information into the shipboard dredge-specific system (DSS) computer to assist in guiding dredge operations. Portions of this Contractor-collected information must be routed to the DQM computer on a real-time basis. Use an RS-232 9600- or 19200-baud serial interface to send standard sensor data to the DQM computer. Configure the serial interface as 8 bits, no parity, and no flow control.

3.2.6 Data Reporting Frequency

Log data as a series of events. Each event will consist of a dataset containing dredge information in accordance with paragraph REQUIREMENTS FOR REPORTED DATA. Each set of measurements (time, position, etc.) will be considered an event. Collect any required information in paragraph REQUIREMENTS FOR REPORTED DATA that is not an averaged variable (draft and ullage) within 1 second of the reported time. Send a data string for an event to the DQM computer every 6 to 12 seconds. This interval must remain constant throughout the contract; do not transmit data strings more than once per every 5 seconds. Collect and compute any averaged variable within this sampling interval.

3.2.7 Data Format

Report data as an extensible Markup Language (W3C standard XML 1.0) document as indicated below. Line breaks and spaces are added for readability, but the carriage return, line feed character combination is added only to delineate records (HOPPER_DREDGING_DATA tag) for actual data transmission.

```
{?xml version="1.0"?}
{HOPPER_DREDGING_DATA version = "2.0"}
  <PLANT_IDENTIFIER>integer string</PLANT_IDENTIFIER>
  {DREDGE_NAME} string32 {/DREDGE_NAME}
  {HOPPER_DATA_RECORD}
    {DATE_TIME} time date string {/DATE_TIME}
    {CONTRACT_NUMBER} string32 {/CONTRACT_NUMBER}
    {LOAD_NUMBER} integer string {/LOAD_NUMBER}
    {VESSEL_X coord_type="LL"} floating point string
  {/VESSEL_X}
    {VESSEL_Y coord_type="LL"} floating point string
  {/VESSEL_Y}
    {PORT_DRAG_X coord_type="LL"} floating point string
  {/PORT_DRAG_X}
    {PORT_DRAG_Y coord_type="LL"} floating point string
  {/PORT_DRAG_Y}
```

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```
        {STBD_DRAG_X coord_type="LL"} floating point string
{/STBD_DRAG_X}
        {STBD_DRAG_Y coord_type="LL"} floating point string
{/STBD_DRAG_Y}
        {HULL_STATUS} OPEN/CLOSED string {/HULL_STATUS}
        {VESSEL_COURSE} floating point string {VESSEL_COURSE}
        {VESSEL_SPEED} floating point string {/VESSEL_SPEED}
        {VESSEL_HEADING} floating point string {/VESSEL_HEADING}
        {TIDE} floating point string {/TIDE}
        {DRAFT_FORE} floating point string {/DRAFT_FORE}
        {DRAFT_AFT} floating point string {/DRAFT_AFT}
        {ULLAGE_FORE} floating point string {/ULLAGE_FORE}
        {ULLAGE_AFT} floating point string {/ULLAGE_AFT}
        {HOPPER_VOLUME} floating point string {/HOPPER_VOLUME}
        {DISPLACEMENT} floating point string {/DISPLACEMENT}
        {EMPTY_DISPLACEMENT} floating point string
{/EMPTY_DISPLACEMENT}
        {DRAGHEAD_DEPTH_PORT} floating point string
{/DRAGHEAD_DEPTH_PORT}
        {DRAGHEAD_DEPTH_STBD} floating point string
{/DRAGHEAD_DEPTH_STBD}
        {PORT_DENSITY} floating point string {/PORT_DENSITY}
        {STBD_DENSITY} floating point string {/STBD_DENSITY}
        {PORT_VELOCITY} floating point string {/PORT_VELOCITY}
        {STBD_VELOCITY} floating point string {/STBD_VELOCITY}
        {PUMP_RPM_PORT} floating point string {/PUMP_RPM_PORT}
        {PUMP_RPM_STBD} floating point string {/PUMP_RPM_STBD}
        {VALVE_1_LOCATION} string32 {VALVE_1_LOCATION}
        {VALVE_1_STATUS} open/closed {/VALVE_1_STATUS}
        {VALVE_1_LATCHED} true/false {/VALVE_1_LATCHED}
        {VALVE_2_LOCATION} string32 {/VALVE_2_LOCATION}
        {VALVE_2_STATUS} open/closed {/VALVE_2_STATUS}
        {VALVE_2_LATCHED} true/false {/VALVE_2_LATCHED}
        {VALVE_3_LOCATION} string32 {/VALVE_3_LOCATION}
        {VALVE_3_STATUS} open/closed {/VALVE_3_STATUS}
        {VALVE_3_LATCHED} true/false {/VALVE_3_LATCHED}
        {VALVE_4_LOCATION} string32 {/VALVE_4_LOCATION}
        {VALVE_4_STATUS} open/closed {/VALVE_4_STATUS}
        {VALVE_4_LATCHED} true/false {/VALVE_4_LATCHED}
        {PUMP_OUT_ON} true/false/unknown string {/PUMP_OUT_ON}
{/HOPPER_DATA_RECORD}

{/HOPPER_DREDGING_DATA}
Carriage Return - ASCII value 13
Line Feed - ASCII value 10
```

Example

```
{?xml version="1.0"?}
{HOPPER_DREDGING_DATA version = "2.0"}
  <PLANT_IDENTIFIER>9999</PLANT_IDENTIFIER>
  {DREDGE_NAME}Essayons{/DREDGE_NAME}
  {HOPPER_DATA_RECORD}
    {DATE_TIME}04/11/2002 13:12:05{/DATE_TIME}
    {CONTRACT_NUMBER}GDSNWP-11-G-0001{/CONTRACT_NUMBER}
    {LOAD_NUMBER}102{/LOAD_NUMBER}
    {VESSEL_X coord_type="LL"}-80.123333{/VESSEL_X}
    {VESSEL_Y coord_type="LL"}10.123345{/VESSEL_Y}
    {PORT_DRAG_X coord_type="LL"}-80.1233371{/PORT_DRAG_X}
```

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```
{PORT_DRAG_Y coord_type="LL"}10.12335{/PORT_DRAG_Y}
{STBD_DRAG_X coord_type="LL"}-80.123339{/STBD_DRAG_X}
{STBD_DRAG_Y coord_type="LL"}10.123347{/STBD_DRAG_Y}
{HULL_STATUS}CLOSED{/HULL_STATUS}
{VESSEL_COURSE}258{/VESSEL_COURSE}
{VESSEL_SPEED}3.4{/VESSEL_SPEED}
{VESSEL_HEADING}302{/VESSEL_HEADING}
{TIDE}-0.1{/TIDE}
{DRAFT_FORE}10.05{/DRAFT_FORE}
{DRAFT_AFT}15.13{/DRAFT_AFT}
{ULLAGE_FORE}10.11{/ULLAGE_FORE}
{ULLAGE_AFT}10.22{/ULLAGE_AFT}
{HOPPER_VOLUME}2555.2{/HOPPER_VOLUME}
{DISPLACEMENT}4444.1{/DISPLACEMENT}
{EMPTY_DISPLACEMENT}2345.0{/EMPTY_DISPLACEMENT}
{DRAGHEAD_DEPTH_PORT}55.10{/DRAGHEAD_DEPTH_PORT}
{DRAGHEAD_DEPTH_STBD}53.21{/DRAGHEAD_DEPTH_STBD}
{PORT_DENSITY}1.02{/PORT_DENSITY}
{STBD_DENSITY}1.03{/STBD_DENSITY}
{PORT_VELOCITY}22.1{/PORT_VELOCITY}
{STBD_VELOCITY}23.3{/STBD_VELOCITY}
{PUMP_RPM_PORT}55{/PUMP_RPM_PORT}
{PUMP_RPM_STBD}54{/PUMP_RPM_STBD}
{VALVE_1_LOCATION}Starboard Dragarm{VALVE_1_LOCATION}
{VALVE_1_STATUS}open{/VALVE_1_STATUS}
{VALVE_1_LATCHED>true{/VALVE_1_LATCHED}
{VALVE_2_LOCATION}Port Dragarm{/VALVE_2_LOCATION}
{VALVE_2_STATUS}closed{/VALVE_2_STATUS}
{VALVE_2_LATCHED>false{/VALVE_2_LATCHED}
{VALVE_3_LOCATION}Port Sea Chest{/VALVE_3_LOCATION}
{VALVE_3_STATUS}closed{/VALVE_3_STATUS}
{VALVE_3_LATCHED>false{/VALVE_3_LATCHED}
{VALVE_4_LOCATION}Starboard Sea Chest{/VALVE_4_LOCATION}
{VALVE_4_STATUS}open{/VALVE_4_STATUS}
{VALVE_4_LATCHED>false{/VALVE_4_LATCHED}
{PUMP_OUT_ON>false{/PUMP_OUT_ON}
{/HOPPER_DATA_RECORD}
{/HOPPER_DREDGING_DATA}
{cr}
{lf}
<?xml version="1.0"?>
  <HOPPER_DREDGING_DATA version = "2.0">
    <PLANT_IDENTIFIER>9999</PLANT_IDENTIFIER>
    {DREDGE_NAME}Essayons{/DREDGE_NAME}
    {HOPPER_DATA_RECORD}
      {DATE_TIME}04/11/2002 13:12:10{/DATE_TIME}
      {CONTRACT_NUMBER}GDSNWP-11-G-0001{/CONTRACT_NUMBER}
      {LOAD_NUMBER}102{/LOAD_NUMBER}
      {VESSEL_X coord_type="LL"}-80.123334{/VESSEL_X}
      {VESSEL_Y coord_type="LL"}10.123346{/VESSEL_Y}
      {PORT_DRAG_X coord_type="LL"}-80.1233372{/PORT_DRAG_X}
      {PORT_DRAG_Y coord_type="LL"}10.12336{/PORT_DRAG_Y}
      {STBD_DRAG_X coord_type="LL"}-80.123340{/STBD_DRAG_X}
      {STBD_DRAG_Y coord_type="LL"}10.123348{/STBD_DRAG_Y}
      {HULL_STATUS}CLOSED{/HULL_STATUS}
      {VESSEL_COURSE}259{/VESSEL_COURSE}
      {VESSEL_SPEED}3.5{/VESSEL_SPEED}
      {VESSEL_HEADING}300{/VESSEL_HEADING}
      {TIDE}-0.1{/TIDE}
```

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```
{DRAFT_FORE}10.00{/DRAFT_FORE}  
{DRAFT_AFT}15.15{/DRAFT_AFT}  
{ULLAGE_FORE}10.15{/ULLAGE_FORE}  
{ULLAGE_AFT}10.20{/ULLAGE_AFT}  
{HOPPER_VOLUME}2555.5{/HOPPER_VOLUME}  
{DISPLACEMENT}4444.0{/DISPLACEMENT}  
{EMPTY_DISPLACEMENT}2345.0{/EMPTY_DISPLACEMENT}  
{DRAGHEAD_DEPTH_PORT}55.15{/DRAGHEAD_DEPTH_PORT}  
{DRAGHEAD_DEPTH_STBD}53.19{/DRAGHEAD_DEPTH_STBD}  
{PORT_DENSITY}1.00{/PORT_DENSITY}  
{STBD_DENSITY}1.01{/STBD_DENSITY}  
{PORT_VELOCITY}22.5{/PORT_VELOCITY}  
{STBD_VELOCITY}23.3{/STBD_VELOCITY}  
{PUMP_RPM_PORT}55{/PUMP_RPM_PORT}  
{PUMP_RPM_STBD}54{/PUMP_RPM_STBD}  
{VALVE_1_LOCATION}Starboard Dragarm{VALVE_1_LOCATION}  
{VALVE_1_STATUS}open{/VALVE_1_STATUS}  
{VALVE_1_LATCHED}true{/VALVE_1_LATCHED}  
{VALVE_2_LOCATION}Port Dragarm{VALVE_2_LOCATION}  
{VALVE_2_STATUS}closed{/VALVE_2_STATUS}  
{VALVE_2_LATCHED>false{/VALVE_2_LATCHED}  
{VALVE_3_LOCATION}Port Sea Chest{/VALVE_3_LOCATION}  
{VALVE_3_STATUS}closed{/VALVE_3_STATUS}  
{VALVE_3_LATCHED>false{/VALVE_3_LATCHED}  
{VALVE_4_LOCATION}Starboard Sea Chest{/VALVE_4_LOCATION}  
{VALVE_4_STATUS}open{/VALVE_4_STATUS}  
{VALVE_4_LATCHED} false{/VALVE_4_LATCHED}  
{PUMP_OUT_ON}false{/PUMP_OUT_ON}  
{/HOPPER_DATA_RECORD}  
{/HOPPER_DREDGING_DATA}  
{cr}  
{lf}
```

3.2.8 Data Reporting

The system must transmit correctly formatted event data XML strings to the DQM database continuously from mobilization until the last USACE post-dredging survey has been accepted. If the Internet connection (paragraph INTERNET ACCESS) is non-operable, perform manual backups from the dredge computer of the XML data string which would have been transmitted to the DQM computer over the serial connection for each day the device is inoperable and submit to the DQM Support Center within 48 hours. This submission does not replace the requirement of correcting the issue affecting the automatic transmission of data. In the event of data transfer, transmission, or hardware failure, maintain a manually recorded disposal log consisting of a series of events. These events are start of dredging, end of dredging, pre-disposal, and post-disposal. Include time stamp (GMT), position (Latitude and Longitude WGS84), draft, ullage, volume, and displacement for each event. Submit disposal logs on a daily basis to the Contracting Officer during the time when the system is not operational.

3.2.9 Contractor Data Backup

Maintain an archive of all data sent to the DQM computer during the dredging contract. The Contracting Officer may require, at no increase in the contract price, that the Contractor provide a copy of these data covering specified time periods. Provide the data, transmitted to the DQM computer, in the XML format with no line breaks between the parameters and

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each record string on a separate line. The naming convention for the files is {dredgename}_{StartYYYYMMddhhmmss}_{EndYYYYMMddhhmmss}.txt. Submit data via storage medium acceptable to the Contracting Officer.

At the end of the dredging contract, contact the DQM Support Center prior to discarding the data. The DQM Support Center will verify that all data has been received and appropriately archived before giving the Contractor discard permission. Record the following information in a separate section at the end of the dredge's onboard copy of the DPIP:

Person who made the call
Date of the call
DQM representative who gave permission to discard

3.3 PERFORMANCE REQUIREMENTS

The Contractor's DQM system must be fully operational at the start of dredging operations and fully certified prior to moving dredge material on the contract (see paragraph NATIONAL DREDGING QUALITY MANAGEMENT PROGRAM CERTIFICATION). To meet contract requirements for operability, in addition to certification, the Contractor's system must provide a data string with all values for all parameters while operating, as described in the specifications. Additionally, all hardware must be compliant with hardware requirements (paragraph COMPUTER REQUIREMENTS). Quality data strings are considered to be those providing values for all parameters reported when operating according to the specification. Make repairs necessary to restore data return compliance within 48 hours. Failure by the Contractor to report the required data within the specified time window for dredge measurements (see paragraph DATA REPORTING FREQUENCY and paragraph DATA REPORTING) will result in withholding of up to 10 percent of the contract progress payment per FAR 52.232-5 Payments under Fixed-Price Construction Contracts.

3.4 LIST OF ITEMS TO BE PROVIDED BY THE CONTRACTOR

DPIP

<https://dqm.usace.army.mil/Certifications/Index.aspx>

DQM System

Sensor instrumentation: paragraph REQUIREMENTS FOR REPORTED DATA
DQM computer: paragraph NATIONAL DREDGING QUALITY MANAGEMENT
PROGRAM SYSTEM REQUIREMENTS

Dredge Data

Event documentation: paragraph DATA REPORTING
Dredge data backups: paragraph CONTRACTOR DATA BACKUP

QA Equipment on the Dredge

Dragarm depth chain
Ullage tape
Refractometer
Water sampling device

-- End of Section --